

NUMBER	TITLE	EXPLANATION
DIVISION 01 – GENERAL REQUIREMENTS		
01 00 00	General Requirements	May be used as Division level Section title.
01 10 00	Summary	
01 11 00	Summary of Work	Includes: description of construction delivered under a single contract. Description of provisions for future construction. May Include: list of products ordered in advance, owner-furnished products, salvaged material and equipment, products delivered to owner, and owner-installed products. Alternate Terms/Abbreviations: purchase contract: procurement contract: purchase order. See: 00 24 00 for procurement scopes. 00 54 13 for agreement form supplementary scope statement. 01 31 00 for coordination of work under a single contract. 01 64 00 for owner-supplied products.
01 11 13	Work Covered by Contract Documents	
01 11 16	Work by Owner	
01 11 19	Purchase Contracts	
01 12 00	Multiple Contract Summary	Includes: description of construction delivered under more than one construction contract; for example, multiple prime, construction management, and design-build. Summary of contracts, related prior construction, related concurrent construction, contract interface, phasing of construction, and temporary facilities and controls responsibility. May Include: list of products ordered in advance, owner-furnished products, salvaged material and equipment, products delivered to owner, and owner-installed products, future construction, and provisions for future construction associated with each contract. See: 00 24 00 for procurement scopes. 00 54 13 for agreement form supplementary scope statement. 01 31 16 for multiple contract coordination.

DIVISION **01**

NUMBER	TITLE	EXPLANATION
01 12 13	Summary of Contracts	
01 12 16	Work Sequence	
01 12 19	Contract Interface	

01 14 00 Work Restrictions

Includes: restrictions that affect construction operations.
Location of construction, occupancy requirements, and use of premises and site.

See: 01 53 00 for temporary construction related to access and use of site.

01 55 00 for temporary vehicular access and parking.

01 56 00 for temporary barriers and enclosures.

01 14 13	Access to Site
01 14 16	Coordination with Occupants
01 14 19	Use of Site
01 14 33	Work in Rights-of-Way

01 18 00 Project Utility Sources

Includes: information concerning utility companies providing permanent services to the project.

See: 01 51 00 for temporary utilities.

01 18 13	Utility Service Connections
	Notes: this is the location to specify information necessary for connecting utilities and where connection is made by utility.

01 20 00 Price and Payment Procedures

01 21 00 Allowances

Includes: administrative and procedural requirements for cash and quantity allowances for products, installation, testing, and contingencies. Time allowances. In a project manual application, other titles affected by allowances should reference this title.

May Include: allowance amounts and descriptions.

See: 00 70 00 for conditions of the contract for allowance requirements.

See Also: 00 24 00 for procurement scopes for allowance amounts and items.

00 40 00 for allowances included in procurement forms.

00 43 21 for allowances procurement form supplement.

00 54 21 for allowances schedule agreement form supplement.

NUMBER	TITLE	EXPLANATION
01 21 13	Cash Allowances	
01 21 16	Contingency Allowances	
01 21 19	Testing and Inspecting Allowances	
01 21 23	Installation Allowances	
01 21 26	Product Allowances	
01 21 29	Quantity Allowances	
01 21 43	Time Allowances	

01 22 00 Unit Prices

Includes: administrative and procedural requirements for unit prices and measurement and payment.

May Include: list and description of unit price items.

See Also: 00 24 00 for procurement scopes for unit price descriptions.

00 40 00 for unit prices included in procurement forms.

00 43 22 for unit price procurement form supplement.

00 54 22 for unit prices schedule agreement form supplement.

01 22 13	Unit Price Measurement
01 22 16	Unit Price Payment

01 23 00 Alternates

Includes: administrative and procedural requirements for alternate bids or proposals. Includes submission and acceptance procedures.

May Include: list and description of each alternate.

Alternate Terms/Abbreviations: alternates: alternatives: bid items.

See Also: 00 24 00 for procurement scopes for alternate descriptions.

00 40 00 for alternates included in procurement forms.

00 43 23 for alternates procurement form supplement.

01 24 00 Value Analysis

Includes: administrative and procedural requirements for value analysis, application for consideration, and consideration of proposals.

May Include: appropriate forms.

Alternate Terms/Abbreviations: value analysis: value management: value improvement.

See: 00 63 00 for modification forms.

01 26 00 for contract modification procedures.

NUMBER	TITLE	EXPLANATION
01 24 13	Value Engineering	
01 25 00	Substitution Procedures	Includes: administrative and procedural requirements for consideration of requests for substitution during the construction stage. May Include: substitution procedures applicable during both procurement and construction stages. Substitution request form (during construction). Alternate Terms/Abbreviations: accepted substitute: approved equal. See: 00 43 25 for substitution request form (during procurement). 00 70 00 for conditions of the contract for substitution requirements. 01 24 00 for value analysis procedures. 01 62 00 for contractor's options in selecting products. See Also: 00 26 00 for procurement substitution procedures. 00 63 25 for substitution request form (during construction).
01 25 13	Product Substitution Procedures	
01 25 16	Execution Substitution Procedures	
01 26 00	Contract Modification Procedures	Includes: administrative and procedural requirements for making clarifications and proposals for change, and modifications to the contract documents. May Include: request for interpretation form, request for information form. Clarification form. Supplemental instruction form. Field order form. Written amendment form. Construction change directive form. Work change directive form. Request for proposal form. Proposal worksheet summary form. Proposal worksheet detail form. Change order request form. Change order form. Alternate Terms/Abbreviations: RFI: request for interpretation: request for information. See: 00 70 00 for conditions of the contract for contract modification requirements. 00 93 00 for record clarifications and proposals. 00 94 00 for record modifications. See Also: 00 63 00 for modification forms.

NUMBER	TITLE	EXPLANATION
01 26 13	Requests for Information	Alternate Terms/Abbreviations: RFI: request for interpretation: request for information.
01 26 19	Clarification Notices	
01 26 33	Minor Changes in the Work	
01 26 36	Supplemental Instructions	
01 26 39	Field Orders	
01 26 43	Amendments	
01 26 46	Construction Change Directives	
01 26 49	Work Change Directives	
01 26 53	Proposal Requests	
01 26 54	Proposal Worksheet Summaries	
01 26 57	Change Order Requests	
01 26 63	Change Orders	
01 29 00	Payment Procedures	Includes: administrative and procedural requirements for processing of schedules of values, progress payments, and final payment. May Include: schedule of values form. Application for payment form. Sales tax form. Consent of surety to reduction of retainage form. Stored material form. See: 00 70 00 for conditions of the contract for payment requirements. See Also: 00 62 00 for payment procedure forms.
01 29 73	Schedule of Values	
01 29 76	Progress Payment Procedures	
01 29 83	Payment Procedures for Testing Laboratory Services	
01 30 00	Administrative Requirements	
01 31 00	Project Management and Coordination	Includes: administrative and procedural requirements for the management and coordination of subcontractors and coordination with other contractors and owners. Describes coordination of various parts of work under a single contract and under multiple contracts. Alternate Terms/Abbreviations: meetings: conferences. See: 01 11 00 for summary of work under a single contract. 01 12 00 for multiple contract summary. 01 71 13 for mobilization requirements. 01 73 00 for execution requirements.

DIVISION 01

NUMBER	TITLE	EXPLANATION
01 31 13	Project Coordination	
01 31 14	Facility Services Coordination	
01 31 16	Multiple Contract Coordination	
01 31 19	Project Meetings	
01 31 19.13	Preconstruction Meetings	
01 31 19.16	Site Mobilization Meetings	
01 31 19.23	Progress Meetings	
01 31 19.33	Preinstallation Meetings	
01 31 23	Project Web Site	
01 31 26	Electronic Communication Protocols	

01 32 00 Construction Progress Documentation

Includes: administrative and procedural requirements for scheduling, recording, and reporting progress.

May Include: construction schedule form. Work plan schedule form. Construction equipment form. Other appropriate forms.

Alternate Terms/Abbreviations: CPM: critical path method.

See: 00 31 21 for survey information provided to bidders or proposers.

00 43 83 for proposed construction schedule form.

00 43 86 for proposed work plan schedule form.

01 71 23 for field engineering during construction.

01 78 29 for final site survey.

See Also: 00 62 83 for construction schedule form.

00 62 86 for work plan schedule form.

01 32 13	Scheduling of Work	
01 32 16	Construction Progress Schedule	
01 32 16.13	Network Analysis Schedules	
01 32 19	Submittals Schedule	
01 32 23	Survey and Layout Data	
01 32 26	Construction Progress Reporting	
01 32 29	Periodic Work Observation	
01 32 33	Photographic Documentation	
01 32 36	Video Monitoring and Documentation	
	Includes: web cameras.	
01 32 43	Purchase Order Tracking	

NUMBER	TITLE	EXPLANATION
01 33 00	Submittal Procedures	Includes: administrative and procedural requirements for processing of submittals during the construction stage. May Include: submittal transmittal form. Construction waste diversion form. Recycled content of materials form. Other appropriate forms. See: 01 43 00 for quality assurance procedures. 01 78 00 for closeout submittals. See Also: 00 62 11 for submittal transmittal form. 00 62 23 for construction waste diversion form. 00 62 34 for recycled content of materials form. 01 45 13 for source quality control procedures. 01 45 16 for field quality control procedures. 01 78 53 for sustainable design closeout documentation.
01 33 13	Certificates	
01 33 16	Design Data	
01 33 19	Field Test Reporting	
01 33 23	Shop Drawings, Product Data, and Samples	
01 33 26	Source Quality Control Reporting	
01 33 29	Sustainable Design Reporting	Notes: Level 4 numbers below to be used in conjunction with 01 33 29 Sustainable Design Reporting.
01 33 29.01	Material Cost Summary Form	
01 33 29.02	Wood-Containing Product List	
01 33 29.03	Metal-Containing Product List	
01 33 29.04	Material Content Form	
01 33 29.05	New Product Source Form	
01 33 29.06	Reused Product Form	
01 33 29.07	Prohibited Content Installer Certification	

NUMBER	TITLE	EXPLANATION
01 35 00	Special Procedures	Includes: administrative and procedural requirements for special project situations; for example, relocations, restoration, historic restoration, alteration, renovation, preservation, and hazardous material abatement. Administrative and procedural requirements for specific facility types; for example, airports, detention facilities, ground transportation facilities, hospitals and medical facilities, industrial and process facilities, nuclear facilities, and shopping centers. May Include: infection control form. Other appropriate forms. Alternate Terms/Abbreviations: alteration procedures: rehabilitation procedures: remodeling procedures: renovation procedures: repair procedures. See: 00 31 26 for existing hazardous material information. 00 73 19 for health and safety requirements. 00 73 63 for security requirements. 02 26 00 for hazardous material assessment. 02 50 00 for site remediation. 02 60 00 for contaminated site material removal. 02 80 00 for facility remediation. XX 01 00 at the beginning of divisions for operation and maintenance titles. XX 01 00 at the beginning of divisions for specific preservation and restoration requirements. See Also: 00 62 19 for infection control form.
01 35 03	Conservation Treatment Procedures	
01 35 13	Special Project Procedures	
01 35 13.13	Special Project Procedures for Airport Facilities	
01 35 13.16	Special Project Procedures for Detention Facilities	
01 35 13.19	Special Project Procedures for Healthcare Facilities	
01 35 13.26	Special Project Procedures for Clean Rooms	
01 35 13.43	Special Project Procedures for Contaminated Sites	
01 35 16	Alteration Project Procedures	
01 35 23	Owner Safety Requirements	
01 35 26	Governmental Safety Requirements	
01 35 29	Health, Safety, and Emergency Response Procedures	
01 35 29.13	Health, Safety, and Emergency Response Procedures for Contaminated Sites	
01 35 33	Infection Control Procedures	
01 35 43	Environmental Procedures	
01 35 43.13	Environmental Procedures for Hazardous Materials	
01 35 43.16	Environmental Procedures for Toxic Materials	
01 35 46	Indoor Air Quality Procedures	
01 35 53	Security Procedures	
01 35 63	Sustainability Certification Project Requirements	
01 35 66	Sustainability Certification Project Procedures	
01 35 73	Delegated Design Procedures	

NUMBER	TITLE	EXPLANATION
01 35 91	Period Treatment Procedures	
01 40 00	Quality Requirements	
01 41 00	Regulatory Requirements	Includes: information required for conformance to regulatory requirements, such as building codes, mechanical codes, electrical codes, zoning, other regulations, and fees applicable to the project as established by federal, state, and local agencies. See Also: 00 31 43 for permit application. 00 70 00 for conditions of the contract regulatory requirements. 00 73 73 for statutory requirements.
01 41 13	Codes	
01 41 16	Laws	
01 41 19	Rules	
01 41 23	Fees	
01 41 26	Permit Requirements	
01 42 00	References	Includes: lists of reference standards applicable to the project. Abbreviations, acronyms, and definitions used in the construction documents. See: 00 23 00 for procurement definitions. 00 71 00 for contracting documents.
01 42 13	Abbreviations and Acronyms	
01 42 16	Definitions	
01 42 19	Reference Standards	

NUMBER	TITLE	EXPLANATION
01 43 00	Quality Assurance	Includes: administrative and procedural requirements for proactive activities to assure the quality of construction before and during the execution of the work. General qualification requirements for participants. Administrative and procedural requirements for field services provided by manufacturer's representatives. General requirements for mockups and field samples constructed, applied, or assembled at the site for review and use as a quality standard. May Include: field services provided by manufacturer's representatives for startup and testing. See: 00 45 13 for bidder's qualifications. 00 45 16 for proposer's qualifications. 01 33 00 for submittal procedures. 01 45 00 for quality control. 01 45 33 for code-required special inspections and procedures. See Also: 01 75 00 for manufacturer's field services during startup and testing.
01 43 13	Manufacturer Qualifications	
01 43 16	Supplier Qualifications	
01 43 19	Fabricator Qualifications	
01 43 23	Installer Qualifications	
01 43 26	Testing and Inspecting Agency Qualifications	
01 43 29	Code-Required Special Inspector Qualifications	
01 43 33	Manufacturer's Field Services	
01 43 36	Field Samples	
01 43 39	Mockups	
01 45 00	Quality Control	Includes: administrative and procedural requirements for reactive activities to evaluate completed activities and elements for conformance with the requirements. Correction of defective construction. See: 01 43 00 for quality assurance. 01 43 26 for testing and inspecting agency qualifications. 01 43 29 for code-required special inspector qualifications.
01 45 13	Source Quality Control Procedures	
01 45 16	Field Quality Control Procedures	
01 45 16.13	Contractor Quality Control	
01 45 23	Testing and Inspecting Services	
01 45 26	Plant Inspection Procedures	
01 45 29	Testing Laboratory Services	
01 45 31	Climate Resiliency Quality Control Procedures	
01 45 33	Code-Required Special Inspections and Procedures	

NUMBER	TITLE	EXPLANATION
01 50 00	Temporary Facilities and Controls	Includes: requirements for installation, maintenance, and removal of temporary utilities, controls, facilities, and construction aids during construction. See: 01 12 00 for temporary facility and control responsibility under multiple contracts. 01 74 00 for cleaning and waste management.
01 51 00	Temporary Utilities	Includes: procedures and requirements for utilities used on site during construction. See: 01 18 00 for project utility sources. 33 00 00 for permanent utilities.
01 51 13	Temporary Electricity	
01 51 16	Temporary Fire Protection	
01 51 19	Temporary Fuel-Oil	
01 51 23	Temporary Heating, Cooling, and Ventilating	
01 51 26	Temporary Lighting	
01 51 29	Temporary Natural Gas	
01 51 33	Temporary Telecommunications	
01 51 36	Temporary Water	
01 52 00	Construction Facilities	Includes: procedures and requirements for facilities used on site during construction. See: 00 70 00 for conditions of the contract safety and sanitation requirements. 10 43 16 for permanent first aid cabinets.
01 52 13	Field Offices and Sheds	
01 52 16	First Aid Facilities	
01 52 19	Sanitary Facilities	
01 53 00	Temporary Construction	Includes: procedures and requirements for temporary access facilities required to accommodate construction or the owner's operations that are removed, reconstructed, or decommissioned upon completion of the construction.
01 53 13	Temporary Bridges	
01 53 16	Temporary Decking	
01 53 19	Temporary Overpasses	
01 53 23	Temporary Ramps	
01 53 26	Temporary Runarounds	

NUMBER	TITLE	EXPLANATION
01 54 00	Construction Aids	Includes: procedures and requirements related to tools and equipment used during construction. May Include: requirements for sharing construction aids for construction delivered under multiple contracts. Does Not Include: safety related procedures or contractor's means and methods. See Also: 01 12 00 for construction aid responsibility under multiple contracts.
01 54 13	Temporary Elevators	
01 54 16	Temporary Hoists	
01 54 19	Temporary Cranes	
01 54 23	Temporary Scaffolding and Platforms	
01 54 26	Temporary Swing Staging	
01 55 00	Vehicular Access and Parking	Includes: procedures and requirements for access and parking facilities required to accommodate construction or the owner's operations that are removed, reconstructed, or decommissioned upon completion of the construction. See: 01 14 13 for access to site restrictions.
01 55 13	Temporary Access Roads	
01 55 16	Haul Routes	
01 55 19	Temporary Parking Areas	
01 55 23	Temporary Roads	
01 55 26	Traffic Control	
01 55 29	Staging Areas	
01 56 00	Temporary Barriers and Enclosures	Includes: procedures and requirements for facilities for protection of public, occupants, and spaces during construction. Does Not Include: safety and health-related procedures for contractor employees and subcontractors. Alternate Terms/Abbreviations: temporary fencing; hoarding.
01 56 13	Temporary Air Barriers	
01 56 16	Temporary Dust Barriers	
01 56 19	Temporary Noise Barriers	
01 56 23	Temporary Barricades	
01 56 26	Temporary Fencing	

NUMBER	TITLE	EXPLANATION
01 56 29	Temporary Protective Walkways	
01 56 33	Temporary Security Barriers	
01 56 36	Temporary Security Enclosures	
01 56 39	Temporary Tree and Plant Protection	
01 57 00	Temporary Controls	Includes: procedures and requirements for environmental controls during construction. See: 31 10 00 for permanent site clearing. 31 20 00 for permanent earth moving. 31 25 00 for permanent erosion and sedimentation controls.
01 57 13	Temporary Erosion and Sediment Control	
01 57 16	Temporary Pest Control	
01 57 19	Temporary Environmental Controls	
01 57 23	Temporary Storm Water Pollution Control	
01 57 26	Site Watering for Dust Control	
01 57 33	Temporary Security	
01 58 00	Project Identification	Includes: signage used during construction to identify the project and the project team. Does Not Include: permanent facility signage.
01 58 13	Temporary Project Signage	
01 58 16	Temporary Interior Signage	
01 60 00	Product Requirements	
01 61 00	Common Product Requirements	Includes: general requirements for products used in construction. New, salvaged, and reused products.
01 61 13	Software Licensing Requirements	
01 62 00	Product Options	Includes: procedures and requirements for contractor's options in selecting products, in general. General rules and procedures for determination of acceptable products. May Include: products form. See: 00 26 00 for procurement substitution procedures. 01 25 00 for substitution procedures. See Also: 00 62 33 for products form.

NUMBER	TITLE	EXPLANATION
01 64 00	Owner-Furnished Products	Includes: requirements for products that are furnished by the owner. Procedures and requirements for coordinating, scheduling, handling, and storing products that are furnished by the owner. May Include: list of products ordered by owner and owner-furnished products. Alternate Terms/Abbreviations: owner-furnished products: owner-supplied products. See Also: 01 11 00 for list of products furnished by owner under single contract. 01 12 00 for list of products furnished by owner under multiple contracts.
01 65 00	Product Delivery Requirements	Includes: requirements for packing, shipping, delivery, and acceptance of products, in general. See: 01 64 00 for acceptance of owner-supplied products.
01 66 00	Product Storage and Handling Requirements	Includes: requirements for storage and handling of products, in general. See: 00 62 79 for stored material form.
01 66 13	Product Storage and Handling Requirements for Hazardous Materials	
01 66 16	Product Storage and Handling Requirements for Toxic Materials	
01 70 00	Execution and Closeout Requirements	
01 71 00	Examination and Preparation	Includes: procedures and general requirements for determining acceptable conditions for installation. Procedures and general requirements for preparing to install, erect, or apply products. See: 00 31 19 for existing condition information provided to bidders or proposers. 00 31 21 for survey information provided to bidders or proposers. 01 31 19 for site mobilization meetings. 01 76 00 for protecting installed construction. 01 78 29 for final site survey. 02 21 00 for surveys performed by the contractor before construction begins.

NUMBER	TITLE	EXPLANATION
01 71 13	Mobilization	
01 71 16	Acceptance of Conditions	
01 71 23	Field Engineering	
01 71 23.13	Construction Layout	
01 71 23.16	Construction Surveying	
01 71 33	Protection of Adjacent Construction	
01 71 36	Non-Destructive Concrete Examination	Includes: concrete x-ray, radar, and ultrasound examination.
01 73 00	Execution	Includes: requirements for installing, applying, or erecting products, in general. New, pre-purchased, salvaged for reuse, and owner-supplied products. Provisions for accommodating owner-installed products. See: 01 31 19 for preinstallation meetings. 02 41 19 for selective structure demolition. 02 42 00 for removal and diversion of construction materials.
01 73 13	Application	
01 73 16	Erection	
01 73 19	Installation	
01 73 23	Bracing and Anchoring	
01 73 26	Existing Products	
01 73 29	Cutting and Patching	
01 74 00	Cleaning and Waste Management	Includes: requirements for maintaining site in clean and neat condition during construction. Final cleaning prior to turning the project over to the owner. See: 00 45 23 for statement of disposal facility. 00 62 23 for construction waste diversion form.
01 74 13	Progress Cleaning	
01 74 16	Site Maintenance	
01 74 19	Construction Waste Management and Disposal	
01 74 23	Final Cleaning	
01 75 00	Starting and Adjusting	Includes: administrative and procedural requirements for the initial checkout, startup, and adjusting; to ensure safe operation during acceptance testing and commissioning. Acceptance testing. See: 01 79 00 for demonstration and training. 01 91 00 for system performance evaluation. 01 92 00 for facility operation. 01 93 00 for facility maintenance.

DIVISION **01**

NUMBER	TITLE	EXPLANATION
01 75 13	Checkout Procedures	
01 75 16	Startup Procedures	
01 76 00	Protecting Installed Construction	Includes: procedures and requirements for protecting installed construction, in general. See: 01 71 33 for protecting adjacent construction.
01 77 00	Closeout Procedures	Includes: administrative and procedural requirements for completion of the work. May Include: substantial and final completion steps. See: 00 65 00 for closeout forms. 01 50 00 for removal of temporary facilities and controls. 01 74 23 for final cleaning. 01 78 00 for closeout submittals. 01 78 29 for final site survey.
01 77 13	Preliminary Closeout Reviews	
01 77 16	Final Closeout Reviews	
01 77 19	Closeout Requirements	

NUMBER	TITLE	EXPLANATION
01 78 00	Closeout Submittals	Includes: administrative and procedural requirements for the processing of submittals at the completion of the project. May Include: construction waste diversion form. Recycled content of materials form. Closeout forms. Certificate of compliance form. Certificate of substantial completion form. Certificate of completion form. Affidavit of payment of debts and claims form. Affidavit of release of liens form. Consent of surety to final payment form. Acceptance certificate form. Final settlement certificate form. Warranty form. Statutory declaration form. Alternate Terms/Abbreviations: extra stock: attic stock. O and M: operation and maintenance. Project record documents: as-built drawings: as-built specifications. Warranties: guarantees. See: 00 31 21 for survey information provided to bidders or proposers. 00 93 00 for record clarifications and proposals. 00 94 00 for record modifications. 01 33 00 for submittals during construction. 01 33 29 for sustainable design reporting. 01 71 23 for field engineering during construction. 01 77 00 for closeout procedures. 01 92 00 for facility operation. 01 93 00 for facility maintenance. See Also: 00 62 23 for construction waste diversion form. 00 62 34 for recycled content of materials form. 00 65 00 for closeout forms.
01 78 13	Completion and Correction List	
01 78 19	Maintenance Contracts	
01 78 23	Operation and Maintenance Data	
01 78 23.13	Operation Data	
01 78 23.16	Maintenance Data	
01 78 23.19	Preventative Maintenance Instructions	
01 78 29	Final Site Survey	
01 78 33	Bonds	
01 78 36	Warranties	
01 78 39	Project Record Documents	
01 78 43	Spare Parts	
01 78 46	Extra Stock Materials	
01 78 53	Sustainable Design Closeout Documentation	

NUMBER	TITLE	EXPLANATION
01 79 00	Demonstration and Training	Includes: administrative and procedural requirements for demonstration of the products and systems of the facility, and training of the owner's operating and maintenance personnel. See: 01 78 23 for submittal of operation and maintenance data.
01 80 00	Performance Requirements	
01 81 00	Facility Performance Requirements	Includes: requirements for elements (systems and assemblies) that perform basic fundamental functions. Organized by functional elements in accordance with <i>UniFormat</i> . Performance or other type requirements for systems and assemblies that include elements within multiple titles, in multiple divisions, or both. Notes: performance and other type requirements that apply to the subject of a single title should be included within the appropriate specification Section.
01 81 13	Sustainable Design Requirements	Notes: to be used with design-build projects. See: 01 35 63 for sustainability certification project requirements. 01 35 66 for sustainability certification project procedures.
01 81 16	Facility Environmental Requirements	
01 81 19	Indoor Air Quality Requirements	
01 82 00	Facility Substructure Performance Requirements	
01 82 13	Foundation Performance Requirements	
01 82 16	Basement Construction Performance Requirements	
01 83 00	Facility Shell Performance Requirements	
01 83 13	Superstructure Performance Requirements	
01 83 16	Exterior Enclosure Performance Requirements	
01 83 19	Roofing Performance Requirements	
01 84 00	Interiors Performance Requirements	
01 84 13	Interior Construction Performance Requirements	
01 84 16	Stairways Performance Requirements	
01 84 19	Interior Finishes Performance Requirements	
01 85 00	Conveying Equipment Performance Requirements	

NUMBER	TITLE	EXPLANATION
01 86 00	Facility Services Performance Requirements	
01 86 13	Fire-Suppression Performance Requirements	
01 86 16	Plumbing Performance Requirements	
01 86 19	HVAC Performance Requirements	
01 86 23	Integrated Automation Performance Requirements	
01 86 26	Electrical Performance Requirements	
01 86 29	Communications Performance Requirements	
01 86 33	Electronic Safety and Security Performance Requirements	
01 87 00	Equipment and Furnishings Performance Requirements	
01 87 13	Equipment Performance Requirements	
01 87 16	Furnishings Performance Requirements	
01 88 00	Other Facility Construction Performance Requirements	
01 88 13	Special Construction Performance Requirements	
01 88 16	Selective Construction Performance Requirements	
01 89 00	Site Construction Performance Requirements	
01 89 13	Site Preparation Performance Requirements	
01 89 16	Site Improvements Performance Requirements	
01 89 19	Site Plumbing Utilities Performance Requirements	
01 89 23	Site HVAC Utilities Performance Requirements	
01 89 26	Site Electrical Utilities Performance Requirements	
01 89 29	Other Site Construction Performance Requirements	
01 90 00	Life Cycle Activities	
01 91 00	Commissioning	
		Includes: administrative and procedural requirements for commissioning facilities and facility systems.
		See: 00 70 00 for conditions of the contract for relationship of commissioning to participants and other construction activities.
		01 75 00 for starting and adjusting equipment prior to commissioning.
		01 79 00 for demonstration and training of the owner's personnel.
01 91 13	General Commissioning Requirements	
01 91 16	Facility Substructure Commissioning	
		Includes: commission of systems and assemblies which have elements in multiple divisions.
		See: XX 08 00 at the beginning of Divisions for the commissioning of systems and assemblies which are included within one Division.
01 91 16.13	Foundation Commissioning	

DIVISION **01**

NUMBER	TITLE	EXPLANATION
01 91 16.53	Basement Construction Commissioning	
01 91 19	Facility Shell Commissioning	
01 91 19.13	Superstructure Commissioning	
01 91 19.43	Exterior Enclosure Commissioning	
01 91 19.73	Roofing Commissioning	
01 91 23	Interiors Commissioning	
01 91 23.13	Interior Construction Commissioning	
01 91 23.43	Stairways Commissioning	
01 91 23.73	Interior Finishes Commissioning	
01 91 26	Integrated System Commissioning	

01 92 00 Facility Operation

Includes: administrative and procedural requirements for operating facility, after commissioning.

See: 01 78 23 for submittal of operation and maintenance data.
XX 01 00 at the beginning of Divisions for the operation of subjects which are included within one Division.

01 92 13 Facility Operation Procedures

01 93 00 Facility Maintenance

Includes: administrative and procedural requirements for maintaining facility, after commissioning.

See: 01 78 19 for submittal of maintenance contracts.
01 78 23 for submittal of operation and maintenance data.
XX 01 00 at the beginning of Divisions for the maintenance of subjects which are included within one Division.

01 93 13 Facility Maintenance Procedures

01 93 16 Recycling Programs

01 94 00 Facility Decommissioning

Includes: administrative and procedural requirements for deactivating a facility or portion of it from operation.

See: 02 40 00 for demolition and structure moving.
02 50 00 for site remediation.
02 60 00 for contaminated site material removal.
02 70 00 for water remediation.
02 80 00 for facility remediation.

01 94 13 Facility Decommissioning Procedures

SPECIFICATIONS GROUP

FACILITY CONSTRUCTION SUBGROUP

DIVISION 02 – EXISTING CONDITIONS

DIVISION 03 – CONCRETE

DIVISION 04 – MASONRY

DIVISION 05 – METALS

DIVISION 06 – WOOD, PLASTICS, AND COMPOSITES

DIVISION 07 – THERMAL AND MOISTURE PROTECTION

DIVISION 08 – OPENINGS

DIVISION 09 – FINISHES

DIVISION 10 – SPECIALTIES

DIVISION 11 – EQUIPMENT

DIVISION 12 – FURNISHINGS

DIVISION 13 – SPECIAL CONSTRUCTION

DIVISION 14 – CONVEYING EQUIPMENT

Division 15 – Reserved for Future Expansion

Division 16 – Reserved for Future Expansion

Division 17 – Reserved for Future Expansion

Division 18 – Reserved for Future Expansion

Division 19 – Reserved for Future Expansion

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NUMBER	TITLE	EXPLANATION
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DIVISION 02 – EXISTING CONDITIONS

02 00 00 Existing Conditions May be used as Division level Section title.

02 01 00 Maintenance of Existing Conditions

Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of existing conditions.

Level 4 Numbering Recommendation:

.51-.59 for maintenance.

.61-.69 for repair.

.71-.79 for rehabilitation.

.81-.89 for replacement.

.91-.99 for restoration.

02 01 50 Maintenance of Site Remediation
 02 01 65 Maintenance of Underground Storage Tank Removal
 02 01 80 Maintenance of Facility Remediation
 02 01 86 Maintenance of Hazardous Waste Drum Handling

02 03 00 Conservation Treatment for Existing Period Conditions

02 03 01 Maintenance of Existing Period Conditions
 02 03 01.19 Mothballing Period Structures

Includes: U.S. Department of the Interior, National Parks Service, Technical Preservation Services, Preservation Brief # 31, Mothballing Historic Buildings, Washington, D.C.: GPO, 1993, <http://www.nps.gov/tps/how-to-preserve/briefs/31-mothballing.htm> (accessed March 20, 2014). "The actual mothballing effort involves controlling the long-term deterioration of the building while it is unoccupied as well as finding methods to protect it from sudden loss by fire or vandalism. This requires securing the building from unwanted entry, providing adequate ventilation to the interior, and shutting down or modifying existing utilities. Once the building is de-activated or secured, the long-term success will depend on periodic maintenance and surveillance monitoring."

02 03 41 Selective Demolition for Period Structures
 02 03 42 Removal and Salvage of Period Construction Materials
 02 03 43 Period Structure Relocating
 02 03 44 Shoring and Support of Period Structures

NUMBER	TITLE	EXPLANATION
02 05 00	Common Work Results for Existing Conditions	Includes: subjects common to multiple titles in Division 02. See: 01 80 00 for performance requirements of subjects common to multiple titles. 31 05 19 for geosynthetics for earthwork. 32 05 19 for geosynthetics for exterior improvements.
02 05 19	Geosynthetics for Existing Conditions	
02 05 19.13	Geotextiles for Existing Conditions	
02 05 19.16	Geomembranes for Existing Conditions	
02 05 19.19	Geogrids for Existing Conditions	
02 06 00	Schedules for Existing Conditions	Includes: schedules of items common to multiple titles in Division 02. Notes: a schedule may be included on drawings, in the project manual, or a project book.
02 06 13	Geotechnical Baseline Report	
02 06 14	Geotechnical Data Report	
02 06 30	Schedules for Subsurface Investigations	
02 06 30.13	Boring or Test Pit Log Schedule	
02 06 50	Schedules for Site Remediation	
02 06 65	Schedules for Underground Storage Tank Removal	
02 06 80	Schedules for Facility Remediation	
02 06 86	Schedules for Hazardous Waste Drum Handling	
02 08 00	Commissioning of Existing Conditions	Includes: commissioning of items common to multiple titles in Division 02. See: 01 91 00 for commissioning of subjects common to multiple Divisions.
02 20 00	Assessment	Includes: assessments of existing conditions conducted by the contractor before construction begins as a project activity. See: 00 31 00 for available project information.

NUMBER	TITLE	EXPLANATION
02 21 00	Surveys	Includes: surveys conducted by the contractor before construction begins as a project activity. See: 00 31 21 for site survey information provided to the bidders, proposers, or contractors before construction begins. 00 31 21 for measured drawing information provided to the bidders or proposers before construction begins. 01 71 23 for field engineering during construction. 01 78 29 for final site survey.
02 21 13	Site Surveys	
02 21 13.13	Boundary and Survey Markers	
02 21 13.23	Archaeological and Historic Surveys	
02 21 16	Measured Drawings	
02 22 00	Existing Conditions Assessment	Includes: assessments of existing conditions conducted by the contractor before construction begins as a project activity. See: 00 31 19 for existing conditions information provided to the bidders, proposers, or contractors before construction begins. 01 71 16 for acceptance of conditions. 13 48 00 for sound, vibration, and seismic control. 44 20 00 for noise pollution control.
02 22 13	Movement and Vibration Assessment	
02 22 16	Acoustic Assessment	
02 22 19	Traffic Assessment	
02 22 23	Accessibility Assessment	
02 24 00	Environmental Assessment	Includes: environmental assessments conducted by the contractor before construction begins as a project activity. See: 00 31 24 for environmental assessment information provided to the bidders, proposers, or contractors before construction begins. 01 35 43 for environmental procedures. 44 10 00 for air pollution control. 44 40 00 for water pollution control equipment.
02 24 13	Natural Environment Assessment	
02 24 13.13	Air Assessment	
02 24 13.43	Water Assessment	
02 24 13.73	Land Assessment	
02 24 23	Chemical Sampling and Analysis of Soils	
02 24 43	Transboundary and Global Environmental Aspects Assessment	

NUMBER	TITLE	EXPLANATION
02 25 00	Existing Material Assessment	Includes: assessments of existing materials conducted by the contractor before construction begins as a project activity. See: 00 31 25 for existing material information provided to the bidders, proposers, or contractors before construction begins. 03 82 00 for concrete boring. 07 10 00 for dampproofing and waterproofing. 07 30 00 for steep slope roofing. 07 50 00 for membrane roofing.
02 25 16	Existing Concrete Assessment	
02 25 16.13	Concrete Assessment Drilling	
02 25 19	Existing Masonry Assessment	
02 25 19.13	Masonry Assessment Drilling	
02 25 23	Existing Metals Assessment	
02 25 23.13	Welding Investigations	
02 25 26	Existing Wood, Plastics, and Composites Assessment	
02 25 29	Existing Thermal and Moisture Protection Assessment	
02 25 29.13	Waterproofing Investigations	
02 25 29.23	Roofing Investigations	
02 26 00	Hazardous Material Assessment	Includes: assessments of existing hazardous materials conducted by the contractor before construction begins as a project activity. Alternate Terms/Abbreviations: PCB: polychlorinated biphenyl. See: 00 31 26 for existing hazardous materials information provided to the bidders, proposers, or contractors before construction begins. 01 35 43 for environmental procedures for hazardous materials. 02 80 00 for hazardous material remediation. 02 82 00 for asbestos remediation. 02 83 00 for lead remediation. 02 84 00 for PCB remediation. 02 86 00 for hazardous waste drum handling. 02 87 13 for mold remediation.
02 26 23	Asbestos Assessment	
02 26 26	Lead Assessment	
02 26 29	Polychlorinated Biphenyl Assessment	
02 26 33	Biological Assessment	
02 26 33.13	Mold Assessment	
02 26 36	Hazardous Waste Drum Assessment	

NUMBER	TITLE	EXPLANATION
02 30 00	Subsurface Investigation	Includes: subsurface investigations and monitoring conducted by the contractor before construction begins as a project activity. Notes: other building elements may be used in lightning protection systems in lieu of air terminals. See: 00 31 00 for available project information. See Also: 26 41 00 for facility lightning protection.
02 31 00	Geophysical Investigations	Includes: geophysical investigations and monitoring conducted by the contractor before construction begins as a project activity. See: 00 31 31 for geophysical data provided to the bidders, proposers, or contractors before construction begins. 13 48 00 for seismic control. 13 52 00 for seismic instrumentation. 26 35 23 for electromagnetic-interference filters. 40 91 16 for electromagnetic process measurement devices. 40 91 16 for magnetic field process measurement devices.
02 31 13	Seismic Investigations	
02 31 16	Gravity Investigations	
02 31 19	Magnetic Investigations	
02 31 23	Electromagnetic Investigations	
02 31 26	Electrical Resistivity Investigations	
02 31 29	Magnetotelluric Investigations	
02 32 00	Geotechnical Investigations	Includes: geotechnical investigations and monitoring conducted by the contractor before construction begins as a project activity. See: 00 31 32 for geotechnical data provided to the bidders, proposers, or contractors before construction begins. 02 71 00 for groundwater treatment. 31 09 13 for groundwater monitoring during construction. 33 11 53 for monitoring wells. 41 66 13 for mobile boring and drilling rigs.
02 32 13	Subsurface Drilling and Sampling	
02 32 16	Material Testing	
02 32 19	Exploratory Excavations	
02 32 23	Geotechnical Monitoring Before Construction	
02 32 23.13	Groundwater Monitoring Before Construction	

NUMBER	TITLE	EXPLANATION
02 40 00	Demolition and Structure Moving	Includes: removal and disposal of structures. Moving or raising structures. See Also: 33 05 61 for concrete manholes. 33 05 63 for concrete vaults and chambers. 33 05 66 for concrete distribution chambers. 33 05 69 for concrete trenchways. 33 05 71 for cleanouts. 33 05 73 for polyethylene manholes. 33 05 76 for fiberglass manholes.
02 41 00	Demolition	Includes: partial or complete removal and disposal of structures, at grade, above grade, and below grade. Hydrodemolition and implosion. May Include: capping and removal of existing utilities. See: 01 35 91 for historic treatment procedures. 01 73 29 for cutting and patching. 02 42 00 for removal and diversion of construction materials. 02 43 00 for structure moving. 02 60 00 for contaminated site material removal. 31 10 00 for site clearing. 31 23 16 for excavation.
02 41 13	Selective Site Demolition	
02 41 13.13	Paving Removal	
02 41 13.23	Utility Line Removal	
02 41 13.33	Railtrack Removal	
02 41 16	Structure Demolition	
02 41 16.13	Building Demolition	
02 41 16.23	Tower Demolition	
02 41 16.33	Bridge Demolition	
02 41 16.43	Dam Demolition	
02 41 19	Selective Demolition	
02 41 19.13	Selective Building Demolition	
02 41 19.16	Selective Interior Demolition	
02 41 19.19	Selective Facility Services Demolition	
02 41 19.33	Selective Bridge Demolition	

NUMBER	TITLE	EXPLANATION
02 42 00	Removal and Diversion of Construction Materials	Includes: salvage (removal, repair, storage) of existing materials and equipment, and its disposal by removal from site. Storage, reinstallation, or delivery to owner. Cleaning and other preparatory work on salvaged materials. See: 01 35 91 for historic treatment procedures. 01 73 26 for existing products. 02 41 00 for demolition and removal. 02 43 00 for structure moving.
02 42 13	Deconstruction of Structures	
02 42 13.13	Deconstruction of Buildings	
02 42 19	Recycling of Construction Materials	Includes: items taken from project location to a recycling facility.
02 42 19.13	Paper Recycling	Includes: corrugated cardboard recycling, aluminum and plastic beverage container recycling.
02 42 19.15	Concrete Recycling	
02 42 19.17	Masonry Recycling	Includes: brick, stone, pavers, clay materials.
02 42 19.19	Metal Recycling	Includes: (nonferrous metals): copper, aluminum, zinc, a/c radiators, electric motors, (ferrous metals): iron, steel, rail, OTM: other track material, rebar, chillers.
02 42 19.23	Wood Recycling	Includes: cabinetry, fixtures, doors, lumber, millwork, pallets, crates.
02 42 19.25	Plastic Recycling	Includes: PET: polyethylene terephthalate, LDPE: low density polyethylene, HDPE: high density polyethylene, PP: polypropylene, PVC: polyvinyl chloride.
02 42 19.29	Solid Surface Recycling	Includes: granite.
02 42 19.31	Insulation Recycling	Includes: batt.
02 42 19.35	Roofing Recycling	Includes: shingles.
02 42 19.39	Window Recycling	
02 42 19.41	Glass Recycling	
02 42 19.43	Gypsum Board Recycling	Includes: acoustical, plaster.

DIVISION 02

NUMBER	TITLE	EXPLANATION
02 42 19.45	Ceiling Tile Recycling	
02 42 19.47	Flooring Recycling	Includes: carpet, underlayment, non-PVC backed tile-carpet, PVC backed tile-carpet, granite tile, vinyl composite tile, ceramic tile, porcelain tile.
02 42 19.49	Paint and Coatings Recycling and Management	Includes: interior/exterior architectural paints, deck coatings, floor paints, elastomeric, primers, sealers, under-coaters, stains, shellacs, lacquers, varnishes, urethanes, waterproofing concrete, roof, wood sealers and repellents. Supplies, thinners, spirits, solvents, additives, tints, field and lawn paints.
02 42 19.51	Paint Remanufacturing	
02 42 19.53	Electronics Recycling	Includes: computers and accessories, data center, office equipment, copiers, audiovisual equipment, entertainment equipment, telecommunications equipment, lab and medical equipment, batteries, UPS: uninterruptible power supplies.
02 42 19.55	Appliance Recycling	
02 42 19.57	Mattress Recycling	
02 42 19.81	Plumbing Fixture Recycling	Includes: sinks, toilets, tubs.
02 42 19.83	HVAC Recycling	Includes: chillers, A/C units, coils.
02 42 19.89	Electrical Recycling	Includes: equipment, wire transformers, boxes, panels, lighting, components, ballasts, bulbs.
02 42 19.91	Yard Waste Recycling	Includes: mulch, bark, compost, stone, tree branches, land clearing debris, stumps, logs.
02 42 19.95	Tire, Rubber Recycling	
02 42 21	Salvage of Construction Materials	Includes: items taken from project location for reuse.
02 42 21.11	Deconstruction Contracting	
02 42 21.15	Concrete Salvage	
02 42 21.17	Masonry Salvage	Includes: brick, stone, pavers.
02 42 21.19	Metal Salvage	
02 42 21.23	Wood Salvage	Includes: cabinetry, fixtures, doors, lumber, millwork, pallets, crates.
02 42 21.29	Solid Surface Salvage	Includes: granite.

NUMBER	TITLE	EXPLANATION
02 42 21.35	Roofing Salvage	Includes: supplies, shingles.
02 42 21.37	Door Salvage	
02 42 21.39	Window Salvage	
02 42 21.41	Glass Salvage	
02 42 21.43	Gypsum Board Salvage	Includes: acoustical, plaster.
02 42 21.45	Ceiling Tile Salvage	
02 42 21.47	Flooring Salvage	Includes: carpet, underlayment, tile-carpet, granite tile, vinyl composite tile, ceramic tile, porcelain tile.
02 42 21.49	Paint and Coatings Salvage	Includes: interior/exterior architectural paints, deck coatings, floor paints, elastomeric, primers, sealers, under-coaters, stains, shellacs, lacquers, varnishes, urethanes, waterproofing concrete, roof, wood sealers and repellents. Supplies, thinners, spirits, solvents, additives, tints, field and lawn paints.
02 42 21.53	Electronics Salvage	Includes: computers and accessories, data center, office equipment, copiers, audiovisual equipment, entertainment equipment, telecommunications equipment, lab and medical equipment, batteries, UPS: uninterruptible power supplies.
02 42 21.55	Appliance Salvage	Includes: washer, dryer, dishwasher, stove, oven, microwave.
02 42 21.57	Restaurant Equipment Salvage	
02 42 21.59	Toilet Room Finishings Salvage	Includes: vanities, shower doors, hardware, partitions.
02 42 21.61	Office Finishings Salvage	Includes: desks, file cabinets, office furniture, cubicles, partitions, systems furniture.
02 42 21.67	Residential Finishings Salvage	Includes: furniture, art, rugs, framed mirrors, décor, lamps.
02 42 21.81	Plumbing Fixture Salvage	Includes: sinks, toilets, tubs, pipe.
02 42 21.83	HVAC Salvage	Includes: working units.
02 42 21.89	Electrical Salvage	Includes: equipment, wire transformers, boxes, panels, lighting, components, ballasts, bulbs.

NUMBER	TITLE	EXPLANATION
02 42 21.91	Yard Waste Salvage	Includes: plant materials, trees, stone.
02 43 00	Structure Moving	Includes: preparation and processes of relocating and raising structures. See: 31 43 00 for concrete raising.
02 43 13	Structure Relocation	
02 43 13.13	Building Relocation	
02 43 16	Structure Raising	
02 43 16.13	Building Raising	
02 50 00	Site Remediation	Includes: remediation technologies for contaminated sites. See: 02 24 23 for chemical sampling and analysis of soils. 02 60 00 for contaminated site material removal. 02 70 00 for water remediation. 02 80 00 for remediation of materials within structures. 31 21 00 for off-gassing mitigation.
02 51 00	Physical Decontamination	Includes: soil decontamination by physical treatment processes. See: 02 52 00 for chemical decontamination. 02 53 00 for thermal decontamination. 02 54 00 for biological decontamination. 02 55 00 for remediation soil stabilization. 02 61 00 for removal and disposal of contaminated soils.
02 51 13	Coagulation and Flocculation Decontamination	
02 51 16	Reverse-Osmosis Decontamination	
02 51 19	Solidification and Stabilization Decontamination	
02 51 23	Mechanical Filtration Decontamination	
02 51 26	Radioactive Decontamination	
02 51 29	Surface Cleaning Decontamination	
02 51 29.13	High-Pressure Water Cleaning Decontamination	
02 51 29.16	Vacuum Sweeping Cleaning Decontamination	
02 51 33	Surface Removal Decontamination	
02 51 33.13	Surface Removal Decontamination by Grinding	
02 51 33.16	Surface Removal Decontamination by Sand Blasting	
02 51 33.19	Surface Removal Decontamination by Ultrasound	

NUMBER	TITLE	EXPLANATION
02 52 00	Chemical Decontamination	Includes: soil decontamination by chemical treatment processes. See: 02 51 00 for physical decontamination. 02 53 00 for thermal decontamination. 02 54 00 for biological decontamination. 02 55 00 for remediation soil stabilization. 02 61 00 for removal and disposal of contaminated soils.
02 52 13	Chemical Precipitation Decontamination	
02 52 16	Ion Change Decontamination	
02 52 19	Neutralization Decontamination	
02 53 00	Thermal Decontamination	Includes: soil decontamination by thermal treatment processes. See: 02 51 00 for physical decontamination. 02 52 00 for chemical decontamination. 02 54 00 for biological decontamination. 02 55 00 for remediation soil stabilization. 02 61 00 for removal and disposal of contaminated soils.
02 53 13	Incineration Decontamination	
02 53 13.13	Remediation of Contaminated Soils and Sludges by Incineration	
02 53 16	Thermal Desorption Decontamination	
02 53 16.13	Remediation of Contaminated Soils by Thermal Desorption	
02 53 19	Vitrification Decontamination	
02 54 00	Biological Decontamination	Includes: soil decontamination by biological treatment processes. See: 02 51 00 for physical decontamination. 02 52 00 for chemical decontamination. 02 53 00 for thermal decontamination. 02 55 00 for remediation soil stabilization. 02 61 00 for removal and disposal of contaminated soils. 02 72 16 for biological water decontamination.
02 54 13	Aerobic Processes Decontamination	
02 54 16	Anaerobic Processes Decontamination	
02 54 19	Bioremediation Decontamination	
02 54 19.13	Bioremediation Using Landfarming	
02 54 19.16	Bioremediation of Soils Using Windrow Composting	
02 54 19.19	Bioremediation Using Bacteria Injection	
02 54 23	Soil Washing Through Separation/Solubilization	
02 54 26	Organic Decontamination	

NUMBER	TITLE	EXPLANATION
02 55 00	Remediation Soil Stabilization	Includes: insitu and exsitu soil stabilization for remediation. See: 02 51 00 for physical decontamination. 02 52 00 for chemical decontamination. 02 53 00 for thermal decontamination. 02 54 00 for biological decontamination. 02 61 00 for removal and disposal of contaminated soils. 31 32 00 for soil stabilization.
02 56 00	Site Containment	Includes: containment of contaminated site materials. See: 31 21 00 for off-gassing mitigation. 31 35 26 for containment barriers.
02 56 13	Waste Containment	
02 56 13.13	Geomembrane Waste Containment	
02 56 19	Gas Containment	
02 56 19.13	Fluid-Applied Gas Barrier	
02 57 00	Sinkhole Remediation	Includes: remediation of underground voids in soluble rock created by groundwater. Alternate Terms/Abbreviations: sinkholes: karst. See Also: 31 23 23 for backfilling. 31 43 13 for pressure grouting. 31 43 16 for compaction grouting.
02 57 13	Sinkhole Remediation by Grouting	
02 57 13.13	Sinkhole Remediation by Compaction Grouting	
02 57 13.16	Sinkhole Remediation by Cap Grouting	
02 57 16	Sinkhole Remediation by Backfilling	
02 58 00	Snow Control	Includes: permanent means to control the movement of snow on the site. See: 01 56 26 for temporary fencing. 32 31 00 for fences and gates.
02 58 13	Snow Fencing	
02 58 16	Snow Avalanche Control	

NUMBER	TITLE	EXPLANATION
02 60 00	Contaminated Site Material Removal	Includes: removal and disposal of contaminated site materials. See: 02 24 23 for chemical sampling and analysis of soils. 02 50 00 for site remediation. 02 70 00 for water remediation. 02 80 00 for remediation of materials within structures. 44 50 00 for solid waste control and reuse.
02 61 00	Removal and Disposal of Contaminated Soils	Includes: removal of toxic and hazardous site materials. Alternate Terms/Abbreviations: PCB: polychlorinate biphenyl. See: 02 51 00 for physical decontamination. 02 52 00 for chemical decontamination. 02 53 00 for thermal decontamination. 02 54 00 for biological decontamination. 02 55 00 for remediation soil stabilization. 02 81 00 for transportation and disposal of hazardous materials within structures. 02 82 00 for remediation of asbestos materials within structures. 02 84 00 for remediation of PCB materials within structures. 31 23 16 for excavation.
02 61 13	Excavation and Handling of Contaminated Material	
02 61 23	Removal and Disposal of Polychlorinate Biphenyl Contaminated Soils	
02 61 26	Removal and Disposal of Asbestos Contaminated Soils	
02 61 29	Removal and Disposal of Organically Contaminated Soils	
02 62 00	Hazardous Waste Recovery Processes	Includes: removal of hazardous and toxic materials from soil. May Include: removal of hazardous and toxic materials from air and water. See: 43 30 00 for gas and liquid purification equipment. 44 10 00 for air pollution control.
02 62 13	Air and Steam Stripping	
02 62 16	Soil Vapor Extraction	
02 62 19	Soil Washing and Flushing	

NUMBER	TITLE	EXPLANATION
02 65 00	Underground Storage Tank Removal	Includes: removal of underground tanks and appurtenances which have been used to store toxic and hazardous materials. See: 02 86 00 for hazardous waste drum handling. 31 23 16 for excavation.
02 70 00	Water Remediation	Includes: remediation of contaminated water. See: 02 50 00 for site remediation. 02 60 00 for contaminated site material removal. 02 80 00 for remediation of materials within structures. 44 40 00 for water pollution control equipment.
02 71 00	Groundwater Treatment	Includes: treatment of contaminated groundwater. See: 02 32 23 for groundwater monitoring before construction. 33 11 53 for monitoring wells.
02 72 00	Water Decontamination	Includes: decontamination technologies for treatment of contaminated water. See: 02 54 00 for biological soil decontamination. 43 30 00 for gas and liquid purification equipment. 44 40 00 for water pollution control equipment.
02 72 13	Chemical Water Decontamination	
02 72 16	Biological Water Decontamination	
02 72 19	Electrolysis Water Decontamination	
02 80 00	Facility Remediation	Includes: remediation technologies for abatement and removal and disposal of contaminated materials within structures. Alternate Terms/Abbreviations: hazardous materials: poisonous materials. See: 00 31 00 for existing hazardous materials information. 02 26 00 for hazardous materials assessment. 02 50 00 for remediation of site materials. 02 60 00 for contaminated site material removal. 02 70 00 for water remediation.

NUMBER	TITLE	EXPLANATION
02 81 00	Transportation and Disposal of Hazardous Materials	Includes: requirements for transporting and disposal of hazardous materials within structures. See: 01 35 43 for environmental procedures for hazardous materials. 02 61 00 for removal and disposal of contaminated site materials. 44 50 00 for solid waste control and reuse.
02 82 00	Asbestos Remediation	Includes: abatement or removal of asbestos material within structures. See: 00 31 26 for existing asbestos information provided to bidders or proposers before construction begins. 02 26 23 for asbestos assessments performed under the contract. 02 61 26 for removal and disposal of asbestos site materials.
02 82 13	Asbestos Abatement	Includes: abatement or removal of asbestos material within structures, and removal of piping with coatings of asbestos-containing material (ACM), removal of ACM-contaminated pipeline coatings or coverings, cutting of pipelines containing, covered, or coated with ACM, and monitoring of work area during cutting and cleaning operations.
02 82 13.13	Glovebag Asbestos Abatement	
02 82 13.16	Precautions for Asbestos Abatement	
02 82 13.19	Asbestos Floor Tile and Mastic Abatement	
02 82 13.33	Asbestos Abatement for Utilities	
02 82 16	Engineering Control of Asbestos Containing Materials	
02 82 33	Removal and Disposal of Asbestos Containing Materials	
02 83 00	Lead Remediation	Includes: abatement or removal of lead material within structures. See: 00 31 26 for existing lead information provided to bidders or proposers before construction begins. 02 26 26 for lead assessments performed under the contract.
02 83 13	Lead Hazard Control Activities	
02 83 19	Lead-Based Paint Remediation	
02 83 19.13	Lead-Based Paint Abatement	
02 83 33	Removal and Disposal of Material Containing Lead	
02 83 33.13	Lead-Based Paint Removal and Disposal	

NUMBER	TITLE	EXPLANATION
02 84 00	Polychlorinate Biphenyl Remediation	Includes: abatement or removal of polychlorinated biphenyl material within structures. Alternate Terms/Abbreviations: PCB: polychlorinate biphenyl. See: 00 31 26 for existing PCB information provided to bidders or proposers before construction begins. 02 26 29 for PCB assessments performed under the contract. 02 61 23 for removal and disposal of PCB site materials.
02 84 16	Handling of Lighting Ballasts and Lamps Containing PCBs and Mercury	
02 84 33	Removal and Disposal of Polychlorinate Biphenyls	
02 86 00	Hazardous Waste Drum Handling	Includes: requirements for handling drums containing material contaminated with toxic and hazardous materials. See: 00 31 26 for hazardous waste drum information provided to the bidders and proposers before construction begins. 02 26 36 for hazardous waste drum assessments performed under the contract. 02 65 00 for removal of underground storage tanks.
02 87 00	Biohazard Remediation	
02 87 13	Mold Remediation	
02 87 13.13	Precautions for Mold Remediation	
02 87 13.16	Mold Remediation Preparation and Containment	
02 87 13.19	Mold Remediation Clearance Air Sampling	
02 87 13.33	Removal and Disposal of Materials with Mold	
02 87 16	Excrement Removal	
02 87 16.13	Bird Excrement Removal	
02 87 16.16	Rodent Excrement Removal	

NUMBER	TITLE	EXPLANATION
DIVISION 03 – CONCRETE		
03 00 00	Concrete	May be used as Division level Section title.

03 01 00 Maintenance of Concrete

Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of concrete. Includes graffiti removal. Includes epoxy-injected crack repair.

Level 4 Numbering Recommendation: Level 4 titles for concrete are provided as examples for user's creation of more detailed titles for other subjects as required. Following Level 4 numbering is recommended:

.51-.59 for maintenance.

.61-.69 for repair.

.71-.79 for rehabilitation.

.81-.89 for replacement.

.91-.99 for restoration.

See: 02 25 16 for existing concrete assessment.

03 30 00 for cleaning new cast-in-place concrete.

03 40 00 for cleaning new precast concrete.

09 96 23 for graffiti-resistant coatings.

32 01 26 for rigid pavement surface treatment.

32 01 29 for rigid pavement rehabilitation.

03 01 10	Maintenance of Concrete Forming and Accessories
03 01 20	Maintenance of Concrete Reinforcing
03 01 23	Maintenance of Stressing Tendons
03 01 30	Maintenance of Cast-in-Place Concrete
03 01 30.51	Cleaning of Cast-in-Place Concrete
03 01 30.61	Resurfacing of Cast-in-Place Concrete
03 01 30.71	Rehabilitation of Cast-in-Place Concrete
03 01 30.72	Strengthening of Cast-in-Place Concrete
03 01 40	Maintenance of Precast Concrete
03 01 40.51	Cleaning of Precast Concrete
03 01 40.61	Resurfacing of Precast Concrete
03 01 40.71	Rehabilitation of Precast Concrete
03 01 40.72	Strengthening of Precast Concrete
03 01 50	Maintenance of Cast Decks and Underlayment
03 01 50.51	Cleaning Cast Decks and Underlayment
03 01 50.61	Resurfacing of Cast Decks and Underlayment
03 01 50.71	Rehabilitation of Cast Decks and Underlayment
03 01 50.72	Strengthening of Cast Decks and Underlayment
03 01 60	Maintenance of Grouting
03 01 70	Maintenance of Mass Concrete
03 01 80	Maintenance of Concrete Cutting and Boring

03 03 00 Conservation Treatment for Period Concrete

DIVISION 03

NUMBER	TITLE	EXPLANATION
03 03 30	Conservation Treatment for Period Cast-in-Place Concrete	
03 03 31	Conservation Treatment for Period Structural Concrete	
03 03 33	Conservation Treatment for Period Architectural Concrete	

03 05 00 Common Work Results for Concrete

Includes: subjects common to multiple titles in Division 03.

See: 01 80 00 for performance requirements of subjects common to multiple titles.

03 05 05	Selective Demolition for Concrete	
	See: 02 41 19 for selective demolition for whole structure demolition.	

03 06 00 Schedules for Concrete

Includes: schedules of items common to multiple titles in Division 03.

Notes: a schedule may be included on drawings, in the project manual, or a project book.

03 06 10	Schedules for Concrete Forming and Accessories	
03 06 20	Schedules for Concrete Reinforcing	
03 06 20.13	Concrete Beam Reinforcing Schedule	
03 06 20.16	Concrete Slab Reinforcing Schedule	
03 06 30	Schedules for Cast-in-Place Concrete	
03 06 30.13	Concrete Footing Schedule	
03 06 30.16	Concrete Column Schedule	
03 06 30.19	Concrete Slab Schedule	
03 06 30.23	Concrete Shaft Schedule	
03 06 30.26	Concrete Beam Schedule	
03 06 40	Schedules for Precast Concrete	
03 06 40.13	Precast Concrete Panel Schedule	
03 06 50	Schedules for Cast Decks and Underlayment	
03 06 60	Schedules for Grouting	
03 06 70	Schedules for Mass Concrete	
03 06 80	Schedules for Concrete Cutting and Boring	

03 08 00 Commissioning of Concrete

Includes: commissioning of items common to multiple titles in Division 03.

See: 01 91 00 for commissioning of subjects common to multiple Divisions.

03 10 00 Concrete Forming and Accessories

NUMBER	TITLE	EXPLANATION
03 11 00	Concrete Forming	Includes: permanent and temporary forms for structural and architectural cast-in-place concrete. Metal pan forms, wood forms, plastic forms, slip forms, and corrugated paper forms for placing concrete. Form liners, void forms. May Include: manufactured joints, anchors, inserts, and waterstops. See: 03 40 00 for precast concrete formwork. 03 51 00 for cast roof deck form boards. 05 30 00 for metal deck forms. 07 95 13 for expansion joint cover assemblies. See Also: 03 15 13 for manufactured joints, anchors, inserts, and waterstops. 03 30 00 for cast-in-place concrete forming. Products: form coatings, form ties.
03 11 13	Structural Cast-in-Place Concrete Forming	
03 11 13.13	Concrete Slip Forming	
03 11 13.16	Concrete Shoring	
03 11 13.19	Falsework	
03 11 16	Architectural Cast-in-Place Concrete Forming	
03 11 16.13	Concrete Form Liners	
03 11 19	Insulating Concrete Forming	
03 11 23	Permanent Stair Forming	Includes: forms specifically designed for the forming of stairways that remain in place permanently.
03 11 26	Permanent Tier Forming	
03 15 00	Concrete Accessories	Includes: anchors, cast-in anchors, inserts, expansion and contraction joints, and waterstops. See: 07 95 13 for expansion joint cover assemblies. See Also: 03 11 00 for manufactured joints, anchors, inserts, and waterstops. 05 05 19 for post-installed concrete anchors.
03 15 13	Waterstops	
03 15 13.13	Non-Expanding Waterstops	
03 15 13.16	Expanding Waterstops	
03 15 13.19	Combination Expanding and Injection Hose Waterstops	
03 15 13.21	Injection Hose Waterstops	
03 15 16	Concrete Construction Joints	
03 15 19	Cast-In Concrete Anchors	

NUMBER	TITLE	EXPLANATION
03 15 21	Termite Barrier	See Also: 10 81 00 for pest control devices.

03 20 00 Concrete Reinforcing

Includes: reinforcement of all types to increase the structural capabilities of concrete.

See: 03 38 00 for post-tensioned concrete.
04 05 19 for masonry reinforcing.

See Also: 03 30 00 for cast-in-place concrete reinforcement.
03 40 00 for precast concrete reinforcement.

03 21 00 Reinforcement Bars

Includes: tension, compression, and temperature-reinforcing steel for concrete. Coated reinforcing steel.

Alternate Terms/Abbreviations: reinforcing steel: steel rebar: reinforcing bars.

Products: bar supports, bar chairs, bar spacers.

03 21 11	Plain Steel Reinforcement Bars
03 21 13	Galvanized Reinforcement Steel Bars
03 21 16	Epoxy-Coated Reinforcement Steel Bars
03 21 19	Stainless Steel Reinforcement Bars
	Includes: corrosion-resistant steel reinforcing bars.

03 21 21	Composite Reinforcement Bars
03 21 21.11	Glass Fiber-Reinforced Polymer Reinforcement Bars
03 21 21.13	Organic Fiber-Reinforced Polymer Reinforcement Bars
03 21 21.16	Carbon Fiber-Reinforced Polymer Reinforcement Bars

03 22 00 Fabric and Grid Reinforcing

Includes: welded wire fabric used as reinforcement for concrete. Coated welded wire fabric.

Alternate Terms/Abbreviations: WWF: welded wire fabric: welded wire mesh: welded wire reinforcement.

03 22 13	Galvanized Welded Wire Fabric Reinforcing
03 22 16	Epoxy-Coated Welded Wire Fabric Reinforcing
03 22 19	Composite Grid Reinforcing

03 23 00 Stressed Tendon Reinforcing

Includes: stressing tendons for reinforcement of concrete.

Products: seven-wire stress-relieved strand, stress-relieved wire, tendon ducts.

NUMBER	TITLE	EXPLANATION
03 24 00	Fibrous Reinforcing	Includes: fibrous reinforcing for reinforcement of concrete. Steel fibers, synthetic fibers. Products: steel fibers, synthetic fibers.
03 25 00	Composite Reinforcing	
03 25 13	Glass Fiber-Reinforced Polymer Reinforcing	
03 25 16	Organic Fiber-Reinforced Polymer Reinforcing	
03 25 19	Carbon Fiber-Reinforced Polymer Reinforcing	
03 30 00	Cast-in-Place Concrete	Includes: aggregate, cement, additives, and pigmented mix. May Include: concrete forming, accessories, and reinforcing. Granular subbase and vapor retarders for concrete slabs-on-grade. CIP concrete tunnel lining. CIP concrete shaft lining. CIP concrete retaining walls. Alternate Terms/Abbreviations: CIP: cast-in-place. Cast-in-place: poured-in-place. See: 03 50 00 for cast decks and underlayment. 32 13 00 for rigid pavement. 41 62 13 for cement mixer trucks. See Also: 03 10 00 for CIP concrete forming and accessories. 03 20 00 for CIP concrete reinforcing. 07 26 00 for vapor retarders for concrete slabs-on-grade. 13 47 13 for cathodic protection. 31 23 00 for granular subbase for concrete slabs-on-grade. 31 35 23 for slope protection with CIP concrete slope paving. 31 74 13 for CIP concrete tunnel lining. 31 75 13 for CIP concrete shaft lining. 32 32 13 for CIP concrete retaining walls.
03 30 53	Miscellaneous Cast-in-Place Concrete	Includes: may be used for projects with small amounts of CIP concrete.
03 31 00	Structural Concrete	Includes: cast-in-place concrete for structural purposes where appearance is not a prime consideration. Cast-in-place concrete beams, columns, girders, joists, slabs, and other concrete framing members. Alternate Terms/Abbreviations: self-compacting concrete: self-consolidating concrete: self-placing concrete: self-leveling concrete.

DIVISION **03**

NUMBER	TITLE	EXPLANATION
03 31 13	Heavyweight Structural Concrete	
03 31 16	Lightweight Structural Concrete	
03 31 19	Shrinkage-Compensating Structural Concrete	
03 31 23	High-Performance Structural Concrete	
03 31 24	Ultra High-Performance Structural Concrete	
03 31 26	Self-Compacting Concrete	
03 33 00	Architectural Concrete	Includes: cast-in-place concrete where appearance is a prime consideration.
03 33 13	Heavyweight Architectural Concrete	
03 33 16	Lightweight Architectural Concrete	
03 34 00	Low Density Concrete	Includes: concrete of low density serving as structural concrete. Alternate Terms/Abbreviations: low density concrete: lightweight concrete. See: 03 52 00 for lightweight concrete roof insulation.
03 35 00	Concrete Finishing	Includes: leveling, floating, troweling, and other treatments of flat concrete, including sealing and hardening. Treatment of formed concrete surfaces to achieve certain surface characteristics. See: 03 53 00 for concrete topping. 03 54 00 for cast underlayment. 09 61 00 for floor treatments. 32 13 13 for exposed aggregate concrete paving. 32 13 16 for stamped concrete paving. Products: admixtures, additives, aggregate, cement, coloring agents, fly ash, accelerators, bonding agents, air entrainment, curing agents, densifiers, hardeners, plasticizers, retarders, sealers.
03 35 13	High-Tolerance Concrete Floor Finishing	
03 35 16	Heavy-Duty Concrete Floor Finishing	Includes: finishes to reinforce high weight bearing, high traffic, high usage areas.
03 35 19	Colored Concrete Finishing	
03 35 23	Exposed Aggregate Concrete Finishing	
03 35 26	Grooved Concrete Surface Finishing	
03 35 29	Tooled Concrete Finishing	
03 35 33	Stamped Concrete Finishing	

NUMBER	TITLE	EXPLANATION
03 35 43	Polished Concrete Finishing	
03 35 43.13	Polished and Dyed Concrete Finishing	
03 35 43.16	Polished and Stained Concrete Finishing	
03 35 46	Concrete Topical Treatments	Includes: concrete treatments such as sealers, hardeners, densifiers, and repellants, applied after curing.
03 37 00	Specialty Placed Concrete	Includes: various methods of specially placing concrete. Concrete placed as dry mix. Alternate Terms/Abbreviations: shotcrete: gunite. See: 03 53 00 for concrete topping. 31 32 33 for shotcrete soil slope stabilization. 31 33 33 for shotcrete rock slope stabilization. 31 52 00 for cofferdams. 31 74 19 for shotcrete tunnel lining. 32 13 16 for roller-compacted concrete paving. 35 53 00 for underwater construction.
03 37 13	Shotcrete	
03 37 16	Pumped Concrete	
03 37 19	Pneumatically Placed Concrete	
03 37 23	Roller-Compacted Concrete	
03 37 26	Underwater Placed Concrete	
03 38 00	Post-Tensioned Concrete	Includes: cast-in-place concrete with stressing tendons and jacking to apply load on tendons. May Include: unstressed concrete reinforcing. See: 03 20 00 for concrete reinforcing. 03 23 00 for stressed tendon reinforcing.
03 38 13	Post-Tensioned Concrete Preparation	
03 38 16	Unbonded Post-Tensioned Concrete	
03 38 19	Bonded Post-Tensioned Concrete	
03 39 00	Concrete Curing	Includes: any method of curing concrete.
03 39 13	Water Concrete Curing	
03 39 16	Sand Concrete Curing	
03 39 23	Membrane Concrete Curing	
03 39 23.13	Chemical Compound Membrane Concrete Curing	
03 39 23.23	Sheet Membrane Concrete Curing	

NUMBER	TITLE	EXPLANATION
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03 40 00 Precast Concrete

Includes: concrete cast in forms, either on or off site, before final placement. Forming, reinforcing, inserts, and accessories.

May Include: precast concrete tunnel lining. Precast concrete shaft lining. Precast concrete retaining walls.

See: 31 35 23 for slope protection with precast concrete slope paving.

31 62 13 for precast concrete driven piles.

See Also: 31 74 16 for precast concrete tunnel lining.

31 75 16 for precast concrete shaft lining.

32 32 16 for precast concrete retaining walls.

03 41 00 Precast Structural Concrete

Includes: precast concrete for structural purposes where appearance is not a prime consideration. Precast concrete beams, columns, girders, joists, slabs, and other concrete framing members.

Alternate Terms/Abbreviations: precast: prestressed.

See: 03 45 00 for precast architectural concrete.

03 47 00 for site-cast concrete.

03 48 00 for precast concrete specialties.

05 55 16 for metal stair nosings.

03 41 13	Precast Concrete Hollow Core Planks
03 41 16	Precast Concrete Slabs
03 41 23	Precast Concrete Stairs
03 41 33	Precast Structural Pretensioned Concrete
03 41 36	Precast Structural Post-Tensioned Concrete

NUMBER	TITLE	EXPLANATION
03 45 00	Precast Architectural Concrete	Includes: precast concrete where appearance is a prime consideration. May Include: unit masonry, stone veneer, and tile used as facings for architectural precast concrete panels. Alternate Terms/Abbreviations: precast: plant-precast: prestressed. See: 03 41 00 for precast structural concrete. 03 47 00 for site-cast concrete. 03 48 00 for precast concrete specialties. 04 72 00 for cast stone masonry. 07 44 00 for faced roof and wall panels. See Also: 04 20 00 for unit masonry facing. 04 40 00 for stone facing. 09 30 00 for tile facing.
03 45 13	Faced Architectural Precast Concrete	
03 45 33	Precast Architectural Pretensioned Concrete	
03 45 36	Precast Architectural Post-Tensioned Concrete	
03 47 00	Site-Cast Concrete	Includes: concrete cast in forms on site before final placement. Alternate Terms/Abbreviations: site-cast: site-precast. See: 03 41 00 for precast structural concrete. 03 45 00 for precast architectural concrete. 03 48 00 for precast concrete specialties.
03 47 13	Tilt-Up Concrete	
03 47 16	Lift-Slab Concrete	

NUMBER	TITLE	EXPLANATION
03 48 00	Precast Concrete Specialties	Includes: miscellaneous precast concrete items. See: 03 41 00 for precast structural concrete. 03 45 00 for precast architectural concrete. 03 47 00 for site-cast concrete. 04 72 00 for cast stone masonry. 07 76 00 for precast concrete roof pavers. 32 14 13 for precast concrete unit paving. 32 16 13 for precast concrete curbs and gutters. 33 05 61 for concrete manholes. 33 05 63 for concrete vaults and chambers. 33 05 66 for concrete distribution chambers. 33 05 69 for concrete trenchways. 33 05 71 for cleanouts. 33 05 73 for polyethylene manholes. 33 05 76 for fiberglass manholes. 33 42 13 for stormwater culverts. 33 71 16 for precast concrete electrical utility poles. 34 71 13 for precast concrete vehicle barriers. See Also: 32 17 13 for parking bumpers.
03 48 13	Precast Concrete Bollards	
03 48 13.11	Precast Concrete Security Bollards	
03 48 16	Precast Concrete Splash Blocks	
03 48 19	Precast Concrete Stair Treads	
03 48 26	Precast Concrete Parking Bumpers	See Also: 32 17 13 for parking bumpers.
03 48 33	Precast Pre-Framed Concrete Panels	Alternate Terms/Abbreviations: studcast concrete: thinshell concrete.
03 48 43	Precast Concrete Trim	
03 49 00	Glass-Fiber-Reinforced Concrete	Includes: products using a cement/aggregate slurry reinforced with glass fibers. Alternate Terms/Abbreviations: GFRC: glass-fiber-reinforced concrete. See: 06 82 00 for glass-fiber-reinforced plastic. 07 44 53 for glass-fiber-reinforced wall cementitious panels. 09 27 13 for glass-fiber-reinforced plaster fabrications.
03 49 13	Glass-Fiber-Reinforced Concrete Column Covers	
03 49 16	Glass-Fiber-Reinforced Concrete Spandrels	

NUMBER	TITLE	EXPLANATION
03 49 43	Glass-Fiber-Reinforced Concrete Trim	
03 50 00	Cast Decks and Underlayment	Includes: poured and prefabricated cast floor and roof decks and fill. Includes form boards, reinforcement, and support members when part of a system. See: 03 34 00 for low density concrete. 03 41 00 for precast concrete decks. 07 22 13 for asphaltic perlite concrete deck.
03 51 00	Cast Roof Decks	Includes: unitized panels of gypsum and cementitious wood fiber roof decking.
03 51 13	Cementitious Wood Fiber Decks	
03 51 16	Gypsum Concrete Roof Decks	
03 52 00	Lightweight Concrete Roof Insulation	Includes: insulating concrete used in roofing assemblies. May Include: corrugated permanent metal forms when part of the cast deck system. See: 03 34 00 for low density concrete. 07 22 13 for asphaltic perlite concrete deck. 07 22 16 for roof board insulation. See Also: 03 11 00 for concrete forming. 05 30 00 for metal deck forms.
03 52 13	Composite Concrete Roof Insulation	
03 52 16	Lightweight Insulating Concrete	
03 52 16.13	Lightweight Cellular Insulating Concrete	
03 52 16.16	Lightweight Aggregate Insulating Concrete	
03 53 00	Concrete Topping	Includes: cellular concrete for floor construction and concrete topping for slabs. See: 03 35 00 for concrete floor finishing. 03 52 00 for lightweight concrete roof insulation. 03 54 00 for cast underlayment.
03 53 13	Emery-Aggregate Concrete Topping	
03 53 16	Iron-Aggregate Concrete Topping	
03 53 19	Concrete Overlayment	Includes: pedestrian or vehicular traffic wearing surface treatments.

NUMBER	TITLE	EXPLANATION
03 54 00	Cast Underlayment	Includes: self-leveling and troweled concrete leveler and underlayment. See: 03 53 00 for concrete topping. 06 16 26 for underlayment.
03 54 13	Gypsum Cement Underlayment	
03 54 16	Hydraulic Cement Underlayment	
03 60 00	Grouting	Includes: shrink-resistant, mortar and fluid-like materials used for filling joints and cavities and setting and anchoring items in masonry and concrete. May Include: grouting of items included elsewhere, such as structural metal, metal fabrications, decorative metals, and equipment. See: 04 05 16 for masonry grouting. 09 30 00 for tile joint grouting. 09 63 00 for masonry flooring joint grouting. 31 73 00 for tunnel grouting. 32 14 00 for unit paving joint grouting. See Also: 05 10 00 for grouting to set structural metal framing. 05 20 00 for grouting to set metal joists. 05 50 00 for grouting to set metal fabrications. 05 70 00 for grouting to set decorative metals. Individual equipment titles for grouting to set equipment.
03 61 00	Cementitious Grouting	Includes: cementitious mixture of low slump for setting and anchoring. See: 03 62 00 for non-shrink grouting. 03 63 00 for epoxy grouting.
03 61 13	Dry-Pack Grouting	
03 62 00	Non-Shrink Grouting	Includes: cementitious mixture with additives to compensate for shrinkage for setting and anchoring. See: 03 61 00 for cementitious grouting. 03 63 00 for epoxy grouting.
03 62 13	Non-Metallic Non-Shrink Grouting	

NUMBER	TITLE	EXPLANATION
03 62 16	Metallic Non-Shrink Grouting	
03 63 00	Epoxy Grouting	Includes: epoxy mixture with stable volume characteristics for setting and anchoring. See: 03 61 00 for cementitious grouting. 03 62 00 for non-shrink grouting. 03 64 23 for epoxy injection grouting.
03 64 00	Injection Grouting	Includes: grouting cracks and other defects in concrete construction. Alternate Terms/Abbreviations: injection grouting: pressure grouting. See: 03 01 00 for concrete rehabilitation and repair. 31 32 23 for pressure grouting soil stabilization. 31 43 13 for pressure grouting for concrete raising. 31 43 16 for compaction grouting for concrete raising.
03 64 23	Epoxy Injection Grouting	
03 70 00	Mass Concrete	Includes: large volume placement of cast-in-place concrete with dimensions large enough to require special construction techniques and procedures to compensate for the generation of heat, volume change, and to minimize cracks. Alternate Terms/Abbreviations: raft foundations: mat foundations. See: 35 71 13 for concrete gravity dams. 35 72 13 for concrete arch dams. 35 74 13 for concrete buttress dams.
03 71 00	Mass Concrete for Raft Foundations	
03 72 00	Mass Concrete for Dams	
03 80 00	Concrete Cutting and Boring	
03 81 00	Concrete Cutting	Includes: cutting or sawing through concrete. See: 03 82 00 for concrete boring.

DIVISION **03**

NUMBER	TITLE	EXPLANATION
03 81 13	Flat Concrete Sawing	
03 81 16	Track Mounted Concrete Wall Sawing	
03 81 19	Wire Concrete Wall Sawing	
03 81 23	Hand Concrete Wall Sawing	
03 81 26	Chain Concrete Wall Sawing	
03 82 00	Concrete Boring	Includes: creating circular hole through concrete with an auger or drill. See: 02 25 16 for assessment drilling of existing concrete. 03 81 00 for concrete cutting.
03 82 13	Concrete Core Drilling	

NUMBER	TITLE	EXPLANATION
DIVISION 04 – MASONRY		
04 00 00	Masonry	May be used as Division level Section title.

04 01 00 Maintenance of Masonry

Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of masonry.

Level 4 Numbering Recommendation: Level 4 titles for masonry are provided as examples for user's creation of more detailed titles for other subjects as required. Following Level 4 numbering is recommended:

.51-.59 for maintenance.

.61-.69 for repair.

.71-.79 for rehabilitation.

.81-.89 for replacement.

.91-.99 for restoration.

See: 02 25 19 for existing masonry assessment.

04 20 00 for cleaning new unit masonry.

04 40 00 for cleaning new stone assemblies.

04 50 00 for cleaning new refractory masonry.

04 60 00 for cleaning new corrosion-resistant masonry.

04 70 00 for cleaning new manufactured masonry.

04 01 20	Maintenance of Unit Masonry
04 01 20.41	Unit Masonry Stabilization
04 01 20.51	Unit Masonry Maintenance
04 01 20.52	Unit Masonry Cleaning
04 01 20.91	Unit Masonry Restoration
04 01 20.93	Testing and Sampling Brick Units for Restoration
04 01 40	Maintenance of Stone Assemblies
04 01 40.51	Stone Maintenance
04 01 40.52	Stone Cleaning
04 01 40.91	Stone Restoration
04 01 50	Maintenance of Refractory Masonry
04 01 60	Maintenance of Corrosion-Resistant Masonry
04 01 70	Maintenance of Manufactured Masonry

04 03 00 Conservation Treatment for Period Masonry

Includes: testing and sampling, dismantling, repairing, replacing, repointing.

May Include: limestone, granite, sandstone.

04 03 01	Maintenance for Period Masonry
04 03 01.13	Period Masonry Cleaning
04 03 05	Common Work Results for Period Masonry
04 03 05.13	Period Masonry Mortaring

DIVISION **04**

NUMBER	TITLE	EXPLANATION
04 03 05.16	Period Masonry Grouting	
04 03 05.19	Period Masonry Anchorage and Reinforcing	
04 03 21	Conservation Treatment for Period Clay Unit Masonry	
04 03 21.19	Clay Brick Conservation Treatment	
04 03 21.21	Terra Cotta Conservation Treatment	
04 03 21.23	Faience Block Conservation Treatment	
04 03 22	Conservation Treatment for Period Concrete Unit Masonry	
04 03 24	Conservation Treatment for Period Adobe Unit Masonry	
04 03 43	Conservation Treatment for Period Stone Masonry	
04 05 00	Common Work Results for Masonry	Includes: subjects common to multiple titles in Division 04. See: 01 80 00 for performance requirements of subjects common to multiple titles.
04 05 05	Selective Demolition for Masonry	See: 02 41 19 for selective demolition for whole structure demolition.
04 05 13	Masonry Mortaring	Includes: masonry mortaring for other titles in Division 04. Products: premixed and colored mortar.
04 05 13.16	Chemical-Resistant Masonry Mortaring	
04 05 13.19	Epoxy Masonry Mortaring	
04 05 13.23	Surface Bonding Masonry Mortaring	
04 05 13.26	Engineered Masonry Mortaring	
04 05 13.29	Refractory Masonry Mortaring	
04 05 13.91	Masonry Restoration Mortaring	
04 05 16	Masonry Grouting	Includes: masonry grouting for other titles in Division 04.
04 05 16.16	Chemical-Resistant Masonry Grouting	
04 05 16.26	Engineered Masonry Grouting	
04 05 19	Masonry Anchorage and Reinforcing	Includes: masonry anchorage and reinforcing for other titles in Division 04. Alternate Terms/Abbreviations: masonry anchors: masonry ties.
04 05 19.13	Continuous Joint Reinforcing	
04 05 19.16	Masonry Anchors	
04 05 19.26	Masonry Reinforcing Bars	
04 05 19.29	Stone Anchors	
04 05 21	Masonry Strengthening	

NUMBER	TITLE	EXPLANATION
04 05 23	Masonry Accessories	Includes: masonry accessories other than those intended to anchor or reinforce for other titles in Division 04. See: 07 62 00 for sheet metal flashing. 07 65 00 for flexible flashing.
04 05 23.13	Masonry Control and Expansion Joints	
04 05 23.16	Masonry Embedded Flashing	
04 05 23.19	Masonry Cavity Drainage, Weepholes, and Vents	
04 06 00	Schedules for Masonry	Includes: schedules of items common to multiple titles in Division 04. Notes: a schedule may be included on drawings, in the project manual, or a project book.
04 06 20	Schedules for Unit Masonry	
04 06 20.13	Masonry Unit Schedule	
04 06 40	Schedules for Stone Assemblies	
04 06 50	Schedules for Refractory Masonry	
04 06 60	Schedules for Corrosion-Resistant Masonry	
04 06 70	Schedules for Manufactured Masonry	
04 08 00	Commissioning of Masonry	Includes: commissioning of items common to multiple titles in Division 04. See: 01 91 00 for commissioning of subjects common to multiple Divisions.
04 20 00	Unit Masonry	Includes: masonry unit construction, load-bearing and non-load-bearing, grouted and non-grouted, reinforced and non-reinforced. May Include: masonry mortaring, masonry grouting, masonry anchorage and reinforcing, and masonry accessories. Unit masonry retaining walls. See: 04 05 13 for masonry mortaring. 04 05 16 for masonry grouting. 04 05 19 for masonry anchorage and reinforcing. 04 05 23 for masonry accessories. 32 32 19 for unit masonry retaining walls.

NUMBER	TITLE	EXPLANATION
04 21 00	Clay Unit Masonry	Includes: load-bearing and non-load-bearing clay masonry. See: 04 27 00 for multi-wythe unit masonry. 04 54 00 for refractory brick masonry. 04 61 00 for chemical-resistant brick masonry. 04 71 00 for manufactured brick masonry. 09 63 13 for brick flooring. 12 11 00 for sculptured brick panels and brick murals. 13 49 16 for lead bricks. 32 14 16 for brick unit paving. Products: face brick, building brick.
04 21 13	Brick Masonry	
04 21 13.13	Brick Veneer Masonry	
04 21 13.23	Surface-Bonded Brick Masonry	
04 21 16	Ceramic Glazed Clay Masonry	
04 21 19	Clay Tile Masonry	
04 21 23	Structural Clay Tile Masonry	
04 21 26	Glazed Structural Clay Tile Masonry	
04 21 29	Terra Cotta Masonry	
04 22 00	Concrete Unit Masonry	Includes: load-bearing and non-load-bearing concrete unit masonry. May Include: integral water repellent. Alternate Terms/Abbreviations: CMU: concrete masonry unit, concrete block: cinder blocks. See: 04 27 00 for multi-wythe unit masonry. 07 19 00 for water repellents. 07 76 00 for precast concrete roof pavers. 31 35 23 for CMU slope protection paving. 31 37 16 for CMU riprap. 32 14 13 for precast concrete unit paving. 32 32 23 for segmental CMU retaining walls. 35 31 29 for CMU revetments, jetties, and groins. 35 42 53 for CMU wall waterway bank protection. 35 43 53 for CMU wall waterway scour protection. Products: concrete brick, pumice block.
04 22 00.13	Concrete Unit Veneer Masonry	
04 22 00.16	Surface-Bonded Concrete Unit Masonry	
04 22 19	Insulated Concrete Unit Masonry	
04 22 23	Architectural Concrete Unit Masonry	
04 22 23.13	Exposed Aggregate Concrete Unit Masonry	

NUMBER	TITLE	EXPLANATION
04 22 23.16	Fluted Concrete Unit Masonry	
04 22 23.19	Molded-Face Concrete Unit Masonry	
04 22 23.23	Prefaced Concrete Unit Masonry	
04 22 23.26	Sound-Absorbing Concrete Unit Masonry	
04 22 23.29	Split-Face Concrete Unit Masonry	
04 22 26	Autoclaved Aerated Concrete Unit Masonry	
04 22 33	Interlocking Concrete Unit Masonry	
04 23 00	Glass Unit Masonry	Includes: glass masonry unit construction. See: 08 60 00 for skylights. 09 76 00 for plastic blocks. See Also: 08 55 13 for tornado-resistant windows. 08 55 23 for blast-resistant windows.
04 23 13	Vertical Glass Unit Masonry	
04 23 16	Glass Unit Masonry Floors	
04 23 19	Glass Unit Masonry Skylights	
04 24 00	Adobe Unit Masonry	Includes: adobe masonry unit construction. See: 04 70 00 for manufactured masonry. 09 25 26 for natural clay plastering.
04 24 13	Site-Cast Adobe Unit Masonry	
04 24 16	Manufactured Adobe Unit Masonry	
04 25 00	Unit Masonry Panels	Includes: masonry assemblies erected as panels. May Include: sculptured brick panels and brick murals. See: 04 42 00 for panelized exterior stone cladding. 07 42 63 for fabricated wall panel assemblies. 12 11 16 for sculptured brick panels and brick murals.
04 25 13	Metal-Supported Unit Masonry Panels	
04 26 00	Single-Wythe Unit Masonry	
04 26 13	Masonry Veneer	

NUMBER	TITLE	EXPLANATION
04 27 00	Multiple-Wythe Unit Masonry	Includes: masonry composed of various masonry units. May Include: non-masonry components such as insulation and flashing. See: 07 21 00 for thermal insulation. 07 60 00 for flashing and sheet metal.
04 27 13	Composite Unit Masonry	
04 27 23	Cavity Wall Unit Masonry	
04 28 00	Concrete Form Masonry Units	Includes: masonry units used as a permanent form for cast-in-place concrete. May Include: non-masonry components such as insulation and flashing. See: 07 21 00 for thermal insulation. 07 60 00 for flashing and sheet metal.
04 28 13	Dry-Stacked, Concrete-Filled Masonry Units	
04 28 23	Mortar-Set, Concrete-Filled Masonry Units	
04 29 00	Engineered Unit Masonry	Includes: unit masonry assemblies with engineered reinforcing members. Alternate Terms/Abbreviations: reinforced unit masonry.

NUMBER	TITLE	EXPLANATION
04 40 00	Stone Assemblies	Includes: natural and cut stone assembly construction. May Include: masonry mortaring, masonry grouting, masonry anchorage and reinforcing, and masonry accessories. See: 09 30 33 for stone tiling. 09 63 40 for stone flooring. 09 75 00 for interior wall stone facing. 10 21 13 for stone toilet compartments. 12 36 40 for stone countertops. 31 37 16 for stone riprap. 32 14 40 for stone paving. 32 15 40 for crushed stone surfacing. 32 16 13 for stone curbs. 35 31 29 for stone seawalls, revetments, jetties, and groins. 35 42 53 for stone wall waterway bank protection. 35 43 53 for stone wall waterway scour protection. See Also: 04 05 13 for masonry mortaring. 04 05 16 for masonry grouting. 04 05 19 for masonry anchorage and reinforcing. 04 05 23 for masonry accessories. Products: stone materials: bluestone, granite, limestone, marble, quartzite, sandstone, serpentine, travertine, and slate. Collected natural stone: excavated stone, field stone, and riverbed stone. Quarried cut stone: cleft-face stone and dimension stone.
04 41 00	Dry-Placed Stone	Includes: stone laid without mortar. See: 35 31 29 for stone revetments, jetties, and groins.
04 42 00	Exterior Stone Cladding	Includes: fabricated stone panels mechanically attached to supporting backup or installed within a supporting system. See: 04 25 00 for unit masonry panels. 04 25 13 for metal-supported unit masonry panels. 07 44 00 for faced roof and wall panels. 09 75 00 for interior stone facing.
04 42 13	Masonry-Supported Stone Cladding	
04 42 16	Steel-Stud-Supported Stone Cladding	
04 42 19	Strongback-Frame-Supported Stone Cladding	
04 42 23	Truss-Supported Stone Cladding	
04 42 26	Grid-System-Supported Stone Cladding	

NUMBER	TITLE	EXPLANATION
04 42 43	Stone Panels for Curtain Walls	
04 43 00	Stone Masonry	Includes: stone set with masonry mortar. See: 09 75 00 for interior stone facing.
04 43 13	Stone Masonry Veneer	
04 43 13.13	Anchored Stone Masonry Veneer	
04 43 13.16	Adhered Stone Masonry Veneer	
04 43 16	Stone Fabrications	
04 50 00	Refractory Masonry	Includes: refractory masonry unit construction used in chimneys and combustion chambers. May Include: masonry mortaring, masonry grouting, masonry anchorage and reinforcing, and masonry accessories. See: 04 05 13 for refractory masonry mortaring. 10 31 00 for manufactured fireplaces. 10 32 00 for fireplace specialties. 23 51 00 for breechings, chimneys, and stacks. 40 47 00 for refractories. See Also: 04 05 13 for masonry mortaring. 04 05 16 for masonry grouting. 04 05 19 for masonry anchorage and reinforcing. 04 05 23 for masonry accessories.
04 51 00	Flue Liner Masonry	
04 52 00	Combustion Chamber Masonry	
04 53 00	Castable Refractory Masonry	
04 54 00	Refractory Brick Masonry	
04 57 00	Masonry Fireplaces	
04 57 33	Modular Masonry Fireplaces	

NUMBER	TITLE	EXPLANATION
04 60 00	Corrosion-Resistant Masonry	Includes: masonry construction resistant to corrosion from chemicals and other elements. May Include: masonry mortaring, masonry grouting, masonry anchorage and reinforcing, and masonry accessories. See: 04 05 13 for chemical-resistant masonry mortaring. 04 05 16 for chemical-resistant masonry grouting. 09 35 00 for chemical-resistant tiling. 09 63 13 for chemical-resistant brick flooring. 09 96 35 for chemical-resistant coatings. See Also: 04 05 13 for masonry mortaring. 04 05 16 for masonry grouting. 04 05 19 for masonry anchorage and reinforcing. 04 05 23 for masonry accessories.
04 61 00	Chemical-Resistant Brick Masonry	
04 62 00	Vitrified Clay Liner Plate	
04 70 00	Manufactured Masonry	Includes: hydraulically compressed veneer masonry units and artificial stone products. Various types of manufactured masonry. Alternate Terms/Abbreviations: manufactured masonry: simulated masonry. Manufactured stone: artificial stone: simulated stone. See: 03 49 00 for glass-fiber-reinforced concrete. 04 24 16 for manufactured adobe unit masonry. 06 61 13 for plastic simulated stone fabrications. 06 82 00 for glass-fiber-reinforced plastic fabrications.
04 71 00	Manufactured Brick Masonry	
04 71 13	Calcium Silicate Manufactured Brick Masonry	
04 72 00	Cast Stone Masonry	
04 73 00	Manufactured Stone Masonry	
04 73 13	Calcium Silicate Manufactured Stone Masonry	

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NUMBER	TITLE	EXPLANATION
DIVISION 05 – METALS		
05 00 00	Metals	May be used as Division level Section title.

05 01 00 Maintenance of Metals

Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of metals.

Level 4 Numbering Recommendation: one Level 4 title for metals is provided as an example for user's creation of more detailed titles for other subjects as required. Following Level 4 numbering is recommended:

.51-.59 for maintenance.

.61-.69 for repair.

.71-.79 for rehabilitation.

.81-.89 for replacement.

.91-.99 for restoration.

See: 02 25 23 for existing metals assessment.

05 01 10	Maintenance of Structural Metal Framing
05 01 20	Maintenance of Metal Joists
05 01 30	Maintenance of Metal Decking
05 01 40	Maintenance of Cold-Formed Metal Framing
05 01 50	Maintenance of Metal Fabrications
05 01 70	Maintenance of Decorative Metal
05 01 70.91	Historic Treatment of Decorative Metal

05 03 00 Conservation Treatment for Period Metals

05 03 12	Conservation Treatment for Period Structural Steel
05 03 12.13	Conservation Treatment for Period Architecturally Exposed Steel
05 03 12.23	Conservation Treatment for Period Structural Steel for Buildings
05 03 51	Conservation Treatment for Period Metal Stairs
05 03 58	Conservation Treatment for Period Column Covers
05 03 70	Conservation Treatment for Period Decorative Metal
05 03 71	Conservation Treatment for Period Decorative Metal Stairs
05 03 73	Conservation Treatment for Period Decorative Metal Railings
05 03 75	Conservation Treatment for Period Decorative Formed Metal
05 03 76	Conservation Treatment for Period Decorative Forged Metal

05 05 00 Common Work Results for Metals

Includes: subjects common to multiple titles in Division 05.

See: 01 80 00 for performance requirements of subjects common to multiple titles.

05 05 05	Selective Demolition for Metals
	See: 02 41 19 for selective demolition for whole structure demolition.

DIVISION 05

NUMBER	TITLE	EXPLANATION
05 05 13	Shop-Applied Coatings for Metal	Includes: coatings applied in the shop or factory. Anodic coatings, fluoropolymer coatings, galvanic coatings, metallic coatings, paint coatings, porcelain enamel coatings, and powdered coatings. Metalizing. See: 06 05 83 for shop-applied wood coatings. 09 90 00 for field-applied coatings. 41 34 13 for anodizing equipment. 41 34 23 for coating equipment. 41 34 36 for electroplating equipment.
05 05 19	Post-Installed Concrete Anchors	See: 03 15 19 for cast-in concrete anchors.
05 05 23	Metal Fastenings	Includes: methods for fastening metals together and for joining metals to other metals, including bolting, brazing, chemical bonding, riveting, soldering, mechanical fasteners, and welding. See: 06 05 23 for wood, plastic, and composite fastenings. 41 36 19 for joining equipment. Products: anchor rods, anchor bolts, carriage bolts.
05 05 53	Security Metal Fastenings	
05 06 00	Schedules for Metals	Includes: schedules of items common to multiple titles in Division 05. Notes: a schedule may be included on drawings, in the project manual, or a project book.
05 06 10	Schedules for Structural Metal Framing	
05 06 10.13	Steel Column Schedule	
05 06 10.16	Steel Beam Schedule	
05 06 20	Schedules for Metal Joists	
05 06 20.13	Steel Joist Schedule	
05 06 30	Schedules for Metal Decking	
05 06 40	Schedules for Cold-Formed Metal Framing	
05 06 50	Schedules for Metal Fabrications	
05 06 70	Schedules for Decorative Metal	
05 08 00	Commissioning of Metals	Includes: commissioning of items common to multiple titles in Division 05. See: 01 91 00 for commissioning of subjects common to multiple divisions.

NUMBER	TITLE	EXPLANATION
05 10 00	Structural Metal Framing	Includes: structural metal elements. Permanent bracing and bridge members. Metal columns, beams, girders, joists, purlins, studs, trusses, and other metal framing. Usually Includes: related fastenings and shop-applied coatings. See: 07 95 53 for joint slide bearings. 13 32 00 for space frames. 13 33 00 for geodesic structures. See Also: 05 05 13 for shop-applied coatings for metal. 05 05 23 for metal fastenings.
05 12 00	Structural Steel Framing	Includes: structural metal framing fabricated from structural steel. Alternate Terms/Abbreviations: BRB: buckling restrained braces. See: 05 16 33 for bridge cabling. 05 40 00 for cold-formed metal framing. 07 95 63 for bridge expansion joint cover assemblies. 09 21 00 for non-structural framing in plaster and gypsum board assemblies. 09 22 16 for non-structural framing. 09 22 26 for suspension systems. 09 53 00 for acoustical ceiling suspension assemblies. 13 34 00 for fabricated engineered structures. 32 34 00 for fabricated bridges. 34 81 00 for bridge machinery. 34 82 00 for bridge specialties.
05 12 13	Architecturally-Exposed Structural Steel Framing	
05 12 16	Fabricated Fire Protected Steel Columns	
05 12 19	Buckling Restrained Braces	Alternate Terms/Abbreviations: BRB: buckling restrained braces.
05 12 23	Structural Steel for Buildings	
05 12 33	Structural Steel for Bridges	
05 13 00	Structural Stainless-Steel Framing	Includes: structural metal framing fabricated from stainless steel.

NUMBER	TITLE	EXPLANATION
05 14 00	Structural Aluminum Framing	Includes: structural metal framing fabricated from aluminum.
05 14 13	Architecturally-Exposed Structural Aluminum Framing	
05 15 00	Wire Rope Assemblies	Includes: non-structural assemblies composed of wire rope.
		Alternate Terms/Abbreviations: wire rope: wire cable.
		See: 05 16 00 for structural cabling. 05 73 16 for wire rope decorative metal railings. 34 14 00 for cable transportation.
05 15 13	Aluminum Wire Rope Assemblies	
05 15 16	Steel Wire Rope Assemblies	
05 15 19	Stainless-Steel Wire Rope Assemblies	
05 16 00	Structural Cabling	See: 05 12 33 for structural steel for bridges. 05 15 00 for non-structural wire rope assemblies. 08 88 23 for cable suspended glazing. 13 31 23 for tensioned fabric structures. 32 34 00 for fabricated bridges. 34 14 00 for cable transportation.
05 16 13	Cable Bow Truss Assemblies	
05 16 33	Bridge Cabling	
05 17 00	Structural Rod Assemblies	
05 19 00	Tension Rod and Cable Truss Assemblies	
05 19 13	Façade Support Truss Assemblies	
05 19 19	Canopy Support Truss Assemblies	
05 20 00	Metal Joists	Includes: metal joists, bridging, anchors, and accessories.
		Usually Includes: related fastenings and shop-applied coatings.
		Alternate Terms/Abbreviations: metal joists: bar joists.
		See: 05 05 13 for shop-applied coatings for metal. 05 05 23 for metal fastenings.

NUMBER	TITLE	EXPLANATION
05 21 00	Steel Joist Framing	Includes: metal joists fabricated from structural steel. Alternate Terms/Abbreviations: steel joists: steel bar joists. See: 05 25 00 for aluminum joist framing. 05 42 00 for cold-formed metal joist framing. 06 17 36 for metal-web wood joists.
05 21 13	Deep Longspan Steel Joist Framing	
05 21 16	Longspan Steel Joist Framing	
05 21 19	Open Web Steel Joist Framing	
05 21 23	Steel Joist Girder Framing	
05 25 00	Aluminum Joist Framing	Includes: metal joists fabricated from aluminum. Alternate Terms/Abbreviations: aluminum joists: aluminum bar joists. See: 05 21 00 for steel joist framing. 05 42 00 for cold-formed metal joist framing. 06 17 36 for metal-web wood joists.
05 30 00	Metal Decking	Includes: metal floor and roof deck of fluted, ribbed, and cellular configurations; composite decks; anchors, closures, and related accessories. Metal decking used as form for concrete or lightweight insulating concrete fill. Usually Includes: related fastenings and shop-applied coatings. See: 07 40 00 for roofing and siding panels. See Also: 03 11 00 for concrete forming. 03 52 00 for metal deck used to support lightweight concrete roof insulation. 05 05 13 for shop-applied coatings for metal. 05 05 23 for metal fastenings.
05 31 00	Steel Decking	Includes: metal deck fabricated from steel. May Include: steel decking used as form for concrete or lightweight insulating concrete fill. See Also: 03 52 00 for corrugated permanent metal forms used to support lightweight concrete roof insulation.

DIVISION **05**

NUMBER	TITLE	EXPLANATION
05 31 13	Steel Floor Decking	
05 31 23	Steel Roof Decking	
05 31 33	Steel Form Decking	
05 33 00	Aluminum Decking	Includes: metal deck fabricated from aluminum. May Include: aluminum decking used as form for concrete or lightweight insulating concrete fill. See: 03 52 00 for corrugated permanent metal forms used to support lightweight concrete roof insulation.
05 33 13	Aluminum Floor Decking	
05 33 23	Aluminum Roof Decking	
05 34 00	Acoustical Metal Decking	Includes: metal deck fabricated to include acoustical insulation. See: 09 51 00 for acoustical ceilings.
05 35 00	Raceway Decking Assemblies	Includes: metal deck fabricated to provide raceway for power and communications wiring. Alternate Terms/Abbreviations: raceway decking: electrified floors: cellular decking. See: 26 05 19 for undercarpet electrical power cables. 26 05 39 for underfloor raceways for electrical systems.
05 36 00	Composite Metal Decking	Includes: decks fabricated from a composite of materials. See: 03 52 00 for corrugated permanent metal forms used to support lightweight concrete roof insulation.
05 36 13	Composite Steel Plate and Elastomer Decking	

NUMBER	TITLE	EXPLANATION
05 40 00	Cold-Formed Metal Framing	Includes: bracing, fasteners, and related accessories for light-gage, load-bearing metal elements. Usually Includes: shop-applied coatings. May Include: framing for plaster and gypsum board systems. See: 05 10 00 for structural metal framing. 09 22 26 for suspension systems. 09 53 00 for acoustical ceiling suspension assemblies. See Also: 05 05 13 for shop-applied coatings for metal. 05 05 23 for metal fastenings. 09 21 00 for non-structural framing in plaster and gypsum board assemblies. 09 22 16 for non-structural metal framing.
05 41 00	Structural Metal Stud Framing	Includes: cold-formed metal framing intended for structural applications. Alternate Terms/Abbreviations: structural: loadbearing. See: 09 21 00 for non-structural metal framing in plaster and gypsum board assemblies. 09 22 16 for non-structural metal framing.
05 42 00	Cold-Formed Metal Joist Framing	Includes: cold-formed metal framing intended for floor, roof, and ceiling joist applications. See: 05 20 00 for metal joists. 05 44 00 for cold-formed metal trusses.
05 42 13	Cold-Formed Metal Floor Joist Framing	
05 42 23	Cold-Formed Metal Roof Joist Framing	
05 43 00	Slotted Channel Framing	Includes: cold-formed metal framing fabricated as slotted channel framing.
05 44 00	Cold-Formed Metal Trusses	Includes: cold-formed metal framing fabricated as trusses. See: 05 42 00 for cold-formed metal joist framing.
05 44 13	Cold-Formed Metal Roof Trusses	

NUMBER	TITLE	EXPLANATION
05 45 00	Metal Support Assemblies	Includes: cold-formed metal support framing for electrical, communications, mechanical, healthcare, and other items. See: 21 05 29 for hangers and supports for fire-suppression piping and equipment. 22 05 29 for hangers and supports for plumbing piping and equipment. 23 05 29 for hangers and supports for HVAC piping and equipment. 25 05 28 for hangers and supports for integrated automation. 26 05 29 for hangers and supports for electrical systems. 27 05 28 for hangers and supports for communications systems. 27 05 28 for pathways for communications systems.
05 45 13	Mechanical Metal Supports	
05 45 16	Electrical Metal Supports	
05 45 19	Communications Metal Supports	
05 45 23	Healthcare Metal Supports	

05 50 00 Metal Fabrications

Includes: metal items fabricated from standard metal shapes and plates that are not classified in other locations.

Usually Includes: related fastenings and shop-applied coatings.

See: 05 70 00 for decorative metals.
07 60 00 for flashing and sheet metal.
10 26 13 for manufactured corner guards.
11 98 12 for detention doors and frames.
12 48 13 for entrance floor mats frames.
32 31 00 for fences and gates.

See Also: 05 05 13 for shop-applied coatings for metal.
05 05 23 for metal fastenings.

NUMBER	TITLE	EXPLANATION
05 51 00	Metal Stairs	Includes: fabricated stairs, catwalks, ladders, and fire escapes of various types of a utilitarian nature. Alternate Terms/Abbreviations: inclined ladders: ships ladders. See: 03 11 23 for permanent stair forming. 03 41 23 for precast concrete stairs. 05 53 00 for metal gratings. 05 54 00 for metal floor plates. 05 55 00 for metal stair treads and nosings. 05 71 00 for decorative metal stairs. 05 71 13 for prefabricated spiral stairs. 06 43 00 for wood stairs and railings. 13 44 00 for modular mezzanines. 14 91 23 for escape chutes.
05 51 13	Metal Pan Stairs	
05 51 16	Metal Floor Plate Stairs	
05 51 17	Alternating Tread Stairs	
05 51 19	Metal Grating Stairs	
05 51 23	Metal Fire Escapes	
05 51 33	Metal Ladders	
05 51 33.13	Vertical Metal Ladders	
05 51 33.16	Inclined Metal Ladders	
05 51 33.23	Alternating Tread Ladders	
05 51 36	Metal Walkways	Includes: walkways and ramps.
05 51 36.13	Metal Catwalks	
05 51 36.16	Metal Ramps	
05 51 36.19	Metal Platforms	Includes: walkways, ramps, and fixed platforms.
05 52 00	Metal Railings	Includes: metal handrails and railings of various types of a utilitarian nature. Alternate Terms/Abbreviations: railings: balustrades. See: 05 73 00 for decorative metal railings. 10 26 16 for protective corridor handrails.
05 52 13	Pipe and Tube Railings	

NUMBER	TITLE	EXPLANATION
05 53 00	Metal Gratings	Includes: metal floor and traffic gratings. See: 05 51 19 for metal grating stairs. 12 48 19 for entrance floor gratings. 32 94 43 for tree grates. 33 44 00 for storm utility water drains.
05 53 13	Bar Gratings	
05 53 16	Plank Gratings	
05 53 19	Expanded Metal Gratings	Includes: expanded metal gratings, metal frames, fasteners, and supports.
05 54 00	Metal Floor Plates	Includes: metal floor plates and plates in paving. See: 05 51 16 for metal floor plate stairs.
05 55 00	Metal Stair Treads and Nosings	Includes: metal stair treads and nosings. See: 03 30 00 for CIP concrete stairs. 03 41 23 for precast concrete stairs. 05 51 00 for metal stairs.
05 55 13	Metal Stair Treads	
05 55 16	Metal Stair Nosings	
05 56 00	Metal Castings	Includes: metal castings of a utilitarian nature not associated with site utilities. See: 05 74 00 for decorative metal castings. 32 94 43 for tree grates. 33 40 00 for storm drainage utilities.
05 58 00	Formed Metal Fabrications	Includes: custom formed metal items for special purposes other than roofing and flashing applications. Custom enclosures for columns and heating and cooling units and piping. See: 05 75 00 for decorative formed metal. 07 60 00 for flashing and sheet metal. 12 36 16 for metal countertops. 23 82 00 for standard enclosures for convection heating and cooling units.

NUMBER	TITLE	EXPLANATION
05 58 13	Column Covers	
05 58 16	Formed Metal Enclosures	
05 58 19	Heating/Cooling Unit Covers	
05 58 23	Formed Metal Guards	

05 59 00 Metal Specialties

Includes: specialties fabricated of standard metal shapes for specific purposes.

Alternate Terms/Abbreviations: detention enclosures: jail cells: penitentiary cells: prison cells.

See: 10 00 00 for specialties.

11 98 12 for detention doors and frames.

11 98 14 for detention door hardware.

11 98 21 for detention windows.

13 42 63 for detention cell modules.

28 52 00 for detention security systems.

28 52 11 for detention security system requirements.

32 34 00 for fabricated bridges.

05 59 13	Metal Balconies
05 59 63	Detention Enclosures

05 70 00 Decorative Metal

Includes: metal items and metal work of a decorative nature. Wrought iron, muntz metal. Anchors and related accessories.

Usually Includes: related fastenings and shop-applied coatings.

Alternate Terms/Abbreviations: decorative metal: ornamental metal. Muntz metal: yellow metal.

See: 05 50 00 for metal fabrications of a utilitarian nature.

12 14 00 for sculptures.

32 31 19 for decorative metal fences and gates.

See Also: 05 05 13 for shop-applied coatings for metal.
05 05 23 for metal fastenings.

05 71 00 Decorative Metal Stairs

Includes: fabricated stairs of various types of a decorative nature.

See: 03 11 23 for permanent stair forming.

03 41 23 for precast concrete stairs.

05 51 00 for metal stairs of a utilitarian nature.

06 43 00 for wood stairs and railings.

NUMBER	TITLE	EXPLANATION
05 71 13	Fabricated Metal Spiral Stairs	
05 73 00	Decorative Metal Railings	Includes: metal handrails and railings of various types of a decorative nature. See: 05 15 00 for wire rope assemblies. 05 52 00 for metal railings of a utilitarian nature. 10 26 16 for protective corridor handrails.
05 73 13	Glazed Decorative Metal Railings	
05 73 16	Wire Rope Decorative Metal Railings	
05 74 00	Decorative Metal Castings	Includes: metal castings of a decorative nature not associated with site utilities. See: 05 56 00 for metal castings of a utilitarian nature. 32 94 43 for tree grates. 33 40 00 for storm drainage utilities.
05 75 00	Decorative Formed Metal	Includes: custom formed metal items of a decorative nature for special purposes other than roofing and flashing applications. See: 05 58 00 for formed metal fabrications. 07 60 00 for flashing and sheet metal. 12 36 16 for metal countertops.
05 76 00	Decorative Forged Metal	Includes: forged metal items of a decorative nature for any purpose.
05 77 00	Decorative Extruded Metal	

NUMBER	TITLE	EXPLANATION
DIVISION 06 – WOOD, PLASTICS, AND COMPOSITES		
06 00 00	Wood, Plastics, and Composites	May be used as Division level Section title.

06 01 00	Maintenance of Wood, Plastics, and Composites	Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of wood, plastics, and composites.
		Level 4 Numbering Recommendation: Level 4 titles for wood, plastics, and composites are provided as examples for user's creation of more detailed titles for other subjects as required. Following Level 4 numbering is recommended: .51-.59 for maintenance. .61-.69 for repair. .71-.79 for rehabilitation. .81-.89 for replacement. .91-.99 for restoration.
		See: 02 25 26 for existing wood, plastics, and composites assessment.
06 01 10	Maintenance of Rough Carpentry	
06 01 10.71	Rough Carpentry Rehabilitation	
06 01 10.91	Rough Carpentry Restoration	
06 01 10.92	Rough Carpentry Preservation	
06 01 20	Maintenance of Finish Carpentry	
06 01 20.71	Finish Carpentry Rehabilitation	
06 01 20.91	Finish Carpentry Restoration	
06 01 20.92	Finish Carpentry Preservation	
06 01 40	Maintenance of Architectural Woodwork	
06 01 40.51	Architectural Woodwork Cleaning	
06 01 40.61	Architectural Woodwork Refinishing	
06 01 40.91	Architectural Woodwork Restoration	
06 01 50	Maintenance of Structural Plastics	
06 01 60	Maintenance of Plastic Fabrications	
06 01 60.51	Plastic Cleaning	
06 01 60.71	Plastic Rehabilitation	
06 01 60.91	Plastic Restoration	
06 01 60.92	Plastic Preservation	
06 01 70	Maintenance of Structural Composites	
06 01 80	Maintenance of Composite Assemblies	
06 01 80.51	Composite Cleaning	
06 01 80.71	Composite Rehabilitation	
06 01 80.91	Composite Restoration	
06 01 80.92	Composite Preservation	

DIVISION 06

NUMBER	TITLE	EXPLANATION
06 03 00	Conservation Treatment for Period Wood	
06 03 01	Maintenance for Period Wood	
06 03 01.31	Period Wood Storage and Protection	
06 03 05	Common Work Results for Period Wood	
06 03 05.73	Long-term Period Wood Treatment	
06 03 05.74	Eradication of Insects in Period Wood	
06 03 05.75	Antiseptic Treatment of Period Wood	
06 03 13	Conservation Treatment of Period Log Construction	
06 03 13.23	Conservation Treatment for Period Heavy Timber	
06 03 20	Conservation Treatment for Period Finish Carpentry	
06 03 20.13	Period Exterior Finish Carpentry	
06 03 20.23	Period Interior Finish Carpentry	
06 03 40	Conservation Treatment for Period Architectural Woodwork	
06 03 40.13	Period Exterior Architectural Woodwork	
06 03 40.23	Period Interior Architectural Woodwork	
06 05 00	Common Work Results for Wood, Plastics, and Composites	
		Includes: subjects common to multiple titles in Division 06.
		See: 01 80 00 for performance requirements of subjects common to multiple titles.
06 05 05	Selective Demolition for Wood, Plastics, and Composites	See: 02 41 19 for selective demolition for whole structure demolition.
06 05 23	Wood, Plastic, and Composite Fastenings	Includes: methods for fastening wood, plastic, and composites together, including nailing, screwing, bolting, metal connectors, and adhesives.
		See: 05 05 23 for metal fastenings.
06 05 73	Wood Treatment	Includes: fire-retardant and preservative wood treatments.
		Alternate Terms/Abbreviations: wood treatment: pressure treatment.
		See: 09 96 43 for fire-retardant coatings.
06 05 73.13	Fire-Retardant Wood Treatment	
06 05 73.33	Preservative Wood Treatment	
06 05 73.91	Long-Term Wood Treatment	
06 05 73.93	Eradication of Insects in Wood	
06 05 73.96	Antiseptic Treatment of Wood	

NUMBER	TITLE	EXPLANATION
06 05 83	Shop-Applied Wood Coatings	Includes: coatings applied in the shop or factory. See: 05 05 13 for shop-applied coatings for metal. 09 90 00 for field-applied coatings.
06 06 00	Schedules for Wood, Plastics, and Composites	Includes: schedules of items common to multiple titles in Division 06. Notes: a schedule may be included on drawings, in the project manual, or a project book.
06 06 10	Schedules for Rough Carpentry	
06 06 10.13	Nailing Schedule	
06 06 10.16	Wood Beam Schedule	
06 06 10.19	Plywood Shear Wall Schedule	
06 06 10.23	Plywood Web Joist Schedule	
06 06 10.26	Wood Truss Schedule	
06 06 20	Schedules for Finish Carpentry	
06 06 40	Schedules for Architectural Woodwork	
06 06 50	Schedules for Structural Plastics	
06 06 60	Schedules for Plastic Fabrications	
06 06 70	Schedules for Structural Composites	
06 06 80	Schedules for Composite Assemblies	
06 08 00	Commissioning of Wood, Plastics, and Composites	Includes: commissioning of items common to multiple titles in Division 06. See: 01 91 00 for commissioning of subjects common to multiple Divisions.
06 10 00	Rough Carpentry	Includes: wood foundations, framing, sheathing, and decking using lumber, timber, and engineered wood products. Blocking and supports to join the members and anchor the framework to other construction. Erection of the units and incorporation of connectors and supports to join the units and anchor the units to other construction. Usually Includes: related fasteners, adhesives, and wood treatment. See: 06 05 23 for wood, plastic, and composite fastenings. 06 05 73 for wood treatment.
06 10 53	Miscellaneous Rough Carpentry	Includes: may be used for projects with a small amount of rough carpentry.

DIVISION **06**

NUMBER	TITLE	EXPLANATION
06 10 63	Exterior Rough Carpentry	
06 11 00	Wood Framing	Includes: mechanically-graded lumber for framing and those engineered lumber products that are available at lumber yards and that are intended to be cut to length in the field. Wood columns, beams, girders, joists, purlins, rafters, studs, bracing, and other wood framing. See: 06 17 00 for shop-fabricated structural wood. 13 32 00 for space frames. 13 33 00 for geodesic structures.
06 11 13	Engineered Wood Products	
06 11 16	Mechanically Graded Lumber	
06 12 00	Structural Panels	Includes: reinforced and unreinforced wood-based structural panels. Structural insulated panels. See: 06 16 00 for sheathing. 07 44 53 for glass-fiber-reinforced cementitious panels. 07 44 56 for mineral-fiber-reinforced cementitious panels.
06 12 13	Cementitious Reinforced Panels	
06 12 16	Stressed Skin Panels	
06 13 00	Heavy Timber Construction	Includes: framing using timber with a minimum dimension of five inches. Alternate Terms/Abbreviations: heavy timber: mill-framed structures. See: 06 15 19 for timber decking. 06 17 00 for shop-fabricated structural wood. 06 18 00 for glue-laminated construction. 31 62 19 for timber piles. 32 32 29 for timber retaining walls. 32 34 00 for fabricated bridges. 32 94 16 for landscape timbers. 34 11 33 for track cross ties. 35 51 13 for floating wood piers.
06 13 13	Log Construction	
06 13 13.91	Period Horizontal Log Work	
06 13 16	Pole Construction	
06 13 23	Heavy Timber Framing	
06 13 26	Heavy Timber Trusses	

NUMBER	TITLE	EXPLANATION
06 13 33	Heavy Timber Pier Construction	
06 14 00	Treated Wood Foundations	Includes: treated wood support structures installed as foundations. See: 31 62 19 for timber piles.
06 15 00	Wood Decking	Includes: flooring and decking using lumber and timber. See: 06 13 00 for heavy timber. 06 18 00 for glue-laminated construction. 06 53 00 for plastic decking. 06 73 00 for composite decking. 09 64 00 for wood flooring.
06 15 13	Wood Floor Decking	
06 15 13.91	Carvel Planking	
06 15 16	Wood Roof Decking	
06 15 19	Timber Decking	
06 15 23	Laminated Wood Decking	
06 15 33	Wood Patio Decking	
06 16 00	Sheathing	Includes: sheathing, subflooring, and underlayment. Alternate Terms/Abbreviations: OSB: oriented strand board; chipboard: flake board. Plywood: sturd-i-floor. Structural-use panels: structural wood panel. See: 03 54 00 for cast underlayment. 06 12 00 for structural panels. 07 25 00 for weather barriers. 09 60 00 for flooring. 09 62 48 for acoustic flooring. See Also: 09 29 00 for gypsum sheathing. Products: boards, structural-use panels, oriented strand board, particleboard, hardboard, fiberboard, and other wood-based construction panels.
06 16 13	Insulating Sheathing	
06 16 23	Subflooring	
06 16 26	Underlayment	
06 16 33	Wood Board Sheathing	
06 16 36	Wood Panel Product Sheathing	
06 16 43	Gypsum Sheathing	

DIVISION **06**

NUMBER	TITLE	EXPLANATION
06 16 53	Moisture-Resistant Sheathing Board	
06 16 63	Cementitious Sheathing	

06 17 00 Shop-Fabricated Structural Wood

Includes: columns, beams, joists, trusses, and other structural units fabricated from wood or wood in combination with metal elements.

Alternate Terms/Abbreviations: LVL: laminated veneer lumber.
PSL: parallel strand lumber.

See: 05 20 00 for metal joists.
06 18 00 for glued-laminated construction.
13 32 00 for space frames.
13 33 00 for geodesic structures.

06 17 13	Laminated Veneer Lumber
06 17 19	Cross-Laminated Timber
06 17 23	Parallel Strand Lumber
06 17 33	Wood I-Joists
06 17 36	Metal-Web Wood Joists
06 17 43	Rim Boards
06 17 53	Shop-Fabricated Wood Trusses

06 18 00 Glued-Laminated Construction

Includes: columns, beams, decking, and other structural units fabricated glue-laminated construction.

Alternate Terms/Abbreviations: GLULAM: glued-laminated construction.

See: 06 15 23 for laminated wood decking.
06 17 13 for laminated veneer lumber.
06 17 23 for parallel strand lumber.

06 18 13	Glued-Laminated Beams
06 18 16	Glued-Laminated Columns

NUMBER	TITLE	EXPLANATION
06 20 00	Finish Carpentry	Includes: finish wood construction that can be purchased ready-made and installed by a finish carpenter, not requiring the specialized skills of architectural woodwork fabrication. Usually Includes: related fasteners, adhesives, and wood treatment. See: 06 40 00 for architectural woodwork. 07 40 00 for wood roofing and siding panels. 10 57 23 for manufactured wood closet and utility shelving. 12 36 19 for wood countertops. See Also: 06 05 23 for wood, plastic, and composite fastenings. 06 05 73 for wood treatment. 06 05 83 for shop-applied wood coatings.
06 20 13	Exterior Finish Carpentry	
06 20 23	Interior Finish Carpentry	
06 22 00	Millwork	Includes: standard pattern wood trim and moldings. See: 06 44 00 for ornamental woodwork. 06 46 00 for shop-fabricated wood trim. 06 48 00 for wood frames.
06 22 13	Standard Pattern Wood Trim	
06 25 00	Prefinished Paneling	Includes: prefinished standard hardboard paneling and other ready-made wood product paneling. See: 06 26 00 for board paneling. 06 42 00 for shop-fabricated wood paneling. 09 74 00 for flexible wood sheets. See Also: 09 78 00 for interior wall paneling. 09 78 13 for metal interior wall paneling.
06 25 13	Prefinished Hardboard Paneling	
06 25 16	Prefinished Plywood Paneling	

NUMBER	TITLE	EXPLANATION
06 26 00	Board Paneling	Includes: standard profile board paneling. See: 06 25 00 for prefinished paneling. 06 42 00 for shop-fabricated wood paneling. 09 74 00 for flexible wood sheets. See Also: 09 78 00 for interior wall paneling. 09 78 13 for metal interior wall paneling.
06 26 13	Profile Board Paneling	
06 40 00	Architectural Woodwork	Includes: shop-fabricated wood, laminate-clad, and simulated wood fabrications. Veneers and flitches. Usually Includes: related fasteners, adhesives, wood treatment, and shop-applied finishes. See: 09 54 26 for suspended wood ceilings. 09 64 00 for wood flooring. 09 90 00 for field-applied coatings. 09 96 43 for fire-retardant coatings. 10 21 00 for laminate-clad compartments and cubicles. 10 51 00 for wood and laminate-clad lockers. 10 82 00 for manufactured wood grilles and screens. See Also: 06 05 23 for wood, plastic, and composite fastenings. 06 05 73 for wood treatment. 06 05 83 for shop-applied wood coatings.
06 40 13	Exterior Architectural Woodwork	
06 40 23	Interior Architectural Woodwork	
06 41 00	Architectural Wood Casework	Includes: shop-fabricated cabinets, cases, and fixtures. Cabinet hardware. May Include: countertops and sinks integral with countertops. Alternate Terms/Abbreviations: PLAM: plastic laminate. See: 06 61 13 for cultured marble fabrications. 06 61 16 for solid surfacing fabrications. 12 30 00 for manufactured casework. See Also: 12 36 00 for countertops.

NUMBER	TITLE	EXPLANATION
06 41 13	Wood-Veneer-Faced Architectural Cabinets	
06 41 16	Plastic-Laminate-Clad Architectural Cabinets	
06 41 93	Cabinet and Drawer Hardware	
06 42 00	Wood Paneling	Includes: shop-fabricated wall paneling fabricated with solid lumber, wood-veneer panel products, flush wood paneling, and plastic laminate. Veneers and flitches. See: 06 25 00 for prefinished paneling. 06 26 00 for standard profile board paneling. 09 74 00 for flexible wood sheets.
06 42 13	Wood Board Paneling	
06 42 14	Stile and Rail Wood Paneling	
06 42 16	Flush Wood Paneling	
06 42 19	Plastic-Laminate-Faced Wood Paneling	
06 43 00	Wood Stairs and Railings	Includes: stairs, stringers, treads, nosings, risers, handrails, railings, and related items. See: 03 41 23 for precast concrete stairs. 05 51 00 for metal stairs. 05 71 00 for decorative metal stairs. 10 26 16 for protective corridor handrails.
06 43 13	Wood Stairs	
06 43 16	Wood Railings	
06 44 00	Ornamental Woodwork	Includes: shop-fabricated ornamental woodwork items of unique design fabricated for specific application. See: 10 74 23 for manufactured cupolas.
06 44 13	Wood Turnings	
06 44 16	Wood Pilasters	
06 44 19	Wood Grilles	
06 44 23	Wood Corbels	
06 44 26	Wood Cupolas	
06 44 29	Wood Finials	
06 44 33	Wood Mantels	
06 44 36	Wood Pediment Heads	
06 44 39	Wood Posts and Columns	

NUMBER	TITLE	EXPLANATION
06 46 00	Wood Trim	Includes: shop-fabricated standing and running trim. See: 06 22 13 for standard pattern wood trim. 10 26 16 for protective corridor handrails.
06 46 13	Wood Door and Window Casings	
06 46 16	Wood Aprons	
06 46 19	Wood Base and Shoe Moldings	
06 46 23	Wood Chair Rails	
06 46 26	Wood Cornices	
06 46 29	Wood Fasciae and Soffits	
06 46 33	Wood Stops, Stools, and Sills	
06 46 91	Splicing of Wooden Components	
06 48 00	Wood Frames	Includes: interior and exterior wood frames for doors, sidelights, transoms, and borrowed lights. Wood veneer frames and stick-built windows. See: 08 14 00 for wood doors. 08 52 00 for manufactured wood windows.
06 48 13	Exterior Wood Door Frames	
06 48 16	Interior Wood Door Frames	
06 48 19	Ornamental Wood Frames	
06 48 23	Stick-Built Wood Windows	
06 48 26	Wood-Veneer Frames	
06 49 00	Wood Screens and Shutters	Includes: shop-fabricated wood framed screens, blinds, and shutters. See: 08 91 00 for wood louvers. 10 82 00 for manufactured wood grilles and screens. 32 35 13 for self-supporting exterior screens and louvers.
06 49 13	Wood Screens	
06 49 16	Wood Blinds	See: 12 21 13 for horizontal louver blinds.
06 49 19	Wood Shutters	See: 12 23 00 for interior shutters.

NUMBER	TITLE	EXPLANATION
06 50 00	Structural Plastics	Includes: plastic lumber and other structural plastics. Usually Includes: related fasteners and adhesives. See: 06 40 00 for plastic laminate. 06 60 00 for plastic fabrications. 06 70 00 for structural composites. 06 80 00 for composite fabrications. See Also: 06 05 23 for wood, plastic, and composite fastenings.
06 51 00	Structural Plastic Shapes and Plates	Includes: plastics in form of framing members intended for structural purposes.
06 51 13	Plastic Lumber	
06 52 00	Plastic Structural Assemblies	Includes: plastics other than framing members intended for structural purposes. See: 07 40 00 for plastic roofing and siding panels.
06 53 00	Plastic Decking	Includes: flooring and decking fabricated of plastic. See: 06 15 00 for wood decking. 06 73 00 for composite decking.
06 53 13	Solid Plastic Decking	

NUMBER	TITLE	EXPLANATION
06 60 00	Plastic Fabrications	Includes: non-structural plastic fabrications formed to various profiles for a variety of purposes. See: 04 73 00 for manufactured stone masonry. 06 50 00 for structural plastics. 06 80 00 for composite fabrications. 07 31 53 for plastic shakes. 07 40 00 for plastic roofing and siding panels. 08 15 00 for plastic doors. 08 53 00 for plastic windows. 08 84 00 for plastic glazing. 09 30 26 for plastic tiling. 09 76 00 for plastic blocks. 10 21 00 for plastic compartments and cubicles. 10 51 26 for plastic lockers. 12 34 00 for manufactured plastic casework. 12 36 23 for plastic countertops. 12 36 61 for simulated stone countertops.
06 61 00	Cast Polymer Fabrications	
06 61 13	Simulated Stone Fabrications	Includes: cultured marble fabrications, quartz surfacing fabrications, simulated granite.
06 61 16	Solid Surfacing Fabrications	
06 62 00	Plastic Foam Fabrications	
06 63 00	Plastic Railings	
06 64 00	Plastic Paneling	
06 64 13	Plastic Lattice Paneling	
06 65 00	Plastic Trim	
06 66 00	Custom Ornamental Simulated Woodwork	

NUMBER	TITLE	EXPLANATION
06 70 00	Structural Composites	Includes: materials composed of various plastics with wood, glass, or carbon fibers formed into a composite material. Composite lumber and other structural composites. Usually Includes: related fasteners and adhesives. See: 06 40 00 for plastic laminate. 06 50 00 for structural plastics. 06 60 00 for plastic fabrications. 06 80 00 for composite fabrications. See Also: 06 05 23 for wood, plastic, and composite fastenings.
06 71 00	Structural Composite Shapes and Plates	Includes: composites in form of framing members for structural purposes. Fiberglass structural shapes. Products: fiberglass structural shapes.
06 71 13	Composite Lumber	
06 72 00	Composite Structural Assemblies	Includes: composites other than framing members intended for structural purposes. See: 07 40 00 for composite roofing and siding panels.
06 72 13	Composite Joist Assemblies	
06 72 23	Composite Stair Assemblies	
06 73 00	Composite Decking	Includes: decking fabricated of composites. See: 06 15 00 for wood decking. 06 53 00 for plastic decking.
06 73 13	Composite Structural Decking	
06 74 00	Composite Gratings	
06 74 13	Fiberglass Reinforced Gratings	

NUMBER	TITLE	EXPLANATION
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06 80 00 Composite Fabrications

Includes: materials composed of various plastics with wood, glass, or carbon fibers formed into a composite material. Non-structural composite fabrications formed to various profiles for a variety of purposes.

See: 03 49 00 for glass-fiber-reinforced concrete.

06 40 00 for plastic laminate.

06 50 00 for structural plastics.

06 60 00 for plastic fabrications.

06 70 00 for structural composites.

07 31 33 for composite rubber shingles.

07 40 00 for composite roofing and siding panels.

07 44 53 for glass-fiber-reinforced wall cementitious panels.

08 16 00 for composite doors.

08 54 00 for composite windows.

09 27 13 for glass-fiber-reinforced plaster fabrications.

06 81 00 Composite Railings

06 81 13 Glass-Fiber-Reinforced Plastic Railings

06 82 00 Composite Trim

06 83 00 Composite Paneling

06 83 13 Resin Composite Paneling

06 83 16 Fiberglass Reinforced Paneling

NUMBER	TITLE	EXPLANATION
DIVISION 07 – THERMAL AND MOISTURE PROTECTION		
07 00 00	Thermal and Moisture Protection	May be used as Division level Section title.

07 01 00 Operation and Maintenance of Thermal and Moisture Protection

Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of thermal and moisture protection.

Level 4 Numbering Recommendation: Level 4 titles for wood, plastics, and composites are provided as examples for user's creation of more detailed titles for other subjects as required.

Following Level 4 numbering is recommended:

.51-.59 for maintenance.

.61-.69 for repair.

.71-.79 for rehabilitation.

.81-.89 for replacement.

.91-.99 for restoration.

See: 02 25 29 for existing thermal and moisture protection assessment.

07 01 10	Maintenance of Dampproofing and Waterproofing
07 01 10.81	Waterproofing Replacement
07 01 20	Maintenance of Thermal Protection
07 01 30	Maintenance of Steep Slope Roofing
07 01 40	Maintenance of Roofing and Siding Panels
07 01 50	Maintenance of Membrane Roofing
07 01 50.13	Roof Moisture Survey
07 01 50.16	Roof Maintenance Program
07 01 50.19	Preparation for Re-Roofing
07 01 50.23	Roof Removal
07 01 50.61	Roof Re-Coating
07 01 50.81	Roof Replacement
07 01 50.91	Roofing Restoration
07 01 60	Maintenance of Flashing and Sheet Metal
07 01 60.71	Flashing and Sheet Metal Rehabilitation
07 01 60.91	Flashing and Sheet Metal Restoration
07 01 60.92	Flashing and Sheet Metal Preservation
07 01 70	Operation and Maintenance of Roof Specialties and Accessories
07 01 80	Maintenance of Fire and Smoke Protection
07 01 90	Maintenance of Joint Protection
07 01 90.71	Joint Sealant Rehabilitation
07 01 90.81	Joint Sealant Replacement

NUMBER	TITLE	EXPLANATION
07 03 00	Conservation Treatment for Period Roofing	
07 03 30	Conservation Treatment for Period Steep Slope Roofing	
07 03 30.13	Conservation Treatment for Period Canvas Roofing	
07 03 30.16	Conservation Treatment for Period Board Roofing	
07 03 31	Conservation Treatment for Period Shingles and Shakes	
07 03 31.16	Conservation Treatment for Period Metal Shingles	
07 03 31.26	Conservation Treatment for Period Slate Shingles	
07 03 31.29	Conservation Treatment for Period Wood Shingles and Shakes	
07 03 32	Conservation Treatment for Period Roof Tiles	
07 03 32.13	Conservation Treatment for Period Clay Roof Tiles	
07 03 32.16	Conservation Treatment for Period Concrete Roof Tiles	
07 03 32.19	Conservation Treatment for Period Metal Roof Tiles	
07 03 33	Conservation Treatment for Period Natural Roof Coverings	
07 03 33.13	Conservation Treatment for Period Sod Roofing	
07 03 40	Conservation Treatment for Period Roofing and Siding Panels	
07 03 41	Conservative Treatment for Period Roof Panels	
07 03 41.13	Conservation Treatment for Period Metal Roof Panels	
07 03 41.23	Conservation Treatment for Period Wood Roof Panels	
07 03 46	Conservation Treatment for Period Wood Siding	
07 03 60	Conservation Treatment of Period Flashing and Sheet Metal May Include: materials such as tern-plate, lead, etc.	
07 03 61	Conservation Treatment for Period Sheet Metal Roofing	
07 03 61.13	Conservation Treatment for Period Standing Seam Sheet Metal Roofing	
07 03 61.16	Conservation Treatment for Period Batten Seam Sheet Metal Roofing	
07 03 61.19	Conservation Treatment for Period Flat Seam Sheet Metal Roofing	
07 05 00	Common Work Results for Thermal and Moisture Protection	
	Includes: subjects common to multiple titles in Division 07.	
	See: 01 80 00 for performance requirements of subjects common to multiple titles.	
07 05 05	Selective Demolition for Thermal and Moisture Protection See: 02 41 19 for selective demolition for whole structure demolition.	
07 05 43	Cladding Support Systems	
07 05 53	Fire and Smoke Assembly Identification	
07 06 00	Schedules for Thermal and Moisture Protection	
	Includes: schedules of items common to multiple titles in Division 07.	
	Notes: a schedule may be included on drawings, in the project manual, or a project book.	
07 06 10	Schedules for Dampproofing and Waterproofing	
07 06 20	Schedules for Thermal Protection	

NUMBER	TITLE	EXPLANATION
07 06 30	Schedules for Steep Slope Roofing	
07 06 40	Schedules for Roofing and Siding Panels	
07 06 50	Schedules for Membrane Roofing	
07 06 60	Schedules for Flashing and Sheet Metal	
07 06 70	Schedules for Roof Specialties and Accessories	
07 06 80	Schedules for Fire and Smoke Protection	
07 06 80.13	Fire Protection Schedule	
07 06 80.16	Firestopping Schedule	
07 06 90	Schedules for Joint Protection	
07 06 90.13	Joint Sealant Schedule	

07 08 00 Commissioning of Thermal and Moisture Protection

Includes: commissioning of items common to multiple titles in Division 07.

See: 01 91 00 for commissioning of subjects common to multiple Divisions.

07 10 00 Dampproofing and Waterproofing

Includes: dampproofing materials used to resist moisture penetration through foundation walls and similar surfaces subject to high humidity, dampness, or direct water contact, but not subject to hydraulic pressures. Impervious waterproofing membranes, coatings, and other materials applied to walls, slabs, decks, and other surfaces subject to hydrostatic pressure or water immersion. Boards and coatings, and drainage media integral to dampproofing and waterproofing.

See: 03 15 13 for waterstops.

07 20 00 for vapor retarders integral with insulation.

07 50 00 for membrane waterproofing.

31 05 16 for drainage aggregate.

07 11 00 Dampproofing

Includes: bituminous and cementitious dampproofing applied in a viscous state or in sheets.

See: 07 13 00 for sheet waterproofing.

07 16 00 for cementitious waterproofing.

07 26 00 for vapor retarders.

07 27 00 for air barriers.

07 11 13	Bituminous Dampproofing
07 11 16	Cementitious Dampproofing
07 11 19	Sheet Dampproofing

NUMBER	TITLE	EXPLANATION
07 12 00	Built-Up Bituminous Waterproofing	Includes: waterproofing systems composed of alternating layers of bituminous sheets and viscous bituminous coatings. See: 02 25 29 for waterproofing investigations. 07 11 13 for bituminous dampproofing. 07 51 00 for built-up bituminous roofing. 07 55 51 for built-up bituminous protected membrane roofing.
07 12 13	Built-Up Asphalt Waterproofing	
07 12 16	Built-Up Coal Tar Waterproofing	
07 13 00	Sheet Waterproofing	Includes: waterproofing formed with sheets of elastomeric, bituminous, modified bituminous, or thermoplastic materials, and depending on properties other than layered application for effectiveness. See: 02 25 29 for waterproofing investigations. 07 52 00 for modified bituminous membrane roofing. 07 53 00 for elastomeric membrane roofing. 07 54 00 for thermoplastic membrane roofing. 07 55 52 for modified bituminous protected membrane roofing. 07 55 53 for elastomeric protected membrane roofing. 07 55 54 for thermoplastic protected membrane roofing.
07 13 13	Bituminous Sheet Waterproofing	
07 13 26	Self-Adhering Sheet Waterproofing	
07 13 52	Modified Bituminous Sheet Waterproofing	
07 13 53	Elastomeric Sheet Waterproofing	
07 13 54	Thermoplastic Sheet Waterproofing	
07 14 00	Fluid-Applied Waterproofing	Includes: waterproofing applied in a viscous state. Alternate Terms/Abbreviations: fluid-applied: liquid-applied. See: 02 25 29 for waterproofing investigations. 07 18 00 for traffic coatings. 07 55 56 for fluid-applied protected membrane roofing. 07 55 56 for hot-applied rubberized asphalt protected membrane roofing. 07 56 00 for fluid-applied roofing.
07 14 13	Hot Fluid-Applied Rubberized Asphalt Waterproofing	
07 14 14	Hot Fluid-Applied Polyurea Waterproofing	
07 14 16	Cold Fluid-Applied Waterproofing	

NUMBER	TITLE	EXPLANATION
07 15 00	Sheet Metal Waterproofing	Includes: waterproofing formed of sheet metal. See: 02 25 29 for waterproofing investigations. 07 11 19 for sheet dampproofing. 07 62 00 for sheet metal flashing and trim.
07 15 13	Sheet Lead Waterproofing	
07 16 00	Cementitious and Reactive Waterproofing	Includes: waterproofing that achieves utility through chemical action. Hydrolithic waterproofing and acrylic-modified cement waterproofing. Exterior insulation cement plaster assemblies. See: 02 25 29 for waterproofing investigations. 07 11 16 for cementitious dampproofing. 07 24 16 for polymer-modified EIFS. 09 97 26 for cementitious coatings.
07 16 13	Polymer Modified Cement Waterproofing	
07 16 16	Crystalline Waterproofing	
07 16 19	Metal Oxide Waterproofing	
07 17 00	Bentonite Waterproofing	Includes: waterproofing formed of bentonite clay. See: 02 25 29 for waterproofing investigations. 31 35 26 for bentonite slurry trenches. 31 56 13 for bentonite slurry walls.
07 17 13	Bentonite Panel Waterproofing	
07 17 16	Bentonite Composite Sheet Waterproofing	
07 18 00	Traffic Coatings	Includes: surface applied waterproofing exposed to weather and suitable for pedestrian or vehicular traffic. Elastomeric, non-slip, polyurethane, silicone rubber, urethane traffic coatings. Alternate Terms/Abbreviations: traffic coatings: traffic-bearing waterproofing. See: 07 14 00 for fluid-applied waterproofing. 07 19 00 for water repellents. 09 61 00 for flooring treatments. 09 67 00 for fluid-applied flooring.
07 18 13	Pedestrian Traffic Coatings	
07 18 16	Vehicular Traffic Coatings	

NUMBER	TITLE	EXPLANATION
07 19 00	Water Repellents	Includes: transparent fluid-applied dampproofing applied to exposed surfaces of masonry, concrete, stucco, and similar materials to resist moisture penetration. Alternate Terms/Abbreviations: fluid-applied: liquid-applied. See: 04 22 00 for CMU with integral water repellent. 07 18 00 for traffic coatings. 09 90 00 for vapor-resistant primers.
07 19 13	Acrylic Water Repellents	
07 19 16	Silane Water Repellents	
07 19 19	Silicone Water Repellents	
07 19 23	Siloxane Water Repellents	
07 19 26	Stearate Water Repellents	
07 19 29	Penetrating Polyester Water Repellents	

07 20 00 Thermal Protection

Includes: insulation applied for thermal protection, elastomeric finish systems, vapor retarders, and air barriers. Cellulose, expanded polystyrene, extruded polystyrene, fiberglass, foamed glass, mineral fiber, perlite, polyisocyanurate, polyurethane, reflective, rock wool, vermiculite insulations.

See: 03 11 19 for insulating concrete forming.
03 52 16 for lightweight insulating concrete.
04 22 19 for insulated concrete unit masonry.
06 16 13 for insulating sheathing.
22 07 00 for plumbing insulation.
23 07 00 for HVAC insulation.
40 42 00 for process piping and equipment insulation.

NUMBER	TITLE	EXPLANATION
07 21 00	Thermal Insulation	Includes: organic and inorganic insulation applied for thermal protection in walls, ceilings, attics, and crawl spaces; under concrete slabs on grade; and at perimeter of foundations. Vapor retarders integral with insulation. Alternate Terms/Abbreviations: blanket insulation: batt insulation. Board insulation: rigid insulation: semi-rigid insulation. Loose fill insulation: fill insulation: granular insulation. Sprayed insulation: spray-on insulation. See: 06 16 13 for insulating sheathing. 07 22 16 for roof board insulation. 09 81 00 for acoustic insulation. See Also: 04 27 00 for insulation integral to multi-wythe unit masonry. 07 26 00 for vapor retarders. 07 27 00 for air barriers.
07 21 13	Board Insulation	
07 21 13.13	Foam Board Insulation	
07 21 13.16	Fibrous Board Insulation	
07 21 13.19	Mineral Board Insulation	
07 21 16	Blanket Insulation	
07 21 19	Foamed-In-Place Insulation	May Include: various insulation materials that require mixing multiple components to create a stable matrix that is different from the original components to contain air or other insulating gas, and is foamed or poured in place. See Also: 07 21 29 sprayed insulation.
07 21 23	Loose-Fill Insulation	
07 21 26	Blown Insulation	
07 21 29	Sprayed Insulation	May Include: various insulation materials such as cellulose or other fibrous products that are mixed with adhesives and sprayed onto surfaces. See Also: 07 21 19 for foamed-in-place insulation.
07 21 53	Reflective Insulation	
07 21 63	Fluid-Applied Insulative Coating	

NUMBER	TITLE	EXPLANATION
07 22 00	Roof and Deck Insulation	Includes: organic or inorganic insulation applied for thermal protection as a part of roofing assemblies and under decks. Tapered roof insulation. Alternate Terms/Abbreviations: board insulation: rigid insulation: semi-rigid insulation. See: 03 52 00 for lightweight concrete roof insulation. 07 50 00 for roof and deck insulation in membrane roofing.
07 22 13	Asphaltic Perlite Concrete Deck	
07 22 16	Roof Board Insulation	
07 24 00	Exterior Insulation and Finish Systems	Includes: both polymer-based and polymer-modified coating and finish applied directly to foam insulation base or directly to other compatible substrate. Alternate Terms/Abbreviations: EIFS: exterior insulation and finish system. Board insulation: rigid insulation: semi-rigid insulation. See: 09 24 23 for cement stucco.
07 24 13	Polymer-Based Exterior Insulation and Finish System	
07 24 16	Polymer-Modified Exterior Insulation and Finish System	Includes: exterior insulation cement plaster assemblies.
07 24 19	Water-Drainage Exterior Insulation and Finish System	
07 24 23	Direct-Applied Finish Systems	See Also: 09 97 00 for special coatings.
07 25 00	Weather Barriers	Includes: materials applied separately in wall, roof, and floor assemblies to resist vapor penetration, primarily to mitigate the consequences of thermal variation. Materials installed at walls, roof-wall connections, perimeter of door and window openings, and similar locations to provide a continuous, impermeable barrier to air infiltration or loss.

NUMBER	TITLE	EXPLANATION
07 26 00	Vapor Retarders	Includes: bituminous, laminated, and plastic vapor retarders. May Include: vapor retarder under concrete slabs on grade. See: 07 21 00 for thermal insulation with integral vapor retarders. 09 29 00 for gypsum board with integral vapor retarders. See Also: 03 30 00 for vapor retarders under concrete slabs on grade.
07 26 13	Above-Grade Vapor Retarders	
07 26 16	Below-Grade Vapor Retarders	
07 26 23	Below-Grade Gas Retarders	Includes: methane, radon, and hydrocarbon retarders.
07 27 00	Air Barriers	Includes: reinforced rubber and sheet metal membranes, plastic sheet, board products with sealed joints, cement parging, and similar products. Alternate Terms/Abbreviations: air barrier: housewrap. Fluid-applied: liquid-applied. Notes: gypsum board, concrete, metal decking, and similar building elements normally located under other titles may also serve as air barriers. See Also: 07 26 00 for vapor barriers that also serve as air barriers.
07 27 13	Modified Bituminous Sheet Air Barriers	
07 27 16	Sheet Metal Membrane Air Barriers	
07 27 19	Plastic Sheet Air Barriers	
07 27 23	Board Product Air Barriers	
07 27 26	Fluid-Applied Membrane Air Barriers	
07 27 36	Sprayed Foam Air Barrier	
07 30 00	Steep Slope Roofing	Includes: roof shingles, shakes, and roof tiles. Underlayment and fastening products and methods. See: 02 25 29 for roofing investigations. 07 62 00 for sheet metal flashing and trim. 07 65 00 for flexible flashing. 07 70 00 for roofing specialties and accessories.
07 30 91	Canvas Roofing	Includes: sloped roofing made of canvas.

NUMBER	TITLE	EXPLANATION
07 31 00	Shingles and Shakes	Includes: lapped roof shingles and shakes. Solar collector shingles. See: 07 32 00 for roof tiles. 07 41 00 for roof panels. 07 46 00 for siding. 26 31 00 for photovoltaic collectors.
07 31 13	Asphalt Shingles	
07 31 13.13	Fiberglass-Reinforced Asphalt Shingles	
07 31 16	Metal Shingles	
07 31 19	Mineral-Fiber Cement Shingles	
07 31 23	Porcelain Enamel Shingles	
07 31 26	Slate Shingles	
07 31 29	Wood Shingles and Shakes	
07 31 29.13	Wood Shingles	
07 31 29.16	Wood Shakes	
07 31 33	Composite Rubber Shingles	
07 31 53	Plastic Shakes	
07 32 00	Roof Tiles	Includes: lapped roof tiles. See: 07 31 00 for shingles and shakes.
07 32 13	Clay Roof Tiles	
07 32 14	Ceramic and Porcelain Roof Tiles	
07 32 16	Concrete Roof Tiles	
07 32 19	Metal Roof Tiles	
07 32 23	Mineral-Fiber Cement Roof Tiles	
07 32 26	Plastic Roof Tiles	
07 32 29	Rubber Tiles/Panels	
07 33 00	Natural Roof Coverings	Includes: organic and special roof coverings. See: 07 55 63 for vegetated protected membrane roofing.
07 33 13	Sod Roofing	
07 33 16	Thatched Roofing	
07 33 63	Vegetated Steep-Slope Roofing	

NUMBER	TITLE	EXPLANATION
07 40 00	Roofing and Siding Panels	Includes: siding, wall panels, and roof panels, with or without integral interior panels or insulation. Fabricated panel assemblies, which may include framing, sheathing, insulation, substrates, interior finish, and exterior finish. May Include: roof and wall panels that are used with a fabricated engineered structure. See: 03 40 00 for precast concrete panels. 04 25 00 for fabricated masonry panels. 07 21 00 for wall insulation. 08 44 00 for glazed curtain wall spandrel and infill panels. 08 90 00 for louvers and vents. 09 21 16 for gypsum board assemblies. See Also: 13 34 00 for roof and wall panels that are part of a fabricated engineered structure.
07 41 00	Roof Panels	Includes: manufactured roof panels and fabricated panel assemblies. Fabricated roof panel assemblies, including other components such as structural framing, insulation, flashing, and interior finish. Aluminum, copper, steel, stainless steel, and terne metal roof panels. Batten seam, flat seam, and standing seam metal roof panels. See: 07 31 00 for shingles and shakes. 07 61 00 for custom fabricated sheet metal roofing.
07 41 13	Metal Roof Panels	
07 41 16	Insulated Metal Roof Panels	
07 41 23	Wood Roof Panels	
07 41 33	Plastic Roof Panels	
07 41 43	Composite Roof Panels	
07 41 63	Fabricated Roof Panel Assemblies	
07 42 00	Wall Panels	Includes: manufactured wall panels and fabricated panel assemblies. Fabricated wall panel assemblies, including other components such as structural framing, insulation, flashing, and interior finish. See: 04 25 00 for fabricated masonry panels. 07 24 00 for exterior insulation and finish systems.
07 42 13	Metal Wall Panels	
07 42 13.13	Formed Metal Wall Panels	
07 42 13.16	Metal Plate Wall Panels	

DIVISION **07**

NUMBER	TITLE	EXPLANATION
07 42 13.19	Insulated Metal Wall Panels	
07 42 13.23	Metal Composite Material Wall Panels	
07 42 23	Wood Wall Panels	
07 42 26	Tile Wall Panels	Includes: tile panels used as a cladding assembly without mortar and grout.
07 42 29	Terra Cotta Wall Panels	Alternate Terms/Abbreviations: terra cotta
07 42 33	Plastic Wall Panels	
07 42 43	Composite Wall Panels	
07 42 46	Cementitious Wall Panels	Includes: panels manufactured from high modulus concrete and fiber-reinforced cement.
07 42 63	Fabricated Wall Panel Assemblies	
07 42 93	Soffit Panels	

07 44 00 Faced Panels

Includes: roof and wall panels coated on the exterior surface with aggregate, porcelain enamel, ceramic tile, or quarry tile. Fabricated panel assemblies with faced-panel exterior finish. Roof and wall panels composed of fiber-reinforced cementitious material. Fabricated-faced panel assemblies, including other components such as structural framing, insulation, flashing, and interior finish.

See: 03 45 13 for faced architectural precast concrete.
06 12 13 for cementitious reinforced panels.

07 44 13	Aggregate Coated Panels
07 44 16	Porcelain Enameled Faced Panels
07 44 19	Tile-Faced Panels
07 44 23	Ceramic-Tile-Faced Panels
07 44 33	Metal Faced Panels
07 44 53	Glass-Fiber-Reinforced Cementitious Panels
07 44 56	Mineral-Fiber-Reinforced Cementitious Panels
07 44 63	Fabricated Faced Panel Assemblies

07 46 00 Siding

Includes: shaped or unshaped boards or panels used as exterior surface of a wall or soffit. Fabricated panel assemblies with exterior siding and other components such as structural framing, insulation, flashing, and interior.

See: 06 16 00 for sheathing.
07 26 00 for vapor retarders.
07 27 00 for air barriers.
07 31 00 for shingles and shakes.

NUMBER	TITLE	EXPLANATION
07 46 16	Aluminum Siding	
07 46 19	Steel Siding	
07 46 21	Zinc Siding	
07 46 23	Wood Siding	
07 46 24	Wood Shingle and Shake Siding	Includes: wood shingle and shake siding, weather-resistive barriers, and flexible flashing.
07 46 24.13	Wood Shingle Siding	
07 46 24.16	Wood Shake Siding	
07 46 26	Hardboard Siding	
07 46 29	Plywood Siding	
07 46 33	Plastic Siding	
07 46 43	Composition Siding	Alternate Terms/Abbreviations: composition siding: wood-fiber-reinforced resin siding.
07 46 46	Fiber-Cement Siding	
07 46 63	Fabricated Panel Assemblies with Siding	

07 50 00 Membrane Roofing

Includes: roofing assemblies applied to the structural substrate, over insulation, or protected with insulation as appropriate to the particular assembly. Composition and elastomeric flashing. Surfacing materials, coatings, and ballast.

May Include: roof insulation, vapor retarders, and air barriers. Walkways, roof drains, and other items integral with roofing assembly. Roof pavers.

Alternate Terms/Abbreviations: membrane roofing: sheet roofing.

See: 02 25 29 for roofing investigations.

07 10 00 for waterproofing.

07 62 00 for sheet metal flashing and trim.

See Also: 07 22 16 for roof board insulation.

07 26 00 for vapor retarders.

07 27 00 for air barriers.

07 72 43 for roof walk boards and walkways.

07 76 00 for roof pavers.

22 14 26 for roof drains.

NUMBER	TITLE	EXPLANATION
07 51 00	Built-Up Bituminous Roofing	Includes: roofing assemblies composed of alternating layers of bituminous sheets and viscous bituminous coatings. See: 07 11 13 for bituminous dampproofing. 07 12 00 for built-up bituminous waterproofing. 07 55 51 for built-up bituminous protected membrane roofing.
07 51 13	Built-Up Asphalt Roofing	
07 51 13.13	Cold-Applied Built-Up Asphalt Roofing	
07 51 16	Built-Up Coal Tar Roofing	
07 51 23	Glass-Fiber-Reinforced Asphalt Emulsion Roofing	
07 52 00	Modified Bituminous Membrane Roofing	Includes: roofing assemblies formed with modified bituminous membranes. Alternate Terms/Abbreviations: APP: atactic polypropylene. SBS: styrene-butadiene-styrene. See: 07 11 13 for bituminous dampproofing. 07 13 52 for modified bituminous sheet waterproofing. 07 55 52 for modified bituminous protected membrane roofing.
07 52 13	Atactic-Polypropylene-Modified Bituminous Membrane Roofing	
07 52 13.11	Cold Adhesive Applied Atactic-Polypropylene-Modified Bituminous Membrane Roofing	
07 52 13.13	Torch-Applied Atactic-Polypropylene-Modified Bituminous Membrane Roofing	
07 52 13.14	Mechanically Fastened Atactic-Polypropylene-Modified Bituminous Membrane Roofing	
07 52 16	Styrene-Butadiene-Styrene Modified Bituminous Membrane Roofing	
07 52 16.11	Cold Adhesive Styrene-Butadiene-Styrene Modified Bituminous Membrane Roofing	
07 52 16.12	Hot-Mopped Styrene-Butadiene-Styrene Modified Bituminous Membrane Roofing	
07 52 16.13	Torch-Applied Styrene-Butadiene-Styrene Modified Bituminous Membrane Roofing	
07 52 16.14	Mechanically Fastened Styrene-Butadiene-Styrene Modified Bituminous Membrane Roofing	
07 53 00	Elastomeric Membrane Roofing	Includes: roofing assemblies formed with non-vulcanized and vulcanized elastomeric membranes. Ballasted, fully-adhered, and mechanically attached installations. Alternate Terms/Abbreviations: CPE: chlorinated polyethylene. CSPE: chlorosulfonate polyethylene. EPDM: ethylene propylene diene monomer. PIB: polyisobutylene. See: 07 13 53 for elastomeric sheet waterproofing. 07 55 53 for elastomeric protected membrane roofing.
07 53 13	Chlorinated-Polyethylene Roofing	
07 53 16	Chlorosulfonate-Polyethylene Roofing	
07 53 23	Ethylene-Propylene-Diene-Monomer Roofing	

NUMBER	TITLE	EXPLANATION
07 53 29	Polyisobutylene Roofing	
07 54 00	Thermoplastic Membrane Roofing	Includes: roofing assemblies formed with reinforced and unreinforced thermoplastic membranes. Ballasted, fully-adhered, and mechanically attached installations. Alternate Terms/Abbreviations: CPA: copolymer alloy. EIP: ethylene interpolymer. PVC: polyvinyl chloride. TPO: thermoplastic polyolefin. NBP: nitrile butadiene polymer. See: 07 13 54 for thermoplastic sheet waterproofing. 07 55 54 for thermoplastic protected membrane roofing.
07 54 13	Copolymer-Alloy Roofing	
07 54 16	Ketone Ethylene Ester Roofing	
07 54 19	Polyvinyl-Chloride Roofing	
07 54 23	Thermoplastic-Polyolefin Roofing	
07 54 26	Nitrile-Butadiene-Polymer Roofing	
07 55 00	Protected Membrane Roofing	Includes: roofing assemblies with protective insulation and other surfacing materials covering membranes applied directly to the structural substrate. Alternate Terms/Abbreviations: fluid-applied: liquid-applied. IRMA: insulated roof membrane assembly; inverted roof membrane assembly. PRM: protected roof membrane. Vegetated protected membrane roofing: landscaped protected membrane roofing: green roof: sustainable design roof. See: 07 12 00 for built-up bituminous waterproofing. 07 13 52 for modified bituminous sheet waterproofing. 07 13 53 for elastomeric sheet waterproofing. 07 13 54 for thermoplastic sheet waterproofing. 07 14 00 for fluid-applied waterproofing. 07 33 63 for vegetated roofing. 07 51 00 for built-up bituminous roofing. 07 52 00 for modified bituminous membrane roofing. 07 53 00 for elastomeric membrane roofing. 07 54 00 for thermoplastic membrane roofing. 07 56 00 for fluid-applied roofing.
07 55 51	Built-Up Bituminous Protected Membrane Roofing	
07 55 52	Modified Bituminous Protected Membrane Roofing	
07 55 53	Elastomeric Protected Membrane Roofing	
07 55 54	Thermoplastic Protected Membrane Roofing	
07 55 56	Fluid-Applied Protected Membrane Roofing	
07 55 56.13	Hot-Applied Rubberized Asphalt Protected Membrane Roofing	
07 55 63	Vegetated Protected Membrane Roofing	

NUMBER	TITLE	EXPLANATION
07 56 00	Fluid-Applied Roofing	Includes: roofing applied in a viscous state. Chlorosulfonated polyethylene (CSPE) and methyl methacrylate (MMA) roofing. Alternate Terms/Abbreviations: fluid-applied: liquid-applied. See: 07 14 00 for fluid-applied waterproofing. 07 55 56 for fluid-applied protected membrane roofing.
07 57 00	Coated Foamed Roofing	Includes: roofing assemblies consisting of spray-applied materials which expand through chemical reaction and an elastomeric coating.
07 57 13	Sprayed Polyurethane Foam Roofing	
07 58 00	Roll Roofing	Includes: roofing formed of surfaced, prepared bituminous sheets.
07 60 00	Flashing and Sheet Metal	Includes: sheet metal roofing, accessories, and trim, and specialties associated with roofing, and metal and other flexible flashings for roofing and wall construction. See: 07 15 00 for sheet metal waterproofing.
07 61 00	Sheet Metal Roofing	Includes: custom fabricated or field-formed roofing composed of a single layer of metal sheets installed over structural substrate. Aluminum, copper, lead, steel, stainless steel, monel, and terne sheet metal roofing. See: 07 41 13 for manufactured metal roof panels.
07 61 13	Standing Seam Sheet Metal Roofing	
07 61 16	Batten Seam Sheet Metal Roofing	
07 61 19	Flat Seam Sheet Metal Roofing	

NUMBER	TITLE	EXPLANATION
07 62 00	Sheet Metal Flashing and Trim	Includes: shop and field-formed accessories and trim. May Include: metal flashing for masonry. See: 07 71 00 for manufactured gutters, downspouts, counterflashing, gravel stops, copings, roof expansion joint covers, and fasciae. See Also: 04 27 13 for metal flashing integral with composite unit masonry. Products: formed gutters, downspouts, counterflashing, scuppers, gravel stops, copings, roof expansion joint covers, pitch pans, plumbing drain flashing, and diverters.
07 62 13	Fabricated Copings	
07 62 16	Fabricated Counterflashing Systems	
07 62 19	Fabricated Gravel Stops and Fasciae	
07 62 23	Fabricated Gutters and Downspouts	
07 62 29	Fabricated Roof Expansion Joints	
07 62 33	Fabricated Scuppers	
07 63 00	Sheet Metal Roofing Specialties	Includes: roof-related protective and ornamental items and roof-related sheet metal items. See: 07 71 00 for manufactured roof specialties.
07 64 00	Sheet Metal Wall Cladding	
07 64 13	Standing Seam Sheet Metal Wall Cladding	
07 64 16	Batten Seam Sheet Metal Wall Cladding	
07 64 19	Flat Seam Sheet Metal Wall Cladding	
07 65 00	Flexible Flashing	Includes: flexible flashings for roof and wall construction. May Include: flexible flashing for masonry. See: 04 27 00 for flexible flashing integral with multi-wythe unit masonry. 07 50 00 for flexible flashing integral with membrane roofing.
07 65 13	Laminated Sheet Flashing	
07 65 16	Modified Bituminous Sheet Flashing	
07 65 19	Plastic Sheet Flashing	
07 65 23	Rubber Sheet Flashing	
07 65 26	Self-Adhering Sheet Flashing	

NUMBER	TITLE	EXPLANATION
07 70 00	Roof and Wall Specialties and Accessories	Includes: standard manufactured components, both formed and extruded. See: 08 60 00 for skylights. 10 74 00 for manufactured exterior specialties.
07 71 00	Roof Specialties	Includes: manufactured specialties serving as roofing perimeter facing, drainage, and expansion control. May Include: manufactured flashing for penetrations. See: 07 62 00 for shop- and field-formed sheet metal accessories and trim. 07 63 00 for sheet metal roofing specialties. 07 77 00 for wall specialties. 07 95 13 for expansion joint cover assemblies. 22 14 26 for roof drains.
07 71 13	Manufactured Copings	
07 71 16	Manufactured Counterflashing Systems	
07 71 19	Manufactured Gravel Stops and Fasciae	
07 71 23	Manufactured Gutters and Downspouts	
07 71 23.13	Gutter Debris Guards	
07 71 26	Reglets	
07 71 29	Manufactured Roof Expansion Joints	
07 71 33	Manufactured Scuppers	
07 72 00	Roof Accessories	Includes: accessories installed on or in roofing other than mechanical or structural items. May Include: roof curbs for mechanical equipment. Alternate Terms/Abbreviations: roof hatch: scuttle. See: 06 10 00 for field-fabricated wood curbs. 07 76 00 for roof pavers. 08 62 00 for unit skylights. 08 91 13 for motorized wall louvers. 08 91 16 for wall installations of motorized large-scale operable ventilation panels. 08 95 13 for soffit vents. 08 95 16 for wall vents. 23 70 00 for roof curbs for rooftop-mounted HVAC equipment.
07 72 13	Manufactured Curbs	
07 72 23	Relief Vents	

NUMBER	TITLE	EXPLANATION
07 72 26	Ridge Vents	
07 72 33	Roof Hatches	
07 72 36	Smoke Vents	
07 72 39	Operable Roof Vents	Includes: motorized large-scale operable ventilation panels.
07 72 43	Roof Walk Boards	
07 72 46	Roof Walkways	
07 72 53	Snow Guards	
07 72 63	Waste Containment Assemblies	
07 72 73	Vegetated Roof Systems	Includes: continuous vegetated roof assemblies, tray-type vegetated roof assemblies, walkway pavers, vegetated mats, and geofoam fill.
07 76 00	Roof Pavers	Includes: pavers and decking over roofing, including pedestal-mounted pavers. May Include: pavers for ballasting of membrane roofing. See: 07 50 00 for stone ballast for membrane roofing. 07 72 46 for roof walkways. See Also: 07 50 00 for pavers for membrane roofing.
07 76 13	Roof Ballast Pavers	
07 76 16	Roof Decking Pavers	
07 77 00	Wall Specialties	Includes: manufactured specialties for wall construction. See: 07 71 00 for roof specialties.
07 80 00	Fire and Smoke Protection	
07 81 00	Applied Fire Protection	Includes: specialized coatings, mineral fiber, and cementitious coverings to provide fire resistance to building components. Fireproof duct wrap. See: 06 05 73 for fire-retardant wood treatment of wood products. 07 82 00 for board fire protection. 09 23 82 for fire protected gypsum plastering. 09 96 43 for fire-retardant coatings. 09 96 46 for intumescent painting.

DIVISION 07

NUMBER	TITLE	EXPLANATION
07 81 13	Cement Aggregate Fire Protection	
07 81 16	Cementitious Fire Protection	
07 81 19	Foamed Magnesium-Oxychloride Fire Protection	
07 81 23	Intumescent Fire Protection	
07 81 26	Magnesium Cement Fire Protection	
07 81 29	Mineral-Fiber Cementitious Fire Protection	
07 81 33	Mineral-Fiber Fire Protection	

07 82 00 Board Fire Protection

Includes: board materials that provide fire resistance to building components.

Does Not Include: fire barriers of gypsum board, plaster, and similar construction for prevention of the spread of fire.

See: 07 81 00 for applied fire protection.
09 23 82 for fire protected gypsum plastering.
09 29 82 for gypsum board fire protection.

07 82 13	Calcium-Silicate Board Fire Protection
07 82 16	Slag-Fiber Board Fire Protection

07 84 00 Firestopping

Includes: materials installed in cavities, around penetrations, and openings in floors, walls, partitions, and other building components to prevent the spread of fire and smoke.
Firestopping for mechanical, electrical, and communications penetrations of fire-resistant rated assemblies.

See: 07 21 00 for thermal insulation.
07 92 00 for joint sealants.
07 95 13 for fire-resistant rated expansion joint cover assemblies.

07 84 13	Penetration Firestopping
07 84 13.13	Penetration Firestopping Mortars
07 84 13.16	Penetration Firestopping Devices
07 84 43	Joint Firestopping
07 84 53	Building Perimeter Firestopping

Alternate Terms/Abbreviations: fire safing.

07 86 00 Smoke Seals

Includes: material installed at the perimeter of openings to prevent the spread of smoke.

NUMBER	TITLE	EXPLANATION
07 87 00	Smoke Containment Barriers	Includes: draftstopping panels suspended from the ceiling or placed in plenum cavities to prevent or to inhibit the spread of smoke. See: 06 10 00 for wood draftstopping. 09 29 00 for gypsum board draftstopping.
07 90 00	Joint Protection	
07 91 00	Preformed Joint Seals	Includes: preformed compression seals and gaskets. See: 07 92 00 for joint sealants. 07 95 13 for expansion joint cover assemblies.
07 91 13	Compression Seals	
07 91 16	Joint Gaskets	
07 91 23	Backer Rods	
07 91 26	Joint Fillers	
07 92 00	Joint Sealants	Includes: elastomeric and rigid joint sealants, and related accessories such as joint fillers and backer rod. Joint sealant to provide acoustical isolation. Nonsag, pourable, mildew-resistant, sanitary, acrylic, bituminous, latex, polysulfide, polyurethane, PVC, rubber, silicone, synthetic, urethane joint sealants. Alternate Terms/Abbreviations: joint sealants: calking: caulking: joint sealers. See: 07 84 43 for fire-resistant joint sealants. 08 85 00 for glazing sealants. 32 13 73 for concrete paving joint sealants. See Also: 09 20 00 for acoustic joint sealants in plaster and gypsum board assemblies.
07 92 13	Elastomeric Joint Sealants	
07 92 16	Rigid Joint Sealants	
07 92 19	Acoustical Joint Sealants	

NUMBER	TITLE	EXPLANATION
07 95 00	Expansion Control	Includes: manufactured expansion joint assemblies, including frames, covers, and gaskets. Fire-resistant barriers for fire-resistant rated expansion control assemblies. See: 03 15 00 for concrete expansion and contraction joints. 05 12 33 for structural steel for bridges. 07 71 29 for roof expansion joints. 07 84 43 for fire-resistant joint sealants. 07 91 00 for preformed joint seals. 07 92 00 for joint sealants. 34 82 00 for bridge specialties.
07 95 13	Expansion Joint Cover Assemblies	
07 95 13.13	Interior Expansion Joint Cover Assemblies	Includes: expansion joint cover assemblies for interior floors, walls, and ceilings. Moisture barriers and attachment devices.
07 95 13.16	Exterior Expansion Joint Cover Assemblies	Includes: expansion joint cover assemblies for exterior walls, soffits, and parapets. Moisture barriers and attachment devices.
07 95 13.19	Parking Deck Expansion Joint Cover Assemblies	Includes: expansion joint cover assemblies for use in exterior slabs subject to vehicular traffic. Moisture barriers and attachment devices.
07 95 53	Joint Slide Bearings	
07 95 63	Bridge Expansion Joint Cover Assemblies	

NUMBER	TITLE	EXPLANATION
DIVISION 08 – OPENINGS		
08 00 00	Openings	May be used as Division level Section title.

08 01 00 Operation and Maintenance of Openings

Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of openings.

Level 4 Numbering Recommendation: Level 4 titles for wood windows are provided as examples for user's creation of more detailed titles for other subjects as required. Following Level 4 numbering is recommended:

.51-.59 for maintenance.

.61-.69 for repair.

.71-.79 for rehabilitation.

.81-.89 for replacement.

.91-.99 for restoration.

08 01 10	Operation and Maintenance of Doors and Frames
08 01 11	Operation and Maintenance of Metal Doors and Frames
08 01 14	Operation and Maintenance of Wood Doors
08 01 15	Operation and Maintenance of Plastic Doors
08 01 16	Operation and Maintenance of Composite Doors
08 01 17	Operation and Maintenance of Integrated Door Opening Assemblies
08 01 30	Operation and Maintenance of Specialty Doors and Frames
08 01 32	Operation and Maintenance of Sliding Glass Doors
08 01 33	Operation and Maintenance of Coiling Doors and Grilles
08 01 34	Operation and Maintenance of Special Function Doors
08 01 35	Operation and Maintenance of Folding Doors and Grilles
08 01 36	Operation and Maintenance of Panel Doors
08 01 39	Operation and Maintenance of Traffic Doors
08 01 40	Operation and Maintenance of Entrances, Storefronts, and Curtain Walls
08 01 41	Operation and Maintenance of Entrances
08 01 42	Operation and Maintenance of Storefronts
08 01 44	Operation and Maintenance of Curtain Walls
08 01 50	Operation and Maintenance of Windows
08 01 51	Operation and Maintenance of Metal Windows
08 01 52	Operation and Maintenance of Wood Windows
08 01 52.61	Wood Window Repairs
08 01 52.71	Wood Window Rehabilitation
08 01 52.81	Wood Window Replacement
08 01 52.91	Wood Window Restoration
08 01 52.93	Historic Treatment of Wood Windows
08 01 53	Operation and Maintenance of Plastic Windows
08 01 54	Operation and Maintenance of Composite Windows
08 01 56	Operation and Maintenance of Special Function Windows
08 01 60	Operation and Maintenance of Skylights and Roof Windows
08 01 61	Operation and Maintenance of Roof Windows
08 01 62	Operation and Maintenance of Unit Skylights

DIVISION **08**

NUMBER	TITLE	EXPLANATION
08 01 63	Operation and Maintenance of Metal-Framed Skylights	
08 01 64	Operation and Maintenance of Plastic-Framed Skylights	
08 01 70	Operation and Maintenance of Hardware	
08 01 71	Operation and Maintenance of Door Hardware	
08 01 74	Operation and Maintenance of Access Control Hardware	
08 01 75	Operation and Maintenance of Window Hardware	
08 01 80	Maintenance of Glazing	
08 01 81	Maintenance of Glass Glazing	
08 01 84	Maintenance of Plastic Glazing	
08 01 88	Maintenance of Special Function Glazing	
08 01 90	Operation and Maintenance of Louvers and Vents	
08 01 91	Operation and Maintenance of Louvers	
08 01 92	Operation and Maintenance of Louvered Equipment Enclosures	
08 01 95	Operation and Maintenance of Vents	
08 03 00	Conservation Treatment for Period Openings	
08 03 11	Conservation Treatment for Period Metal Doors and Frames	
08 03 11.23	Conservation Treatment for Period Bronze Doors and Frames	
08 03 14	Conservation Treatment for Period Wood Doors	
08 03 14.23	Conservation Treatment for Period Metal-Faced Wood Doors	
	Alternate Terms/Abbreviations: Kalamein doors.	
08 03 14.33	Conservation Treatment for Period Stile and Rail Wood Doors	
08 03 30	Conservation Treatment for Period Specialty Doors and Frames	
08 03 34	Conservation Treatment for Period Special Function Doors	
08 03 34.59	Conservation Treatment for Period Vault Doors	
08 03 50	Conservation Treatment for Period Windows	
08 03 51	Conservation Treatment for Period Metal Windows	
08 03 51.16	Conservation Treatment for Period Bronze Windows	
08 03 51.23	Conservation Treatment for Period Steel Windows	
08 03 52	Conservation Treatment for Period Wood Windows	
08 03 70	Conservation Treatment for Period Hardware	
08 03 71	Conservation Treatment for Period Door Hardware	
08 03 75	Conservation Treatment for Period Window Hardware	
08 03 90	Conservation Treatment for Period Louvers and Vents	
08 03 91	Conservation Treatment for Period Louvers	
08 03 95	Conservation Treatment for Period Vents	
08 05 00	Common Work Results for Openings	
	Includes: subjects common to multiple Sections in Division 08.	
	See: 01 80 00 for performance requirements of subjects common to multiple titles.	
08 05 05	Selective Demolition for Openings	
	See: 02 41 19 for selective demolition for whole structure demolition.	

NUMBER	TITLE	EXPLANATION
08 06 00	Schedules for Openings	Includes: schedules of items common to multiple Sections in Division 08. Notes: a schedule may be included on drawings, in the project manual, or a project book.
08 06 10	Door Schedule	
08 06 10.13	Door Type Schedule	
08 06 10.16	Frame Type Schedule	
08 06 40	Schedules for Entrances, Storefronts, and Curtain Walls	
08 06 41	Entrance Schedule	
08 06 42	Storefront Schedule	
08 06 50	Window Schedule	
08 06 60	Skylight Schedule	
08 06 70	Hardware Schedule	
08 06 71	Door Hardware Schedule	See Also: 28 06 10 for schedules for access control
08 06 80	Glazing Schedule	
08 06 90	Louver and Vent Schedule	
08 08 00	Commissioning of Openings	Includes: commissioning of items common to multiple Sections in Division 08. See: 01 91 00 for commissioning of subjects common to multiple Divisions.
08 10 00	Doors and Frames	Includes: doors and door assemblies for a variety of purposes and types of operation. Fire-resistant rated doors and frames. Usually Includes: glazing accessories in doors, louvers integral to door construction, and louvers in fire-resistant rated doors. See: 08 30 00 for specialty doors. 08 42 00 for entrances. See Also: 08 80 00 for glazing. 08 91 26 for door louvers.

NUMBER	TITLE	EXPLANATION
08 11 00	Metal Doors and Frames	Includes: hinged or pivoting doors, fixed panels within frames, and frames manufactured of metal. Muntz metal. Formed and extruded metal frames for doors, sidelights, transoms, borrowed lights, and other openings. Alternate Terms/Abbreviations: hollow metal doors and frames: steel doors and frames. Muntz metal: yellow metal. See: 08 12 00 for metal frames only. 08 13 00 for metal doors only. 08 32 13 for sliding aluminum-framed glass doors. 08 41 00 for metal-framed entrances and storefronts. 08 71 00 for door hardware.
08 11 13	Hollow Metal Doors and Frames	
08 11 13.13	Standard Hollow Metal Doors and Frames	
08 11 13.16	Custom Hollow Metal Doors and Frames	
08 11 16	Aluminum Doors and Frames	
08 11 19	Stainless-Steel Doors and Frames	
08 11 23	Bronze Doors and Frames	
08 11 63	Metal Screen and Storm Doors and Frames	Includes: doors and screens manufactured of metal used for storm or thermal protection or for screening, usually as adjuncts to other doors. See: 08 14 66 for wood screen doors and frames. 08 14 69 for wood storm doors and frames. 08 15 66 for plastic screen doors and frames. 08 15 69 for plastic storm doors and frames.
08 11 63.13	Steel Screen and Storm Doors and Frames	
08 11 63.23	Aluminum Screen and Storm Doors and Frames	
08 11 66	Metal Screen Doors and Frames	
08 11 66.13	Steel Screen Doors and Frames	
08 11 66.23	Aluminum Screen Doors and Frames	
08 11 69	Metal Storm Doors and Frames	
08 11 69.13	Steel Storm Doors and Frames	
08 11 69.23	Aluminum Storm Doors and Frames	
08 11 73	Sliding Metal Fire Doors	Includes: fire-resistant rated doors manufactured of metal that operate by sliding. See: 08 13 73 for non-fire-rated sliding metal doors. 08 32 00 for sliding glass doors. 08 35 13 for folding fire doors. 08 42 29 for sliding automatic entrances.

NUMBER	TITLE	EXPLANATION
08 11 74	Sliding Metal Grilles	Includes: grilles manufactured or formed of metal that operate by sliding. See: 08 33 00 for coiling doors and grilles. 08 35 16 for folding grilles.
08 12 00	Metal Frames	Includes: formed and extruded metal frames for doors, sidelights, transoms, borrowed lights, and other openings. Alternate Terms/Abbreviations: hollow metal frames: steel frames. See: 08 11 00 for metal doors and frames. 08 13 00 for metal doors. 08 41 00 for metal-framed entrances and storefronts.
08 12 13	Hollow Metal Frames	
08 12 13.13	Standard Hollow Metal Frames	
08 12 13.53	Custom Hollow Metal Frames	
08 12 16	Aluminum Frames	
08 12 16.13	Fire-Rated Aluminum Frames	
08 12 19	Stainless-Steel Frames	
08 12 23	Bronze Frames	
08 13 00	Metal Doors	Includes: hinged or pivoting doors and fixed panels manufactured of metal. Alternate Terms/Abbreviations: hollow metal doors: steel doors. See: 08 11 00 for metal doors and frames. 08 11 63 for metal storm and screen doors and frames. 08 11 73 for sliding metal firedoors. 08 12 00 for metal frames. 08 32 00 for sliding metal-framed glass doors. 08 42 00 for metal-framed entrances.
08 13 13	Hollow Metal Doors	
08 13 13.13	Standard Hollow Metal Doors	
08 13 13.53	Custom Hollow Metal Doors	
08 13 16	Aluminum Doors	
08 13 16.13	Aluminum Terrace Doors	
08 13 19	Stainless-Steel Doors	
08 13 23	Bronze Doors	
08 13 73	Sliding Metal Doors	
08 13 76	Bifolding Metal Doors	

NUMBER	TITLE	EXPLANATION
08 14 00	Wood Doors	Includes: hinged, sliding, or pivoting doors manufactured of wood with solid or hollow cores, flush panel or stile and rail, with wood, plastic laminate, and metal veneers. Alternate Terms/Abbreviations: metal-faced doors: kalamein doors. Stile and rail wood doors: paneled wood doors. See: 06 48 00 for wood door frames. 08 12 00 for metal frames for wood doors. 08 15 00 for plastic doors. 08 16 00 for composite doors. 08 32 19 for sliding wood-framed glass doors.
08 14 13	Carved Wood Doors	
08 14 16	Flush Wood Doors	
08 14 23	Clad Wood Doors	
08 14 23.13	Metal-Faced Wood Doors	
08 14 23.16	Plastic-Laminate-Faced Wood Doors	
08 14 23.17	Plastic-Faced Wood Doors	
08 14 23.19	Molded-Hardboard-Faced Wood Doors	
08 14 29	Prefinished Wood Doors	
08 14 33	Stile and Rail Wood Doors	
08 14 33.13	Wood Terrace Doors	
08 14 66	Wood Screen Doors	Includes: doors and screens manufactured of wood used for storm or thermal protection or for screening, usually as adjuncts to other doors. See: 08 11 63 for metal storm and screen doors and frames. 08 15 66 for plastic screen doors and frames. 08 15 69 for plastic storm doors and frames. 08 32 19 for sliding wood-framed glass doors.
08 14 69	Wood Storm Doors	
08 14 73	Sliding Wood Doors	
08 14 76	Bifolding Wood Doors	
08 15 00	Plastic Doors	Includes: hinged, sliding, or pivoting doors manufactured of plastic. See: 08 11 63 for metal storm and screen doors and frames. 08 12 00 for metal frames for plastic doors. 08 14 00 for wood doors. 08 16 00 for composite doors. 08 32 16 for sliding plastic-framed glass doors.
08 15 13	Laminated Plastic Doors	
08 15 16	Solid Plastic Doors	

NUMBER	TITLE	EXPLANATION
08 15 66	Plastic Screen Doors	Includes: doors and screens manufactured of plastic used for storm or thermal protection or for screening, usually as adjuncts to other doors. See: 08 11 63 for metal storm and screen doors and frames. 08 14 66 for wood screen doors and frames. 08 14 69 for wood storm doors and frames.
08 15 69	Plastic Storm Doors	
08 15 73	Sliding Plastic Doors	
08 15 76	Bifolding Plastic Doors	
08 16 00	Composite Doors	Includes: hinged, sliding, or pivoting doors manufactured of composite materials. See: 08 12 00 for metal frames for composite doors. 08 14 00 for wood doors. 08 14 23 for clad wood doors. 08 15 00 for plastic doors.
08 16 13	Fiberglass Doors	
08 16 73	Sliding Composite Doors	
08 16 76	Bifolding Composite Doors	
08 17 00	Integrated Door Opening Assemblies	Includes: assemblies, manufactured of a variety of materials, complete with doors, frames, and hardware. Alternate Terms/Abbreviations: integrated door opening assemblies: preassembled door assemblies: packaged door assemblies: pre-hung doors. See: 08 34 00 for special function doors. 08 34 73 for sound control door assemblies. 08 39 00 for pressure-resistant doors.
08 17 13	Integrated Metal Door Opening Assemblies	
08 17 23	Integrated Wood Door Opening Assemblies	
08 17 33	Integrated Plastic Door Opening Assemblies	
08 17 43	Integrated Composite Door Opening Assemblies	
08 30 00	Specialty Doors and Frames	Includes: doors and door assemblies for a variety of special functions and applications involving a variety of operating methods. Fire-resistant rated doors and frames. Hardware, controls, operators, and drive mechanisms.

NUMBER	TITLE	EXPLANATION
08 31 00	Access Doors and Panels	Includes: interior and exterior doors and panels in walls, ceilings and floors to provide access to concealed spaces. See: 07 72 33 for roof hatches. 23 33 33 for duct-mounted access doors.
08 31 13	Access Doors and Frames	
08 31 13.53	Security Access Doors and Frames	
08 31 16	Access Panels and Frames	
08 32 00	Sliding Glass Doors	Includes: fully glazed doors and frames of any material with sliding operation. See: 08 13 73 for sliding metal doors. 08 14 73 for sliding wood doors. 08 15 73 for sliding plastic doors. 08 42 29 for sliding automatic entrance doors. 08 43 29 for sliding storefronts.
08 32 13	Sliding Aluminum-Framed Glass Doors	
08 32 16	Sliding Plastic-Framed Glass Doors	
08 32 19	Sliding Wood-Framed Glass Doors	
08 33 00	Coiling Doors and Grilles	Includes: doors and grilles of any material that open by coiling. Rolling fire-resistant rated doors and shutters. Alternate Terms/Abbreviations: coiling doors and grilles: roll-up doors and grilles: rolling doors and grilles. See: 08 11 74 for sliding metal grilles. 08 35 16 for folding grilles.
08 33 13	Coiling Counter Doors	
08 33 16	Coiling Counter Grilles	
08 33 23	Overhead Coiling Doors	
08 33 23.13	Overhead Rapid Coiling Doors	
08 33 26	Overhead Coiling Grilles	
08 33 33	Side Coiling Doors	
08 33 36	Side Coiling Grilles	
08 33 43	Overhead Coiling Smoke Curtains	
08 33 44	Overhead Coiling Fire Curtains	

NUMBER	TITLE	EXPLANATION
08 34 00	Special Function Doors	Includes: doors, gates, and frames of any material with specific characteristics for a special function. See: 08 17 00 for packaged door assemblies. 08 38 00 for traffic doors. 08 39 00 for pressure-resistant doors. 13 21 00 for controlled environment rooms for doors integral part of manufactured room construction.
08 34 13	Cold Storage Doors	
08 34 16	Hangar Doors	
08 34 19	Industrial Doors	
08 34 33	Lightproof Doors	Includes: doors with specific characteristics to exclude light when closed. Doors intended to allow use without light penetration for film developing or similar operations.
08 34 36	Darkroom Doors	
08 34 46	Radio-Frequency-Interference Shielding Doors	Includes: doors with specific characteristics to exclude radio-frequency-interference when closed. Alternate Terms/Abbreviations: RFI: radio-frequency-interference. See: 08 56 46 for radio-frequency-interference shielding windows.
08 34 49	Radiation Shielding Doors and Frames	Includes: doors with specific characteristics to exclude radiation when closed. See: 08 56 49 for radiation shielding windows. 08 88 49 for radiation-resistant glazing. 13 49 00 for radiation protection.
08 34 49.13	Neutron Shielding Doors and Frames	
08 34 53	Security Doors and Frames	Includes: doors, frames, and gates with specific characteristics for high security applications. See: 08 31 13 for security access doors and frames. 08 39 53 for blast-resistant doors. 08 42 33 for security revolving door entrances. 08 56 53 for security windows. 08 56 56 for security window screens. 08 71 53 for security door hardware. 08 88 53 for security glazing. 28 10 00 for access control

DIVISION **08**

NUMBER	TITLE	EXPLANATION
08 34 56	Security Gates	
08 34 58	File Room Doors and Frames	
08 34 59	Vault Doors and Day Gates	Includes: doors and gates with specific characteristics to provide security for a vault. See: 13 27 00 for factory assembled vaults.
08 34 73	Sound Control Door Assemblies	Includes: door and frame assemblies with specific characteristics to exclude sound when closed. Alternate Terms/Abbreviations: sound control door assemblies: sound retardant door assemblies. See: 08 17 00 for integrated door opening assemblies. 08 56 73 for sound control windows.
08 34 73.13	Metal Sound Control Door Assemblies	Includes: metal sound-control doors and frames, sound-control seals, thresholds, hardware, and accessories.
08 34 73.16	Wood Sound Control Door Assemblies	Includes: wood sound-control doors and frames, sound-control seals, thresholds, hardware, and accessories.
08 35 00	Folding Doors and Grilles	Includes: doors and grilles that open by folding as an accordion or as a set of panels. May be fire-resistant rated. See: 08 10 00 for bifolding doors. 08 11 73 for sliding metal fire doors. 08 11 74 for sliding metal grilles. 08 33 00 for coiling doors and grilles. 08 36 00 for panel doors. 10 22 16 for folding gates. 10 22 23 for portable partitions, screens, and panels.
08 35 13	Folding Doors	
08 35 13.13	Accordion Folding Doors	
08 35 13.23	Accordion Folding Fire Doors	
08 35 13.33	Panel Folding Doors	
08 35 16	Folding Grilles	
08 35 16.13	Accordion Folding Grilles	

NUMBER	TITLE	EXPLANATION
08 36 00	Panel Doors	Includes: panelized doors that open guided by movement of panels in sections or single panel. Alternate Terms/Abbreviations: panel doors: garage doors. See: 08 35 13 for panel folding doors. 10 22 23 for portable partitions, screens, and panels.
08 36 13	Sectional Doors	
08 36 16	Single-Panel Doors	
08 36 19	Multi-Leaf Vertical Lift Doors	
08 36 23	Telescoping Vertical Lift Doors	
08 38 00	Traffic Doors	Includes: doors with specific characteristics for heavy traffic, including vehicular traffic. See: 08 42 29 for automatic entrances. 08 42 33 for revolving door entrances. 11 13 00 for loading dock equipment.
08 38 13	Flexible Strip Doors	
08 38 16	Flexible Traffic Doors	
08 38 19	Rigid Traffic Doors	
08 39 00	Pressure-Resistant Doors	Includes: doors with specific characteristics to withstand and maintain differential pressure. See: 08 17 00 for packaged door assemblies. 08 34 00 for special function doors. 08 42 39 for pressure-resistance entrances. 08 55 00 for pressure-resistant windows. 08 88 39 for pressure-resistant glazing. 13 21 00 for controlled environment rooms for doors integral part of manufactured room construction.
08 39 13	Airtight Doors	
08 39 19	Watertight Doors	
08 39 53	Blast-Resistant Doors	

NUMBER	TITLE	EXPLANATION
08 40 00	Entrances, Storefronts, and Curtain Walls	Includes: entrances, storefronts, and curtain walls fabricated from formed and extruded metals and glass components for exterior and interior applications. Doors of various operating methods, framing for doors, sidelights, panels, and transoms. Shop glazing and shop applied hardware. May Include: perimeter joint sealant. See: 06 48 00 for wood frames. 08 11 00 for metal doors and frames. 08 32 00 for sliding metal-framed glass doors. 08 70 00 for field installed hardware. 08 80 00 for field installed glazing. See Also: 07 92 00 for joint sealants.
08 41 00	Entrances and Storefronts	Includes: entrances and storefronts of various materials for exterior and interior applications. See: 08 42 00 for entrances. 08 43 00 for storefronts.
08 41 13	Aluminum-Framed Entrances and Storefronts	Includes: entrances and storefronts of metal-framed glass construction for exterior and interior applications. Muntz metal. Alternate Terms/Abbreviations: muntz metal: yellow metal. See: 08 43 13 for aluminum-framed storefronts. 08 43 16 for bronze-framed storefronts. 08 43 19 for stainless steel-framed storefronts. 08 44 13 for glazed aluminum curtain walls. 08 44 16 for glazed bronze curtain walls. 08 44 19 for glazed stainless steel curtain walls.
08 41 13.13	Fire-Rated Aluminum-Framed Entrances and Storefronts	
08 41 16	Bronze-Framed Entrances and Storefronts	
08 41 19	Stainless-Steel-Framed Entrances and Storefronts	
08 41 23	Steel-Framed Entrances and Storefronts	
08 41 26	All-Glass Entrances and Storefronts	Includes: entrances and storefronts formed of glass panels without metal framing. See: 08 44 26 for structural glass curtain walls.
08 41 26.13	Point Supported All-Glass Storefronts	

NUMBER	TITLE	EXPLANATION
08 42 00	Entrances	Includes: entrances of various operating methods for exterior and interior applications. See: 08 41 00 for entrances and storefronts. 08 43 00 for storefronts.
08 42 13	Aluminum-Framed Entrances	Includes: exterior and interior manual-swing aluminum entrance door and frame units, thresholds, hardware, glazing, and accessories for doors.
08 42 26	All-Glass Entrances	Includes: entrances formed of glass panels without metal framing.
08 42 29	Automatic Entrances	Includes: entrances of any material with integral opening and closing operators and actuating devices, other than revolving doors. Alternate Terms/Abbreviations: automatic entrances: biparting entrances. See: 08 71 13 for automatic door operators for use with doors other than automatic entrances.
08 42 29.13	Folding Automatic Entrances	
08 42 29.23	Sliding Automatic Entrances	
08 42 29.33	Swinging Automatic Entrances	
08 42 33	Revolving Door Entrances	Includes: entrances of any material that are operated by revolving action. See: 08 71 53 for security door hardware. 08 88 53 for security glazing. 28 10 00 for access control.
08 42 33.13	Security Revolving Door Entrances	
08 42 36	Balanced Door Entrances	Includes: entrances of any material that employ pivots to allow operation with minimum force.
08 42 39	Pressure-Resistant Entrances	Includes: entrances intended to withstand and maintain differential pressure. See: 08 39 00 for pressure-resistance doors. 08 55 00 for pressure-resistant windows. 08 88 39 for pressure-resistant glazing.

NUMBER	TITLE	EXPLANATION
08 42 43	Intensive Care Unit/Critical Care Unit Entrances	Includes: entrances with specific characteristics for intensive care or critical care applications. Alternate Terms/Abbreviations: ICU/CCU: intensive care unit/critical care unit.
08 43 00	Storefronts	Includes: one- or two-story openings infilled with glass and other materials for exterior and interior applications. Alternate Terms/Abbreviations: storefronts: mallfronts. See: 08 41 00 for entrances and storefronts. 08 42 00 for entrances.
08 43 11	Timber-Framed Storefronts	
08 43 13	Aluminum-Framed Storefronts	Includes: storefronts of metal-framed glass construction for exterior and interior applications. See: 08 44 13 for glazed aluminum curtain walls. 08 44 16 for glazed bronze curtain walls. 08 44 19 for glazed stainless steel curtain walls.
08 43 13.13	Fire-Rated Aluminum Storefronts	
08 43 16	Bronze-Framed Storefronts	
08 43 19	Stainless-Steel-Framed Storefronts	
08 43 23	Steel-Framed Storefronts	
08 43 26	All-Glass Storefronts	Includes: storefronts formed of glass panels without metal framing. See: 08 44 26 for structural glass curtain walls.
08 43 27	Channel Glass Storefronts	
08 43 29	Sliding Storefronts	Includes: storefronts of any material that open through sliding action of the panels.
08 44 00	Curtain Wall and Glazed Assemblies	Includes: multi-story cladding fabricated from glass and other materials for exterior applications. Framing members, sash, glazing, and sealants for complete enclosure. Insulated and non-insulated spandrel and infill panels. Louvers integral to curtain wall. Fixed sun screens integral to curtain wall. See: 04 42 43 for stone panels for curtain walls. 14 84 13 for window washing scaffolding. 14 84 23 for window washing hoists.

NUMBER	TITLE	EXPLANATION
08 44 11	Glazed Timber Curtain Walls	
08 44 12	Glazed Composite Curtain Wall	
08 44 12.13	Glazed Fiberglass Curtain Wall	
08 44 13	Glazed Aluminum Curtain Walls	Includes: multi-story cladding of metal-framed glass construction for exterior applications.
		See: 08 41 13 for aluminum-framed entrances and storefronts. 08 41 16 for bronze-framed entrances and storefronts. 08 41 19 for stainless steel-framed entrances and storefronts.
08 44 16	Glazed Bronze Curtain Walls	
08 44 18	Glazed Steel Curtain Walls	
08 44 19	Glazed Stainless-Steel Curtain Walls	
08 44 23	Structural Sealant Glazed Curtain Walls	
08 44 26	Structural Glass Curtain Walls	Includes: structural glass walls and tensile curtain walls.
		See: 08 41 26 for all-glass entrances and storefronts.
08 44 26.13	Glass Framed Point Supported Glass Curtain Walls	
08 44 26.16	Metal Framed Point Supported Glass Curtain Walls	
08 44 26.23	Cable Framed Point Supported Glass Curtain Walls	
08 44 33	Sloped Glazing Assemblies	Includes: glazed curtain walls in which the glazed panels are primarily on a sloped plane.
		See: 08 43 26 for all glass storefronts. 13 34 13 for glazed structures.
08 45 00	Translucent Wall and Roof Assemblies	Includes: translucent panel assemblies.
08 45 13	Structured-Polycarbonate-Panel Assemblies	
08 45 23	Fiberglass-Sandwich-Panel Assemblies	
08 46 00	Window Wall Assemblies	

NUMBER	TITLE	EXPLANATION
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08 50 00 Windows

Includes: windows of any material, both operating and non-operating. Fixed, single-hung, double-hung, sliding, projected, hopper, awning, casement, and jalousie windows. Operating hardware, screens, and other accessories. Factory glazing. Louver blinds integral to window construction.

May Include: window sills and stools. Perimeter joint sealant.

See: 08 01 50 for replacement windows.

08 12 00 for metal sidelight, transom, and borrowed light frames.

08 40 00 for entrances, storefronts, and curtain walls.

08 61 00 for roof windows.

08 80 00 for field-installed glazing.

10 74 43 for below-grade egress assemblies.

10 74 46 for window wells.

13 34 13 for glazed structures.

See Also: 05 58 00 for formed metal sills and stools.

06 46 00 for wood sills, stools, and casings.

07 92 00 for joint sealants.

08 51 00 Metal Windows

Includes: windows framed with any metal.

See: 08 43 00 for metal-framed storefronts.

08 51 13	Aluminum Windows
08 51 16	Bronze Windows
08 51 19	Stainless-Steel Windows
08 51 23	Steel Windows
08 51 23.13	Hot-Rolled Steel Windows

Includes: hot-rolled steel windows, glazing, and hardware.

08 51 23.23	Cold-Rolled Steel Windows
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Includes: cold-rolled steel windows, glazing, and hardware.

08 51 66	Metal Window Screens
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Includes: windows and screens framed with metal used for storm or thermal protection or for screening, usually as adjuncts to other windows.

See: 08 52 66 for wood window screens.

08 52 69 for wood storm windows.

08 53 66 for vinyl window screens.

08 53 69 for vinyl storm windows.

08 54 66 for fiberglass window screens.

08 54 69 for fiberglass storm windows.

NUMBER	TITLE	EXPLANATION
08 51 69	Metal Storm Windows	
08 52 00	Wood Windows	Includes: windows framed with wood. See: 06 48 00 for custom wood framed windows.
08 52 13	Metal-Clad Wood Windows	
08 52 16	Plastic-Clad Wood Windows	
08 52 66	Wood Window Screens	Includes: windows and screens framed with wood used for storm or thermal protection or for screening, usually as adjuncts to other windows. See: 08 51 66 for metal window screens. 08 51 69 for metal storm windows. 08 53 66 for vinyl window screens. 08 53 69 for vinyl storm windows. 08 54 66 for fiberglass window screens. 08 54 69 for fiberglass storm windows.
08 52 69	Wood Storm Windows	
08 53 00	Plastic Windows	Includes: windows framed with plastic. See: 08 51 66 for metal window screens. 08 51 69 for metal storm windows. 08 52 66 for wood window screens. 08 52 69 for wood storm windows. 08 54 66 for fiberglass window screens. 08 54 69 for fiberglass storm windows.
08 53 13	Vinyl Windows	
08 53 66	Vinyl Window Screens	Includes: windows and screens framed with vinyl used for storm or thermal protection or for screening, usually as adjuncts to other windows.
08 53 69	Vinyl Storm Windows	
08 54 00	Composite Windows	Includes: windows framed with composite materials. See: 08 52 13 for metal-clad wood windows. 08 52 16 for plastic-clad wood windows.
08 54 13	Fiberglass Windows	

NUMBER	TITLE	EXPLANATION
08 54 66	Fiberglass Window Screens	Includes: windows and screens framed with fiberglass used for storm or thermal protection or for screening, usually as adjuncts to other windows. See: 08 51 66 for metal window screens. 08 51 69 for metal storm windows. 08 52 66 for wood window screens. 08 52 69 for wood storm windows. 08 53 66 for vinyl window screens. 08 53 69 for vinyl storm windows.
08 54 69	Fiberglass Storm Windows	
08 54 73	Vinyl-Wood Composite Windows	
08 55 00	Pressure-Resistant Windows	Includes: windows with special characteristics to withstand and maintain differential pressure. See: 08 39 00 for pressure-resistant doors. 08 42 39 for pressure-resistant entrances. 08 88 39 for pressure-resistant glazing.
08 55 13	Tornado-Resistant Windows	See Also: 04 23 00 for glass unit masonry.
08 55 23	Blast-Resistant Windows	See Also: 04 23 00 for glass unit masonry.
08 56 00	Special Function Windows	Includes: windows with special characteristics for a special function. May Include: accessories such as pivoting turntables, voice transmitters, and deal trays.
08 56 19	Pass Windows	May Include: accessories such as pivoting turntables, voice transmitters, and deal trays.
08 56 46	Radio-Frequency-Interference Shielding Windows	Includes: windows with special characteristics to exclude radio-frequency interference. Alternate Terms/Abbreviations: RFI: radio frequency interference. See: 08 34 46 for radio-frequency-interference shielding doors.

NUMBER	TITLE	EXPLANATION
08 56 49	Radiation Shielding Windows	Includes: windows with special characteristics to exclude radiation. See: 08 34 49 for radiation shielding doors. 08 88 49 for radiation-resistant glazing. 13 49 00 for radiation protection.
08 56 53	Security Windows	Includes: windows and window screens with special characteristics for high security applications. Alternate Terms/Abbreviations: service and teller windows: cashier windows. See: 08 88 53 for security glazing. 28 20 00 for video surveillance.
08 56 56	Security Window Screens	
08 56 59	Service and Teller Window Units	
08 56 73	Sound Control Windows	Includes: windows with specific characteristics to exclude sound. Alternate Terms/Abbreviations: sound control windows: sound retardant windows. See: 08 34 73 for sound control doors.
08 56 88	Interior Insulating Windows	
08 60 00	Roof Windows and Skylights	Includes: glazed openings located above the interior space, fixed or operating. Shop glazing, operating hardware, screens, and other accessories. See: 04 23 19 for glass unit masonry skylights. 08 44 33 for sloped glazed assemblies. 08 80 00 for field-installed glazing. 13 34 13 for glazed structures.
08 61 00	Roof Windows	Includes: operable and non-operable roof windows framed of any material.
08 61 13	Metal Roof Windows	
08 61 16	Wood Roof Windows	

NUMBER	TITLE	EXPLANATION
08 62 00	Unit Skylights	Includes: skylights without framing except at the perimeter of the glazing. Alternate Terms/Abbreviations: skylight: ceiling domes.
08 62 13	Domed Unit Skylights	
08 62 16	Pyramidal Unit Skylights	
08 62 19	Vaulted Unit Skylights	
08 62 23	Tubular Skylights	
08 63 00	Metal-Framed Skylights	Includes: skylights with any metal framing.
08 63 13	Domed Metal-Framed Skylights	
08 63 16	Pyramidal Metal-Framed Skylights	
08 63 19	Vaulted Metal-Framed Skylights	
08 63 23	Ridge Metal-Framed Skylights	
08 63 53	Motorized Metal-Framed Skylights	
08 64 00	Plastic-Framed Skylights	Includes: skylights with plastic framing.
08 65 00	Glazed Canopies	
08 65 13	Suspended Glazed Canopies	Includes: custom constructed canopy assemblies. See Also: 10 73 00 for protective covers.
08 67 00	Skylight Protection and Screens	Includes: screens and other protection for skylights.
08 70 00	Hardware	Includes: hardware and gaskets for doors and windows not supplied as part of the manufactured item. Electronic, magnetic, hydraulic, and pneumatic components for operating, controlling, and monitoring doors and windows.
08 71 00	Door Hardware	Includes: hardware for the operation of doors, including hanging, latching, locking, controlling, and automatic operation. Hinges, pivots, sliding and folding door hardware, latches, locks, cylinders, exit devices, closers, holders, self-closing hinges, push plates, kick plates, other door trim, weatherstripping, seals, gaskets, and thresholds. Alternate Terms/Abbreviations: door hardware: finish hardware, builder's hardware. Exit devices: panic devices.

NUMBER	TITLE	EXPLANATION
08 71 13	Automatic Door Operators	
08 71 53	Security Door Hardware	Includes: hardware with specific characteristics for high security applications.
08 74 00	Non-Integrated Access Control Hardware	
08 74 13	Non-Integrated Card Key Access Control Hardware	
08 74 16	Non-Integrated Keypad Access Control Hardware	
08 74 19	Non-Integrated Biometric Identity Access Control Hardware	
08 75 00	Window Hardware	
		Includes: hardware for the operation of windows. Weatherstripping, seals, and gaskets for windows.
08 75 13	Automatic Window Equipment	
08 75 16	Window Operators	
08 78 00	Special Function Hardware	
		Includes: mechanical and electro-mechanical hardware intended for a specialized purpose, which establishes a determining characteristic.
08 79 00	Hardware Accessories	
		Includes: accessories for doors and windows not included in other titles.
08 79 13	Key Storage Equipment	
08 80 00	Glazing	
		Includes: glass and plastic glazing for doors, sidelights, transoms, borrowed lights, entrances, storefronts, curtain walls, windows, skylights, and glazed structures.
		Does Not Include: glazing integral with manufactured items such as casework, food service equipment, and telephone enclosures.
		See: 08 10 00 for doors and frames. 08 30 00 for shop glazing of entrances, storefronts, and curtain walls. 08 50 00 for factory glazing of windows. 08 60 00 for shop glazing of roof windows and skylights. 13 34 13 for glazed structures.

NUMBER	TITLE	EXPLANATION
08 81 00	Glass Glazing	Includes: transparent and translucent glass glazing for general and special purpose applications. Alternate Terms/Abbreviations: low-e: low emissivity. See: 04 23 00 for glass unit masonry. 05 73 13 for glazed decorative metal railings. 12 17 00 for etched and stained art glass. Products: bent glass, chemically strengthened glass, coated glass, composite glass, fire-resistant rated glass, float glass, heat-strengthened glass, impact resistant glass, insulating glass, low emissivity glass, laminated glass, rolled glass, spandrel glass, tempered glass, and wired glass.
08 81 13	Decorative Glass Glazing	
08 81 23	Exterior Glass Glazing	
08 81 26	Interior Glass Glazing	
08 83 00	Mirrors	Includes: unframed mirrors of any material. See: 08 88 33 for transparent mirrored glazing. 10 28 00 for metal-framed mirrors as toilet or bath accessory. 10 86 00 for security domes and mirrors. 12 40 00 for mirrors as part of furnishings.
08 83 13	Mirrored Glass Glazing	
08 83 16	Mirrored Plastic Glazing	
08 84 00	Plastic Glazing	Includes: transparent and translucent plastic glazing for general and special purpose applications. Insulating, acrylic, polycarbonate plastic glazing. See: 08 88 00 for special function glazing. 09 76 00 for plastic blocks. 12 17 00 for etched and stained art glass. Products: insulating, acrylic, polycarbonate plastic glazing.
08 84 13	Decorative Plastic Glazing	

NUMBER	TITLE	EXPLANATION
08 85 00	Glazing Accessories	Includes: accessories to install glazing. See: 08 87 00 for glazing surface films. Products: glazing compounds, setting blocks, shims, tape, splines, clips, glazing sealant, and other accessories.
08 87 00	Glazing Surface Films	Includes: films applied to surface of glazing to change performance or appearance of glazing. See: 08 56 53 for security windows. 08 88 53 for security glazing. 10 71 13 for exterior sun control devices.
08 87 13	Solar Control Films	
08 87 23	Safety and Security Films	
08 87 23.13	Safety Films	
08 87 23.16	Security Films	
08 87 26	Bird Control Film	
08 87 33	Decorative Films	
08 88 00	Special Function Glazing	Includes: glazing with specific characteristics for a special function or application. See: 05 16 00 for structural cabling. 13 31 23 for tensioned fabric structures with cable supports.
08 88 13	Fire-Rated Glazing	
08 88 19	Hurricane-Resistant Glazing	
08 88 23	Cable Suspended Glazing	
08 88 33	Transparent Mirrored Glazing	
08 88 36	Switchable Glass	
08 88 36.16	Electronically Controlled Switchable Glass	
08 88 39	Pressure-Resistant Glazing	Includes: glazing with specific characteristics to withstand and maintain differential pressure. See: 08 39 00 for pressure-resistant doors. 08 42 39 for pressure-resistant entrances. 08 55 00 for pressure-resistant windows.

NUMBER	TITLE	EXPLANATION
08 88 49	Radiation-Resistant Glazing	Includes: glazing with specific characteristics to exclude radiation. See: 08 34 49 for radiation shielding doors. 08 56 49 for radiation shielding windows. 13 49 00 for radiation protection.
08 88 53	Security Glazing	Includes: glazing with specific characteristics for high security applications. Alternate Terms/Abbreviations: ballistics-resistant: bullet-resistant. See: 08 34 53 for security doors and frames. 08 39 53 for blast-resistant doors. 08 42 33 for security revolving door entrances. 08 56 53 for security windows. 28 10 00 for access control.
08 88 56	Ballistics-Resistant Glazing	
08 90 00	Louvers and Vents	Includes: items and enclosures for ventilation that are not an integral part of mechanical equipment. See: 10 82 00 for grilles and screens.
08 91 00	Louvers	Includes: louvers of any material. See: 07 42 00 for louvers integral with wall panels. 08 44 00 for louvers integral with curtain walls. 23 37 00 for louvers integral with HVAC ducts.
08 91 13	Motorized Wall Louvers	
08 91 16	Operable Wall Louvers	
08 91 19	Fixed Louvers	
08 91 26	Door Louvers	Includes: louvers in doors which are not integral to door construction. May Include: louvers in fire-resistant rated doors. See: 08 10 00 for louvers integral with door construction. See Also: 08 10 00 for louvers in fire-resistant rated doors.

NUMBER	TITLE	EXPLANATION
08 92 00	Louvered Equipment Enclosures	Includes: louvers used as visual screening of equipment. See: 10 82 13 for exterior grilles and screens. 23 37 23 for penthouse ventilators.
08 95 00	Vents	Includes: vents of any material. Alternate Terms/Abbreviations: wall vents: brick vents. See: 07 72 00 for roof vents.
08 95 13	Soffit Vents	
08 95 16	Wall Vents	Alternate Terms/Abbreviations: block and brick vents.
08 95 33	Explosion Vents	
08 95 43	Flood Vents	

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NUMBER	TITLE	EXPLANATION
DIVISION 09 – FINISHES		
09 00 00	Finishes	May be used as Division level Section title.

09 01 00 Maintenance of Finishes

Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of finishes.

Level 4 Numbering Recommendation: Level 4 titles for finishes are provided as examples for user's creation of more detailed titles for other subjects as required. Following Level 4 numbering is recommended:

.51-.59 for maintenance.

.61-.69 for repair.

.71-.79 for rehabilitation.

.81-.89 for replacement.

.91-.99 for restoration.

See: 02 83 33 for lead-based paint removal and disposal.

09 01 20	Maintenance of Plaster and Gypsum Board
09 01 20.91	Plaster Restoration
09 01 30	Maintenance of Tiling
09 01 30.91	Tile Restoration
09 01 50	Maintenance of Ceilings
09 01 50.91	Ceiling Restoration
09 01 60	Maintenance of Flooring
09 01 60.91	Flooring Restoration
09 01 70	Maintenance of Wall Finishes
09 01 70.91	Wall Finish Restoration
09 01 80	Maintenance of Acoustic Treatment
09 01 90	Maintenance of Painting and Coating
09 01 90.51	Paint Cleaning
09 01 90.52	Maintenance Repainting
09 01 90.53	Maintenance Coatings
09 01 90.61	Repainting
09 01 90.91	Paint Restoration
09 01 90.92	Coating Restoration
09 01 90.93	Paint Preservation

09 03 00 Conservation Treatment of Period Finishes

09 03 21	Conservation Treatment for Period Plaster Assemblies
	May Include: requirements for gypsum plastering, Portland cement plastering, veneer plastering, and plaster fabrications and details.
09 03 25	Conservation Treatment for Period Plastering
09 03 25.23	Conservation Treatment for Period Lime-Based Plaster
09 03 30	Conservation Treatment for Period Tile

DIVISION 09

NUMBER	TITLE	EXPLANATION
09 03 50	Conservation Treatment for Period Ceilings	
09 03 51	Conservation Treatment for Period Suspended Ceilings	
09 03 56	Conservation Treatment for Textured Ceilings	
09 03 56.16	Conservation Treatment for Period Metal-Panel Textured Ceilings	
09 03 60	Conservation Treatment for Period Flooring	
09 03 63	Conservation Treatment for Period Masonry Flooring	
09 03 63.13	Conservation Treatment for Period Brick Flooring	
09 03 63.40	Conservation Treatment for Period Stone Flooring	
09 03 64	Conservation Treatment for Period Wood Flooring	
09 03 64.29	Conservation Treatment for Period Wood Strip and Plank Flooring	
09 03 65	Conservation Treatment for Period Resilient Flooring	
09 03 65.16	Conservation Treatment for Period Resilient Sheet Flooring	
09 03 66	Conservation Treatment for Period Terrazzo Flooring	
09 03 70	Conservation Treatment for Period Wall Finishes	
09 03 72	Conservation Treatment for Period Wall Coverings	
09 03 90	Conservation Treatment for Period Painting and Coating	
09 03 91	Conservation Treatment for Period Painted Surfaces	May Include: repainting of existing interior and exterior surfaces, analysis of historic coating components, and matching faux finishes.
09 03 91.13	Conservation Treatment for Period Exterior Painting	
09 03 91.23	Conservation Treatment for Period Interior Painting	

09 05 00 Common Work Results for Finishes

Includes: subjects common to multiple titles in Division 09.

See: 01 80 00 for performance requirements of subjects common to multiple titles.

09 05 05	Selective Demolition for Finishes	See: 02 41 19 for selective demolition for whole structure demolition.
09 05 13	Common Finishes	
09 05 61	Common Work Results for Flooring Preparation	
09 05 61.13	Moisture Vapor Emission Control	
09 05 71	Acoustic Underlayment	

09 06 00 Schedules for Finishes

Includes: schedules of items common to multiple titles in Division 09.

Notes: a schedule may be included on drawings, in the project manual, or a project book.

09 06 00.13	Room Finish Schedule	
09 06 20	Schedules for Plaster and Gypsum Board	
09 06 30	Schedules for Tiling	
09 06 50	Schedules for Ceilings	

NUMBER	TITLE	EXPLANATION
09 06 60	Schedules for Flooring	
09 06 70	Schedules for Wall Finishes	
09 06 80	Schedules for Acoustical Treatment	
09 06 90	Schedules for Painting and Coating	
09 06 90.13	Paint Schedule	

09 08 00 Commissioning of Finishes

Includes: commissioning of items common to multiple titles in Division 09.

See: 01 91 00 for commissioning of subjects common to multiple Divisions.

09 20 00 Plaster and Gypsum Board

Includes: non-structural metal support assemblies for plaster and gypsum board. Exterior and interior veneer plaster, acoustical plaster, plaster partitions, and plaster fireproofing. Exterior and interior gypsum board, gypsum board assemblies, accessories, and trim.

May Include: acoustical insulation and sealants.

See: 05 40 00 for structural cold-formed metal framing.

06 11 00 for wood framing.

07 24 00 for exterior insulation and finish systems.

See Also: 07 92 19 for acoustic sealants.

09 81 00 for acoustic insulation.

09 21 00 Plaster and Gypsum Board Assemblies

Includes: plaster, lath, board, non-structural framing, accessories, and trim for plaster and gypsum board assemblies.

09 21 13	Plaster Assemblies
09 21 16	Gypsum Board Assemblies
09 21 16.23	Gypsum Board Shaft Wall Assemblies
09 21 16.33	Gypsum Board Area Separation Wall Assemblies

NUMBER	TITLE	EXPLANATION
09 22 00	Supports for Plaster and Gypsum Board	Includes: non-structural framing systems, ceiling suspension systems, and furring systems for plaster and gypsum board. Lathing, accessories, and trim for plaster. Base for veneer plaster. See: 05 10 00 for structural metal framing. 05 40 00 for cold-formed structural metal framing. 06 11 00 for wood framing. 07 40 00 for metal framing with roofing and siding panels. 08 44 00 for metal framing for curtain walls. 09 21 00 for metal framing for plaster and gypsum board assemblies. 09 53 00 for acoustical ceiling suspension assemblies. 13 34 00 for metal supports for fabricated engineered structures. 23 32 00 for air plenums and chases.
09 22 13	Metal Furring	
09 22 13.13	Metal Channel Furring	
09 22 13.23	Resilient Channel Furring	
09 22 16	Non-Structural Metal Framing	
09 22 16.13	Non-Structural Metal Stud Framing	
09 22 26	Suspension Systems	
09 22 26.23	Metal Suspension Systems	
09 22 26.33	Plastic Suspension Systems	
09 22 36	Lath	
09 22 36.13	Gypsum Lath	
09 22 36.23	Metal Lath	
09 22 39	Veneer Plaster Base	
09 23 00	Gypsum Plastering	Includes: gypsum plaster fireproofing. May Include: non-structural metal framing, ceiling suspension, and furring systems. See: 07 82 00 for board fire protection. See Also: 09 22 00 for metal supports for plaster.
09 23 13	Acoustical Gypsum Plastering	
09 23 82	Fire Protected Gypsum Plastering	

NUMBER	TITLE	EXPLANATION
09 24 00	Cement Plastering	Includes: interior and exterior plaster (stucco) finishes. May Include: non-structural metal framing, ceiling suspension, and furring systems. See: 07 24 00 for exterior insulation and finish systems. See Also: 09 22 00 for non-structural metal supports for plaster.
09 24 13	Acoustical Cement Plastering	
09 24 23	Cement Stucco	
09 24 33	Cement Parging	
09 25 00	Other Plastering	Includes: acrylic and lime-based plastering. Venetian plastering. May Include: non-structural metal framing, ceiling suspension, and furring systems. See Also: 09 22 00 for non-structural metal supports for plaster.
09 25 13	Acrylic Plastering	
09 25 13.13	Acrylic Plaster Finish	
09 25 23	Lime Based Plastering	
09 25 26	Natural Clay Plastering	Includes: adobe finishes.
09 26 00	Veneer Plastering	Includes: gypsum plaster capable of thin application over veneer plaster base. May Include: non-structural metal framing, ceiling suspension, and furring systems. See Also: 09 22 00 for non-structural metal supports for plaster.
09 26 13	Gypsum Veneer Plastering	
09 27 00	Plaster Fabrications	Includes: glass-fiber-reinforced plaster and simulated plaster fabrications. Alternate Terms/Abbreviations: GFRP: glass-fiber-reinforced plaster. See: 06 82 00 for glass-fiber-reinforced plastic.

NUMBER	TITLE	EXPLANATION
09 27 13	Glass-Fiber-Reinforced Gypsum Fabrications	
09 27 23	Simulated Plaster Fabrications	
09 28 00	Backing Boards and Underlayments	Includes: gypsum and cementitious backing board for other finishes. See: 09 30 00 for tiling.
09 28 13	Cementitious Backing Boards	
09 28 16	Glass-Mat Faced Gypsum Backing Boards	
09 28 19	Fiberglass Gypsum Backing Boards	
09 29 00	Gypsum Board	Includes: unfinished and prefinished gypsum board over metal and wood framing; accessories and trim; joint taping and finishing. May Include: non-structural metal framing, ceiling suspension, and furring systems. Gypsum and cementitious backing board for other finishes. Alternate Terms/Abbreviations: gypsum board: gypsum wallboard: drywall. See: 05 40 00 for cold-formed structural metal framing. 06 11 00 for wood framing. 07 82 00 for board fire protection. 09 56 13 for gypsum-panel textured ceilings. 10 22 19 for demountable gypsum board partitions. See Also: 06 16 43 for gypsum sheathing. 09 22 00 for metal supports for gypsum board. 09 28 00 for backing boards and underlayments.
09 29 82	Gypsum Board Fire Protection	

NUMBER	TITLE	EXPLANATION
09 30 00	Tiling	Includes: manufactured surfacing units of impervious, vitreous, semi-vitreous, and non-vitreous materials; glazed, unglazed, conductive, abrasive, and textured surfaces. Related mortar, adhesives, grout, trim, and accessories. Related antifracture membranes. Tile towel bar holders, toilet paper holders, soap dishes, and other tile accessories. Tiling by tile material. May Include: tile murals. See: 03 45 13 for tile faced architectural precast concrete. 07 32 00 for roof tiles. 07 44 19 for tile faced panels. 09 28 00 for tile backing boards. 09 51 23 for acoustical tile ceilings. 09 63 40 for stone flooring. 09 66 16 for terrazzo floor tile. 09 75 00 for stone facing. See Also: 12 11 00 for tile murals. Products: ceramic tile: porcelain mosaic tile, glazed wall tile, monocuttura tile, and bicottura tile. Paver tile: glazed and unglazed porcelain paver ceramic tile and natural clay paver ceramic tile. Stone tile: natural cut agglomerate tile.
09 30 13	Ceramic Tiling	Includes: ceramic tile: porcelain mosaic tile, glazed wall tile, monocuttura tile, and bicottura tile.
09 30 16	Quarry Tiling	Includes: paver tile: glazed and unglazed porcelain paver ceramic tile and natural clay paver ceramic tile.
09 30 19	Paver Tiling	
09 30 23	Glass Mosaic Tiling	Includes: stone tile: natural cut agglomerate tile.
09 30 26	Plastic Tiling	
09 30 29	Metal Tiling	
09 30 33	Stone Tiling	
09 30 36	Concrete Tiling	
09 30 39	Brick Tiling	

NUMBER	TITLE	EXPLANATION
09 31 00	Thin-Set Tiling	Includes: installation of tile without a thick mortar bed by use of dry-set mortars, latex-modified mortars, epoxy mortars, furan resin mortars, and adhesives. See: 09 32 00 for mortar-bed tiling. 09 33 00 for conductive tiling. 09 34 00 for waterproofing-membrane tiling. 09 35 00 for chemical-resistant tiling.
09 31 13	Thin-Set Ceramic Tiling	
09 31 16	Thin-Set Quarry Tiling	
09 31 19	Thin-Set Paver Tiling	
09 31 23	Thin-Set Glass Mosaic Tiling	
09 31 26	Thin-Set Plastic Tiling	
09 31 29	Thin-Set Metal Tiling	
09 31 33	Thin-Set Stone Tiling	
09 31 36	Thin-Set Concrete Tiling	
09 32 00	Mortar-Bed Tiling	Includes: installation of tile with thick mortar bed. Welded wire fabric reinforcing. Alternate Terms/Abbreviations: WWF: welded wire fabric, welded wire mesh: welded wire reinforcement. See: 03 22 00 for fabric and grid reinforcing. 09 31 00 for thin-set tiling. 09 33 00 for conductive tiling. 09 34 00 for waterproofing-membrane tiling. 09 63 00 for masonry flooring with thick mortar bed.
09 32 13	Mortar-Bed Ceramic Tiling	
09 32 16	Mortar-Bed Quarry Tiling	
09 32 19	Mortar-Bed Paver Tiling	
09 32 23	Mortar-Bed Glass Mosaic Tiling	
09 32 26	Mortar-Bed Plastic Tiling	
09 32 29	Mortar-Bed Metal Tiling	
09 32 33	Mortar-Bed Stone Tiling	
09 32 36	Mortar-Bed Concrete Tiling	

NUMBER	TITLE	EXPLANATION
09 33 00	Conductive Tiling	Includes: installation of electrically conductive tiles, which are static conductive to prevent spark discharge. See: 09 31 00 for thin-set tiling. 09 32 00 for mortar-bed tiling. 09 34 00 for waterproofing-membrane tiling. 09 35 00 for chemical-resistant tiling. 09 61 36 for static-resistant flooring treatment. 09 65 33 for conductive resilient flooring. 09 65 36 for static-control resilient flooring. 09 66 33 for conductive terrazzo flooring. 09 67 13 for conductive elastomeric liquid flooring.
09 33 13	Conductive Ceramic Tiling	
09 33 16	Conductive Quarry Tiling	
09 33 19	Conductive Paver Tiling	
09 33 23	Conductive Glass Mosaic Tiling	
09 33 26	Conductive Plastic Tiling	
09 33 29	Conductive Metal Tiling	
09 33 33	Conductive Stone Tiling	
09 33 36	Conductive Concrete Tiling	
09 34 00	Waterproofing-Membrane Tiling	Includes: tiling with integral waterproofing. See: 07 10 00 for separate waterproofing membrane. 09 31 00 for thin-set tiling. 09 32 00 for mortar-bed tiling. 09 33 00 for conductive tiling. 09 35 00 for chemical-resistant tiling.
09 34 13	Waterproofing-Membrane Ceramic Tiling	
09 34 16	Waterproofing-Membrane Quarry Tiling	
09 34 19	Waterproofing-Membrane Paver Tiling	
09 34 23	Waterproofing-Membrane Glass Mosaic Tiling	
09 34 26	Waterproofing-Membrane Plastic Tiling	
09 34 29	Waterproofing-Membrane Metal Tiling	
09 34 33	Waterproofing-Membrane Stone Tiling	
09 34 36	Waterproofing-Membrane Concrete Tiling	

NUMBER	TITLE	EXPLANATION
09 35 00	Chemical-Resistant Tiling	Includes: tiling with chemical-resistant tile, mortar, and grout. See: 04 61 00 for chemical-resistant brick masonry. 09 31 00 for thin-set tiling. 09 32 00 for mortar-bed tiling. 09 33 00 for conductive tiling. 09 34 00 for waterproofing-membrane tiling. 09 62 35 for acid-resistant flooring. 09 63 13 for chemical-resistant brick flooring. 09 96 35 for chemical-resistant coatings.
09 35 13	Chemical-Resistant Ceramic Tiling	
09 35 16	Chemical-Resistant Quarry Tiling	
09 35 19	Chemical-Resistant Paver Tiling	
09 35 23	Chemical-Resistant Glass Mosaic Tiling	
09 35 26	Chemical-Resistant Plastic Tiling	
09 35 29	Chemical-Resistant Metal Tiling	
09 35 33	Chemical-Resistant Stone Tiling	
09 35 36	Chemical-Resistant Concrete Tiling	

09 50 00 Ceilings

Includes: ceiling panels, units, and materials manufactured as finished ceiling surfaces. Ceiling suspension assemblies.

See: 23 32 33 for air-distribution ceiling plenums.

09 51 00 Acoustical Ceilings

Includes: suspended and directly applied tiles and panels with specific characteristics for acoustical purposes.

May Include: ceiling suspension assemblies.

See: 05 34 00 for acoustical metal decking.
09 23 13 for acoustical gypsum plastering.
09 58 00 for integrated ceiling assemblies.
09 62 48 for acoustic flooring.
09 80 00 for acoustic treatment.
09 83 16 for acoustic ceiling coating.
13 21 48 for sound-conditioned rooms.
13 48 00 for sound, vibration, and seismic control.

See Also: 09 53 00 for acoustical ceiling suspension assemblies.

09 51 13	Acoustical Panel Ceilings	
09 51 14	Acoustical Fabric-Faced Panel Ceilings	See: 09 54 46 for fabric-wrapped ceiling panels.

NUMBER	TITLE	EXPLANATION
09 51 23	Acoustical Tile Ceilings	
09 51 26	Acoustical Wood Ceilings	
09 51 33	Acoustical Metal Pan Ceilings	
09 51 33.13	Acoustical Snap-in Metal Pan Ceilings	
09 51 53	Direct-Applied Acoustical Ceilings	
09 53 00	Acoustical Ceiling Suspension Assemblies	Includes: suspension systems for acoustical ceilings. See: 09 22 26 for suspension systems. 09 51 00 for acoustical ceilings.
09 53 13	Curved Profile Ceiling Suspension Assemblies	
09 53 23	Metal Acoustical Ceiling Suspension Assemblies	
09 53 33	Plastic Acoustical Ceiling Suspension Assemblies	
09 54 00	Specialty Ceilings	Includes: specialty ceiling panels, units, and materials manufactured as finished ceiling surfaces. Acoustical provisions are of secondary consideration. Ceiling suspension assemblies. May Include: acoustical provisions. See: 05 34 00 for acoustical metal decking. 08 83 00 for mirrors. 09 51 00 for acoustical ceilings. 09 56 00 for textured ceilings. 09 57 00 for special function ceilings. 09 58 00 for integrated ceiling assemblies. 09 77 00 for special wall surfacing. 12 05 13 for fabrics. 26 51 00 for interior lighting.
09 54 13	Open Metal Mesh Ceilings	
09 54 16	Luminous Ceilings	
09 54 19	Mirror Panel Ceilings	
09 54 21	Metal Pan Ceilings	
09 54 23	Linear Metal Ceilings	
09 54 26	Suspended Wood Ceilings	
09 54 29	Suspended Plastic Ceilings	
09 54 33	Decorative Panel Ceilings	
09 54 36	Suspended Decorative Grids	
09 54 43	Stretched-Fabric Ceiling Systems	
09 54 46	Fabric-Wrapped Ceiling Panels	
09 54 53	Fiberglass Reinforced Panel Ceilings	Alternate Terms/Abbreviations: FRP.

NUMBER	TITLE	EXPLANATION
09 56 00	Textured Ceilings	Includes: specialty ceilings characterized by a textured surface. See: 09 29 00 for gypsum board. 09 54 00 for specialty ceilings. 09 94 13 for textured finishing.
09 56 13	Gypsum-Panel Textured Ceilings	
09 56 16	Metal-Panel Textured Ceilings	
09 56 19	Plastic-Panel Textured Ceilings	
09 57 00	Special Function Ceilings	Includes: ceiling assemblies with an additional special function. See: 08 34 53 for security doors and frames. 09 54 00 for specialty ceilings.
09 57 53	Security Ceiling Assemblies	
09 58 00	Integrated Ceiling Assemblies	Includes: ceiling assemblies with integrated lighting, HVAC, and plumbing provisions. Alternate Terms/Abbreviations: integrated ceiling: ventilating ceiling. See: 09 51 00 for acoustical ceilings. 09 54 00 for specialty ceilings. 23 30 00 for HVAC air distribution. 26 51 00 for interior lighting.
09 60 00	Flooring	Includes: finished floor coverings primarily intended for interior locations, set over structural support.
09 61 00	Flooring Treatment	Includes: coatings and surfacings for finished flooring, applied to provide a specific performance characteristic. See: 03 35 13 for high-tolerance concrete floor finishing. 03 35 16 for heavy-duty concrete floor finishing. 03 54 00 for cast underlayment. 07 18 00 for traffic coatings. 09 33 00 for conductive tiling. 09 65 33 for conductive resilient flooring. 09 65 36 for static-control resilient flooring. 09 66 33 for conductive terrazzo flooring. 09 67 13 for conductive elastomeric liquid flooring.

NUMBER	TITLE	EXPLANATION
09 61 13	Slip-Resistant Flooring Treatment	
09 61 19	Concrete Floor Staining	
09 61 36	Static-Resistant Flooring Treatment	
09 62 00	Specialty Flooring	Includes: heavy-duty and other specialty flooring.
		See: 09 35 00 for chemical-resistant tiling.
		09 51 00 for acoustical ceilings.
		09 63 13 for chemical-resistant brick flooring.
		09 64 00 for wood flooring.
		09 64 66 for wood athletic flooring.
		09 65 66 for resilient athletic flooring.
		09 67 66 for fluid-applied athletic flooring.
		09 72 13 for cork wall coverings.
		09 80 00 for acoustic treatment.
		09 96 35 for chemical-resistant coatings.
		13 28 66 for demountable athletic surfaces.
09 62 13	Asphaltic Plank Flooring	
09 62 16	Concrete Plank Flooring	
09 62 19	Laminate Flooring	
09 62 26	Leather Flooring	
09 62 29	Cork Flooring	
09 62 35	Acid-Resistant Flooring	
09 62 48	Acoustic Flooring	
09 62 53	Synthetic Turf Flooring	See: 32 18 13 for synthetic grass surfacing.
09 62 63	Metal Flooring	
09 62 63.13	Aluminum Flooring	
09 62 63.16	Stainless Steel Flooring	
09 62 83	Structural Glass Flooring	
09 63 00	Masonry Flooring	Includes: fired clay unit masonry and natural, simulated, and cast stone flooring. Flagstone, granite, marble, and slate stone flooring.
		See: 04 20 00 for unit masonry.
		04 40 00 for stone assemblies.
		04 73 00 for manufactured stone masonry.
		09 30 33 for cut natural stone tiling.
		09 35 00 for chemical-resistant tiling.
		09 62 35 for acid-resistant flooring.
		09 75 00 for stone facing.
		09 96 35 for chemical-resistant coatings.
		32 14 00 for unit paving.
09 63 13	Brick Flooring	

NUMBER	TITLE	EXPLANATION
09 63 13.35	Chemical-Resistant Brick Flooring	
09 63 40	Stone Flooring	
09 63 43	Composition Stone Flooring	
09 64 00	Wood Flooring	Includes: flooring composed of wood with various types of backing. Wood sleeper and subfloor installation, adhesive installation. Alternate Terms/Abbreviations: athletic flooring: gymnasium flooring: sports flooring. See: 06 15 00 for wood decking. 09 64 36 for bamboo flooring. 09 65 66 for resilient athletic flooring. 09 67 66 for fluid-applied athletic flooring. 11 66 00 for athletic equipment. 13 28 66 for demountable athletic surfaces.
09 64 16	Wood Block Flooring	
09 64 19	Wood Composition Flooring	
09 64 23	Wood Parquet Flooring	
09 64 23.13	Acrylic-Impregnated Wood Parquet Flooring	
09 64 29	Wood Strip and Plank Flooring	
09 64 33	Laminated Wood Flooring	
09 64 36	Bamboo Flooring	
09 64 53	Resilient Wood Flooring Assemblies	
09 64 66	Wood Athletic Flooring	
09 65 00	Resilient Flooring	Includes: resilient tile and sheet flooring. Resilient base, resilient stair treads and risers, resilient stair nosings, resilient edging, and transitions for carpet. Alternate Terms/Abbreviations: athletic flooring: gymnasium flooring: sports flooring. VCT: vinyl composition tile. See: 09 33 00 for conductive tiling. 09 61 36 for static-resistant flooring treatment. 09 64 66 for wood athletic flooring. 09 66 33 for conductive terrazzo flooring. 09 67 13 for conductive elastomeric liquid flooring. 09 67 66 for fluid-applied athletic flooring. 13 28 66 for demountable athletic surfaces. Products: linoleum, rubber, vinyl, vinyl composition floor coverings.
09 65 13	Resilient Base and Accessories	
09 65 13.13	Resilient Base	

NUMBER	TITLE	EXPLANATION
09 65 13.23	Resilient Stair Treads and Risers	
09 65 13.26	Resilient Stair Nosings	
09 65 13.33	Resilient Accessories	
09 65 13.36	Resilient Carpet Transitions	
09 65 16	Resilient Sheet Flooring	
09 65 16.23	Vinyl Sheet Flooring	
09 65 16.33	Rubber Sheet Flooring	
09 65 16.43	PVC-Free Sheet Flooring	
09 65 19	Resilient Tile Flooring	
09 65 19.19	Vinyl Composition Tile Flooring	
09 65 19.23	Vinyl Tile Flooring	
09 65 19.33	Rubber Tile Flooring	
09 65 19.43	Polyester Composition Tile Flooring	
09 65 33	Conductive Resilient Flooring	
09 65 36	Static-Control Resilient Flooring	
09 65 36.13	Static-Dissipative Resilient Flooring	
09 65 36.16	Static-Resistant Resilient Flooring	
09 65 43	Linoleum Flooring	
09 65 66	Resilient Athletic Flooring	
09 66 00	Terrazzo Flooring	Includes: cast-in-place, sand-cushion, monolithic, bonded, and adhesively bonded portland cement terrazzo; poured-in-place epoxy, polyester, and conductive plastic matrix terrazzo; and precast terrazzo. Accessories and finish sealers.
09 66 13	Portland Cement Terrazzo Flooring	
09 66 13.13	Sand Cushion Terrazzo Flooring	
09 66 13.16	Monolithic Terrazzo Flooring	
09 66 13.19	Bonded Terrazzo Flooring	
09 66 13.23	Palladina Terrazzo Flooring	
09 66 13.26	Rustic Terrazzo Flooring	
09 66 13.33	Structural Terrazzo Flooring	
09 66 16	Terrazzo Floor Tile	Includes: precast terrazzo.
09 66 16.13	Portland Cement Terrazzo Floor Tile	
09 66 16.16	Plastic-Matrix Terrazzo Floor Tile	
09 66 23	Resinous Matrix Terrazzo Flooring	Includes: thin-set terrazzo.
		See: 09 67 00 for fluid-applied flooring.
09 66 23.13	Polyacrylate Modified Cementitious Terrazzo Flooring	
09 66 23.16	Epoxy-Resin Terrazzo Flooring	
09 66 23.19	Polyester-Resin Terrazzo Flooring	
09 66 33	Conductive Terrazzo Flooring	See: 09 33 00 for conductive tiling. 09 61 36 for static-resistant flooring treatment. 09 65 33 for conductive resilient flooring. 09 67 13 for conductive elastomeric liquid flooring.

NUMBER	TITLE	EXPLANATION
09 66 33.13	Conductive Epoxy-Resin Terrazzo	
09 66 33.16	Conductive Polyester-Resin Terrazzo Flooring	
09 66 33.19	Conductive Plastic-Matrix Terrazzo Flooring	

09 67 00 **Fluid-Applied Flooring**

Includes: flooring applied in a viscous state.

Alternate Terms/Abbreviations: fluid-applied: liquid-applied.
Fluid-applied flooring: seamless flooring. Athletic flooring:
gymnasium flooring: sports flooring.

See: 07 18 00 for traffic coatings.
09 33 00 for conductive tiling.
09 61 36 for static-resistant flooring treatment.
09 64 66 for wood athletic flooring.
09 65 00 for resilient flooring.
09 65 33 for conductive resilient flooring.
09 65 66 for resilient athletic flooring.
09 66 23 for resinous matrix terrazzo flooring.
09 66 33 for conductive terrazzo flooring.
09 96 53 for elastomeric coatings.
13 28 66 for demountable athletic surfaces.

Products: epoxy, epoxy novolac, latex mastic, methacrylate acrylic, polyester, polyurethane, urethane, vinyl ester fluid-applied flooring.

09 67 13	Elastomeric Liquid Flooring
09 67 13.33	Conductive Elastomeric Liquid Flooring
09 67 16	Epoxy-Marble Chip Flooring
09 67 19	Magnesium-Oxychloride Flooring
09 67 23	Resinous Flooring
09 67 26	Quartz Flooring
09 67 66	Fluid-Applied Athletic Flooring

09 68 00 **Carpeting**

Includes: carpet, carpet cushions, and accessories. Direct-glue-down installation, double-glue-down installation, carpet with attached-cushion installation, preapplied adhesive installation, hook-and-loop installation, stretch-in installation.

May Include: wall carpeting.

Alternate Terms/Abbreviations: sheet carpeting: roll carpeting: broadloom.

See: 12 48 00 for rugs and mats.

See Also: 09 69 00 for carpet tile with access flooring.
09 73 00 for wall carpeting.

NUMBER	TITLE	EXPLANATION
09 68 13	Tile Carpeting	
09 68 16	Sheet Carpeting	
09 69 00	Access Flooring	Includes: free-standing, elevated accessible floor assembly forming an underfloor cavity that may be used for utility services. Accessories and floor panel finish. May Include: air outlets and inlets integral with floor panels. Loose-laid or glued-down carpet tile. Alternate Terms/Abbreviations: access flooring: computer floor: elevated floor: raised floor: free access floors: pedestal floors. See: 23 30 00 for HVAC air distribution. 23 32 36 for air-distribution floor plenums. See Also: 09 68 13 for loose-laid or glued-down carpet tile. 23 37 00 for air outlets and inlets.
09 69 13	Rigid-Grid Access Flooring	
09 69 16	Snap-on Stringer Access Flooring	
09 69 19	Stringerless Access Flooring	
09 69 33	Low-Profile Fixed Height Access Flooring	
09 69 53	Access Flooring Accessories	
09 69 56	Access Flooring Stairs and Stringers	

09 70 00 Wall Finishes

Includes: wall finishes applied over solid substrates.

See: 09 23 00 for gypsum plastering.
09 24 00 for cement plastering.
09 29 00 for gypsum board.
09 30 00 for tiling.
09 83 13 for acoustic wall coating.
09 84 00 for acoustic room components.
09 90 00 for painting and coating.

NUMBER	TITLE	EXPLANATION
09 72 00	Wall Coverings	Includes: vinyl-coated fabrics, cork, wallpaper, and textile wall coverings. See: 06 25 00 for prefinished paneling. 06 26 00 for board paneling. 06 42 00 for custom wood paneling. 09 62 29 for cork flooring. 09 73 00 for wall carpeting. 09 74 00 for flexible wood sheets. 10 11 43 for visual display wall panels. 10 11 46 for visual display fabrics. 12 05 13 for fabrics.
09 72 13	Cork Wall Coverings	
09 72 16	Vinyl-Coated Fabric Wall Coverings	
09 72 16.13	Flexible Vinyl Wall Coverings	
09 72 16.16	Rigid-Sheet Vinyl Wall Coverings	
09 72 19	Textile Wall Coverings	
09 72 23	Wallpapering	
09 73 00	Wall Carpeting	Includes: carpet for use on walls. See: 09 68 00 for carpeting. 10 11 46 for visual display fabrics.
09 74 00	Flexible Wood Sheets	Includes: wood wall covering and flexible wood veneers. See: 06 25 00 for prefinished paneling. 06 26 00 for board paneling. 06 42 00 for custom wood paneling.
09 74 13	Wood Wall Coverings	
09 74 16	Flexible Wood Veneers	
09 75 00	Stone Facing	Includes: dimension stone facing for interior walls. See: 04 40 00 for exterior stone assemblies. 09 63 40 for stone flooring.
09 75 13	Stone Wall Facing	
09 75 19	Stone Trim	
09 75 19.13	Stone Base	
09 75 19.23	Stone Window Stools	
09 75 23	Simulated Stone Wall Facing	

NUMBER	TITLE	EXPLANATION
09 76 00	Plastic Blocks	Includes: plastic blocks intended to simulate glass block, for interior partitions. See: 04 23 00 for glass unit masonry.
09 77 00	Special Wall Surfacing	Includes: manufactured interior and exterior decorative wall surface products. See: 09 54 00 for specialty ceilings. 12 05 13 for fabrics.
09 77 13	Stretched-Fabric Wall Systems	
09 77 23	Fabric-Wrapped Panels	
09 77 26	Surfacing Films	
09 77 33	Terrazzo Wall Finish	Includes: tile and cast-in-place terrazzo.
09 77 53	Vegetated Wall Systems	
09 78 00	Interior Wall Paneling	See: 06 25 00 for prefinished paneling. 06 26 00 for board paneling. 06 42 00 for custom wood paneling. See Also: 06 25 00 for prefinished paneling. 06 26 00 for board paneling.
09 78 13	Metal Interior Wall Paneling	See: 06 25 00 for prefinished paneling. 06 26 00 for board paneling. 06 42 00 for custom wood paneling.
09 78 16	Stone-Faced Interior Wall Paneling	
09 78 19	Cementitious Interior Wall Paneling	
09 78 23	Phenolic Interior Wall Paneling	
09 78 26	Plastic Interior Wall Paneling	

NUMBER	TITLE	EXPLANATION
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09 80 00 **Acoustic Treatment**

Includes: acoustical sound-absorbing units, acoustical barriers, acoustic insulation.

See: 04 22 23 for sound-absorbing concrete unit masonry.
 05 34 00 for acoustical metal decking.
 07 92 19 for acoustical joint sealant.
 09 23 13 for acoustical gypsum plastering.
 09 51 00 for acoustical ceilings.
 09 62 48 for acoustic flooring.
 13 21 48 for sound-conditioned rooms.
 13 48 00 for sound, vibration, and seismic control.
 44 20 00 for noise pollution control.

09 81 00 **Acoustic Insulation**

Includes: insulation to provide acoustical isolation.

May Include: acoustic insulation for plaster and gypsum board assemblies.

Alternate Terms/Abbreviations: blanket insulation: batt insulation. Board insulation: rigid insulation: semi-rigid insulation. Sprayed insulation: spray-on insulation.

See: 07 21 00 for thermal insulation.

See Also: 09 20 00 for acoustic insulation for plaster and gypsum board.

09 81 13	Acoustic Board Insulation
09 81 16	Acoustic Blanket Insulation
09 81 29	Sprayed Acoustic Insulation

09 83 00 **Acoustic Finishes**

Includes: coatings and coverings in which the primary function is acoustical absorption.

See: 09 23 13 for acoustical gypsum plastering.
 12 22 00 for window curtains and drapes.

09 83 13	Acoustic Wall Coating
09 83 16	Acoustic Ceiling Coating
09 83 22	Acoustic Drapery

NUMBER	TITLE	EXPLANATION
09 84 00	Acoustic Room Components	Includes: sound absorbing, reflecting, and diffusing wall and ceiling units and accessories. Usually Includes: suspension systems for hanging acoustical units. See: 09 23 13 for acoustical gypsum plastering. 09 83 00 for acoustic finishes.
09 84 13	Fixed Sound-Absorptive Panels	
09 84 14	Acoustic Stretched-Fabric Wall Systems	See: 09 77 13 for stretched-fabric wall systems.
09 84 16	Fixed Sound-Reflective Panels	
09 84 23	Moveable Sound-Absorptive Panels	
09 84 26	Moveable Sound-Reflective Panels	
09 84 33	Sound-Absorbing Wall Units	
09 84 36	Sound-Absorbing Ceiling Units	
09 90 00	Painting and Coating	Includes: stains, transparent finishes, special coatings, varnishes, lacquers, primers, fillers, paint removers, waxes, and preparation of surfaces. Painting of mechanical, electrical, and communications equipment other than for identification. See: 10 14 00 for signage. 21 05 53 for identification for fire-suppression piping and equipment. 22 05 53 for identification for plumbing piping and equipment. 23 05 53 for identification for HVAC piping and equipment. 25 05 53 for identification for integrated automation. 26 05 53 for identification for electrical systems. 27 05 53 for identifications for communications systems. 28 05 53 for identification for electronic safety and security.
09 91 00	Painting	Includes: painting with opaque finishes. Priming, sealing, and finish painting. May Include: pavement marking painting. See Also: 32 17 23 for pavement markings. Products: acrylic latex, alkyd, aluminum, latex, enamel paints. Block fillers.
09 91 13	Exterior Painting	
09 91 23	Interior Painting	

NUMBER	TITLE	EXPLANATION
09 93 00	Staining and Transparent Finishing	Includes: coating with transparent and semi-transparent finishes.
09 93 13	Exterior Staining and Finishing	Includes: coating of surfaces such as wood, composites, concrete, and masonry.
09 93 13.13	Exterior Staining	
09 93 13.53	Exterior Finishing	
09 93 23	Interior Staining and Finishing	Includes: coating of surfaces such as wood, composites, concrete, and masonry.
09 93 23.13	Interior Staining	
09 93 23.53	Interior Finishing	
09 94 00	Decorative Finishing	Includes: textured and other applied finishes other than paint or high-performance coatings. See: 09 56 00 for textured specialty ceilings. 09 91 00 for painting. 09 96 00 for high-performance coatings. 09 96 63 for textured plastic coatings. 12 11 00 for murals. 12 12 00 for wall decorations.
09 94 13	Textured Finishing	
09 94 16	Faux Finishing	
09 94 19	Multicolor Interior Finishing	
09 94 23	Gilding	Includes: application of metallic leaf onto a surface.

NUMBER	TITLE	EXPLANATION
09 96 00	High-Performance Coatings	Includes: high-performance, heavy industrial, glazed resinous, epoxy polyester, polyurethane, polyvinyl fluoride, chlorinated rubber, heat-resistant, and corrosion-resistant coatings. Zinc-rich primers. Fire-retardant paint. See: 04 61 00 for chemical-resistant brick masonry. 06 05 73 for fire-retardant wood treatment. 07 18 00 for traffic coatings. 07 19 00 for water repellents. 07 81 00 for applied fire protection. 07 81 23 for intumescent fire protection. 09 35 00 for chemical-resistant tiling. 09 62 35 for acid-resistant flooring. 09 63 13 for chemical-resistant brick flooring. 09 67 00 for fluid-applied flooring. 09 67 13 for elastomeric liquid flooring. 09 94 13 for textured finishing. 32 17 23 for pavement markings. 33 01 30 for relining sewers.
09 96 13	Abrasion-Resistant Coatings	
09 96 23	Graffiti-Resistant Coatings	
09 96 26	Marine Coatings	
09 96 33	High-Temperature-Resistant Coatings	
09 96 35	Chemical-Resistant Coatings	
09 96 43	Fire-Retardant Coatings	
09 96 46	Intumescent Painting	
09 96 53	Elastomeric Coatings	
09 96 56	Epoxy Coatings	
09 96 59	High-Build Glazed Coatings	
09 96 63	Textured Plastic Coatings	
09 96 66	Aggregate Wall Coatings	
09 97 00	Special Coatings	Includes: special field-applied coatings for steel, concrete, masonry, and on other materials. Storage tank coatings. Cementitious coatings. See: 07 19 00 for water repellents. 07 24 23 for direct-applied finish systems. 21 41 00 for storage tanks for fire-suppression water. 22 12 00 for facility potable-water storage tanks. 23 13 00 for facility fuel-storage tanks. 33 16 00 for water utility storage tanks. 33 56 00 for fuel storage tanks. 41 52 19 for material storage tanks.
09 97 13	Steel Coatings	
09 97 13.13	Interior Steel Coatings	

DIVISION 09

NUMBER	TITLE	EXPLANATION
09 97 13.23	Exterior Steel Coatings	
09 97 13.24	Steel Water Storage Tank Painting	Includes: preparing, priming, and painting of steel elevated water storage tank for potable water use.
09 97 23	Concrete and Masonry Coatings	
09 97 23.24	Concrete Water Storage Tank Painting	Includes: preparing and painting of concrete water storage tanks for potable water use.
09 97 26	Cementitious Coatings	
09 97 26.13	Interior Cementitious Coatings	
09 97 26.23	Exterior Cementitious Coatings	
09 97 35	Dry Erase Coatings	Includes: special coatings on other materials.
		See Also: 10 11 00 for visual display units.

DIVISION 10 – SPECIALTIES

10 00 00 Specialties

May be used as Division level
Section title.

10 01 00 Operation and Maintenance of Specialties

Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of specialties.

Level 4 Numbering Recommendation: following numbering is recommended for the creation of Level 4 titles:

.51-.59 for maintenance.

.61-.69 for repair.

.71-.79 for rehabilitation.

.81-.89 for replacement.

.91-.99 for restoration.

10 01 10	Operation and Maintenance of Information Specialties
10 01 20	Operation and Maintenance of Interior Specialties
10 01 30	Operation and Maintenance of Fireplaces and Stoves
10 01 40	Operation and Maintenance of Safety Specialties
10 01 50	Operation and Maintenance of Storage Specialties
10 01 70	Operation and Maintenance of Exterior Specialties
10 01 80	Operation and Maintenance of Other Specialties

10 05 00 Common Work Results for Specialties

Includes: subjects common to multiple titles in Division 10.

See: 01 80 00 for performance requirements of subjects common to multiple titles.

10 05 05 Selective Demolition for Specialties

See: 02 41 19 for selective demolition for whole structure demolition.

10 06 00 Schedules for Specialties

Includes: schedules of items common to multiple titles in Division 10.

Notes: a schedule may be included on drawings, in the project manual, or a project book.

10 06 10	Schedules for Information Specialties
10 06 10.13	Exterior Signage Schedule
10 06 10.16	Interior Signage Schedule
10 06 20	Schedules for Interior Specialties
10 06 20.13	Toilet, Bath, and Laundry Accessory Schedule
10 06 30	Schedules for Fireplaces and Stoves
10 06 40	Schedules for Safety Specialties

DIVISION **10**

NUMBER	TITLE	EXPLANATION
10 06 50	Schedules for Storage Specialties	
10 06 70	Schedules for Exterior Specialties	
10 06 80	Schedules for Other Specialties	
10 08 00	Commissioning of Specialties	Includes: commissioning of items common to multiple titles in Division 10. See: 01 91 00 for commissioning of subjects common to multiple Divisions.
10 10 00	Information Specialties	
10 11 00	Visual Display Units	Includes: chalkboards, markerboards, tackboards, and other visual display systems; framing systems and accessories. See: 10 13 00 for directories. 11 52 13 for projection screens. 12 35 59 for display casework. See Also: 09 97 35 for dry erase coatings.
10 11 13	Chalkboards	Includes: chalkboards made from slate, composition board, painted material, porcelain-enameled steel, and aluminum. Alternate Terms/Abbreviations: chalkboards: blackboards.
10 11 13.13	Fixed Chalkboards	
10 11 13.23	Modular-Support-Mounted Chalkboards	
10 11 13.33	Rail-Mounted Chalkboards	
10 11 13.43	Portable Chalkboards	
10 11 16	Markerboards	Includes: markerboards make from painted material, porcelain-enameled steel, and aluminum. Markerboards with integral photocopy units and electronic interface with computers. Alternate Terms/Abbreviations: markerboards: whiteboards.
10 11 16.13	Fixed Markerboards	
10 11 16.23	Modular-Support-Mounted Markerboards	
10 11 16.33	Rail-Mounted Markerboards	
10 11 16.43	Portable Markerboards	
10 11 16.53	Electronic Markerboards	

NUMBER	TITLE	EXPLANATION
10 11 23	Tackboards	Includes: display boards with mounting surfaces of cork, composition board, and fabric-covered material. Alternate Terms/Abbreviations: tackboards: bulletin boards: cork boards.
10 11 23.13	Fixed Tackboards	
10 11 23.23	Modular-Support-Mounted Tackboards	
10 11 23.33	Rail-Mounted Tackboards	
10 11 23.43	Portable Tackboards	
10 11 33	Sliding Visual Display Units	Includes: manual and motor-operated units with fixed rear display panel and multiple sliding display panels in front.
10 11 33.13	Horizontal-Sliding Visual Display Units	
10 11 33.23	Vertical-Sliding Visual Display Units	
10 11 36	Visual Display Conference Units	Includes: cabinet units with integral display surfaces, projection screens, and flip charts. See: 11 52 13 for projection screens.
10 11 39	Visual Display Rails	Includes: rail assemblies with mounting surfaces.
10 11 43	Visual Display Wall Panels	Includes: custom-fabricated chalkboard, markerboard, and tackboard panels for full-height and wall applications. See: 09 72 00 for wall coverings.
10 11 46	Visual Display Fabrics	Includes: plastic-film-covered fabrics for use with markers and magnets. See: 09 72 00 for wall coverings. 09 73 00 or wall carpeting.
10 12 00	Display Cases	Includes: manufactured cases for display of items. See: 11 22 00 for refrigerated display equipment. 11 59 00 for exhibit equipment. 12 35 59 for display casework. 14 73 59 for display turntables. 26 55 59 for display lighting.

NUMBER	TITLE	EXPLANATION
10 13 00	Directories	Includes: directories for buildings and facilities. Self-contained computerized directories, which are not part of an integrated system. See: 27 42 00 for electronic digital systems.
10 13 13	Electronic Directories	
10 13 23	Illuminated Directories	
10 14 00	Signage	Includes: directional items, letters, signage, and plaques used in establishing identity or communication. Integral lighted units, neon signage, and other types of illuminated signage. Self-contained electronic message signage, which is not part of an integrated system. See: 01 58 00 for temporary project identification. 10 11 43 for visual display wall panels. 10 11 46 for visual display fabrics. 10 18 00 for informational kiosks. 10 83 00 for flags and banners. 21 05 53 for identification for fire-suppression piping and equipment. 22 05 53 for identification for plumbing piping and equipment. 23 05 53 for identification for HVAC piping and equipment. 25 05 53 for identification for integrated automation. 26 05 53 for identification for electrical systems. 26 52 13 for exit signs. 27 05 53 for identification for communication systems. 27 42 00 for electronic digital systems. 28 05 53 for identification for electronic safety and security. 32 17 26 for tactile warning surfaces. 34 41 13 for traffic signals. 34 42 13 for railway signals. 34 43 13 for airfield signals. 35 10 00 for waterway and marine signals. 35 59 23 for buoys. See Also: 26 52 13 for emergency and exit lighting.
10 14 16	Plaques	
10 14 19	Dimensional Letter Signage	
10 14 23	Panel Signage	
10 14 23.13	Engraved Panel Signage	
10 14 26	Post and Panel/Pylon Signage	
10 14 33	Illuminated Panel Signage	
10 14 43	Photoluminescent Signage	
10 14 53	Traffic Signage	
10 14 53.13	Transportation Reference Markers	

NUMBER	TITLE	EXPLANATION
10 14 63	Electronic Message Signage	
10 14 64	Audible Signage	
10 14 66	Floating Signage	
10 14 67	Tactile Signage	
10 17 00	Telephone Specialties	
		Includes: manufactured telephone enclosures, stalls, booths, and alcoves for installation of telephones and telephone directories.
		See: 10 25 16 for modular service walls. 27 20 00 for data communications. 27 30 00 for voice communications. 27 32 13 for telephone sets. 27 40 00 for audio-video communications.
10 17 13	Telephone Directory Units	
10 17 16	Telephone Enclosures	
10 17 16.13	Telephone Stalls	
10 17 16.16	Telephone Alcoves	
10 17 19	Telephone Shelving	
10 18 00	Informational Kiosks	
		Includes: open air shelter directories, signage, or other information devices.
		See: 10 13 00 for directories. 10 14 00 for signage. 10 73 00 for protective covers. 10 74 23 for cupolas.
10 20 00	Interior Specialties	
10 21 00	Compartments and Cubicles	
		Includes: manufactured compartments and cubicles that are enclosures, dividers, screens, and curtains for toilets, urinals, showers, dressing compartments, and hospital cubicles, including hardware and accessories. Ceiling-hung, floor-supported, overhead-braced, and wall-supported.
10 21 13	Toilet Compartments	
		Includes: compartments appropriate for toilet rooms.
		See: 10 21 16 for shower and dressing compartments. 10 28 13 for toilet accessories. 22 40 00 for plumbing fixtures.
10 21 13.13	Metal Toilet Compartments	

DIVISION **10**

NUMBER	TITLE	EXPLANATION
10 21 13.14	Stainless-Steel Toilet Compartments	Includes: stainless-steel toilet enclosures, entrance screens, urinal screens, anchors, hardware, and accessories.
10 21 13.16	Plastic-Laminate-Clad Toilet Compartments	
10 21 13.17	Phenolic-Core Toilet Compartments	Includes: phenolic-core toilet enclosures, entrance screens, urinal screens. Anchors, hardware, and accessories.
10 21 13.19	Plastic Toilet Compartments	
10 21 13.23	Particleboard Toilet Compartments	
10 21 13.40	Stone Toilet Compartments	
10 21 16	Shower and Dressing Compartments	Includes: compartments appropriate for shower and dressing purposes. See: 10 21 13 for toilet compartments. 10 28 13 for toilet accessories. 10 28 19 for residential tub and shower doors. 22 40 00 for plumbing fixtures.
10 21 16.13	Metal Shower and Dressing Compartments	
10 21 16.14	Stainless-Steel Shower and Dressing Compartments	Includes: stainless-steel shower and dressing compartments and shower receptors, anchors, hardware, and accessories.
10 21 16.16	Plastic-Laminate-Clad Shower and Dressing Compartments	
10 21 16.17	Phenolic-Core Shower and Dressing Compartments	Includes: phenolic-core shower and dressing compartments and shower receptors, anchors, hardware, and accessories.
10 21 16.19	Plastic Shower and Dressing Compartments	
10 21 16.23	Particleboard Shower and Dressing Compartments	
10 21 16.40	Stone Shower and Dressing Compartments	
10 21 23	Cubicle Curtains and Track	Includes: healthcare treatment, dressing room, welding, workshop curtains, tracks, and other hardware. See: 10 25 13 for patient bed service walls. 11 57 00 for vocational shop equipment. 11 70 00 for healthcare equipment.
10 21 23.13	Cubicle Curtains	
10 21 23.16	Cubicle Track and Hardware	
10 21 26	Room Darkening Curtains and Tracks	Alternate Terms/Abbreviations: blackout curtains. See: 12 22 13 for window draperies.
10 22 00	Partitions	Includes: fixed, demountable, and operable fabricated partitions used as enclosures, dividers, and partitions.

NUMBER	TITLE	EXPLANATION
10 22 13	Wire Mesh Partitions	Includes: chain link partitions, and partitions fabricated of wire mesh. Alternate Terms/Abbreviations: wire mesh: woven wire.
10 22 14	Expanded Metal Partitions	
10 22 16	Folding Gates	Includes: folding gates serving as a security or traffic directing partition when closed. See: 08 34 56 for security gates. 08 35 16 for folding grilles.
10 22 19	Demountable Partitions	Includes: partition systems that are connected to permanent construction and are designed to be relocated without damage. Alternate Terms/Abbreviations: demountable partitions: relocatable partitions.
10 22 19.13	Demountable Metal Partitions	
10 22 19.23	Demountable Wood Partitions	
10 22 19.33	Demountable Plastic Partitions	
10 22 19.43	Demountable Composite Partitions	
10 22 19.53	Demountable Gypsum Partitions	
10 22 19.63	Demountable Glass Partitions	
10 22 23	Portable Partitions, Screens, and Panels	Includes: wall screens, movable panel systems, and partitions that are not connected to permanent construction. See: 10 82 23 for interior grilles and screens connected to permanent construction. 12 46 26 for decorative screens.
10 22 23.13	Wall Screens	
10 22 23.23	Movable Panel Systems	
10 22 33	Accordion Folding Partitions	
10 22 36	Coiling Partitions	
10 22 39	Folding Panel Partitions	
10 22 39.13	Folding Glass-Panel Partitions	Includes: manually operated, glass-panel partitions, closures, hardware, seals, suspension systems, tracks, carriers, and accessories.
10 22 43	Sliding Partitions	

NUMBER	TITLE	EXPLANATION
10 25 00	Service Walls	Includes: wall assemblies and wall-mounted units incorporating services such as medical services, drinking fountains, clocks, fire hose and extinguishing cabinets, telephones, and waste receptacles. See: 10 17 00 for telephone specialties. 10 44 00 for fire protection specialties. 11 73 00 for patient care equipment. 22 47 00 for drinking fountains and water coolers.
10 25 13	Patient Bed Service Walls	
10 25 16	Modular Service Walls	
10 26 00	Wall and Door Protection	Includes: manufactured protective devices for walls and doors made of metal, plastic, rubber, and other finish materials. See: 05 50 00 for corner guards fabricated from rolled metal sections or bent plate. 05 52 00 for metal railings. 05 73 00 for decorative metal railings. 06 43 16 for wood railings. 06 46 23 for wood chair rails. Products: crash rails, rub strips, wall guards, door guards, column guards.
10 26 13	Corner Guards	
10 26 16	Bumper Guards	
10 26 16.13	Bumper Rails	
10 26 16.16	Protective Corridor Handrails	
10 26 23	Protective Wall Covering	Products: crash rails, rub strips, wall guards, door guards, column guards.
10 26 23.13	Impact Resistant Wall Protection	
10 26 23.16	Fiberglass Reinforced Protective Wall Covering	
10 26 33	Door and Frame Protection	
10 26 41	Ballistics Resistant Panels	

NUMBER	TITLE	EXPLANATION
10 28 00	Toilet, Bath, and Laundry Accessories	Includes: manufactured items for use in conjunction with toilets, baths, and laundries. See: 08 83 00 for mirrors other than framed mirrors. 09 30 00 for tile accessories. 12 30 00 for manufactured casework. Products: grab bars, towel dispensers, soap dispensers, deodorizers, napkin/tampon dispensers and disposal units, toilet paper dispensers, waste disposal units, robe hooks, framed mirrors, air hand and hair dryers, and other similar public toilet room accessories.
10 28 13	Toilet Accessories	Includes: accessories associated with public toilet rooms. Accessories mounted on or recessed in partitions and compartments. Usually Includes: janitor's closet and other similar accessories. Alternate Terms/Abbreviations: toilet accessories: rest room accessories. See: 10 21 13 for toilet compartments. 10 21 16 for shower and dressing compartments. Products: grab bars, towel dispensers, soap dispensers, deodorizers, napkin/tampon dispensers and disposal units, toilet paper dispensers, waste disposal units, robe hooks, framed mirrors, air hand and hair dryers, and other similar public toilet room accessories.
10 28 13.13	Commercial Toilet Accessories	
10 28 13.19	Healthcare Toilet Accessories	
10 28 13.53	Security Toilet Accessories	
10 28 13.63	Detention Toilet Accessories	
10 28 16	Bath Accessories	Includes: accessories associated with residential bathroom use. May Include: shower and tub doors for residential bathroom use. See: 10 43 16 for first aid cabinets. 12 44 16 for shower curtains. See Also: 10 28 19 for tub and shower doors. Products: towel bars and rings, robe hooks, shower curtain rods, medicine cabinets, framed mirrors, toilet paper dispensers and other similar residential bath accessories.

DIVISION **10**

NUMBER	TITLE	EXPLANATION
10 28 16.13	Residential Bath Accessories	
10 28 19	Tub and Shower Enclosures	Includes: shower and tub doors for residential bathroom use.
		See: 10 21 16 for shower and dressing compartments. 12 44 16 for shower curtains.
10 28 19.16	Shower Doors	
10 28 19.19	Tub Doors	
10 28 23	Laundry Accessories	Includes: laundry room accessories for residential use.
10 28 23.13	Built-In Ironing Boards	
10 28 23.16	Clothes Drying Racks and Lines	
10 28 26	Hygiene and Custodial Accessories	Includes: items such as hand sanitizers and other non-toilet accessories.
		See: 10 28 13 for toilet accessories.

10 30 00 Fireplaces and Stoves

Includes: manufactured fireplaces, stoves, chimneys, dampers, and specialties required for fireplaces and stoves.

See: 04 50 00 for refractory masonry.

10 31 00 Manufactured Fireplaces

Includes: stand-alone fireplace units regardless of heat source.

Usually Includes: fireplace accessories.

See: 04 57 00 for field-fabricated masonry fireplaces.
23 51 00 for breechings, chimneys, and stacks.

See Also: 10 32 00 for fireplace specialties.

10 31 13	Manufactured Fireplace Chimneys
10 31 16	Manufactured Fireplace Forms

10 32 00 Fireplace Specialties

Includes: components and accessories for fireplaces.

See: 04 57 00 for masonry fireplaces.
10 31 00 for manufactured fireplaces.

Products: andirons, ash dumps, fireplace bellows, fire log baskets, fireplace tools.

10 32 13	Fireplace Dampers
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NUMBER	TITLE	EXPLANATION
10 32 16	Fireplace Inserts	
10 32 19	Fireplace Screens	
10 32 23	Fireplace Doors	
10 32 26	Fireplace Water Heaters	
10 35 00	Stoves	Includes: wood, pellet, corn, and coal stoves. Stove pipe. Franklin stoves.
10 35 13	Heating Stoves	
10 35 23	Cooking Stoves	
10 40 00	Safety Specialties	
10 41 00	Emergency Access and Information Cabinets	Includes: accessories to aid emergency access and firefighting.
		Alternate Terms/Abbreviations: emergency key cabinet: lock box.
		See: 08 79 00 for hardware accessories.
10 41 13	Fire Department Plan Cabinets	
10 41 16	Emergency Key Cabinets	
10 43 00	Emergency Aid Specialties	Includes: accessories for emergency aid, including automatic external defibrillators and emergency medical oxygen units.
		Alternate Terms/Abbreviations: AED: automated external defibrillator. EMO: emergency medical oxygen.
		See: 10 28 16 for medicine cabinets.
		28 49 17 for electronic personal safety emergency aid devices.
10 43 13	Defibrillator Cabinets	
10 43 16	First Aid Cabinets	
10 43 21	Accessibility Evacuation Chairs	
10 43 31	Respiration Equipment	
10 43 33	Breathing Air Replenishment Systems	Alternate Terms/Abbreviations: BARS. FFAS: firefighter air systems.

NUMBER	TITLE	EXPLANATION
10 44 00	Fire Protection Specialties	Includes: firefighting devices and storage cabinets, except items or devices connected to a fire-suppression system. See: 10 25 16 for modular service walls. 21 12 00 for fire-suppression standpipes. 21 12 16 for fire-suppression hose reels. 21 12 19 for fire-suppression hose racks. 21 12 23 for fire-suppression hose valves.
10 44 13	Fire Protection Cabinets	See: 21 12 00 for fire-suppression standpipes.
10 44 13.53	Security Fire Extinguisher Cabinets	
10 44 16	Fire Extinguishers	See: 21 12 00 for fire-suppression standpipes.
10 44 16.13	Portable Fire Extinguishers	
10 44 16.16	Wheeled Fire Extinguisher Units	
10 44 19	Fire Blankets	
10 44 43	Fire Extinguisher Accessories	
10 45 00	Photoluminescent Exit Specialties	
10 50 00	Storage Specialties	
10 51 00	Lockers	Includes: manufactured storage facilities providing temporary security of contents; related hardware and locking devices. See: 06 41 00 for custom-fabricated wood lockers. 10 22 13 for wire mesh partitions. 10 57 00 for wardrobe and closet specialties. 12 30 00 for manufactured casework. 12 50 00 for furniture. Products: athletic lockers, school lockers.
10 51 13	Metal Lockers	
10 51 13.13	Coin-Operated Metal Lockers	
10 51 16	Wood Lockers	
10 51 23	Plastic-Laminate-Clad Lockers	
10 51 26	Plastic Lockers	
10 51 26.13	Recycled Plastic Lockers	
10 51 29	Phenolic Lockers	
10 51 33	Glass Lockers	
10 51 43	Wire Mesh Storage Lockers	
10 51 53	Locker Room Benches	

NUMBER	TITLE	EXPLANATION
10 55 00	Postal Specialties	Includes: manufactured items associated with the collection and distribution of mail that is physically sent and delivered. See: 11 29 00 for postal, packaging, and shipping equipment. 41 21 23 for postal conveyors. 41 24 46 for postal sorting machines.
10 55 13	Central Mail Delivery Boxes	
10 55 13.13	Cluster Box Units	
10 55 16	Mail Collection Boxes	
10 55 19	Receiving Boxes	
10 55 23	Mail Boxes	
10 55 23.13	Apartment Mail Boxes	
10 55 23.16	Mail Box Directories	
10 55 23.19	Mail Box Key Keepers	
10 55 26	Parcel Lockers	
10 55 33	Data Distribution Boxes	
10 55 36	Package Depositories	
10 55 91	Mail Chutes	
10 56 00	Storage Assemblies	Includes: manufactured storage assemblies for general storage. Manual and motorized mobile and compact storage systems. See: 10 57 23 for closet and utility shelving. 11 51 23 for library stack systems. 12 35 50 for educational/library casework. 12 56 51 for library furniture. 12 59 00 for shelving integral with systems furniture. 41 53 00 for industrial storage equipment and systems.
10 56 13	Metal Storage Shelving	
10 56 13.13	End-Panel-Support Metal Storage Shelving	
10 56 13.16	Post-and-Shelf Metal Storage Shelving	
10 56 13.19	Post-and-Beam Metal Storage Shelving	
10 56 13.23	Cantilever Metal Storage Shelving	
10 56 16	Fabricated Wood Storage Shelving	
10 56 17	Wall-Mounted Standards and Shelving	
10 56 19	Plastic Storage Shelving	
10 56 19.13	Recycled Plastic Storage Shelving	
10 56 23	Wire Storage Shelving	
10 56 26	Mobile Storage Shelving	
10 56 26.13	Manual Mobile Storage Shelving	
10 56 26.23	Motorized Mobile Storage Shelving	
10 56 29	Storage Racks	
10 56 29.13	Flow Storage Racks	
10 56 29.16	Pallet Storage Racks	
10 56 29.19	Movable-Shelf Storage Racks	
10 56 29.23	Stacker Storage Racks	

DIVISION **10**

NUMBER	TITLE	EXPLANATION
10 56 29.26	Cantilever Storage Racks	
10 56 29.29	Drive-In Storage Racks	
10 56 29.33	Drive-Through Storage Racks	
10 56 29.43	Wine Storage Racks	
10 56 33	Mercantile Storage Assemblies	

10 57 00 Wardrobe and Closet Specialties

Includes: manufactured hat and coat racks, and other specialties for storage of clothing. Manufactured shelving for closets and utility rooms.

See: 06 20 00 for finish carpentry shelving.
10 51 00 for lockers.
10 56 00 for general storage assemblies.
11 21 33 for checkroom equipment.

10 57 13	Hat and Coat Racks	
10 57 16	Boot Racks	
10 57 23	Closet and Utility Shelving	
10 57 23.13	Wire Closet and Utility Shelving	
10 57 23.16	Plastic-Laminate-Clad Closet and Utility Shelving	
10 57 23.19	Wood Closet and Utility Shelving	
10 57 33	Closet and Utility Shelving Hardware	
10 57 36	Closet Doors	
10 57 36.13	Sliding Closet Doors	
10 57 36.16	Bifold Closet Doors	

10 70 00 Exterior Specialties

10 71 00 Exterior Protection

Includes: manufactured items to provide sun control, privacy, security, insulation, and storm protection on exterior of openings; fixed, movable, manually and electrically operated, and automatically controlled devices.

See: 06 49 19 for wood screens and exterior wood shutters.
10 73 13 for awnings.
10 73 16 for canopies.
10 82 13 for exterior grilles and screens.
12 20 00 for interior window treatments.
12 23 00 for interior shutters.

10 71 13	Exterior Sun Control Devices	
	See: 08 44 00 for fixed sun screens integral with curtain walls. 12 26 00 for interior daylighting devices.	
10 71 13.13	Exterior Shutters	
10 71 13.19	Rolling Exterior Shutters	
10 71 13.23	Coiling Exterior Shutters	
10 71 13.26	Decorative Exterior Shutters	

NUMBER	TITLE	EXPLANATION
10 71 13.29	Side-Hinged Exterior Shutters	
10 71 13.43	Fixed Sun Screens	
10 71 13.53	Exterior Daylighting Devices	
10 71 16	Storm Panels	
10 71 16.13	Demountable Storm Panels	
10 71 16.16	Movable Storm Panels	
10 71 19	Flood Barriers	
10 71 19.16	Removable Flood Barriers	

10 73 00 Protective Covers

Includes: manufactured items to provide weather protection or shelter. Gazebos.

Alternate Terms/Abbreviations: transportation stop shelters: bus stop shelters. Car shelters: carports.

See: 10 18 00 for informational kiosks.

10 71 13 for exterior sun control devices.

10 71 16 for storm panels.

13 34 00 for fabricated engineered structures.

34 70 00 for transportation construction and equipment.

34 77 00 for transportation equipment.

See Also: 08 65 13 for suspended glazed canopies.

10 73 13	Awnings
10 73 13.13	Metal Awnings
10 73 13.23	Fabric Awnings
10 73 13.36	Point Supported Glass Awnings
10 73 16	Canopies
10 73 16.13	Metal Canopies
10 73 16.23	Fabric Canopies
10 73 16.33	Glazed Canopies
10 73 16.36	Point Supported Glass Canopies
10 73 23	Car Shelters
10 73 26	Walkway Coverings
10 73 26.13	Metal Walkway Coverings
10 73 26.23	Fabric Walkway Coverings
10 73 26.33	Glazed Walkway Coverings
10 73 26.36	Point Supported Glass Walkway Coverings
10 73 29	Smoking Shelters
10 73 33	Marquees
10 73 43	Transportation Stop Shelters

NUMBER	TITLE	EXPLANATION
10 74 00	Manufactured Exterior Specialties	Includes: complete fabrications of metal, wood, and fiberglass, including all accessories and appurtenances. Alternate Terms/Abbreviations: below-grade egress assemblies: basement window egress assemblies: basement window emergency escape. Window wells: areaways. See: 06 44 26 for custom wood cupolas. 08 50 00 for windows. 10 18 00 for informational kiosks. 11 91 00 for religious equipment. 26 41 00 for lightning protection.
10 74 13	Exterior Clocks	
10 74 23	Cupolas	
10 74 26	Spires	
10 74 29	Steeples	
10 74 33	Weathervanes	
10 74 43	Below-Grade Egress Assemblies	
10 74 46	Window Wells	
10 74 49	Modular Landing and Ramp Systems	
10 75 00	Flagpoles	Includes: complete flagpole fabrications of metal, wood, and fiberglass, including all accessories and appurtenances. May Include: flags and banners. See: 10 83 00 for flags and banners.
10 75 13	Automatic Flagpoles	
10 75 16	Ground-Set Flagpoles	
10 75 19	Nautical Flagpoles	
10 75 23	Wall-Mounted Flagpoles	
10 75 26	Roof-Mounted Flagpoles	Includes: roof-mounted flagpoles made from aluminum, copper alloy (bronze), stainless steel, and fiberglass, and bases and fittings.
10 75 29	Plaza-Mounted Flagpoles	

NUMBER	TITLE	EXPLANATION
10 80 00	Other Specialties	See: 03 48 00 for precast concrete specialties. 05 59 00 for metal specialties. 07 71 00 for roof specialties. 07 77 00 for wall specialties. 32 17 00 for paving specialties. 34 82 00 for bridge specialties. 35 59 00 for marine specialties.
10 81 00	Pest Control Devices	Includes: mechanical, electrical, physical, and chemical repellent systems, and protective devices. See: 31 31 00 for soil treatment. 31 31 13 for rodent control soil treatment. 31 31 16 for termite control soil treatment. See Also: 03 15 21 for termite barrier.
10 81 13	Bird Control Devices	
10 81 16	Insect Control Devices	
10 81 19	Rodent Control Devices	
10 82 00	Grilles and Screens	Includes: grilles and screens of wood, metal, plastic, and other materials used for a variety of purposes. See: 06 49 19 for wood screens and exterior wood shutters. 08 92 00 for louvered equipment enclosures. 10 22 23 for portable partitions, screens, and panels not connected to permanent construction. 10 71 13 for exterior sun control devices. 12 46 26 for decorative screens. 32 35 13 for self-supporting exterior screens and louvers.
10 82 13	Exterior Grilles and Screens	
10 82 19	Exterior Sound Screens	
10 82 23	Interior Grilles and Screens	
10 83 00	Flags and Banners	See: 10 14 00 for signage. 10 75 00 for flagpoles. See Also: 10 75 00 for flags and banners.
10 83 13	Flags	
10 83 16	Banners	

DIVISION 10

NUMBER	TITLE	EXPLANATION
10 84 00	Gas Lighting	See: 26 50 00 for lighting.
10 84 13	Exterior Gas Lighting	See: 26 50 00 for lighting.
10 84 16	Interior Gas Lighting	See: 26 50 00 for lighting.
10 86 00	Security Mirrors and Domes	Includes: physical accessories for security surveillance. See: 28 10 00 for access control. 28 20 00 for video surveillance. 28 30 00 for security detection, alarm, and monitoring.
10 88 00	Scales	Includes: fixed scales. See: 11 21 43 for weighing and wrapping equipment. 34 77 36 for baggage scales. 41 14 36 for weigh scales.

NUMBER	TITLE	EXPLANATION
DIVISION 11 – EQUIPMENT		
11 00 00	Equipment	May be used as Division level Section title. See: 25 31 00 for integrated automation instrumentation and terminal devices for facility equipment. 25 51 00 for integrated automation control of facility equipment. 25 91 00 for integrated automation control sequences for facility equipment. 34 75 00 for roadway equipment. 34 76 00 for railway equipment. 34 77 00 for transportation equipment. 41 00 00 for material processing and handling equipment. 42 00 00 for process heating, cooling, and drying equipment. 43 00 00 for process gas and liquid handling, purification, and storage equipment. 44 00 00 for pollution and waste control equipment. 45 00 00 for industry-specific manufacturing equipment. 46 00 00 for water and wastewater equipment. 48 00 00 for electrical power generation equipment.

11 01 00 Operation and Maintenance of Equipment

Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of equipment.

Level 4 Numbering Recommendation: one Level 4 title for equipment is provided as an example for user's creation of more detailed titles for other subjects as required. Following Level 4 numbering is recommended:

- .51-.59 for maintenance.
- .61-.69 for repair.
- .71-.79 for rehabilitation.
- .81-.89 for replacement.
- .91-.99 for restoration.

DIVISION **11**

NUMBER	TITLE	EXPLANATION
11 01 10	Operation and Maintenance of Vehicle and Pedestrian Equipment	
11 01 20	Operation and Maintenance of Commercial Equipment	
11 01 30	Operation and Maintenance of Residential Equipment	
11 01 40	Operation and Maintenance of Foodservice Equipment	
11 01 50	Operation and Maintenance of Educational and Scientific Equipment	
11 01 56	Operation and Maintenance of Observatory Equipment	
11 01 60	Operation and Maintenance of Entertainment and Recreation Equipment	
11 01 70	Operation and Maintenance of Healthcare Equipment	
11 01 80	Operation and Maintenance of Facility Maintenance and Operation Equipment	
11 01 90	Operation and Maintenance of Other Equipment	

11 05 00 Common Work Results for Equipment

Includes: subjects common to multiple titles in Division 11.

See: 01 80 00 for performance requirements of subjects common to multiple titles.

11 05 05	Selective Demolition for Equipment	See: 02 41 19 for selective demolition for whole structure demolition.
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11 05 13	Common Motor Requirements for Equipment	
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11 06 00 Schedules for Equipment

Includes: schedules of items common to multiple titles in Division 11.

Notes: a schedule may be included on drawings, in the project manual, or a project book.

11 06 10	Schedules for Vehicle and Pedestrian Equipment	
11 06 20	Schedules for Commercial Equipment	
11 06 30	Schedules for Residential Equipment	
11 06 40	Schedules for Foodservice Equipment	
11 06 40.13	Foodservice Equipment Schedule	
11 06 50	Schedules for Educational and Scientific Equipment	
11 06 60	Schedules for Entertainment and Recreation Equipment	
11 06 70	Schedules for Healthcare Equipment	
11 06 70.13	Healthcare Equipment Schedule	
11 06 80	Schedules for Facility Maintenance and Operation Equipment	
11 06 90	Schedules for Other Equipment	
11 06 90.11	Schedules for Detention Equipment	

11 08 00 Commissioning of Equipment

Includes: commissioning of items common to multiple titles in Division 11.

See: 01 91 00 for commissioning of subjects common to multiple Divisions.

NUMBER	TITLE	EXPLANATION
11 10 00	Vehicle and Pedestrian Equipment	Includes: equipment associated with vehicles and pedestrians.
11 11 00	Vehicle Service Equipment	Includes: equipment associated with vehicular service. Alternate Terms/Abbreviations: vehicle service equipment: garage equipment: service station equipment. See: 14 45 00 for vehicle lifts. 14 71 11 for vehicle turntables. 23 10 00 for facility fuel systems. 23 35 16 for engine exhaust systems. 33 50 00 for fuel distribution utilities. 41 22 00 for cranes and hoists.
11 11 13	Compressed-Air Vehicle Service Equipment	
11 11 19	Vehicle Lubrication Equipment	
11 11 23	Tire-Changing Equipment	
11 11 26	Vehicle-Washing Equipment	
11 11 36	Vehicle Charging Equipment	
11 11 36.13	Commercial Electric Vehicle Charging Unit	
11 11 36.16	Residential Electric Vehicle Charging Unit	
11 12 00	Parking Control Equipment	Includes: equipment associated with the control of movement of vehicle parking. Self-contained equipment to collect fares for vehicles that is not part of a centralized transportation ticketing and fare collection system. See: 13 34 23 for fabricated control booths. 28 30 00 for security detection, alarm, and monitoring. 28 31 00 for intrusion detection. 28 31 11 for building intrusion detection. 34 41 00 for roadway signaling and control equipment. 34 52 16 for transportation vehicle ticketing equipment. 34 52 26 for transportation vehicle fare collection equipment. 34 52 33 for transportation vehicle fare gates. 34 71 13 for vehicle barriers. 34 71 19 for vehicle delineators.
11 12 13	Parking Key and Card Control Units	
11 12 16	Parking Ticket Dispensers	
11 12 23	Parking Meters	
11 12 26	Parking Fee Collection Equipment	
11 12 26.13	Parking Fee Coin Collection Equipment	
11 12 33	Parking Gates	
11 12 33.13	Lift Arm Parking Gates	
11 12 33.16	Sliding Parking Gates	

DIVISION **11**

NUMBER	TITLE	EXPLANATION
11 12 33.19	Swinging Parking Gates	
11 13 00	Loading Dock Equipment	Includes: equipment associated with service docks and for the loading and unloading of vehicles. May Include: signals for truck restraints. See: 08 38 00 for traffic doors. 14 45 00 for vehicle lifts. 26 50 00 for lighting. 41 22 00 for cranes and hoists.
11 13 13	Loading Dock Bumpers	
11 13 16	Loading Dock Seals and Shelters	
11 13 16.13	Loading Dock Seals	
11 13 16.23	Loading Dock Shelters	
11 13 16.33	Loading Dock Rail Shelters	
11 13 19	Stationary Loading Dock Equipment	
11 13 19.13	Loading Dock Levelers	
11 13 19.23	Stationary Loading Dock Lifts	
11 13 19.26	Loading Dock Truck Lifts	
11 13 19.33	Loading Dock Truck Restraints	
11 13 23	Portable Dock Equipment	
11 13 23.13	Portable Dock Lifts	
11 13 23.16	Portable Dock Ramps	
11 13 23.19	Portable Dock Bridges	
11 13 23.23	Portable Dock Platforms	
11 13 26	Loading Dock Lights	
11 14 00	Pedestrian Control Equipment	Includes: equipment associated with the control of movement of pedestrians. Self-contained equipment to collect fares for pedestrians that is not part of a centralized transportation ticketing and fare collection system. Pedestrian access control. See: 28 10 00 for access control. 28 20 00 for video surveillance. 34 41 00 for roadway signaling and control equipment. 34 54 00 for transportation passenger fare collection. 34 54 16 for transportation passenger ticketing equipment. 34 54 26 for transportation passenger fare collection equipment. 34 54 33 for transportation passenger fare gates.
11 14 13	Pedestrian Gates	
11 14 13.16	Rotary Gates	
11 14 13.19	Turnstiles	
11 14 16	Pedestrian Control Devices	Includes: permanent and portable retractable belt barriers, stanchions, and other queuing, pedestrian flow, and crowd control devices.

NUMBER	TITLE	EXPLANATION
11 14 19	Portable Posts and Railings	
11 14 26	Pedestrian Fare Collection Equipment	
11 14 26.13	Pedestrian Coin Fare Collection Equipment	
11 14 43	Pedestrian Detection Equipment	
11 14 43.13	Electronic Detection and Counting Systems	
11 14 53	Pedestrian Security Equipment	
11 15 00	Security Control Equipment	
		Includes: security kiosks and guard shacks.
		See Also: 28 10 00 for access control. 28 30 00 for security detection, alarm, and monitoring.
11 20 00	Commercial Equipment	
		Includes: equipment associated with commercial facilities.
		See: 27 51 23 for commercial intercommunications and program systems.
11 21 00	Retail and Service Equipment	
		Includes: fabricated, built-in, or free-standing equipment used in retail and service facilities. Coin-operated equipment for dispensing various foods, drinks, stamps, money, tickets, and other items. Fabricated manual or automated equipment used where wearing apparel and luggage are checked or stored on a temporary basis. Display cases that include equipment.
		See: 06 41 00 for architectural wood casework. 10 12 00 for display cases. 11 22 00 for refrigerated display equipment. 11 29 00 for postal, packaging, and shipping equipment. 11 46 00 for food dispensing equipment. 12 35 59 for display casework. 14 92 00 for pneumatic tube systems.
11 21 13	Cash Registers and Checking Equipment	
11 21 23	Vending Equipment	
11 21 23.13	Merchandise Vending Machines	
11 21 23.16	Reverse Vending Machines	
		Includes: bottle and can return machines.
11 21 23.19	Video Rental Vending Machines	
11 21 23.36	Token Vending Machines	
11 21 24	Money Changing Machines	
11 21 33	Checkroom Equipment	
11 21 43	Weighing and Wrapping Equipment	
11 21 53	Barber and Beauty Shop Equipment	

DIVISION **11**

NUMBER	TITLE	EXPLANATION
11 21 63	Refrigerated Display Equipment	Includes: display cases that include refrigeration. See: 10 12 00 for display cases. 12 35 59 for display cases. See Also: 11 41 00 for food storage equipment.
11 21 73	Commercial Laundry and Dry Cleaning Equipment	Includes: equipment for commercial laundry and dry cleaning operations, including coin-operated equipment. See: 10 28 23 for residential laundry accessories. 14 91 33 for laundry and linen chutes. 41 21 23 for piece material conveyors. 42 32 00 for industrial material dryers.
11 21 73.13	Dry Cleaning Equipment	
11 21 73.16	Drying and Conditioning Equipment	
11 21 73.19	Finishing Equipment	
11 21 73.23	Commercial Ironing Equipment	
11 21 73.26	Commercial Washers and Extractors	
11 21 73.33	Coin-Operated Laundry Equipment	
11 21 73.43	Hanging Garment Conveyors	
11 21 83	Photo Processing Equipment	Includes: photographic-film-processing equipment and other products for darkroom use. See: 08 34 33 for lightproof doors. 08 34 36 for darkroom doors. 12 35 00 for specialty casework.
11 22 00	Banking Equipment	
11 22 13	Vault Equipment	Includes: fabricated, built-in, or free-standing equipment specifically designed for money or valuables. See: 08 34 59 for vault doors and day gates. 13 27 00 for vaults. 23 30 00 for HVAC air distribution.
11 22 13.13	Safe Deposit Boxes	
11 22 13.16	Safes	
11 22 13.23	Vault Ventilators	
11 22 13.26	Time Capsules	

NUMBER	TITLE	EXPLANATION
11 22 16	Teller and Service Equipment	Includes: fabricated, built-in, or free-standing equipment specifically designed for handling and transfer of money and other high-security items. Alternate Terms/Abbreviations: ATM: automatic teller machine. See: 08 34 53 for security doors and frames. 08 56 53 for security windows. 12 35 17 for bank casework. 13 27 00 for vaults. 14 92 00 for pneumatic tube systems.
11 22 16.13	Teller Equipment Systems	
11 22 16.16	Automatic Banking Systems	
11 22 16.19	Money Handling Equipment	
11 22 16.23	Money Depositories	
11 22 16.26	Money Cart Pass-Through	
11 22 16.29	Package Transfer Units	
11 22 16.33	Deal Trays	
11 25 00	Hospitality Equipment	Includes: specialized equipment for the purpose of registering, admitting, and controlling rooms and other information at hotels, motels, hospitals, and other similar facilities. See: 06 41 00 for architectural wood casework. 12 30 00 for manufactured casework. 12 54 00 for hospitality furniture. 13 42 25 for hospitality unit modules. 14 72 00 for hospitality turntables. 27 41 16 for integrated audio-video systems and equipment for restaurants and bars.
11 25 13	Registration Equipment	
11 28 00	Office Equipment	Includes: data and paper processing equipment. Drafting machines, plotters, carto-stereographs. Alternate Terms/Abbreviations: facsimiles: fax machines. See: 12 51 00 for office furniture. 12 59 00 for open office systems furniture. 14 92 00 for pneumatic tube systems. 25 11 19 for integrated automation operator workstations. 27 22 00 for data communications hardware. 27 24 00 for data communications peripheral data equipment. 27 32 29 for facsimiles and modems. 27 33 26 for facsimile servers. 27 41 16 for integrated audio-video systems and equipment for conference rooms and board rooms.

DIVISION **11**

NUMBER	TITLE	EXPLANATION
11 28 13	Computers	
11 28 16	Printers	
11 28 23	Copiers	

11 29 00 Postal, Packaging, and Shipping Equipment

Includes: equipment for normal mailing, packaging, shipping, and delivery operations for professional, commercial, and institutional applications.

Does Not Include: equipment for mailing, packaging, shipping, and delivery operations of post office or products of process engineering operations.

See: 10 55 00 for postal specialties.

11 21 43 for weighing and wrapping equipment.

41 21 23 for postal conveyors.

41 24 46 for postal sorting machines.

41 40 00 for container processing and packaging for processing operations.

41 43 00 for shipping packaging for process operations.

11 29 23	Packaging Equipment
11 29 33	Shipping Equipment
11 29 55	Postal Equipment

11 30 00 Residential Equipment

Includes: built-in or free-standing residential appliances and other components specifically for residential use.

May Include: residential food waste disposers. Residential ceiling fans.

See Also: 11 82 26 for facility waste compactors and destructors.

22 41 26 for residential disposers.

23 34 00 for ceiling fans.

Products: refrigerators, cooktops, ranges, exhaust hoods, ovens, microwave ovens, dishwashers, trash compactors, clothes washers, clothes dryers, ironing boards, and other kitchen and laundry appliances and equipment.

11 30 13	Residential Appliances
11 30 13.13	Residential Kitchen Appliances
11 30 13.23	Residential Laundry Appliances
11 30 33	Retractable Stairs
11 30 34	Residential Ceiling Fans

NUMBER	TITLE	EXPLANATION
11 32 00	Unit Kitchens	Includes: manufactured units incorporating kitchen plumbing fixtures, appliances, casework, and countertops. See: 11 30 13 for residential appliances. 12 35 30 for residential casework.
11 32 13	Metal Unit Kitchens	
11 32 16	Wood Unit Kitchens	
11 32 19	Plastic-Laminate-Clad Unit Kitchens	
11 40 00	Foodservice Equipment	Includes: equipment used for food storage, preparation, delivery and serving, cooking, dispensing, and clean-up in commercial and institutional kitchens, bars, and back bars. Alternate Terms/Abbreviations: foodservice equipment: kitchen equipment. See: 12 30 00 for foodservice casework. 12 36 00 for foodservice countertops. 22 10 00 for plumbing piping and pumps. 22 42 00 for commercial plumbing fixtures. 23 38 13 for commercial-kitchen hoods.
11 41 00	Foodservice Storage Equipment	Includes: refrigerated cases. Fabricated walk-in coolers and freezers. May Include: refrigerated display equipment. See: 11 53 23 for laboratory refrigerators. 13 21 26 for cold storage rooms. See Also: 11 22 00 for refrigerated display equipment.
11 41 13	Refrigerated Food Storage Cases	
11 41 23	Walk-In Coolers	
11 41 26	Walk-In Freezers	
11 41 33	Foodservice Shelving	
11 42 00	Food Preparation Equipment	Includes: equipment used for food preparation other than cooking and baking. Food blenders, chopping blocks, butcher's blocks, food grinders, juicers, knife sharpeners, food mixers, peelers, meat saws, food slicers.
11 42 13	Food Preparation Appliances	

DIVISION **11**

NUMBER	TITLE	EXPLANATION
11 42 16	Food Preparation Surfaces	
11 43 00	Food Delivery Carts and Conveyors	May Include: carts, conveyors and material handling units dedicated to transporting food. See Also: 14 10 00 for cart lifts. 41 21 23 for piece material conveyors.
11 43 13	Food Delivery Carts	
11 43 16	Food Delivery Conveyors	
11 44 00	Food Cooking Equipment	Includes: equipment used to cook and bake food. Barbeque grilles, broilers, deep well cookers, French fryers, griddles, grilles, kettles, microwave ovens, rotisseries. May Include: ventilation hoods with fire-suppression systems. See: 42 15 00 for industrial ovens. See Also: 23 38 13 for commercial-kitchen hoods.
11 44 13	Commercial Ranges	
11 44 16	Commercial Ovens	
11 46 00	Food Dispensing Equipment	Includes: equipment used to dispense food. Bread toasters, warming cabinets, coffee urns, cold dish tables, food warmers, glass fillers, salad tables. Food, dish, and beverage dispensers. See: 11 21 23 for vending equipment.
11 46 13	Bar Equipment	
11 46 16	Service Line Equipment	
11 46 19	Soda Fountain Equipment	
11 46 23	Coffee and Espresso Equipment	
11 46 83	Ice Machines	Includes: equipment used to produce, store, and dispense ice.
11 48 00	Foodservice Cleaning and Disposal Equipment	Includes: equipment used to clean foodservice equipment, utensils, and tableware. Food waste disposers, drain boards, garbage scuppers, pot sinks. See Also: 22 42 26 for commercial disposers.
11 48 13	Commercial Dishwashers	

NUMBER	TITLE	EXPLANATION
11 50 00	Educational and Scientific Equipment	Includes: equipment associated with libraries, educational facilities, laboratories, planetariums, observatories, and museums. Alternate Terms/Abbreviations: educational equipment: school equipment. See: 27 41 16 for integrated audio-video systems and equipment for classrooms. 27 51 23 for education intercommunication and program systems.
11 51 00	Library Equipment	Includes: equipment associated with libraries. See: 12 35 50 for educational/library casework. 12 56 51 for library furniture. 13 44 00 for modular mezzanines. 14 10 00 for book lifts.
11 51 13	Automated Book Storage and Retrieval Systems	
11 51 16	Book Depositories	
11 51 19	Book Theft Protection Equipment	
11 51 23	Library Stack Systems	
11 51 23.13	Metal Library Shelving	
11 52 00	Audio-Visual Equipment	Includes: projection screens, projectors, players, recorders and other audio-visual equipment. See: 10 11 36 for visual display conference units. 11 28 00 for office equipment. 12 56 52 for audio-visual furniture. 27 40 00 for audio-video communications. 27 51 23 for education intercommunication and program systems.
11 52 13	Projection Screens	
11 52 13.13	Fixed Projection Screens	
11 52 13.16	Portable Projection Screens	
11 52 13.19	Rear Projection Screens	
11 52 16	Projectors	
11 52 16.13	Movie Projectors	
11 52 16.16	Slide Projectors	
11 52 16.19	Overhead Projectors	
11 52 16.23	Opaque Projectors	
11 52 16.26	Video Projectors	
11 52 19	Players and Recorders	

DIVISION **11**

NUMBER	TITLE	EXPLANATION
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11 52 23	Audio-Visual Equipment Supports	Includes: operable and fixed a/v equipment support.
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11 53 00 Laboratory Equipment

Includes: equipment for scientific use. Laboratory fume hoods.

See: 12 35 53 for laboratory casework, tops and sinks, and service fixtures mounted on laboratory tops.

12 36 53 for laboratory countertops.

12 56 53 for laboratory furniture.

13 21 00 for controlled environment rooms.

22 45 00 for emergency plumbing fixtures.

22 61 00 for compressed-air systems for laboratory facilities.

22 62 00 for vacuum systems for laboratory facilities.

22 63 00 for gas systems for laboratory facilities.

22 66 53 for laboratory chemical waste and vent piping.

22 67 00 for processed water systems for laboratory facilities.

23 30 00 for fume hood exhaust ducts.

See Also: 23 38 16 for fume hoods connected to HVAC systems.

Products: glove boxes.

11 53 13	Laboratory Fume Hoods
11 53 13.13	Recirculating Laboratory Fume Hoods
11 53 16	Laboratory Incubators
11 53 17	Laboratory Equipment Washers
11 53 19	Laboratory Sterilizers
11 53 23	Laboratory Refrigerators
11 53 26	Laboratory Freezers
11 53 29	Laboratory Controlled-Environment Cabinets
11 53 33	Emergency Safety Appliances
11 53 43	Service Fittings and Accessories
11 53 53	Biological Safety Cabinets

11 55 00 Planetarium Equipment

Includes: projectors and related equipment unique to planetariums.

See: 13 23 00 for planetariums.

11 55 13	Planetarium Projectors
11 55 16	Planetarium Pendulums

11 56 00 Observatory Equipment

Includes: telescopes and related equipment unique to observatories.

See: 13 34 56 for observatories.

NUMBER	TITLE	EXPLANATION
11 56 13	Telescopes	
11 56 16	Telescope Mounts	
11 56 19	Telescope Drive Mechanisms	
11 56 23	Telescope Domes	

11 57 00 Vocational Shop Equipment

Includes: equipment related to vocational shops.

See: 10 21 23 for shop cubicles and cubicle curtains and track.
 12 57 00 for industrial furniture.
 23 35 13 for sawdust collection systems.
 23 35 16 for engine exhaust systems.
 40 17 00 for welding and cutting gases piping.
 41 22 00 for cranes and hoists.
 41 34 23 for spray painting booths.

11 59 00 Exhibit Equipment

Includes: equipment for exhibits and displays.

See: 10 12 00 for display cases.
 11 22 00 for refrigerated display equipment.
 12 35 59 for display casework.
 14 73 00 for exhibit turntables.
 14 73 59 for display turntables.
 26 55 59 for display lighting.

11 60 00 Entertainment and Recreation Equipment

11 61 00 Broadcast, Theater, and Stage Equipment

Includes: equipment for support of broadcast, theatrical, instrumental, or voice programs. Cycloramas, entertainment ticket dispensers, scenery and flats, tormentors.

May Include: stage lighting systems and controls.

Alternate Terms/Abbreviations: theater equipment: auditorium equipment.

See: 05 51 36 for catwalks.
 12 61 00 for fixed audience seating for theaters.
 14 43 13 for stage and orchestra lifts.
 14 74 61 for stage turntables.
 27 41 16 for integrated audio-video systems and equipment for theaters.

See Also: 26 09 61 for theatrical lighting controls.
 26 55 61 for theatrical lighting.
 26 55 83 for broadcast lighting.

DIVISION **11**

NUMBER	TITLE	EXPLANATION
11 61 13	Acoustical Shells	
11 61 23	Folding and Portable Stages	
11 61 33	Rigging Systems and Controls	
11 61 33.11	Lighting Rigging Systems	
11 61 33.16	Scenery Rigging Systems	
11 61 33.19	Curtain Systems	
11 61 43	Stage Curtains	
11 62 00	Musical Equipment	Includes: musical instruments, including fabricated and field-assembled instruments. See: 11 61 00 for theater and stage equipment. 11 91 00 for religious equipment.
11 62 13	Bells	
11 62 16	Carillons	
11 62 19	Organs	
11 66 00	Athletic Equipment	Includes: manufactured equipment intended for use in interior athletic and exercise activities. Alternate Terms/Abbreviations: athletic equipment: sports equipment. Gymnasium: fieldhouse. See: 09 64 66 for wood athletic flooring. 09 65 66 for resilient athletic flooring. 09 67 66 for fluid-applied athletic flooring. 11 67 00 for recreational equipment. 11 67 33 for recreational climbing walls. 11 68 23 for exterior court athletic equipment. 11 68 33 for athletic field equipment. 11 68 43 for exterior scoreboards. 12 63 00 for arena seating. 13 24 66 for athletic rooms. 13 28 00 for athletic and recreational special construction. 13 28 66 for demountable athletic surfaces.
11 66 13	Exercise Equipment	
11 66 23	Gymnasium Equipment	
11 66 23.13	Basketball Equipment	
11 66 23.23	Volleyball Equipment	
11 66 23.33	Interior Tennis Equipment	
11 66 23.43	Interior Track and Field Equipment	
11 66 23.53	Wall Padding	
11 66 23.56	Mat Storage	
11 66 43	Interior Scoreboards	
11 66 53	Gymnasium Dividers	
11 66 53.13	Batting/Golf Cages	

NUMBER	TITLE	EXPLANATION
11 67 00	Recreational Equipment	Includes: manufactured equipment intended for use in recreational activities. Curling rinks, pistol and rifle range equipment.
		Alternate Terms/Abbreviations: pool tables: billiard tables.
		See: 11 66 00 for athletic equipment.
11 67 13	Bowling Alley Equipment	
11 67 23	Shooting Range Equipment	
11 67 33	Climbing Walls	
11 67 43	Table Games Equipment	
11 67 43.13	Pool Tables	
11 67 43.23	Ping-Pong Tables	
11 67 53	Game Room Equipment	
11 67 53.13	Video Games	
11 67 53.23	Pinball Machines	
11 68 00	Play Field Equipment and Structures	Includes: manufactured equipment intended for use in exterior athletic and recreational activities.
		See: 11 66 00 for interior athletic equipment.
		11 66 43 for interior scoreboards.
		11 67 00 for recreational equipment.
		12 63 00 for stadium seating.
		13 11 00 for swimming pools.
		13 14 00 for aquatic park structures.
		13 31 13 for air-supported fabric structures.
		13 34 16 for fabricated engineered grandstands and bleachers.
		26 55 68 for athletic field lighting.
		32 18 00 for exterior athletic and recreational surfacing.
		32 31 13 for recreational and tennis court fences and gates.
		32 31 13 for chain link backstops for softball and baseball.
11 68 13	Playground Equipment	
11 68 16	Play Structures	
11 68 23	Exterior Court Athletic Equipment	
11 68 23.13	Exterior Basketball Equipment	
11 68 23.23	Exterior Volleyball Equipment	
11 68 23.33	Tennis Equipment	
11 68 33	Athletic Field Equipment	
11 68 33.13	Football Field Equipment	
11 68 33.23	Soccer and Field Hockey Equipment	
11 68 33.33	Baseball Field Equipment	
11 68 33.43	Track and Field Equipment	
11 68 43	Exterior Scoreboards	
11 70 00	Healthcare Equipment	

NUMBER	TITLE	EXPLANATION
11 71 00	Medical Sterilizing Equipment	Includes: specialized equipment for healthcare facilities. Film illuminators, fluoroscopes, Hubbard tubs, radioisotopic equipment, surgical equipment. Alternate Terms/Abbreviations: healthcare equipment: medical equipment: hospital equipment. See: 05 45 23 for healthcare metal supports. 10 25 13 for patient bed service walls. 12 35 70 for healthcare casework. 13 19 00 for kennels and animal shelters. 13 49 00 for radiation protection. 14 92 00 for pneumatic tube systems. 22 43 00 for healthcare plumbing fixtures. 22 61 00 for compressed-air systems for healthcare facilities. 22 62 00 for vacuum systems for healthcare facilities. 22 63 00 for gas systems for healthcare facilities. 22 66 70 for healthcare chemical waste and vent piping. 22 67 00 for processed water systems for healthcare facilities. 26 55 70 for healthcare lighting. 27 52 00 for healthcare communications and monitoring systems. 28 41 00 for radiation detection and alarm.
11 72 00	Examination and Treatment Equipment	
11 72 13	Examination Equipment	
11 72 53	Treatment Equipment	
11 73 00	Patient Care Equipment	
11 73 13	Psychiatric Seclusion Room Padding	
11 74 00	Dental Equipment	
11 75 00	Optical Equipment	
11 76 00	Operating Room Equipment	
11 77 00	Radiology Equipment	
11 78 00	Mortuary Equipment	Includes: specialized equipment for the handling and preparation of cadavers. Autopsy equipment, dissecting equipment, embalming tables. See: 12 35 00 for specialty casework. 13 21 26 for cold storage rooms.

NUMBER	TITLE	EXPLANATION
11 78 13	Mortuary Refrigerators	
11 78 16	Crematorium Equipment	
11 78 19	Mortuary Lifts	
11 79 00	Therapy Equipment	Includes: equipment intended for therapeutic activities. Hydrotherapy equipment, teletherapy equipment. See: 11 66 00 for athletic equipment. 11 67 00 for recreational equipment. 13 17 23 for therapeutic pools.
11 80 00	Facility Maintenance and Operation Equipment	Includes: equipment for the collection, handling, and disposal of materials at a single facility. See: 23 35 13 for sawdust collection systems. 44 11 16 for dust collectors.
11 81 00	Facility Maintenance Equipment	Includes: fabricated, built-in, and field-assembled equipment for maintenance operations. See: 14 84 13 for window washing scaffolding. 14 84 23 for window washing hoists. 41 66 23 for mobile sweepers/vacuums. 41 67 00 for plant maintenance equipment.
11 81 13	Floor and Wall Cleaning Equipment	
11 81 16	Housekeeping Carts	
11 81 19	Vacuum Cleaning Systems	
11 81 23	Façade Access Equipment	
11 81 23.13	Window Washing Systems	
11 81 29	Facility Fall Protection	Includes: anchorage and tie off for fall protection.
11 82 00	Facility Solid Waste Handling Equipment	Includes: packaged equipment involving the collection, shredding, compaction, and removal or incineration of trash and other solid waste at a facility. Alternate Terms/Abbreviations: waste compactors: refuse compactors: trash compactors. See: 14 91 82 for trash chutes. 44 50 00 for site-constructed solid waste control and reuse for multiple facilities.

DIVISION **11**

NUMBER	TITLE	EXPLANATION
11 82 13	Solid Waste Bins	
11 82 19	Packaged Incinerators	
11 82 23	Recycling Equipment	
11 82 26	Facility Waste Compactors	
11 82 29	Composting Equipment	
11 82 33	Facility Waste Shredders	See: 44 51 00 for solid waste control equipment.
11 82 36	Facility Waste Balers	See: 44 51 00 for solid waste control equipment.

11 90 00 Other Equipment

11 91 00 Religious Equipment

Includes: fabricated, built-in, or free-standing religious related equipment.

Alternate Terms/Abbreviations: religious equipment: ecclesiastical equipment.

See Also: 10 74 29 for spires and steeples.

11 62 00 for musical equipment.

12 19 00 for religious artwork.

12 35 91 for religious casework.

12 67 00 for pews and benches.

11 91 13 Baptisteries

11 92 00 Agricultural Equipment

Includes: equipment related to land cultivating, raising crops, and feeding, breeding, and raising of livestock.

See: 11 93 00 for horticultural equipment.

13 19 00 for kennels and animal shelters.

32 31 00 for fences and gates.

32 86 00 for agricultural irrigation.

45 15 00 for food manufacturing equipment.

11 92 13 Milkers
 11 92 16 Stock Feeders
 11 92 19 Stock Waterers
 11 92 23 Agricultural Waste Clean-Up Equipment

NUMBER	TITLE	EXPLANATION
11 93 00	Horticultural Equipment	Includes: equipment related to cultivating flowers, fruits, vegetables, or ornamental plants. See: 11 92 00 for agricultural equipment. 13 34 13 for greenhouses. 32 86 00 for agricultural irrigation. 32 90 00 for planting. 32 91 13 for soil preparation. 32 96 00 for transplanting. 45 15 00 for food manufacturing equipment.
11 93 13	Hydroponic Growing Systems	
11 93 16	Seeders	
11 93 19	Transplanters	
11 93 23	Potting Machines	
11 93 26	Flat Fillers	
11 93 29	Baggers	
11 93 33	Soil Mixers	
11 94 00	Veterinary Equipment	
11 95 00	Arts and Crafts Equipment	
11 95 13	Kilns	
11 97 00	Security Equipment	Includes: fabricated, built-in, or free-standing equipment specifically designed for secure operations. See: 08 34 53 for security doors and frames. 08 56 53 for security windows. 08 71 53 for security door hardware. 10 28 13 for security toilet accessories. 10 44 13 for security fire extinguisher cabinets. 10 86 00 for security mirrors and domes. 26 55 53 for security lighting. 28 10 00 for access control.
11 97 13	Deal Drawers	
11 97 16	Gun Ports	
11 97 23	Valuable Material Storage	

NUMBER	TITLE	EXPLANATION
11 98 00	Detention Equipment	Includes: fabricated, built-in, or free-standing equipment specifically designed for detention facilities. Alternate Terms/Abbreviations: detention equipment: jail equipment: penitentiary equipment: prison equipment. See: 05 59 63 for detention enclosures. 10 28 13 for detention toilet accessories. 11 98 12 for detention doors and frames. 11 98 14 for detention door hardware. 11 98 21 for detention windows. 12 55 00 for detention furniture. 13 42 63 for detention cell modules. 26 55 63 for detention lighting. 27 51 23 for detention intercommunications and program systems. 28 52 00 for detention security systems. 28 52 11 for detention security system requirements.
11 98 11	Detention Equipment Requirements	Notes: this Section is not intended to duplicate or supplant Division 01 General Requirements content.
11 98 12	Detention Doors and Frames	Includes: doors and frames with specific characteristics for detention applications.
11 98 13	Detention Pass-Through Doors	
11 98 14	Detention Door Hardware	
11 98 16	Detention Equipment Fasteners	
11 98 19	Detention Room Padding	
11 98 21	Detention Windows	
11 98 23	Detention Window Screens	

DIVISION 12 – FURNISHINGS

12 00 00 Furnishings

May be used as Division level
Section title.

12 01 00 Operation and Maintenance of Furnishings

Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of furnishings.

Level 4 Numbering Recommendation: following numbering is recommended for the creation of Level 4 titles:

.51-.59 for maintenance.

.61-.69 for repair.

.71-.79 for rehabilitation.

.81-.89 for replacement.

.91-.99 for restoration.

12 01 10	Operation and Maintenance of Art
12 01 20	Operation and Maintenance of Window Treatments
12 01 30	Operation and Maintenance of Casework
12 01 40	Operation and Maintenance of Furnishings and Accessories
12 01 50	Operation and Maintenance of Furniture
12 01 60	Operation and Maintenance of Multiple Seating
12 01 90	Operation and Maintenance of Other Furnishings

12 05 00 Common Work Results for Furnishings

Includes: subjects common to multiple titles in Division 12.

See: 01 80 00 for performance requirements of subjects common to multiple titles.

12 05 05	Selective Demolition for Furnishings
	See: 02 41 19 for selective demolition for whole structure demolition.

12 05 13	Fabrics
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12 06 00 Schedules for Furnishings

Includes: schedules of items common to multiple titles in Division 12.

Notes: a schedule may be included on drawings, in the project manual, or a project book.

12 06 10	Schedules for Art
12 06 20	Schedules for Window Treatments
12 06 20.13	Window Treatment Schedule
12 06 30	Schedules for Casework

DIVISION 12

NUMBER	TITLE	EXPLANATION
12 06 30.13	Manufactured Casework Schedule	
12 06 40	Schedules for Furnishings and Accessories	
12 06 40.13	Furnishings Schedule	
12 06 50	Schedules for Furniture	
12 06 60	Schedules for Multiple Seating	
12 06 90	Schedules for Other Furnishings	

12 08 00 Commissioning of Furnishings

Includes: commissioning of items common to multiple titles in Division 12.

See: 01 91 00 for commissioning of subjects common to multiple divisions.

12 10 00 Art

Includes: interior and exterior art objects.

See: 05 70 00 for decorative metals.

06 44 00 for ornamental woodwork.

06 66 00 for custom ornamental simulated woodwork.

09 27 00 for plaster fabrications.

12 11 00 Murals

Includes: murals produced with media other than building products.

See Also: 04 25 00 for sculptured brick panels and brick murals.

09 30 00 for tile murals.

12 14 00 for sculptures.

12 14 23 for relief art.

12 11 13	Photo Murals
12 11 16	Sculptured Brick Panels
12 11 23	Brick Murals
12 11 26	Ceramic Tile Murals
12 11 33	Trompe L'Oeil

12 12 00 Wall Decorations

Includes: flat artwork attached to walls.

See: 09 72 00 for wall coverings.

09 77 00 for special wall surfacing.

09 94 00 for decorative finishing.

12 12 13	Commissioned Paintings
12 12 16	Framed Paintings
12 12 19	Framed Prints
12 12 23	Tapestries

NUMBER	TITLE	EXPLANATION
12 12 26	Wall Hangings	
12 14 00	Sculptures	Includes: free-standing sculptures. See: 12 11 16 for sculptured brick panels.
12 14 13	Carved Sculptures	
12 14 16	Cast Sculptures	
12 14 19	Constructed Sculptures	
12 14 23	Relief Art	
12 17 00	Art Glass	Includes: artistically designed glass windows and glass screens. See: 08 81 13 for decorative glass glazing. 08 84 13 for decorative plastic glazing.
12 17 13	Etched Glass	
12 17 16	Stained Glass	
12 19 00	Religious Art	Includes: artwork with a religious theme. See: 11 91 00 for religious equipment. 12 35 91 for religious casework.
12 20 00	Window Treatments	
12 21 00	Window Blinds	Includes: interior window coverings, associated, hardware, and controls. Alternate Terms/Abbreviations: horizontal louver blinds: venetian blinds: mini-blinds. Roll-down blinds: roller blinds. Roller window shades: roll-up window shades. See: 06 49 19 for wood screens and exterior wood shutters. 08 50 00 for windows with integral louver blinds. 08 87 13 for solar control films. 09 83 22 for acoustic drapery. 10 71 00 for exterior protection. 10 71 13 for exterior shutters.
12 21 13	Horizontal Louver Blinds	See: 06 49 16 for wood blinds.

DIVISION **12**

NUMBER	TITLE	EXPLANATION
12 21 13.13	Metal Horizontal Louver Blinds	
12 21 13.23	Wood Horizontal Louver Blinds	
12 21 13.33	Plastic Horizontal Louver Blinds	
12 21 16	Vertical Louver Blinds	
12 21 16.13	Metal Vertical Louver Blinds	
12 21 16.23	Wood Vertical Louver Blinds	
12 21 16.33	Plastic Vertical Louver Blinds	
12 21 23	Roll-Down Blinds	
12 21 26	Black-Out Blinds	
12 22 00	Curtains and Drapes	
12 22 13	Draperies	
12 22 16	Drapery Track and Accessories	
12 23 00	Interior Shutters	
		Includes: manufactured shutters
		See: 06 49 19 for wood shutters.
12 23 13	Wood Interior Shutters	
12 24 00	Window Shades	
12 24 13	Roller Window Shades	
12 24 16	Pleated Window Shades	
12 24 16.13	Z-Pleated Window Shades	
12 24 16.16	Cellular Shades	
12 24 16.19	Roman Shades	
12 25 00	Window Treatment Operating Hardware	
12 25 09	Window Treatment Control System	
12 25 13	Motorized Drapery Rods	
12 26 00	Interior Daylighting Devices	
		See: 10 71 13 for exterior sun control devices.
12 30 00	Casework	
		Includes: stock modular cabinets and other casework units of any material. Countertops. Sinks integral with countertops. Laboratory countertops with integral sinks and fixtures and accessories mounted on countertops.
		Alternate Terms/Abbreviations: casework: cabinets: cabinetry.
		See: 06 41 00 for architectural wood casework.
		10 51 00 for lockers.
		10 56 00 for storage assemblies.
		10 57 23 for closet and utility shelving.
		11 51 23 for library stack systems.
		12 50 00 for movable furniture.

NUMBER	TITLE	EXPLANATION
12 31 00	Manufactured Metal Casework	Includes: casework manufactured of metal or surfaced in metal. May Include: countertops and sinks and lavatories integral with countertops. See Also: 12 36 00 for countertops.
12 31 16	Manufactured Metal Sandwich Panel Casework	
12 32 00	Manufactured Wood Casework	Includes: casework manufactured of wood or faced with wood veneer or plastic laminate. May Include: countertops and sinks and lavatories integral with countertops. Alternate Terms/Abbreviations: PLAM: plastic laminate. See: 12 34 16 for solid plastic casework. See Also: 12 36 00 for countertops.
12 32 13	Manufactured Wood-Veneer-Faced Casework	
12 32 16	Manufactured Plastic-Laminate-Clad Casework	
12 34 00	Manufactured Plastic Casework	Includes: casework manufactured of plastic. May Include: countertops and sinks and lavatories integral with countertops. See: 12 32 16 for plastic-laminate-clad manufactured casework. See Also: 12 36 00 for countertops.
12 34 16	Manufactured Solid-Plastic Casework	
12 34 19	Manufactured Solid Surface Casework	

NUMBER	TITLE	EXPLANATION
12 35 00	Specialty Casework	Includes: manufactured casework characterized by application or use. Laboratory countertops with integral sinks and fixtures and accessories mounted on countertops. May Include: countertops and sinks and lavatories integral with countertops. Alternate Terms/Abbreviations: healthcare casework: medical casework. See: 10 12 00 for display cases. 11 50 00 for educational and scientific equipment. 11 51 00 for library equipment. 11 53 00 for laboratory equipment. 11 59 00 for exhibit equipment. 11 70 00 for healthcare equipment. 11 91 00 for religious equipment. 12 50 00 for movable furniture. 12 56 51 for movable study carrels. 12 59 00 for open office systems furniture. 13 42 43 for dormitory unit modules. 22 43 00 for healthcare plumbing fixtures. 22 60 00 for healthcare and laboratory systems. See Also: 12 36 00 for countertops.
12 35 17	Bank Casework	
12 35 25	Hospitality Casework	
12 35 30	Residential Casework	
12 35 30.13	Kitchen Casework	
12 35 30.23	Bathroom Casework	
12 35 30.43	Dormitory Casework	
12 35 33	Utility Room Casework	
12 35 36	Mailroom Casework	
12 35 39	Commercial Kitchen Casework	
12 35 50	Educational/Library Casework	
12 35 50.13	Educational Casework	
12 35 50.53	Library Casework	
12 35 50.56	Built-In Study Carrels	
12 35 53	Laboratory Casework	
12 35 53.13	Metal Laboratory Casework	
12 35 53.16	Plastic-Laminate-Clad Laboratory Casework	
12 35 53.19	Wood Laboratory Casework	
12 35 53.23	Solid-Plastic Laboratory Casework	
12 35 59	Display Casework	
12 35 70	Healthcare Casework	
12 35 70.13	Hospital Casework	
12 35 70.16	Nurse Station Casework	
12 35 70.19	Exam Room Casework	
12 35 70.74	Dental Casework	

NUMBER	TITLE	EXPLANATION
12 35 83	Performing Arts Casework	
12 35 91	Religious Casework	
12 36 00	Countertops	Includes: countertops of any material.
		Alternate Terms/Abbreviations: PLAM: plastic laminate.
		See: 06 61 13 for simulated stone fabrications.
		09 30 00 for tile countertops.
		11 53 00 for laboratory equipment.
		12 56 53 for laboratory furniture.
		See Also: 06 41 00 for countertops with architectural wood casework.
		06 60 00 for plastic fabrications.
		12 31 00 for countertops with manufactured metal casework.
		12 32 00 for countertops with manufactured wood casework.
		12 34 00 for countertops with manufactured plastic casework.
		12 35 00 for countertops with manufactured specialty casework.
		12 35 53 for countertops with laboratory casework.
		22 40 00 for sinks and lavatories.
		22 60 00 for laboratory systems.
12 36 13	Concrete Countertops	
12 36 16	Metal Countertops	
12 36 19	Wood Countertops	
12 36 23	Plastic Countertops	
12 36 23.13	Plastic-Laminate-Clad Countertops	
12 36 26	Glass Countertops	
12 36 40	Stone Countertops	
12 36 53	Laboratory Countertops	
12 36 61	Simulated Stone Countertops	
12 36 61.13	Cultured Marble Countertops	
12 36 61.16	Solid Surfacing Countertops	
12 36 61.19	Quartz Agglomerate Countertops	
12 36 71	Paper-Composite Countertops	

12 40 00 Furnishings and Accessories

Includes: interior furnishings and accessories not connected to permanent construction.

Alternate Terms/Abbreviations: ceramics: chinaware.

See: 10 28 16 for shower curtain rods.

10 28 19 for tub and shower doors.

12 93 00 for site furnishings.

NUMBER	TITLE	EXPLANATION
12 41 00	Office Accessories	
12 41 13	Desk Accessories	
12 42 00	Table Accessories	
12 42 13	Ceramics	
12 42 16	Flatware	
12 42 16.13	Silverware	
12 42 19	Hollowware	
12 42 23	Glassware	
12 42 26	Table Linens	
12 42 26.13	Napery	
12 43 00	Portable Lamps	
12 43 13	Lamps	
12 43 13.13	Desk Lamps	
12 43 13.16	Table Lamps	
12 43 13.19	Floor Lamps	
12 44 00	Bath Furnishings	
12 44 13	Bath Linens	
12 44 13.13	Bath Mats	
12 44 13.16	Bath Towels	
12 44 16	Shower Curtains	
12 45 00	Bedroom Furnishings	
12 45 13	Bed Linens	
12 45 13.13	Blankets	
12 45 13.16	Comforters	
12 45 16	Pillows	
12 46 00	Furnishing Accessories	
		Includes: interior furnishing accessories not attached to permanent construction.
		See: 10 22 23 for portable partitions, screens, and panels.
		10 28 13 for built-in toilet room waste disposal units.
		12 93 23 for site trash and litter receptacles.
		27 53 13 for clock systems.
12 46 13	Ash Receptacles	
12 46 16	Bowls	
12 46 19	Clocks	
12 46 23	Decorative Crafts	
12 46 26	Decorative Screens	
12 46 29	Vases	
12 46 33	Waste Receptacles	

NUMBER	TITLE	EXPLANATION
12 48 00	Rugs and Mats	Includes: special floor surfaces at entrances and other loose mats and rugs. Alternate Terms/Abbreviations: entrance mats: vestibule mats. See: 05 53 00 for metal gratings. 09 60 00 for flooring. 09 65 00 for resilient flooring. 09 68 00 for carpeting.
12 48 13	Entrance Floor Mats and Frames	
12 48 13.13	Entrance Floor Mats	
12 48 13.16	Entrance Floor Mat Frames	
12 48 16	Entrance Floor Grilles	
12 48 19	Entrance Floor Gratings	
12 48 23	Entrance Floor Grids	
12 48 26	Entrance Tile	
12 48 43	Floor Mats	
12 48 43.13	Chair Mats	
12 48 43.23	Anti-Fatigue Floor Mats	
12 48 43.26	Fall-Cushioning Floor Mats	
12 48 53	Rugs	
12 48 53.13	Runners	
12 48 53.16	Oriental Rugs	
12 50 00	Furniture	
12 51 00	Office Furniture	Includes: movable interior furniture for office applications. See: 10 22 23 for portable partitions, screens, and panels. 10 56 00 for storage assemblies. 11 28 00 for office equipment. 12 30 00 for manufactured casework. 12 52 23 for office seating. 12 59 00 for systems furniture. 13 22 00 for office shelters and booths. 27 41 23 for audio-video furniture and accessories. 41 53 00 for industrial storage equipment and systems.
12 51 16	Case Goods	
12 51 16.13	Metal Case Goods	
12 51 16.16	Wood Case Goods	
12 51 16.19	Plastic-Laminate-Clad Case Goods	
12 51 19	Filing Cabinets	
12 51 19.13	Lateral Filing Cabinets	
12 51 19.16	Vertical Filing Cabinets	
12 51 23	Office Tables	
12 51 83	Custom Office Furniture	

NUMBER	TITLE	EXPLANATION
12 52 00	Seating	See: 11 70 00 for healthcare equipment. 12 35 70 for healthcare casework. 12 51 00 for office furniture. 12 58 16 for residential chairs. 12 59 00 for systems furniture. 12 60 00 for multiple seating. 12 62 00 for portable audience seating.
12 52 13	Chairs	
12 52 19	Upholstered Seating	
12 52 23	Office Seating	
12 52 70	Healthcare Seating	
12 52 83	Custom Seating	
12 53 00	Retail Furniture	Includes: movable furniture used for retail applications. See: 10 12 00 for display cases. 11 21 00 for mercantile and service equipment.
12 53 83	Custom Retail Furniture	
12 54 00	Hospitality Furniture	Includes: movable furniture used for hospitality applications. See: 11 25 00 for hospitality equipment. 12 35 25 for hospitality casework. 12 64 00 for booths and tables for restaurants. 13 42 25 for hospitality unit modules.
12 54 13	Hotel and Motel Furniture	
12 54 16	Restaurant Furniture	
12 54 83	Custom Hospitality Furniture	
12 55 00	Detention Furniture	Includes: movable furniture used for detention applications. Alternate Terms/Abbreviations: detention furniture: jail furniture: penitentiary furniture: prison furniture. See: 05 59 63 for detention enclosures. 10 28 13 for detention toilet accessories. 11 98 12 for detention doors and frames. 11 98 14 for detention door hardware. 11 98 21 for detention windows. 13 42 63 for detention cell modules. 26 55 63 for detention lighting.

NUMBER	TITLE	EXPLANATION
12 55 13	Detention Bunks	
12 55 16	Detention Desks	
12 55 19	Detention Stools	
12 55 23	Detention Tables	
12 55 26	Detention Safety Clothes Hooks	
12 55 83	Custom Detention Furniture	
12 55 86	Detention Control Room Furniture	

12 56 00 Institutional Furniture

Includes: movable furniture used for institutional applications.

Alternate Terms/Abbreviations: classroom furniture:
educational furniture: school furniture. Healthcare furniture:
medical furniture: hospital furniture.

See: 11 50 00 for educational and scientific equipment.

11 51 00 for library equipment.

11 52 00 for audio-visual equipment.

11 53 00 for laboratory equipment.

11 70 00 for healthcare equipment.

12 35 50 for educational/library casework.

12 35 53 for laboratory casework.

12 35 70 for healthcare casework.

12 36 53 for laboratory countertops.

12 68 13 for pedestal tablet arm chairs.

13 42 43 for dormitory unit modules.

27 41 23 for audio-video furniture and accessories.

12 56 23	Religious Furniture
12 56 33	Classroom Furniture
12 56 33.13	Fixed Classroom Tables
12 56 33.23	Classroom Computer Furniture
12 56 39	Lecterns
12 56 43	Dormitory Furniture
12 56 51	Library Furniture
12 56 51.13	Book Shelves
12 56 51.16	Study Carrels
12 56 51.19	Index Card File Cabinets
12 56 52	Audio-Visual Furniture
12 56 53	Laboratory Furniture
12 56 70	Healthcare Furniture
12 56 83	Custom Institutional Furniture
12 56 86	Institutional Control Room Furniture

12 57 00 Industrial Furniture

Includes: movable furniture used for industrial applications.

See: 10 21 23 for shop cubicle curtains and track.

11 57 00 for vocational shop equipment.

DIVISION **12**

NUMBER	TITLE	EXPLANATION
12 57 13	Welding Benches	
12 57 16	Welding Screens	
12 57 19	Laser Containment Screens	
12 57 83	Custom Industrial Furniture	
12 57 86	Industrial Control Room Furniture	

12 58 00 Residential Furniture

Includes: movable furniture used for residential applications.

See: 11 30 00 for residential equipment.
 12 35 30 for residential casework.
 12 40 00 for furnishings and accessories.
 12 52 13 for chairs.
 12 62 00 for portable audience seating.
 13 42 33 for apartment unit modules.

12 58 13	Couches and Loveseats
12 58 13.13	Futons
12 58 16	Residential Chairs
12 58 16.13	Reclining Chairs
12 58 19	Dining Tables and Chairs
12 58 23	Coffee Tables
12 58 26	Entertainment Centers
12 58 29	Beds
12 58 29.13	Daybeds
12 58 33	Dressers
12 58 33.13	Armoires
12 58 36	Nightstands
12 58 83	Custom Residential Furniture

12 59 00 Systems Furniture

Includes: open office systems furniture, components, accessories, and integrated utilities.

See: 09 69 33 for low-profile fixed height access flooring.
 10 22 23 for portable partitions, screens, and panels.
 11 28 00 for office equipment.
 12 51 00 for office furniture.

12 59 13	Panel-Hung Component System Furniture
12 59 16	Free-Standing Component System Furniture
12 59 19	Beam System Furniture
12 59 23	Desk System Furniture
12 59 83	Custom Systems Furniture

NUMBER	TITLE	EXPLANATION
12 60 00	Multiple Seating	Includes: portable, fixed, and telescoping seating for theaters, auditoriums, lecture halls, stadiums, arenas, gymnasiums, religious facilities, restaurants, and other facilities where multiple seating is required. Fixed booths and fixed combination seat and table modules. See: 11 61 00 for theater and stage equipment. 12 52 13 for portable chairs used with movable tables and used in classrooms and restaurants.
12 61 00	Fixed Audience Seating	Includes: fixed seating for theaters, auditoriums, lecture halls, and other facilities. See: 11 61 00 for theater and stage equipment.
12 61 13	Upholstered Audience Seating	
12 61 16	Molded-Plastic Audience Seating	
12 62 00	Portable Audience Seating	Includes: interlocking, folding, stacking, and other seating for public assembly. See: 12 52 13 for chairs. 12 58 16 for residential chairs.
12 62 13	Folding Chairs	
12 62 16	Interlocking Chairs	
12 62 19	Stacking Chairs	
12 62 23	Portable Bleachers	
12 63 00	Stadium and Arena Seating	Includes: fixed benches and seats for stadiums, arenas, gymnasiums, and other facilities. Alternate Terms/Abbreviations: stadium/arena bench seating: bleachers. See: 12 66 00 for telescoping stands. 13 34 16 for fabricated engineered grandstands and bleachers.
12 63 13	Stadium and Arena Bench Seating	
12 63 23	Stadium and Arena Seats	
12 64 00	Booths and Tables	Includes: booths and tables for restaurants and other facilities. See: 12 54 16 for restaurant furniture.

DIVISION **12**

NUMBER	TITLE	EXPLANATION
12 64 13	Booths	
12 64 16	Tables	
12 64 23	Banquette Seating	
12 65 00	Multiple-Use Fixed Seating	
12 66 00	Telescoping Stands	Includes: bleachers and platforms that retract to save floor space when not in use. See: 12 63 00 for stadium and arena seating. 13 34 16 for fabricated engineered grandstands and bleachers.
12 66 13	Telescoping Bleachers	
12 66 23	Telescoping Chair Platforms	
12 67 00	Pews and Benches	See: 11 91 00 for religious equipment.
12 67 13	Pews	
12 67 23	Benches	
12 68 00	Seat and Table Assemblies	See: 12 56 33 for classroom furniture.
12 68 13	Pedestal Tablet Arm Chairs	
12 90 00	Other Furnishings	
12 92 00	Interior Planters and Artificial Plants	Includes: interior planters and artificial plants. See: 32 90 00 for planting. 32 93 00 for live plants. 32 94 33 for planters. See Also: 32 94 33 for interior planters.
12 92 13	Interior Artificial Plants	
12 92 33	Interior Planters	
12 92 43	Interior Landscaping Accessories	

NUMBER	TITLE	EXPLANATION
12 93 00	Interior Public Space Furnishings	Includes: furniture and fixtures for interior public and transitional spaces such as malls. See: 12 60 00 for multiple seating. 12 68 00 for interior seat and table assemblies. 12 92 00 for interior planters and artificial plants. 32 90 00 for planting. 32 93 00 for live plants. 32 94 33 for planters.
12 93 13	Bicycle Racks	
12 93 14	Bicycle Lockers	
12 93 23	Trash and Litter Receptacles	

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NUMBER	TITLE	EXPLANATION
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DIVISION 13 – SPECIAL CONSTRUCTION

13 00 00

Special Construction

May be used as Division level
Section title.

13 01 00

Operation and Maintenance of Special Construction

Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of special construction.

Level 4 Numbering Recommendation: Level 4 titles for special construction are provided as examples for user's creation of more detailed titles for other subjects as required. Following Level 4 numbering is recommended:

.51-.59 for maintenance.

.61-.69 for repair.

.71-.79 for rehabilitation.

.81-.89 for replacement.

.91-.99 for restoration.

13 01 10

Operation and Maintenance of Special Facility Components

13 01 11

Operation and Maintenance of Swimming Pools

13 01 12

Operation and Maintenance of Fountains

13 01 13

Operation and Maintenance of Aquariums

13 01 14

Operation and Maintenance of Amusement Park Structures and Equipment

13 01 18

Operation and Maintenance of Ice Rinks

13 01 20

Operation and Maintenance of Special Purpose Rooms

13 01 21

Operation and Maintenance of Controlled Environment Rooms

13 01 23

Operation and Maintenance of Planetariums

13 01 30

Operation and Maintenance of Special Structures

13 01 40

Operation and Maintenance of Integrated Construction

13 01 49

Operation and Maintenance of Radiation Protection

13 01 50

Operation and Maintenance of Special Instrumentation

13 01 51

Operation and Maintenance of Stress Instrumentation

13 01 52

Operation and Maintenance of Seismic Instrumentation

13 01 53

Operation and Maintenance of Meteorological Instrumentation

13 05 00

Common Work Results for Special Construction

Includes: subjects common to multiple titles in Division 13.

See: 01 80 00 for performance requirements of subjects common to multiple titles.

13 05 05

Selective Demolition for Special Construction

See: 02 41 19 for selective demolition for whole structure demolition.

NUMBER	TITLE	EXPLANATION
13 06 00	Schedules for Special Construction	Includes: schedules of items common to multiple titles in Division 13. Notes: a schedule may be included on drawings, in the project manual, or a project book.
13 06 10	Schedules for Special Facility Components	
13 06 20	Schedules for Special Purpose Rooms	
13 06 30	Schedules for Special Structures	
13 06 40	Schedules for Integrated Construction	
13 06 50	Schedules for Special Instrumentation	
13 08 00	Commissioning of Special Construction	Includes: commissioning of items common to multiple titles in Division 13. See: 01 91 00 for commissioning of subjects common to multiple Divisions.
13 08 10	Commissioning of Special Facility Components	
13 08 11	Commissioning of Swimming Pools	
13 08 12	Commissioning of Fountains	
13 08 13	Commissioning of Aquariums	
13 08 14	Commissioning of Amusement Park Structures and Equipment	
13 08 18	Commissioning of Ice Rinks	
13 08 20	Commissioning of Special Purpose Rooms	
13 08 21	Commissioning of Controlled Environment Rooms	
13 08 23	Commissioning of Planetariums	
13 08 30	Commissioning of Special Structures	
13 08 40	Commissioning of Integrated Construction	
13 08 50	Commissioning of Special Instrumentation	
13 10 00	Special Facility Components	
13 11 00	Swimming Pools	Includes: fabricated, or pre-engineered components, field assembled for athletic or recreational purposes. Diving boards, lifeguard chairs, floats, swimming pool ladders, underwater speakers. See: 13 14 13 for water slides. 13 14 16 for water generating equipment. 13 17 00 for lap swimming pools. 13 17 00 for tubs and pools. 22 11 00 for facility water distribution. 22 13 00 for facility sanitary sewerage. 22 51 00 for swimming pool plumbing systems. 26 55 29 for underwater lighting.

NUMBER	TITLE	EXPLANATION
13 11 13	Below-Grade Swimming Pools	
13 11 23	On-Grade Swimming Pools	
13 11 33	Elevated Swimming Pools	
13 11 43	Recirculating Gutter Systems	
13 11 46	Swimming Pool Accessories	
13 11 49	Swimming Pool Cleaning Equipment	
13 11 53	Movable Pool Bulkheads	
13 11 56	Movable Pool Floors	

13 12 00 Fountains

Includes: decorative or functional manufactured fountains.

See: 22 11 00 for facility water distribution.
 22 13 00 for facility sanitary sewerage.
 22 52 00 for fountain plumbing systems.
 26 55 29 for underwater lighting.

13 12 13	Exterior Fountains
13 12 23	Interior Fountains

13 13 00 Aquariums

Includes: tubs, bowls, and other containers for containing and displaying of aquatic animals and plants.

See: 22 11 00 for facility water distribution.
 22 13 00 for facility sanitary sewerage.
 26 55 29 for underwater lighting.

13 14 00 Amusement Park Structures and Equipment

Includes: fabricated or pre-engineered components, field-assembled for sport or recreational purposes.

See: 13 34 00 for fabricated engineered structures.
 22 11 00 for facility water distribution.
 22 13 00 for facility sanitary sewerage.
 26 55 29 for underwater lighting.

13 14 13	Water Slides
13 14 16	Wave Generating Equipment
13 14 23	Amusement Park Rides

13 15 00 Specialty Element Construction

Includes: specialty fabrications and scenery for themed environments.

NUMBER	TITLE	EXPLANATION
13 17 00	Tubs and Pools	Includes: fabricated, or pre-engineered components, field assembled for therapeutic or recreational purposes. Lap swimming pools. See: 11 79 00 for therapy equipment. 13 11 00 for swimming pools. 13 24 16 for saunas. 13 24 26 for steam baths. 22 11 00 for facility water distribution. 22 13 00 for facility sanitary sewerage. 26 55 29 for underwater lighting. See Also: 22 41 00 for residential plumbing fixtures. 22 43 00 for healthcare plumbing fixtures.
13 17 13	Hot Tubs	
13 17 23	Therapeutic Pools	
13 17 33	Whirlpool Tubs	
13 18 00	Ice Rinks	Includes: equipment, piping, floor systems, and surrounding enclosures. May Include: freezing equipment. See: 13 28 16 for safety netting. See Also: 23 60 00 for freezing equipment.
13 18 13	Ice Rink Floor Systems	
13 18 16	Ice Rink Dasher Boards	
13 19 00	Kennels and Animal Shelters	Includes: facilities for housing and providing healthcare for animals. See: 10 22 13 for wire mesh partitions. 11 70 00 for animal healthcare equipment. 11 92 00 for agricultural equipment. 32 31 00 for fences and gates.
13 19 13	Kennel Enclosures and Gates	
13 19 16	Kennel Feeding Devices	
13 19 19	Animal Washing Tubs	

NUMBER	TITLE	EXPLANATION
13 20 00	Special Purpose Rooms	Includes: fabricated and pre-engineered rooms designed and constructed to meet specific performance requirements of environmental control, dimensional tolerances, sound, fire protection, and other special conditions. Racquetball and squash court rooms.
13 21 00	Controlled Environment Rooms	See: 08 34 73 for sound control door assemblies. 11 41 23 for food service walk-in coolers. 11 41 26 for food service walk-in freezers. 11 53 00 for laboratory equipment. 11 70 00 for healthcare equipment. 13 24 00 for special activity rooms. 13 26 00 for fabricated rooms. 13 30 00 for special structures. 13 48 00 for sound, vibration, and seismic control. 44 20 00 for noise pollution control.
13 21 13	Clean Rooms	
13 21 16	Hyperbaric Rooms	
13 21 23	Insulated Rooms	
13 21 26	Cold Storage Rooms	
13 21 26.13	Walk-in Coolers	
13 21 26.16	Walk-in Freezers	
13 21 29	Constant Temperature Rooms	
13 21 48	Sound-Conditioned Rooms	
13 22 00	Office Shelters and Booths	See: 11 28 00 for office equipment. 12 51 00 for office furniture.
13 23 00	Planetariums	See: 11 55 00 for planetarium equipment. 13 34 56 for observatories.
13 24 00	Special Activity Rooms	See: 11 66 00 for athletic equipment. 13 17 13 for hot tubs. 13 21 00 for controlled environment rooms. 13 26 00 for fabricated rooms. 13 28 33 for athletic and recreational court walls. 13 28 66 for demountable athletic surfaces.
13 24 16	Saunas	
13 24 26	Steam Baths	
13 24 66	Athletic Rooms	

NUMBER	TITLE	EXPLANATION
13 26 00	Fabricated Rooms	See: 13 21 00 for controlled environment rooms. 13 23 00 for special activity rooms.
13 26 13	Storm Shelter Rooms	Alternate Terms/Abbreviations: tornado shelter rooms
13 27 00	Vaults	See: 08 34 59 for vault doors and day gates.
13 27 16	Modular Fire Vaults	
13 27 53	Security Vaults	
13 27 53.13	Modular Concrete Security Vaults	
13 27 53.16	Modular Metal-Clad Laminated Security Vaults	
13 28 00	Athletic and Recreational Special Construction	Includes: special construction for athletic and recreational activities that are directly related to the adjacent construction. See: 09 64 66 for wood athletic flooring. 09 65 66 for resilient athletic flooring. 09 67 66 for fluid-applied athletic flooring. 13 14 00 for amusement park structures and equipment. 13 24 66 for athletic rooms.
13 28 13	Indoor Soccer Boards	
13 28 16	Safety Netting	
13 28 19	Arena Football Boards	
13 28 26	Floor Sockets	
13 28 33	Athletic and Recreational Court Walls	
13 28 66	Demountable Athletic Surfaces	
13 30 00	Special Structures	Includes: structures of special shapes and support systems not included in other Divisions. See: 03 31 00 for structural concrete. 03 41 00 for precast structural concrete. 05 10 00 for structural metal framing. 06 11 00 for wood framing. 06 50 00 for structural plastics. 06 70 00 for structural composites.

NUMBER	TITLE	EXPLANATION
13 31 00	Fabric Structures	Includes: structures of flexible membrane material supported by air pressure, tensioned cables, or other framing system for temporary or permanent uses. See: 05 16 00 for structural cabling. 08 88 23 for cable suspended glazing.
13 31 13	Air-Supported Fabric Structures	
13 31 13.13	Single-Walled Air-Supported Structures	
13 31 13.16	Multiple-Walled Air-Supported Structures	
13 31 23	Tensioned Fabric Structures	
13 31 33	Framed Fabric Structures	
13 32 00	Space Frames	Includes: three-dimensional structural frame of any material capable of transmitting loads in three dimensions.
13 32 13	Metal Space Frames	
13 32 23	Wood Space Frames	
13 33 00	Geodesic Structures	Includes: dome-shaped structure constructed of straight members of any material that form a triangular gridwork. See: 13 34 23 for fabricated dome structures.
13 33 13	Geodesic Domes	
13 34 00	Fabricated Engineered Structures	Includes: fabricated, pre-engineered structures assembled on temporary or permanent foundations. See: 08 44 33 for sloped glazing assemblies. 10 73 23 for car shelters. 10 73 26 for walkway coverings. 10 73 43 for transportation stop shelters. 11 12 00 for parking control equipment. 11 56 00 for observatory equipment. 11 93 00 for horticultural equipment. 12 63 00 for stadium and arena seating. 12 66 00 for telescoping stands. 13 11 00 for swimming pools. 13 14 00 for amusement park structures and equipment. 13 23 00 for planetariums. 13 26 00 for fabricated rooms. 13 33 13 for geodesic domes. 33 72 00 for utility substations.

DIVISION **13**

NUMBER	TITLE	EXPLANATION
13 34 13	Glazed Structures	
13 34 13.13	Greenhouses	
13 34 13.16	Solariums	
13 34 13.19	Swimming Pool Enclosures	
13 34 13.23	Sunrooms	
13 34 13.26	Conservatories	
13 34 16	Grandstands and Bleachers	
13 34 16.13	Grandstands	
13 34 16.53	Bleachers	
13 34 18	Post Frame Building Systems	
		Alternate Terms/Abbreviations: pole barn.
13 34 19	Metal Building Systems	
13 34 23	Fabricated Structures	
13 34 23.13	Portable and Mobile Buildings	
13 34 23.14	Fabricated Classroom Buildings	
13 34 23.16	Fabricated Control Booths	
13 34 23.19	Fabricated Dome Structures	
13 34 23.23	Fabricated Substation Control Rooms	
13 34 23.26	Fabricated Cold Storage Buildings	
13 34 56	Observatories	
13 35 00	Rammed Earth Construction	
13 35 13	Rammed Earth Walls	
13 35 13.13	Traditional Rammed Earth Walls	
13 35 13.23	Stabilized Insulated Rammed Earth Walls	
13 36 00	Towers	
		Includes: fabricated and pre-engineered structures of a vertical nature for a variety of purposes.
		See: 33 71 13 for electrical transmission towers.
		33 81 13 for communication transmission towers.
		33 81 16 for antenna towers.
		34 73 23 for manufactured airfield control towers.
13 36 13	Metal Towers	
13 36 13.13	Steel Towers	
13 36 23	Wood Towers	
13 40 00	Integrated Construction	
		Includes: special construction that integrates multiple construction components into a fabricated module or assembly.

NUMBER	TITLE	EXPLANATION
13 42 00	Building Modules and Components	Includes: fabricated and pre-engineered modules for site assembly by stacking or linking; facility services components. Alternate Terms/Abbreviations: detention cells: jail cells: penitentiary cells: prison cells. See: 05 59 63 for fabricated detention enclosures. 11 25 00 for hospitality equipment. 11 30 00 for residential equipment. 12 35 25 for hospitality casework. 12 35 30 for residential casework.
13 42 13	Bathroom Unit Modules	
13 42 25	Hospitality Unit Modules	
13 42 33	Apartment Unit Modules	
13 42 43	Dormitory Unit Modules	
13 42 46	Education Unit Modules	
13 42 63	Detention Cell Modules	
13 42 63.13	Precast-Concrete Detention Cell Modules	
13 42 63.16	Steel Detention Cell Modules	
13 42 73	Integrated Interior Modules	
13 42 73.11	Integrated Floor Assemblies	
13 42 73.13	Integrated Wall Assemblies	
13 42 73.15	Integrated Ceiling Assemblies	
13 44 00	Modular Mezzanines	Includes: fabricated and pre-engineered mezzanine modules complete with structural support, flooring, railings, stairs, and other accessories. See: 11 51 23 for library stack systems. 41 53 26 for industrial mezzanine storage systems.
13 47 00	Facility Protection	
13 47 13	Cathodic Protection	Includes: equipment, controls, and installation for cathodic protection of structures and underground metal construction and piping. Active and passive protection systems. Galvanic protection. See: 40 46 42 for cathodic process corrosion protection.
13 47 13.13	Cathodic Protection for Underground and Submerged Piping	Includes: active and passive protection systems.
13 47 13.16	Cathodic Protection for Underground Storage Tanks	Includes: active and passive protection systems.
13 47 13.19	Cathodic Protection for Concrete Reinforcing	Includes: active and passive protection systems.

NUMBER	TITLE	EXPLANATION
13 48 00	Sound, Vibration, and Seismic Control	Includes: fabricated, pre-engineered components for the reduction of sound and vibration within a structure. See: 02 22 13 for movement and vibration assessment. 02 22 16 for acoustic assessment. 08 34 73 for sound control door assemblies. 09 51 00 for acoustical ceilings. 09 62 48 for acoustic flooring. 09 80 00 for acoustic treatment. 13 21 48 for sound-conditioned rooms. 13 52 00 for seismic instrumentation. 21 05 48 for vibration and seismic controls for fire-suppression piping and equipment. 22 05 48 for vibration and seismic control for plumbing piping and equipment. 23 05 48 for vibration and seismic control for HVAC. 23 32 48 for acoustical air plenums. 25 05 48 for vibration and seismic controls for integrated automation. 26 05 48 for vibration and seismic controls for electrical systems. 27 05 48 for vibration and seismic controls for communication systems. 28 05 48 for vibration and seismic controls for electronic safety and security. 44 20 00 for noise pollution control.
13 48 13	Manufactured Sound and Vibration Control Components	
13 48 23	Fabricated Sound and Vibration Control Assemblies	
13 48 53	Manufactured Seismic Control Components	
13 48 63	Fabricated Seismic Control Assemblies	

NUMBER	TITLE	EXPLANATION
13 49 00	Radiation Protection	Includes: fabricated and pre-engineered radiation shielding components and structures for protection against radio frequency, x-ray, nuclear radiation, electromagnetic fields, and high energy magnetic pulse protection. Alternate Terms/Abbreviations: EMI: electromagnetic interference. RFI: radio frequency interference. See: 08 34 46 for RFI shielding doors. 08 34 49 for radiation shielding doors and frames. 08 56 46 for RFI shielding windows. 08 56 49 for radiation shielding windows. 08 88 49 for radiation-resistant glazing. 09 22 36 for lath. 26 35 23 for EMI filters. 26 35 46 for RFI filters. 28 41 00 for radiation detection and alarm. 40 91 16 for electromagnetic process measurement devices.
13 49 13	Integrated X-Ray Shielding Assemblies	
13 49 16	Modular X-Ray Shielding Rooms	
13 49 23	Integrated RFI/EMI Shielding Assemblies	
13 49 26	Modular RFI/EMI Shielding Rooms	Alternate Terms/Abbreviations: radiation shielding: shielding.
13 50 00	Special Instrumentation	Includes: instrumentation for measuring and recording phenomena such as stresses in structures, solar and wind energy, and effects of earthquakes. See: 02 31 13 for seismic investigations. 13 48 00 for sound, vibration, and seismic control. 23 56 00 for solar energy heating equipment. 26 31 00 for photovoltaic collectors. 26 32 23 for wind energy equipment. 34 43 23 for airfield weather observation equipment. 40 91 00 for primary process measurement devices. 40 91 19 for stress/strain process measurement devices. 42 12 23 for solar process heaters. 42 13 26 for solar industrial radiation heat exchangers. 48 14 00 for solar energy electrical power generation equipment. 48 15 00 for wind energy electrical power generation equipment. 48 19 19 for electrical power generation solar tracking equipment.
13 51 00	Stress Instrumentation	

DIVISION **13**

NUMBER	TITLE	EXPLANATION
13 52 00	Seismic Instrumentation	
13 53 00	Meteorological Instrumentation	
13 53 13	Solar Instrumentation	
13 53 23	Wind Instrumentation	

NUMBER	TITLE	EXPLANATION
DIVISION 14 – CONVEYING EQUIPMENT		
14 00 00	Conveying Equipment	May be used as Division level Section title. See: 25 32 00 for integrated automation instrumentation and terminal devices for conveying equipment. 25 52 00 for integrated automation control of conveying equipment. 25 92 00 for integrated automation control sequences for conveying equipment.

14 01 00	Operation and Maintenance of Conveying Equipment	Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of conveying equipment. Level 4 Numbering Recommendation: Level 4 titles for conveying equipment are provided as examples for user's creation of more detailed titles for other subjects as required. Following Level 4 numbering is recommended: .51-.59 for maintenance. .61-.69 for repair. .71-.79 for rehabilitation. .81-.89 for replacement. .91-.99 for restoration.
14 01 10	Operation and Maintenance of Dumbwaiters	
14 01 10.71	Dumbwaiter Rehabilitation	
14 01 20	Operation and Maintenance of Elevators	
14 01 20.71	Elevator Rehabilitation	
14 01 30	Operation and Maintenance of Escalators and Moving Walks	
14 01 30.71	Escalators and Moving Walks Rehabilitation	
14 01 40	Operation and Maintenance of Lifts	
14 01 40.71	Lift Rehabilitation	
14 01 70	Operation and Maintenance of Turntables	
14 01 80	Operation and Maintenance of Scaffolding	
14 01 90	Operation and Maintenance of Other Conveying Equipment	

14 05 00	Common Work Results for Conveying Equipment	Includes: subjects common to multiple titles in Division 14. See: 01 80 00 for performance requirements of subjects common to multiple titles.
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DIVISION 14

NUMBER	TITLE	EXPLANATION
14 05 05	Selective Demolition for Conveying Equipment	See: 02 41 19 for selective demolition for whole structure demolition.
14 06 00	Schedules for Conveying Equipment	Includes: schedules of items common to multiple titles in Division 14. Notes: a schedule may be included on drawings, in the project manual, or a project book.
14 06 10	Schedules for Dumbwaiters	
14 06 20	Schedules for Elevators	
14 06 20.13	Elevator Equipment Schedule	
14 06 30	Schedules for Escalators and Moving Walks	
14 06 40	Schedules for Lifts	
14 06 40.13	Lift Schedule	
14 06 70	Schedules for Turntables	
14 06 80	Schedules for Scaffolding	
14 06 90	Schedules for Other Conveying Equipment	
14 08 00	Commissioning of Conveying Equipment	Includes: commissioning of items common to multiple titles in Division 14. See: 01 91 00 for commissioning of subjects common to multiple Divisions.
14 08 10	Commissioning of Dumbwaiters	
14 08 20	Commissioning of Elevators	
14 08 30	Commissioning of Escalators and Moving Walks	
14 08 40	Commissioning of Lifts	
14 08 70	Commissioning of Turntables	
14 08 80	Commissioning of Scaffolding	
14 10 00	Dumbwaiters	Includes: packaged or field-assembled, motorized or hand-operated units, including cabs, doors, controls, safety equipment, hoistway equipment, and dumbwaiter machinery. Applications such as book lifts and cart lifts. See: 11 43 13 for food delivery carts. 11 51 00 library equipment.
14 11 00	Manual Dumbwaiters	
14 12 00	Electric Dumbwaiters	

NUMBER	TITLE	EXPLANATION
14 14 00	Hydraulic Dumbwaiters	
14 20 00	Elevators	Includes: passenger, freight, residential, and service elevators of all types, including cars, entrances, controls, safety equipment, hoistway equipment, and elevator machinery. See: 01 54 13 for temporary elevators. 14 42 00 for wheelchair lifts. 27 32 23 for elevator telephones.
14 21 00	Electric Traction Elevators	Includes: elevators that operate by electric traction. Usually Includes: elevator cabs. May Include: elevator entrances and elevator equipment and controls. See: 14 26 00 for limited-use/limited-application elevators. See Also: 14 27 00 for custom elevator cabs. 14 28 00 for entrances and elevator equipment and controls.
14 21 13	Electric Traction Freight Elevators	
14 21 23	Electric Traction Passenger Elevators	
14 21 23.13	Machine Room Electric Traction Passenger Elevators	
14 21 23.16	Machine Room-Less Electric Traction Passenger Elevators	Alternate Terms/Abbreviations: MRL.
14 21 33	Electric Traction Residential Elevators	
14 21 43	Electric Traction Service Elevators	
14 24 00	Hydraulic Elevators	Includes: elevators that operate by hydraulic action. Usually Includes: elevator cabs. May Include: elevator entrances and elevator equipment and controls. See: 14 26 00 for limited-use/limited-application elevators. See Also: 14 27 00 for custom elevator cabs. 14 28 00 for entrances and elevator equipment and controls.
14 24 13	Hydraulic Freight Elevators	
14 24 23	Hydraulic Passenger Elevators	

NUMBER	TITLE	EXPLANATION
14 24 33	Hydraulic Residential Elevators	
14 24 43	Hydraulic Service Elevators	
14 25 00	Rack and Pinion Elevators	
14 26 00	Limited-Use/Limited-Application Elevators	Includes: packaged elevators for limited uses or applications. See: 14 21 00 for electric traction elevators. 14 24 00 for hydraulic elevators.
14 27 00	Custom Elevator Cabs and Doors	Includes: elevator cars and elevator car finishes for electric traction and hydraulic elevators. See Also: 14 21 00 for electric traction elevators. 14 24 00 for hydraulic elevators. Products: smoke containment elevator hoistway doors.
14 27 13	Custom Elevator Cab Finishes	
14 27 16	Custom Elevator Doors	
14 28 00	Elevator Equipment and Controls	Includes: elevator entrances, controls, safety equipment, hoistway equipment, and elevator machinery for electric traction and hydraulic elevators. See Also: 14 21 00 for electric traction elevators. 14 24 00 for hydraulic elevators.
14 28 16	Elevator Controls	
14 28 19	Elevator Equipment	
14 28 19.13	Elevator Safety Equipment	
14 28 19.16	Elevator Hoistway Equipment	
14 30 00	Escalators and Moving Walks	Includes: passenger conveying systems composed of moving belts or treads installed in horizontal or inclined position, including associated components, hardware, controls, safety equipment, and related items. See: 34 77 13 for passenger boarding bridges.
14 31 00	Escalators	
14 32 00	Moving Walks	

NUMBER	TITLE	EXPLANATION
14 33 00	Moving Ramps	
14 33 13	Motorized Ramps	
14 33 16	Powered Ramps	
14 40 00	Lifts	
		Includes: vertical or inclined lifts and related equipment for a variety of purposes.
		Alternate Terms/Abbreviations: people lifts: personnel hoists. Stage lifts: rising stages. Vehicle lifts: garage lifts: service station lifts.
		See: 11 11 00 for vehicle service equipment. 11 13 19 for stationary loading dock equipment. 11 61 00 for theater and stage equipment. 11 78 19 for mortuary lifts. 14 10 00 for dumbwaiters. 14 71 11 for vehicle turntables. 14 74 61 for stage turntables. 14 83 00 for elevating platforms. 34 13 00 for funiculars. 34 14 00 for cable transportation.
14 41 00	People Lifts	
14 41 13	Counterbalanced People Lifts	
14 41 16	Endless-Belt People Lifts	
14 41 19	Stairway Chairlifts	
		Includes: stairway chairlifts, support structure, and accessories.
14 42 00	Wheelchair Lifts	
14 42 13	Inclined Wheelchair Lifts	
14 42 16	Vertical Wheelchair Lifts	
14 43 00	Platform Lifts	
14 43 13	Orchestra Lifts	
14 43 16	Stage Lifts	
14 44 00	Sidewalk Lifts	
14 45 00	Vehicle Lifts	
14 45 13	Vehicle Service Lifts	
14 45 23	Vehicle Parking Lifts	
14 46 00	Material Lifts	
		Includes: vertical reciprocating lifts.
		See Also: 41 23 23 for lifts.

NUMBER	TITLE	EXPLANATION
14 70 00	Turntables	Includes: structural turntables for various applications. See: 11 11 00 for vehicle service equipment. 11 25 00 for hospitality equipment. 11 59 00 for exhibit equipment. 11 61 00 for theater and stage equipment. 14 43 16 for stage lifts. 14 45 00 for vehicle lifts. 41 21 36 for piece material turntables.
14 71 00	Industrial Turntables	
14 71 11	Vehicle Turntables	
14 72 00	Hospitality Turntables	
14 72 25	Restaurant Turntables	
14 73 00	Exhibit Turntables	
14 73 59	Display Turntables	
14 74 00	Entertainment Turntables	
14 74 61	Stage Turntables	
14 80 00	Scaffolding	
14 81 00	Suspended Scaffolding	Includes: scaffolding, when part of the completed project. See: 01 54 23 for temporary scaffolding and platforms.
14 81 13	Beam Scaffolding	
14 81 16	Carriage Scaffolding	
14 81 19	Hook Scaffolding	
14 82 00	Rope Climbers	See: 34 13 00 for funiculars. 34 14 00 for cable transportation.
14 82 13	Manual Rope Climbers	
14 82 16	Powered Rope Climbers	
14 83 00	Elevating Platforms	Includes: fixed elevating platforms to provide a movable elevated working platform for people and materials. See: 14 43 00 for platform lifts. 41 66 16 for industrial mobile lifts and cherry pickers.

NUMBER	TITLE	EXPLANATION
14 83 13	Telescoping Platform Lifts	
14 83 13.13	Electric and Battery Telescoping Platform Lifts	
14 83 13.16	Pneumatic Telescoping Platform Lifts	
14 83 16	Scissor Lift Platforms	
14 83 19	Multi-Axis Platform Lifts	
14 84 00	Powered Scaffolding	
		See: 41 22 23 for hoists.
14 84 13	Window Washing Scaffolding	
14 84 23	Window Washing Hoists	
14 90 00	Other Conveying Equipment	
14 91 00	Facility Chutes	
		Includes: chutes which support the operation of the building or structure.
		Does Not Include: chutes which support process engineering operations.
		Alternate Terms/Abbreviations: trash chutes: refuse chutes: rubbish chutes: waste chutes.
		See: 05 51 23 for metal fire escapes.
		10 55 91 for mail chutes.
		11 82 00 for solid waste handling equipment.
		40 30 00 for solid and mixed materials piping and chutes.
		40 32 00 bulk materials piping and chutes.
14 91 13	Coal Chutes	
14 91 23	Escape Chutes	
14 91 33	Laundry and Linen Chutes	
14 91 82	Trash Chutes	
14 92 00	Pneumatic Tube Systems	
		See: 11 21 00 for mercantile and service equipment.
		11 28 00 for office equipment.
		11 70 00 for healthcare equipment.
		41 12 19 for pneumatic conveyors.
14 93 00	Slide Pole Systems	
14 93 13	Firehouse Slide Poles	

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SPECIFICATIONS GROUP

FACILITY SERVICES SUBGROUP

Division 20 – Reserved for Future Expansion

DIVISION 21 – FIRE SUPPRESSION

DIVISION 22 – PLUMBING

DIVISION 23 – HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

Division 24 – Reserved for Future Expansion

DIVISION 25 – INTEGRATED AUTOMATION

DIVISION 26 – ELECTRICAL

DIVISION 27 – COMMUNICATIONS

DIVISION 28 – ELECTRONIC SAFETY AND SECURITY

Division 29 – Reserved for Future Expansion

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NUMBER	TITLE	EXPLANATION
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DIVISION 21 – FIRE SUPPRESSION

21 00 00 Fire Suppression

May be used as Division level Section title.

See: 02 41 19 for selective demolition of existing fire-suppression systems.
 03 30 00 for cast-in-place concrete equipment bases.
 07 84 00 for firestopping.
 07 92 00 for joint sealants.
 08 31 00 for access doors and panels.
 09 91 00 for field painting.
 10 44 00 for fire protection specialties.
 31 23 33 for trenching and backfilling.

21 01 00 Operation and Maintenance of Fire Suppression

Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of fire-suppression systems.

Level 4 Numbering Recommendation: following numbering is recommended for the creation of Level 4 titles:

.51-.59 for maintenance.
 .61-.69 for repair.
 .71-.79 for rehabilitation.
 .81-.89 for replacement.
 .91-.99 for restoration.

21 01 10	Operation and Maintenance of Water-Based Fire-Suppression Systems
21 01 20	Operation and Maintenance of Fire-Extinguishing Systems
21 01 30	Operation and Maintenance of Fire-Suppression Equipment

NUMBER	TITLE	EXPLANATION
21 05 00	Common Work Results for Fire Suppression	Includes: subjects common to multiple titles in Division 21. See: 01 80 00 for performance requirements of subjects common to multiple titles. 05 45 13 for mechanical metal supports. 13 48 00 for sound, vibration, and seismic control. 22 05 29 for hangers and supports for plumbing piping and equipment. 22 05 33 for heat tracing for plumbing piping. 22 05 48 for vibration and seismic controls for plumbing piping and equipment. 22 05 53 for identification for plumbing piping and equipment. 23 05 29 for hangers and supports for HVAC piping and equipment. 23 05 33 for heat tracing for HVAC piping. 23 05 48 for vibration and seismic controls for HVAC piping and equipment. 23 05 53 for identification for HVAC piping and equipment. 40 41 00 for process piping and equipment heat tracing.
21 05 05	Selective Demolition for Fire Suppression	See: 02 41 19 for selective demolition for whole structure demolition.
21 05 13	Common Motor Requirements for Fire-Suppression Equipment	
21 05 16	Expansion Fittings and Loops for Fire-Suppression Piping	
21 05 17	Sleeves and Sleeve Seals for Fire-Suppression Piping	Includes: penetration methods into and through fire-rated and unrated walls, ceilings, and floors. Sealing of conduit penetrations against water is included. Excludes firestopping requirements.
21 05 19	Meters and Gauges for Fire-Suppression Systems	
21 05 23	General-Duty Valves for Water-Based Fire-Suppression Piping	
21 05 29	Hangers and Supports for Fire-Suppression Piping and Equipment	
21 05 33	Heat Tracing for Fire-Suppression Piping	
21 05 48	Vibration and Seismic Controls for Fire-Suppression Piping and Equipment	
21 05 48.13	Vibration Controls for Fire-Suppression Piping and Equipment	Includes: vibration-isolation supports and hangers; vibration-isolation equipment bases.
21 05 53	Identification for Fire-Suppression Piping and Equipment	
21 06 00	Schedules for Fire Suppression	Includes: schedules of items common to multiple titles in Division 21. Notes: a schedule may be included on drawings, in the project manual, or a project book.

NUMBER	TITLE	EXPLANATION
21 06 10	Schedules for Water-Based Fire-Suppression Systems	
21 06 20	Schedules for Fire-Extinguishing Systems	
21 06 30	Schedules for Fire-Suppression Equipment	
21 07 00	Fire Suppression Systems Insulation	Includes: thermal insulation of fire-suppression piping and equipment. See: 07 21 00 for thermal insulation. 22 07 00 for plumbing insulation. 23 07 00 for HVAC insulation. 40 42 00 for process piping and equipment insulation.
21 07 16	Fire-Suppression Equipment Insulation	
21 07 19	Fire-Suppression Piping Insulation	
21 08 00	Commissioning of Fire Suppression	Includes: commissioning of items common to multiple titles in Division 21. See: 01 91 00 for commissioning of subjects common to multiple Divisions.
21 09 00	Instrumentation and Control for Fire-Suppression Systems	Includes: instrumentation and control associated with fire-suppression systems. See: 13 50 00 for special instrumentation. 22 09 00 for instrumentation and control for plumbing. 23 09 00 for instrumentation and control for HVAC. 25 33 00 for integrated automation instrumentation and terminal devices for fire-suppression systems. 25 53 00 for integrated automation control of fire-suppression systems. 25 93 00 for integrated automation control sequences for fire-suppression systems. 28 46 00 for fire detection and alarm. 33 09 00 for instrumentation and control for utilities.
21 10 00	Water-Based Fire-Suppression Systems	Includes: systems that use water for fire extinguishing and suppression. See: 21 30 00 for fire pumps. 28 46 00 for fire detection and alarm.

NUMBER	TITLE	EXPLANATION
21 11 00	Facility Fire-Suppression Water-Service Piping	Includes: piping systems for fire suppression for a structure. Fire hydrants closely associated with a structure. May Include: fire-suppression water service from the structure to the utility water service line. See: 21 30 00 for fire pumps. 33 14 16 for site water utility distribution piping. 33 14 19 for valves and hydrants for water utility service.
21 11 16	Facility Fire Hydrants	
21 11 19	Fire-Department Connections	
21 12 00	Fire-Suppression Standpipes	Includes: Siamese connections, piping, and other components, standpipe, and fire hose connections. See: 10 44 00 for fire protection specialties. 10 44 13 for fire protection cabinets. 10 44 16 for fire extinguishers.
21 12 13	Fire-Suppression Hoses and Nozzles	
21 12 16	Fire-Suppression Hose Reels	
21 12 19	Fire-Suppression Hose Racks	
21 12 23	Fire-Suppression Hose Valves	
21 13 00	Fire-Suppression Sprinkler Systems	Includes: sprinkler systems using water for fire extinguishing and suppression. See: 21 21 00 for carbon-dioxide fire-extinguishing systems. 21 22 00 for clean-agent fire-extinguishing systems. 21 23 00 for wet-chemical fire-extinguishing systems. 21 24 00 for dry-chemical fire-extinguishing systems.
21 13 13	Wet-Pipe Sprinkler Systems	Includes: piping, accessories, and other components charged with water to suppress fire.
21 13 16	Dry-Pipe Sprinkler Systems	Includes: piping, accessories, and other components prepared to suppress fire upon being charged with water by the opening of a sprinkler head or by means of an electrically operated valve.
21 13 19	Preaction Sprinkler Systems	
21 13 23	Combined Dry-Pipe and Preaction Sprinkler Systems	

NUMBER	TITLE	EXPLANATION
21 13 26	Deluge Fire-Suppression Sprinkler Systems	Includes: piping, accessories, and other components prepared to suppress fire through open sprinkler heads upon being charged with water by means of an electrically operated valve.
21 13 29	Water Spray Fixed Systems	
21 13 36	Antifreeze Sprinkler Systems	Includes: piping, accessories, and other components prepared to suppress fire with use of antifreeze to prevent freezing of water in system.
21 13 39	Foam-Water Systems	Includes: piping, accessories, and other components prepared to suppress fire using aqueous film-forming foam.
21 16 00	Fire-Suppression Pressure Maintenance Pumps	Alternate Terms/Abbreviations: pressure maintenance pumps: jockey pumps: make-up pumps.
21 20 00	Fire-Extinguishing Systems	Includes: systems that use other than water for fire extinguishing and suppression. See: 23 38 00 for ventilation hoods with fire-suppression systems. 28 46 00 for fire detection and alarm.
21 21 00	Carbon-Dioxide Fire-Extinguishing Systems	Includes: systems that deliver carbon dioxide for fire extinguishing and suppression. See: 21 13 00 for water fire-suppression sprinkler systems.
21 21 13	Carbon-Dioxide Fire-Extinguishing Piping	
21 21 13.13	High-Pressure, Carbon-Dioxide Fire-Extinguishing Systems	Includes: pipes, fittings, and specialties.
21 21 13.16	Low-Pressure, Carbon-Dioxide Fire-Extinguishing Systems	Includes: pipes, fittings, and specialties.
21 21 16	Carbon-Dioxide Fire-Extinguishing Equipment	
21 22 00	Clean-Agent Fire-Extinguishing Systems	Includes: systems that deliver clean agents for fire extinguishing and suppression. See: 21 13 00 for water fire-suppression sprinkler systems.

NUMBER	TITLE	EXPLANATION
21 22 13	Clean-Agent Fire-Extinguishing Piping	
21 22 16	Clean-Agent Fire-Extinguishing Equipment	
21 23 00	Wet-Chemical Fire-Extinguishing Systems	
		Includes: systems that deliver wet chemicals for fire extinguishing and suppression.
		See: 21 13 00 for water fire-suppression sprinkler systems.
21 23 13	Wet-Chemical Fire-Extinguishing Piping	
21 23 16	Wet-Chemical Fire-Extinguishing Equipment	
21 24 00	Dry-Chemical Fire-Extinguishing Systems	
		Includes: systems that deliver dry chemicals for fire extinguishing and suppression.
		See: 21 13 00 for water fire-suppression sprinkler systems.
21 24 13	Dry-Chemical Fire-Extinguishing Piping	
21 24 16	Dry-Chemical Fire-Extinguishing Equipment	
21 30 00	Fire Pumps	
		Includes: pumps associated with water-based fire-suppression systems.
		See: 21 10 00 for water-based fire-suppression systems.
		21 11 00 for facility fire-suppression water-service piping.
		21 12 00 for fire-suppression standpipes.
		22 11 23 for domestic water pumps.
		22 51 16 for swimming pool pumps.
		22 52 16 for fountain pumps.
		23 21 23 for hydronic pumps.
21 31 00	Centrifugal Fire Pumps	
21 31 13	Electric-Drive, Centrifugal Fire Pumps	
21 31 16	Diesel-Drive, Centrifugal Fire Pumps	
21 32 00	Vertical-Turbine Fire Pumps	
21 32 13	Electric-Drive, Vertical-Turbine Fire Pumps	
21 32 16	Diesel-Drive, Vertical-Turbine Fire Pumps	
21 33 00	Positive-Displacement Fire Pumps	
21 33 13	Electric-Drive, Positive-Displacement Fire Pumps	
21 33 16	Diesel-Drive, Positive-Displacement Fire Pumps	
21 34 00	Fire Pump Accessories	
21 34 13	Pressure Maintenance Pumps	

NUMBER	TITLE	EXPLANATION
21 40 00	Fire-Suppression Water Storage	
21 41 00	Storage Tanks for Fire-Suppression Water	Includes: tanks closely associated with a structure for the storage of water for fire extinguishing and suppression for a structure. See: 09 97 13 for storage tank coatings. 22 12 00 for facility potable-water storage tanks. 23 13 00 for facility fuel-storage tanks. 31 00 00 for earthwork. 33 16 00 for water utility storage tanks. 33 56 00 for fuel storage tanks. 35 00 00 for waterway and marine construction. 43 41 13 for gas and liquid pressure vessels.
21 41 13	Pressurized Storage Tanks for Fire-Suppression Water	
21 41 16	Elevated Storage Tanks for Fire-Suppression Water	
21 41 19	Roof-Mounted Storage Tanks for Fire-Suppression Water	
21 41 23	Ground Suction Storage Tanks for Fire-Suppression Water	
21 41 26	Underground Storage Tanks for Fire-Suppression Water	
21 41 29	Storage Tanks for Fire-Suppression Water Additives	

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NUMBER	TITLE	EXPLANATION
DIVISION 22 – PLUMBING		
22 00 00	Plumbing	May be used as Division level Section title. See: 02 41 19 for selective demolition of existing plumbing systems. 03 30 00 for cast-in-place concrete equipment bases. 07 84 00 for firestopping. 07 92 00 for joint sealants. 08 31 00 for access doors and panels. 09 91 00 for field painting. 31 23 33 for trenching and backfilling.

22 01 00 Operation and Maintenance of Plumbing

Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of plumbing.

Level 4 Numbering Recommendation: following numbering is recommended for the creation of Level 4 titles:

.51-.59 for maintenance.

.61-.69 for repair.

.71-.79 for rehabilitation.

.81-.89 for replacement.

.91-.99 for restoration.

22 01 10	Operation and Maintenance of Plumbing Piping and Pumps
22 01 10.16	Video Piping Inspections
22 01 10.51	Plumbing Piping Cleaning
22 01 10.61	Plumbing Piping Repairs
22 01 10.62	Plumbing Piping Relining
22 01 30	Operation and Maintenance of Plumbing Equipment
22 01 40	Operation and Maintenance of Plumbing Fixtures
22 01 50	Operation and Maintenance of Pool and Fountain Plumbing Systems
22 01 60	Operation and Maintenance of Laboratory and Healthcare Systems

NUMBER	TITLE	EXPLANATION
22 05 00	Common Work Results for Plumbing	Includes: subjects common to multiple titles in Division 22. See: 01 80 00 for performance requirements of subjects common to multiple titles. 05 45 13 for mechanical metal supports. 13 48 00 for sound, vibration, and seismic control. 21 05 29 for hangers and supports for fire-suppression piping and equipment. 21 05 33 for heat tracing for fire-suppression piping. 21 05 48 for vibration and seismic controls for fire-suppression piping and equipment. 21 05 53 for identification for fire-suppression piping and equipment. 23 05 29 for hangers and supports for HVAC piping and equipment. 23 05 33 for heat tracing for HVAC piping. 23 05 48 for vibration and seismic controls for HVAC. 23 05 53 for identification for HVAC piping and equipment. 40 41 00 for process piping and equipment heat tracing.
22 05 05	Selective Demolition for Plumbing	See: 02 41 19 for selective demolition for whole structure demolition.
22 05 13	Common Motor Requirements for Plumbing Equipment	
22 05 16	Expansion Fittings and Loops for Plumbing Piping	
22 05 17	Sleeves and Sleeve Seals for Plumbing Piping	Includes: penetration methods into and through fire-rated and unrated walls, ceilings, and floors. Sealing of conduit penetrations against water is included. Excludes firestopping requirements.
22 05 19	Meters and Gauges for Plumbing Piping	
22 05 23	General-Duty Valves for Plumbing Piping	
22 05 29	Hangers and Supports for Plumbing Piping and Equipment	
22 05 33	Heat Tracing for Plumbing Piping	
22 05 48	Vibration and Seismic Controls for Plumbing Piping and Equipment	
22 05 48.13	Vibration Controls for Plumbing Piping and Equipment	Includes: vibration-isolation supports and hangers; vibration-isolation equipment bases.
22 05 53	Identification for Plumbing Piping and Equipment	
22 05 73	Facility Drainage Manholes	
22 05 76	Facility Drainage Piping Cleanouts	

NUMBER	TITLE	EXPLANATION
22 06 00	Schedules for Plumbing	Includes: schedules of items common to multiple titles in Division 22. Notes: a schedule may be included on drawings, in the project manual, or a project book.
22 06 10	Schedules for Plumbing Piping and Pumps	
22 06 10.13	Plumbing Pump Schedule	
22 06 12	Schedules for Facility Potable Water Storage	
22 06 15	Schedules for General Service Compressed-Air Equipment	
22 06 30	Schedules for Plumbing Equipment	
22 06 30.13	Domestic Water Heater Schedule	
22 06 40	Schedules for Plumbing Fixtures	
22 06 40.13	Plumbing Fixture Schedule	
22 06 50	Schedules for Pool and Fountain Plumbing Systems	
22 06 60	Schedules for Laboratory and Healthcare Systems	
22 07 00	Plumbing Insulation	Includes: thermal insulation of plumbing piping and equipment. See: 07 21 00 for thermal insulation. 21 07 00 for fire-suppression insulation. 23 07 00 for HVAC insulation. 40 42 00 for process piping and equipment insulation.
22 07 16	Plumbing Equipment Insulation	
22 07 19	Plumbing Piping Insulation	
22 08 00	Commissioning of Plumbing	Includes: commissioning of items common to multiple titles in Division 22. See: 01 91 00 for commissioning of subjects common to multiple Divisions.

NUMBER	TITLE	EXPLANATION
22 09 00	Instrumentation and Control for Plumbing	Includes: instrumentation and control associated with plumbing. See: 13 50 00 for special instrumentation. 21 09 00 for instrumentation and control for fire-suppression systems. 22 51 23 for swimming pool equipment controls. 22 52 23 for fountain equipment controls. 23 09 00 for instrumentation and control for HVAC. 25 34 00 for integrated automation instrumentation and terminal devices for plumbing. 25 54 00 for integrated automation control of plumbing. 25 94 00 for integrated automation control sequences for plumbing. 33 09 00 for instrumentation and control for utilities.
22 09 63	Medical Gas Alarms	See Also: 28 42 00 for gas detection and alarm.
22 10 00	Plumbing Piping	
22 11 00	Facility Water Distribution	Includes: piping, equipment, valves, and specialties associated with domestic water distribution within and under structures. Circulating pumps. May Include: water service from the structure to the utility water service line. See: 21 30 00 for fire pumps. 22 13 29 for sanitary sewerage pumps. 22 14 29 for sump pumps. 22 51 16 for swimming pool pumps. 22 52 16 for fountain pumps. 22 67 00 for healthcare and laboratory processed water systems. 23 12 00 for facility fuel pumps. 23 21 00 for hydronic piping and pumps. 23 22 00 for steam and condensate piping and pumps. 32 80 00 for irrigation. 33 11 00 for groundwater sources. 33 14 00 for water utility transmission and distribution. 40 23 00 for water process piping.
22 11 13	Facility Water Distribution Piping	
22 11 16	Domestic Water Piping	
22 11 17	Gray-Water Piping	Includes: pipe materials, fittings, and specialties.

NUMBER	TITLE	EXPLANATION
22 11 19	Domestic Water Piping Specialties	
22 11 23	Domestic Water Pumps	
22 11 23.13	Domestic-Water Packaged Booster Pumps	
22 11 23.23	Domestic-Water In-Line Pumps	Includes: close-coupled, horizontally or vertically mounted, in-line centrifugal domestic water pumps; separately coupled, horizontally mounted, in-line centrifugal domestic water pumps.
22 11 23.26	Close-Coupled, Horizontally Mounted, In-Line Centrifugal Domestic-Water Pumps	
22 11 23.29	Close-Coupled, Vertically Mounted, In-Line Centrifugal Domestic-Water Pumps	
22 11 23.33	Separately Coupled, In-Line Centrifugal Domestic-Water Pumps	
22 11 23.36	Separately Coupled, Horizontally Mounted, In-Line Centrifugal Domestic-Water Pumps	
22 11 23.43	Domestic-Water, Base-Mounted Pumps	Includes: close-coupled, horizontally or vertically mounted, in-line centrifugal domestic water pumps; separately coupled, horizontally mounted, in-line centrifugal domestic water pumps.
22 11 63	Gray-Water Pumps	Includes: close-coupled, horizontally or vertically mounted, in-line centrifugal domestic water pumps; separately coupled, horizontally mounted, in-line centrifugal domestic water pumps.
22 12 00	Facility Potable-Water Storage Tanks	Includes: tanks for storage of potable water serving a facility and located within, on, under, or closely associated with a structure. See: 09 97 13 for steel storage tank coatings. 21 41 00 for facility storage tanks for fire-suppression water. 23 13 00 for facility fuel-storage tanks. 31 00 00 for earthwork. 33 16 00 for site water utility storage tanks. 33 56 00 for site fuel storage tanks. 35 00 00 for waterway and marine construction. 41 52 19 for material storage tanks. 43 41 13 for gas and liquid pressure vessels.
22 12 13	Facility Roof-Mounted, Potable-Water Storage Tanks	
22 12 16	Facility Elevated, Potable-Water Storage Tanks	
22 12 19	Facility Ground-Mounted, Potable-Water Storage Tanks	
22 12 21	Facility Underground Potable-Water Storage Tanks	
22 12 23	Facility Indoor Potable-Water Storage Tanks	
22 12 23.13	Facility Steel, Indoor Potable-Water Storage Pressure Tanks	
22 12 23.16	Facility Steel, Indoor Potable-Water Storage Non-Pressure Tanks	
22 12 23.23	Facility Plastic, Indoor Potable-Water Storage Pressure Tanks	
22 12 23.26	Facility Plastic, Indoor Potable-Water Storage Non-Pressure Tanks	

NUMBER	TITLE	EXPLANATION
22 13 00	Facility Sanitary Sewerage	Includes: sanitary waste and vent piping systems, with piping, floor drains, valves, piping specialties, interceptors, separators, and other components within and under structures. May Include: sanitary piping from the structure to the utility sanitary sewer. Alternate Terms/Abbreviations: FOG: fats, oils, and grease. See: 21 30 00 for fire pumps. 22 11 23 for domestic water pumps. 22 14 29 for sump pumps. 22 51 16 for swimming pool pumps. 22 52 16 for fountain pumps. 22 66 00 for healthcare and laboratory chemical waste systems. 23 12 00 for facility fuel pumps. 23 21 00 for hydronic piping and pumps. 23 22 00 for steam and condensate piping and pumps. 33 31 00 for site sanitary sewerage piping. 33 32 00 for wastewater utility pumping stations. 33 34 00 for onsite wastewater disposal
22 13 13	Facility Sanitary Sewers	
22 13 16	Sanitary Waste and Vent Piping	
22 13 19	Sanitary Waste Piping Specialties	
22 13 19.13	Sanitary Drains	
22 13 19.23	Fats, Oils, and Grease Disposal Systems	
22 13 19.26	Grease Removal Devices	
22 13 19.33	Backwater Valves	
22 13 19.36	Air-Admittance Valves	
22 13 23	Sanitary Waste Interceptors	
22 13 26	Sanitary Waste Separators	
22 13 29	Sanitary Sewerage Pumps	
22 13 29.13	Wet-Pit-Mounted, Vertical Sewerage Pumps	
22 13 29.16	Submersible Sewerage Pumps	
22 13 29.23	Sewerage Pump Reverse-Flow Assemblies	
22 13 29.33	Sewerage Pump Basins and Pits	
22 13 33	Packaged, Submersible Sewerage Pump Units	
22 13 36	Packaged, Wastewater Pump Units	
22 13 43	Facility Packaged Sewage Pumping Stations	
22 13 43.13	Facility Dry-Well Packaged Sewage Pumping Stations	
22 13 43.16	Facility Wet-Well Packaged Sewage Pumping Stations	
22 13 63	Facility Gray Water Tanks	

NUMBER	TITLE	EXPLANATION
22 14 00	Facility Storm Drainage	Includes: storm drainage piping, drains, and other components within, under, or closely associated with a structure. May Include: storm drainage piping from the structure to the utility storm sewer. Storm drains for areas closely associated with a structure such as courtyards, plazas, and loading dock areas. See: 07 50 00 for roof drains integral with membrane roofing. 07 71 23 for gutters and downspouts. 22 13 29 for sanitary sewerage pumps. 31 00 00 for earthwork. 33 41 00 for subdrainage. 33 42 00 for stormwater conveyance. 33 44 00 for site storm water drains. 33 44 11 for field-erected stormwater pumping stations. 33 44 13 for packaged stormwater pumping stations. 35 00 00 for waterway and marine construction. 43 20 00 for liquid handling equipment.
22 14 13	Facility Storm Drainage Piping	
22 14 16	Rainwater Leaders	
22 14 19	Sump Pump Discharge Piping	
22 14 23	Storm Drainage Piping Specialties	
22 14 26	Facility Storm Drains	
22 14 26.13	Roof Drains	
22 14 26.16	Facility Area Drains	
22 14 26.19	Facility Trench Drains	
22 14 29	Sump Pumps	
22 14 29.13	Wet-Pit-Mounted, Vertical Sump Pumps	
22 14 29.16	Submersible Sump Pumps	
22 14 29.19	Sump-Pump Basins and Pits	
22 14 33	Packaged, Pedestal Drainage Pump Units	
22 14 36	Packaged, Submersible, Drainage Pump Units	
22 14 53	Rainwater Storage Tanks	
22 14 63	Facility Storm-Water Retention Tanks	Includes: tanks, fittings, and accessories.
22 15 00	General Service Compressed-Air Systems	Includes: pipe and fittings compressed air serving general service requirements. Piping, valves, piping specialties, pumps, compressors, and other components. See: 22 61 00 for compressed-air systems for laboratory and healthcare facilities. 23 09 43 for pneumatic control system for HVAC. 40 12 00 for compressed air process piping.
22 15 13	General Service Compressed-Air Piping	

DIVISION 22

NUMBER	TITLE	EXPLANATION
22 15 16	General Service Compressed-Air Valves	
22 15 19	General Service Packaged Air Compressors and Receivers	
22 15 19.13	General Service Packaged Reciprocating Air Compressors	
22 15 19.16	General Service Packaged Liquid-Ring Air Compressors	
22 15 19.19	General Service Packaged Rotary-Screw Air Compressors	
22 15 19.23	General Service Packaged Sliding-Vane Air Compressors	

22 30 00 Plumbing Equipment

22 31 00 Domestic Water Softeners

Includes: equipment for the softening of domestic water.

Alternate Terms/Abbreviations: residential: household.

See: 22 10 00 for plumbing piping and pumps.
23 25 00 for HVAC water treatment.
44 40 00 for water pollution control equipment.

22 31 13	Residential Domestic Water Softeners
22 31 16	Commercial Domestic Water Softeners

22 32 00 Domestic Water Filtration Equipment

Includes: equipment for the filtering of domestic water.

See: 22 10 00 for plumbing piping and pumps.
22 51 19 for swimming pool water treatment equipment.
22 52 19 for fountain water treatment equipment.
23 25 00 for HVAC water treatment.
44 40 00 for water pollution control equipment.

22 32 13	Domestic-Water Bag-Type Filters
22 32 16	Domestic-Water Freestanding Cartridge Filters
22 32 19	Domestic-Water Off-Floor Cartridge Filters
22 32 23	Domestic-Water Carbon Filters
22 32 26	Domestic-Water Sand Filters
22 32 26.13	Domestic-Water Circulating Sand Filters
22 32 26.16	Domestic-Water Multimedia Sand Filters
22 32 26.19	Domestic-Water Greensand Filters

NUMBER	TITLE	EXPLANATION
22 33 00	Electric Domestic Water Heaters	Includes: electric equipment to heat domestic water.
		Alternate Terms/Abbreviations: residential: household.
		See: 22 10 00 for plumbing piping and pumps. 22 34 00 for fuel-fired domestic water heaters. 22 35 00 for domestic water heat exchangers. 23 56 00 for solar energy heating equipment. 26 31 00 for photovoltaic collectors. 42 12 23 for solar process heaters. 42 13 26 for industrial solar radiation heat exchangers. 48 14 00 for solar energy electrical power generation equipment.
22 33 13	Instantaneous Electric Domestic Water Heaters	
22 33 13.13	Flow-Control, Instantaneous Electric Domestic Water Heaters	
22 33 13.16	Thermostat-Control, Instantaneous Electric Domestic Water Heaters	
22 33 30	Residential, Electric Domestic Water Heaters	
22 33 30.13	Residential, Small-Capacity Electric Domestic Water Heaters	
22 33 30.16	Residential, Storage Electric Domestic Water Heaters	
22 33 30.23	Residential, Collector-to-Tank, Solar-Electric Domestic Water Heaters	
22 33 30.26	Residential, Collector-to-Tank, Heat-Exchanger-Coil, Solar-Electric Domestic Water Heaters	
22 33 33	Light-Commercial Electric Domestic Water Heaters	
22 33 36	Commercial Domestic Water Electric Booster Heaters	
22 33 36.13	Commercial Domestic Water Electric Booster Heaters	
22 33 36.16	Commercial Storage Electric Domestic Water Heaters	
22 34 00	Fuel-Fired Domestic Water Heaters	Includes: fuel-fired equipment to heat domestic water.
		Alternate Terms/Abbreviations: residential: household.
		See: 22 10 00 for plumbing piping and pumps. 22 33 00 for electric domestic water heaters. 22 35 00 for domestic water heat exchangers. 23 51 00 for breechings, chimneys, and stacks.
22 34 13	Instantaneous, Tankless, Gas Domestic Water Heaters	
22 34 30	Residential Gas Domestic Water Heaters	
22 34 30.13	Residential, Atmospheric, Gas Domestic Water Heaters	
22 34 30.16	Residential, Direct-Vent, Gas Domestic Water Heaters	
22 34 30.19	Residential, Power-Vent, Gas Domestic Water Heaters	
22 34 36	Commercial Gas Domestic Water Heaters	
22 34 36.13	Commercial, Atmospheric, Gas Domestic Water Heaters	
22 34 36.16	Commercial, Power-Burner, Gas Domestic Water Heaters	
22 34 36.19	Commercial, Power-Vent, Gas Domestic Water Heaters	
22 34 36.23	Commercial, High-Efficiency, Gas Domestic Water Heaters	
22 34 36.26	Commercial, Coil-Type, Finned-Tube, Gas Domestic Water Heaters	
22 34 36.29	Commercial, Grid-Type, Finned-Tube, Gas Domestic Water Heaters	

DIVISION 22

NUMBER	TITLE	EXPLANATION
22 34 46	Oil-Fired Domestic Water Heaters	
22 34 46.13	Large-Capacity, Oil-Fired Domestic Water Heaters	
22 34 56	Dual-Fuel-Fired Domestic Water Heaters	
22 35 00	Domestic Water Heat Exchangers	
		Includes: equipment to heat domestic water by means of heat exchange.
		See: 22 10 00 for plumbing piping and pumps. 22 33 00 for electric domestic water heaters. 22 34 00 for fuel-fired domestic water heaters. 23 57 00 for heat exchangers for HVAC.
22 35 13	Instantaneous Domestic Water Heat Exchangers	
22 35 13.13	Heating-Fluid-in-Coil, Instantaneous Domestic Water Heat Exchangers	
22 35 13.16	Domestic-Water-in-Coil, Instantaneous Domestic Water Heat Exchangers	
22 35 13.19	Heating-Fluid-in-U-Tube-Coil, Instantaneous Domestic Water Heat Exchangers	
22 35 23	Circulating, Domestic Water Heat Exchangers	
22 35 23.13	Circulating, Compact Domestic Water Heat Exchangers	
22 35 23.16	Circulating, Storage Domestic Water Heat Exchangers	
22 35 29	Noncirculating, Domestic Water Heat Exchangers	
22 35 29.13	Noncirculating, Compact Domestic Water Heat Exchangers	
22 35 29.16	Noncirculating, Storage Domestic Water Heat Exchangers	
22 35 36	Domestic Water Brazed-Plate Heat Exchangers	
22 35 39	Domestic Water Frame-and-Plate Heat Exchangers	
22 35 43	Domestic Water Heat Reclaimers	
22 36 00	Domestic Water Preheaters	
22 36 13	Solar Domestic Water Preheaters	
22 36 23	Geothermal Domestic Water Preheaters	
22 40 00	Plumbing Fixtures	
22 41 00	Residential Plumbing Fixtures	
		Includes: shower enclosures integral with bathtubs and shower receptors.
		See: 10 28 19 for shower and tub doors. 12 35 30 for residential casework. 12 36 00 for countertops. 12 36 00 for sinks and lavatories integral with countertops. 13 17 13 for hot tubs. 13 17 33 for whirlpool tubs.
		See Also: 12 36 00 for sinks and lavatories integral with countertops.
22 41 13	Residential Water Closets, Urinals, and Bidets	
22 41 13.13	Residential Water Closets	

NUMBER	TITLE	EXPLANATION
22 41 13.16	Residential Urinals	
22 41 13.19	Residential Bidets	
22 41 16	Residential Lavatories and Sinks	
22 41 16.13	Residential Lavatories	
22 41 16.16	Residential Sinks	
22 41 19	Residential Bathtubs	
22 41 23	Residential Showers	
22 41 26	Residential Disposers	
22 41 36	Residential Laundry Trays	
22 41 39	Residential Faucets, Supplies, and Trim	

22 42 00 Commercial Plumbing Fixtures

See: 10 21 13 for toilet compartments.
 10 21 16 for shower and dressing compartments.
 11 40 00 for foodservice equipment.
 11 53 43 for laboratory service fittings and accessories.
 12 30 00 for casework.
 12 36 00 for countertops.
 12 36 00 for sinks and lavatories integral with countertops.
 13 17 13 for hot tubs.
 13 17 33 for whirlpool tubs.

See Also: 11 48 00 for commercial kitchen food waste disposers.

22 42 13	Commercial Water Closets, Urinals, and Bidets
22 42 13.13	Commercial Water Closets
22 42 13.16	Commercial Urinals
22 42 16	Commercial Lavatories and Sinks
22 42 16.13	Commercial Lavatories
22 42 16.16	Commercial Sinks
22 42 19	Commercial Bathtubs
22 42 23	Commercial Showers
22 42 26	Commercial Disposers
22 42 29	Shampoo Bowls
22 42 33	Wash Fountains
22 42 36	Commercial Laundry Trays
22 42 39	Commercial Faucets
22 42 43	Flushometers

22 43 00 Healthcare Plumbing Fixtures

Alternate Terms/Abbreviations: healthcare: medical: hospital.

See: 11 53 43 for laboratory service fittings and accessories.
 11 70 00 for healthcare equipment.
 12 35 70 for healthcare casework.
 12 36 00 for countertops.
 12 36 00 for sinks and lavatories integral with countertops.
 13 17 23 for therapeutic pools.
 13 17 33 for whirlpool tubs.

DIVISION 22

NUMBER	TITLE	EXPLANATION
22 43 13	Healthcare Water Closets	
22 43 16	Healthcare Sinks	
22 43 19	Healthcare Bathtubs	
22 43 23	Healthcare Showers	
22 43 39	Healthcare Faucets	
22 43 43	Healthcare Plumbing Fixture Flushometers	

22 45 00 **Emergency Plumbing Fixtures**

Includes: eyewashes.

Alternate Terms/Abbreviations: emergency showers: drench showers: safety showers.

See: 11 53 00 for laboratory equipment.
11 53 43 for laboratory service fittings and accessories.
11 57 00 for vocational shop equipment.

22 45 13	Emergency Showers
22 45 16	Eyewash Equipment
22 45 19	Self-Contained Eyewash Equipment
22 45 23	Personal Eyewash Equipment
22 45 26	Eye/Face Wash Equipment
22 45 29	Hand-Held Emergency Drench Hoses
22 45 33	Combination Emergency Fixture Units
22 45 36	Emergency Fixture Water-Tempering Equipment

22 46 00 **Security Plumbing Fixtures**

See: 08 31 13 for security access doors and frames.
08 34 53 for security doors and frames.
08 39 53 for blast-resistant doors.
08 42 33 for security revolving door entrances.
08 56 53 for security windows.
08 56 56 for security window screens.
08 71 53 for security door hardware.
08 88 53 for security glazing.
26 55 53 for security lighting.

22 46 13	Security Water Closets and Urinals
22 46 13.13	Security Water Closets
22 46 13.16	Security Urinals
22 46 16	Security Lavatories and Sinks
22 46 16.13	Security Lavatories
22 46 16.16	Security Sinks
22 46 19	Security Showers
22 46 39	Security Faucets, Supplies, and Trim
22 46 43	Security Plumbing Fixture Flushometers
22 46 53	Security Plumbing Fixture Supports

22 47 00 **Drinking Fountains and Water Coolers**

See: 10 25 16 for modular service walls.

NUMBER	TITLE	EXPLANATION
22 47 13	Drinking Fountains	
22 47 16	Pressure Water Coolers	
22 47 19	Water-Station Water Coolers	
22 47 23	Remote Water Coolers	

22 50 00 Pool and Fountain Plumbing Systems

22 51 00 Swimming Pool Plumbing Systems

Includes: piping, main drains, related system specialties, filters, pumps, heaters, and treatment equipment specific to swimming pools.

See: 13 11 00 for fabricated swimming pools.

13 13 00 for aquariums.

13 14 13 for water slides.

13 14 16 for wave generating equipment.

22 11 23 for domestic water pumps.

22 32 00 for domestic water filtration equipment.

22 51 13	Swimming Pool Piping
22 51 16	Swimming Pool Pumps
22 51 19	Swimming Pool Water Treatment Equipment
22 51 23	Swimming Pool Equipment Controls

22 52 00 Fountain Plumbing Systems

Includes: piping, main drains, related system specialties, filters, pumps, and treatment equipment specific to fountains.

See: 13 12 00 for fountains.

22 11 23 for domestic water pumps.

22 32 00 for domestic water filtration equipment.

22 52 13	Fountain Piping
22 52 16	Fountain Pumps
22 52 19	Fountain Water Treatment Equipment
22 52 19.13	Electronic Fountain Water Conditioning Equipment
22 52 23	Fountain Equipment Controls

NUMBER	TITLE	EXPLANATION
22 60 00	Gas and Vacuum Systems for Laboratory and Healthcare Facilities	Includes: compressed air systems, vacuum systems, gas systems, chemical waste systems, and processed water systems for laboratory and healthcare purposes. Alternate Terms/Abbreviations: healthcare: medical: hospital. See: 11 53 00 for laboratory equipment. 11 70 00 for healthcare equipment. 11 74 00 for dental equipment. 12 35 53 for laboratory casework. 12 35 70 for healthcare casework. 12 35 70 for dental casework. 12 36 00 for countertops.
22 61 00	Compressed-Air Systems for Laboratory and Healthcare Facilities	Includes: compressed air systems for laboratory and healthcare purposes. Piping, valves, piping specialties, pumps, compressors, and other components. Alternate Terms/Abbreviations: healthcare: medical: hospital. See: 22 15 00 for general service compressed-air systems. 40 12 00 for compressed air process piping.
22 61 13	Compressed-Air Piping for Laboratory and Healthcare Facilities	
22 61 13.53	Laboratory Compressed-Air Piping	
22 61 13.70	Healthcare Compressed-Air Piping	
22 61 13.74	Dental Compressed-Air Piping	
22 61 19	Compressed-Air Equipment for Laboratory and Healthcare Facilities	
22 61 19.53	Laboratory Compressed-Air Equipment	
22 61 19.70	Healthcare Compressed-Air Equipment	
22 61 19.74	Dental Compressed-Air Equipment	
22 62 00	Vacuum Systems for Laboratory and Healthcare Facilities	Includes: vacuum systems for laboratory and medical purposes. Piping, valves, piping specialties, pumps, and other components. Alternate Terms/Abbreviations: healthcare: medical: hospital. See: 40 18 00 for vacuum systems process piping.
22 62 13	Vacuum Piping for Laboratory and Healthcare Facilities	
22 62 13.53	Laboratory Vacuum Piping	
22 62 13.70	Healthcare, Surgical Vacuum Piping	
22 62 13.74	Dental Vacuum Piping	

NUMBER	TITLE	EXPLANATION
22 62 19	Vacuum Equipment for Laboratory and Healthcare Facilities	
22 62 19.53	Laboratory Vacuum Equipment	
22 62 19.70	Healthcare Vacuum Equipment	
22 62 19.74	Dental Vacuum and Evacuation Equipment	
22 62 23	Waste Anesthesia-Gas Piping	
22 63 00	Gas Systems for Laboratory and Healthcare Facilities	
		Includes: gas systems for laboratory and healthcare purposes. Piping, valves, piping specialties, pumps, and other components. Nitrogen gas, nitrous-oxide, and oxygen gas systems.
		Alternate Terms/Abbreviations: healthcare: medical: hospital.
		See: 23 11 23 for facility natural-gas piping. 40 13 00 for inert gases process piping. 40 16 00 for specialty and high-purity gases piping. 40 16 36 for nitrous oxide gas process piping.
22 63 13	Gas Piping for Laboratory and Healthcare Facilities	
22 63 13.53	Laboratory Gas Piping	
22 63 13.70	Healthcare Gas Piping	
22 63 19	Gas Storage Tanks for Laboratory and Healthcare Facilities	
22 63 19.53	Laboratory Gas Storage Tanks	
22 63 19.70	Healthcare Gas Storage Tanks	
22 66 00	Chemical-Waste Systems for Laboratory and Healthcare Facilities	
		Includes: chemical-waste systems for laboratory and healthcare purposes. Piping, piping specialties, and other components.
		Alternate Terms/Abbreviations: healthcare: medical: hospital.
		See: 22 13 00 for facility sanitary sewerage.
22 66 53	Laboratory Chemical-Waste and Vent Piping	
22 66 70	Healthcare Chemical-Waste and Vent Piping	
22 66 83	Chemical-Waste Tanks	
22 66 83.13	Chemical-Waste Dilution Tanks	
22 66 83.16	Chemical-Waste Neutralization Tanks	

NUMBER	TITLE	EXPLANATION
22 67 00	Processed Water Systems for Laboratory and Healthcare Facilities	Includes: processed water systems for laboratory and healthcare purposes. Piping, valves, piping specialties, pumps, and other components. Alternate Terms/Abbreviations: healthcare: medical: hospital. DI: deionized. RO: reverse osmosis. See: 22 11 00 for facility water distribution. 40 23 00 for water process piping. 40 23 13 for deionized water process piping. 40 23 16 for distilled water process piping. 40 23 33 for reverse osmosis water process piping.
22 67 13	Processed Water Piping for Laboratory and Healthcare Facilities	
22 67 13.13	Distilled-Water Piping	
22 67 13.16	Reverse-Osmosis Water Piping	
22 67 13.19	Deionized-Water Piping	
22 67 19	Processed Water Equipment for Laboratory and Healthcare Facilities	
22 67 19.13	Distilled-Water Equipment	
22 67 19.16	Reverse-Osmosis Water Equipment	
22 67 19.19	Deionized-Water Equipment	

NUMBER	TITLE	EXPLANATION
DIVISION 23 – HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)		
23 00 00	Heating, Ventilating, and Air Conditioning (HVAC)	May be used as Division level Section title. See: 02 41 19 for selective structure demolition of existing HVAC systems. 03 30 00 for cast-in-place concrete equipment bases. 07 84 00 for firestopping. 07 92 00 for joint sealants. 08 31 00 for access doors and panels. 09 91 00 for field painting. 31 23 33 for trenching and backfilling.

23 01 00 Operation and Maintenance of HVAC Systems

Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of HVAC systems.

Level 4 Numbering Recommendation: Level 4 titles for HVAC systems are provided as examples for user's creation of more detailed titles for other subjects as required. Following Level 4 numbering is recommended:

.51-.59 for maintenance.

.61-.69 for repair.

.71-.79 for rehabilitation.

.81-.89 for replacement.

.91-.99 for restoration.

23 01 10	Operation and Maintenance of Facility Fuel Systems
23 01 20	Operation and Maintenance of HVAC Piping and Pumps
23 01 30	Operation and Maintenance of HVAC Air Distribution
23 01 30.51	HVAC Air-Distribution System Cleaning
23 01 50	Operation and Maintenance of Central Heating Equipment
23 01 60	Operation and Maintenance of Central Cooling Equipment
23 01 60.71	Refrigerant Recovery/Recycling
23 01 70	Operation and Maintenance of Central HVAC Equipment
23 01 80	Operation and Maintenance of Decentralized HVAC Equipment
23 01 90	Diagnostic Systems for HVAC

NUMBER	TITLE	EXPLANATION
23 05 00	Common Work Results for HVAC	Includes: subjects common to multiple titles in Division 23. See: 01 80 00 for performance requirements of subjects common to multiple titles. 05 45 13 for mechanical metal supports. 13 48 00 for sound, vibration, and seismic control. 21 05 29 for hangers and supports for fire-suppression piping and equipment. 21 05 33 for heat tracing for fire-suppression piping. 21 05 48 for vibration and seismic controls for fire-suppression piping and equipment. 21 05 53 for identification for fire-suppression piping and equipment. 22 05 29 for hangers and supports for plumbing piping and equipment. 22 05 33 for heat tracing for plumbing piping. 22 05 48 for vibration and seismic controls for plumbing piping and equipment. 22 05 53 for identification for plumbing piping and equipment. 40 41 00 for process piping and equipment heat tracing.
23 05 05	Selective Demolition for Heating, Ventilating, and Air Conditioning (HVAC)	See: 02 41 19 for selective demolition for whole structure demolition.
23 05 13	Common Motor Requirements for HVAC Equipment	
23 05 16	Expansion Fittings and Loops for HVAC Piping	
23 05 17	Sleeves and Sleeve Seals for HVAC Piping	Includes: penetration methods into and through fire-rated and unrated walls, ceilings, and floors. Sealing of conduit penetrations against water is included. Does Not Include: firestopping requirements.
23 05 19	Meters and Gauges for HVAC Piping	
23 05 23	General-Duty Valves for HVAC Piping	
23 05 29	Hangers and Supports for HVAC Piping and Equipment	
23 05 33	Heat Tracing for HVAC Piping	
23 05 48	Vibration and Seismic Controls for HVAC	
23 05 48.13	Vibration Controls for HVAC	Includes: vibration-isolation supports and hangers; vibration-isolation equipment bases.
23 05 53	Identification for HVAC Piping and Equipment	
23 05 63	Anti-Microbial Coatings for HVAC Ducts and Equipment	
23 05 66	Anti-Microbial Ultraviolet Emitters for HVAC Ducts and Equipment	
23 05 93	Testing, Adjusting, and Balancing for HVAC	

NUMBER	TITLE	EXPLANATION
23 06 00	Schedules for HVAC	Includes: schedules of items common to multiple titles in Division 23.
		Notes: a schedule may be included on drawings, in the project manual, or a project book.
23 06 10	Schedules for Facility Fuel Service Systems	
23 06 20	Schedules for HVAC Piping and Pumps	
23 06 20.13	Hydronic Pump Schedule	
23 06 30	Schedules for HVAC Air Distribution	
23 06 30.13	HVAC Fan Schedule	
23 06 30.16	Air Terminal Unit Schedule	
23 06 30.19	Air Outlet and Inlet Schedule	
23 06 30.23	HVAC Air Cleaning Device Schedule	
23 06 50	Schedules for Central Heating Equipment	
23 06 50.13	Heating Boiler Schedule	
23 06 60	Schedules for Central Cooling Equipment	
23 06 60.13	Refrigerant Condenser Schedule	
23 06 60.16	Packaged Water Chiller Schedule	
23 06 70	Schedules for Central HVAC Equipment	
23 06 70.13	Indoor, Central-Station Air-Handling Unit Schedule	
23 06 70.16	Packaged Outdoor HVAC Equipment Schedule	
23 06 80	Schedules for Decentralized HVAC Equipment	
23 06 80.13	Decentralized Unitary HVAC Equipment Schedule	
23 06 80.16	Convection Heating and Cooling Unit Schedule	
23 06 80.19	Radiant Heating and Cooling Unit Schedule	
23 07 00	HVAC Insulation	Includes: thermal insulation of HVAC ducts, piping, and equipment.
		See: 07 21 00 for thermal insulation. 21 07 00 for fire-suppression insulation. 22 07 00 for plumbing insulation. 40 42 00 for process piping and equipment insulation.
23 07 13	Duct Insulation	
23 07 16	HVAC Equipment Insulation	
23 07 19	HVAC Piping Insulation	
23 08 00	Commissioning of HVAC	Includes: commissioning of items common to multiple titles in Division 23.
		See: 01 91 00 for commissioning of subjects common to multiple Divisions.

NUMBER	TITLE	EXPLANATION
23 09 00	Instrumentation and Control for HVAC	Includes: items for operating and controlling HVAC equipment. Various types of controls and the sequence of operation describing the equipment operation. See: 13 50 00 for special instrumentation. 21 09 00 for instrumentation and control for fire-suppression systems. 22 09 00 for instrumentation and control for plumbing. 23 05 93 for testing, adjusting, and balancing for HVAC. 25 35 00 for integrated automation instrumentation and terminal devices for HVAC. 25 55 00 for integrated automation control of HVAC. 25 95 00 for integrated automation control sequences for HVAC.
23 09 13	Instrumentation and Control Devices for HVAC	Includes: piping, tubing, wiring, control panels, thermostats, aquastats, timers, recording and alarm devices, related to HVAC equipment. May Include: instrumentation and control devices as part of integrated automation. See Also: 25 35 00 for integrated automation instrumentation and terminal devices for HVAC. 25 55 00 for integrated automation control of HVAC.
23 09 13.13	Actuators and Operators	
23 09 13.23	Sensors and Transmitters	
23 09 13.33	Control Valves	
23 09 13.43	Control Dampers	
23 09 23	Direct-Digital Control System for HVAC	Includes: HVAC control equipment where control is done by a digital process using computers. May Include: direct-digital control system as part of integrated automation. Alternate Terms/Abbreviations: DDC: direct digital control. See: 40 94 00 for digital process controllers. See Also: 25 35 00 for integrated automation instrumentation and terminal devices for HVAC. 25 55 00 for integrated automation control of HVAC.
23 09 23.11	Control Valves	Includes: control valves and actuators for DDC systems.

NUMBER	TITLE	EXPLANATION
23 09 23.12	Control Dampers	Includes: control dampers and actuators for DDC systems.
23 09 23.13	Energy Meters	Includes: thermal and electric power energy meters that connect to DDC systems.
23 09 23.14	Flow Instruments	Includes: airflow sensors, switches, and transmitters; liquid flow meters, sensors, switches, and transmitters.
23 09 23.16	Gas Instruments	Includes: carbon-dioxide sensors, transmitters, and monitoring systems; oxygen monitoring systems; VOC sensors and transmitters. Alternate Terms/Abbreviations: VOC: volatile organic compound.
23 09 23.17	Level Instruments	Includes: liquid-level switches, sensors, and transmitters.
23 09 23.18	Leak Detection Instruments	Includes: point- and cable-type leak-detection switches.
23 09 23.19	Moisture Instruments	Includes: moisture switches, sensors, and transmitters.
23 09 23.21	Motion Instruments	Includes: PIR, Ultrasonic, and dual-technology motion sensors. Alternate Terms/Abbreviations: PIR: post incident report.
23 09 23.22	Position Instruments	Includes: position limit switches for use in DDC systems for HVAC.
23 09 23.23	Pressure Instruments	Includes: air and liquid pressure sensors, switches, and transmitters.
23 09 23.24	Speed Instruments	Includes: speed switches for DDC controls for HVAC.
23 09 23.27	Temperature Instruments	Includes: air temperature sensors, switches, RTD transmitters; liquid and steam temperature sensors, switches, and transmitters. Alternate Terms/Abbreviations: RTD: resistance temperature device.

DIVISION **23**

NUMBER	TITLE	EXPLANATION
23 09 23.33	Vibration Instruments	Includes: vibration switches and transmitters connected to DDC systems for HVAC.
23 09 23.43	Weather Stations	Includes: weather stations connected to DDC for HVAC to help maintain air quality and temperature control in light of outside weather conditions.
23 09 33	Electric and Electronic Control System for HVAC	Includes: HVAC control equipment where the devices are powered with electricity. See: 40 93 00 for process analog controllers/recorders.
23 09 43	Pneumatic Control System for HVAC	Includes: HVAC control equipment where the devices are powered with compressed air. See: 22 15 00 for general service compressed-air systems. 40 93 00 for process analog controllers/recorders.
23 09 53	Pneumatic and Electric Control System for HVAC	Includes: HVAC control equipment where the devices are powered with electricity and compressed air. See: 22 15 00 for general service compressed-air systems. 40 93 00 for process analog controllers/recorders.
23 09 93	Sequence of Operations for HVAC Controls	Includes: description of how the HVAC equipment is intended to operate.
23 09 93.11	Sequence of Operation for HVAC DDC	Includes: descriptions of operational sequences for HVAC control system.

23 10 00 Facility Fuel Systems

Includes: fuel systems associated with a structure.

See: 11 11 00 for vehicle service equipment.
26 32 13 for engine generators.
28 42 00 for gas detection and alarm.
28 43 00 for fuel oil detection and alarm.
33 50 00 for site fuel distribution utilities.

NUMBER	TITLE	EXPLANATION
23 11 00	Facility Fuel Piping	Includes: fuel piping, valves, piping specialties, and other components within, under, or closely associated with a structure. See: 23 24 00 for internal-combustion engine piping. 33 52 00 for hydrocarbon transmission and distribution. 40 21 00 for liquid fuels process piping. 40 21 23 for fuel oils process piping.
23 11 13	Facility Fuel-Oil Piping	
23 11 16	Facility Gasoline Piping	
23 11 23	Facility Natural-Gas Piping	
23 11 26	Facility Liquefied-Petroleum Gas Piping	
23 12 00	Facility Fuel Pumps	Includes: fuel pumps within or closely associated with a structure. See: 33 52 00 for hydrocarbon transmission and distribution.
23 12 13	Facility Fuel-Oil Pumps	
23 12 16	Facility Gasoline Dispensing Pumps	
23 12 26	Facility Liquefied-Petroleum Gas Pumps	
23 13 00	Facility Fuel-Storage Tanks	Includes: fuel tanks under or closely associated with a structure. See: 09 97 13 for storage tank coatings. 33 56 00 for site fuel storage tanks. 41 52 19 for material storage tanks. 43 41 13 for gas and liquid pressure vessels.
23 13 13	Facility Underground Fuel-Oil, Storage Tanks	
23 13 13.13	Double-Wall Steel, Underground Fuel-Oil, Storage Tanks	
23 13 13.16	Composite, Steel, Underground Fuel-Oil, Storage Tanks	
23 13 13.19	Jacketed, Steel, Underground Fuel-Oil, Storage Tanks	
23 13 13.23	Glass-Fiber-Reinforced-Plastic, Underground Fuel-Oil, Storage Tanks	
23 13 13.33	Fuel-Oil Storage Tank Pumps	
23 13 23	Facility Aboveground Fuel-Oil, Storage Tanks	
23 13 23.13	Vertical, Steel, Aboveground Fuel-Oil, Storage Tanks	
23 13 23.16	Horizontal, Steel, Aboveground Fuel-Oil, Storage Tanks	
23 13 23.19	Containment-Dike, Steel, Aboveground Fuel-Oil, Storage Tanks	
23 13 23.23	Insulated, Steel, Aboveground Fuel-Oil, Storage Tanks	
23 13 23.26	Concrete-Vaulted, Steel, Aboveground Fuel-Oil, Storage Tanks	
23 13 26	Facility Aboveground Liquefied-Petroleum Gas Storage Tanks	

NUMBER	TITLE	EXPLANATION
23 20 00	HVAC Piping and Pumps	Includes: HVAC piping, pumps, and water treatment for a structure. See: 23 05 93 for testing, adjusting, and balancing for HVAC. 23 09 00 for instrumentation and control for HVAC. 33 60 00 for site hydronic and steam energy utilities.
23 21 00	Hydronic Piping and Pumps	Includes: chilled condenser and hot water piping, valves, piping specialties, pumps, and other components within a structure. See: 21 30 00 for fire pumps. 22 11 23 for domestic water pumps. 33 61 00 for site hydronic energy distribution.
23 21 13	Hydronic Piping	
23 21 13.13	Underground Hydronic Piping	
23 21 13.23	Aboveground Hydronic Piping	
23 21 13.33	Ground-Loop Heat-Pump Piping	
23 21 16	Hydronic Piping Specialties	
23 21 23	Hydronic Pumps	
23 21 23.13	In-Line Centrifugal Hydronic Pumps	
23 21 23.16	Base-Mounted, Centrifugal Hydronic Pumps	
23 21 23.19	Vertical-Mounted, Double-Suction Centrifugal Hydronic Pumps	
23 21 23.23	Vertical-Turbine Hydronic Pumps	
23 21 29	Automatic Condensate Pump Units	
23 22 00	Steam and Condensate Piping and Pumps	Includes: steam and condensate piping, valves, piping specialties and other components within a structure. See: 33 63 00 for site steam energy distribution. 40 11 00 for steam process piping.
23 22 13	Steam and Condensate Heating Piping	
23 22 13.13	Underground Steam and Condensate Heating Piping	
23 22 13.23	Aboveground Steam and Condensate Heating Piping	
23 22 16	Steam and Condensate Heating Piping Specialties	
23 22 23	Steam Condensate Pumps	
23 22 23.13	Electric-Driven Steam Condensate Pumps	
23 22 23.23	Pressure-Powered Steam Condensate Pumps	
23 23 00	Refrigerant Piping	Includes: refrigerant piping, valves, piping specialties, refrigerant, and other components within a structure. See: 28 44 00 for refrigerant detection and alarm.

NUMBER	TITLE	EXPLANATION
23 23 13	Refrigerant Piping Valves	
23 23 16	Refrigerant Piping Specialties	
23 23 19	Refrigerant Safety Relief Valve Discharge Piping	
23 23 23	Refrigerants	
23 24 00	Internal-Combustion Engine Piping	Includes: piping for internal-combustion engines other than fuel piping. See: 23 11 00 for facility fuel piping. 23 51 00 for breechings, chimneys, and stacks. 26 32 13 for engine generators. 40 15 00 for combustion system gas piping.
23 24 13	Internal-Combustion Engine Remote-Radiator Coolant Piping	
23 24 16	Internal-Combustion Engine Exhaust Piping	
23 25 00	HVAC Water Treatment	Includes: water treatment equipment for hot water, chilled water, steam, and condenser water systems. One-shot feeders, automatic chemical feed systems, manual chemical feed systems, and chemicals. See: 22 31 00 for domestic water softeners. 22 32 00 for domestic water filtration equipment.
23 25 13	Water Treatment for Closed-Loop Hydronic Systems	
23 25 16	Water Treatment for Open Hydronic Systems	
23 25 19	Water Treatment for Steam System Feedwater	
23 25 23	Water Treatment for Humidification Steam System Feedwater	
23 25 26	Electronic Water Conditioning for HVAC Systems	
23 25 33	HVAC Makeup-Water Filtration Equipment	Includes: water softeners, RO equipment, multimedia filters, self-cleaning strainers, bag-and cartridge-type filters, and centrifugal separators.
23 30 00	HVAC Air Distribution	Includes: ducts, duct accessories, fans, volume control equipment, and air inlets and outlets. See: 23 05 93 for testing, adjusting, and balancing for HVAC. 23 09 00 for instrumentation and control for HVAC.

NUMBER	TITLE	EXPLANATION
23 31 00	HVAC Ducts and Casings	Includes: duct materials and fabrication, including duct hangers and supports. See: 23 07 13 for duct insulation. 23 33 46 for flexible ducts. 23 33 53 for duct liners.
23 31 13	Metal Ducts	
23 31 13.13	Rectangular Metal Ducts	
23 31 13.16	Round and Flat-Oval Spiral Ducts	
23 31 13.19	Metal Duct Fittings	
23 31 16	Nonmetal Ducts	
23 31 16.13	Fibrous-Glass Ducts	
23 31 16.16	Thermoset Fiberglass-Reinforced Plastic Ducts	
23 31 16.19	PVC Ducts	
23 31 16.26	Concrete Ducts	
23 31 19	HVAC Casings	
23 32 00	Air Plenums and Chases	Includes: plenums and chases formed by general construction work results used for HVAC air distribution. See: 09 50 00 for suspended ceilings. 09 69 00 for access flooring. 13 48 00 for sound, vibration, and seismic control. 23 31 00 for HVAC ducts and casings.
23 32 13	Fabricated, Metal Air Plenums	
23 32 33	Air-Distribution Ceiling Plenums	
23 32 36	Air-Distribution Floor Plenums	
23 32 39	Air-Distribution Wall Plenums	
23 32 43	Air-Distribution Chases Formed by General Construction	
23 32 48	Acoustical Air Plenums	
23 33 00	Air Duct Accessories	Includes: dampers, duct access panel and test holes, flexible duct connectors, turning vanes and extractors, and sound attenuators. Balancing, back draft, fire, smoke, and combination fire and smoke dampers. See: 23 07 13 for duct insulation. 23 09 00 for instrumentation and control for HVAC. 23 31 00 for HVAC ducts and casings.
23 33 13	Dampers	
23 33 13.13	Volume-Control Dampers	
23 33 13.16	Fire Dampers	
23 33 13.19	Smoke-Control Dampers	
23 33 13.23	Backdraft Dampers	

NUMBER	TITLE	EXPLANATION
23 33 19	Duct Silencers	
23 33 23	Turning Vanes	
23 33 33	Duct-Mounting Access Doors	
23 33 38	Duct Security Bars	
23 33 43	Flexible Connectors	
23 33 46	Flexible Ducts	
23 33 53	Duct Liners	
23 34 00	HVAC Fans	Includes: various types of fans for air distribution. Ceiling fans and exhaust fans. Alternate Terms/Abbreviations: air curtains: air doors. See: 23 31 00 for HVAC ducts and casings. 23 38 00 for ventilation hoods.
23 34 13	Axial HVAC Fans	
23 34 16	Centrifugal HVAC Fans	
23 34 23	HVAC Power Ventilators	
23 34 33	Air Curtains	
23 34 39	High-Volume, Low-Speed Propeller Fans	Includes: fans, motors, gearboxes, and drives. Alternate Terms/Abbreviations: commercial ceiling fans.
23 35 00	Special Exhaust Systems	Includes: exhaust systems with a special specific purpose. Sawdust collection includes inlets, ducts, fans, and collection bins or cyclones. Exhaust systems for vehicle repair facilities, vehicle and engine test facilities, engine-generator exhaust systems, and engine-driven fire-pump exhaust systems. See: 23 40 00 for HVAC air cleaning devices. 44 11 16 for industrial dust collectors.
23 35 13	Dust Collection Systems	
23 35 13.13	Sawdust Collection Systems	
23 35 16	Engine Exhaust Systems	
23 35 16.13	Positive-Pressure Engine Exhaust Systems	
23 35 16.16	Mechanical Engine Exhaust Systems	
23 35 33	Listed Kitchen Ventilation Exhaust System	Includes: fire-rated kitchen ventilation systems intended chiefly for commercial applications. See Also: 23 31 00 for HVAC ducts and casings for non-rated kitchen ventilation systems. 23 38 13 for listed commercial-kitchen hoods.

NUMBER	TITLE	EXPLANATION
23 36 00	Air Terminal Units	Includes: air terminal units involving heat transfer and forced air supply. Alternate Terms/Abbreviations: VAV: variable air volume.
23 36 13	Constant-Air-Volume Units	
23 36 16	Variable-Air-Volume Units	
23 37 00	Air Outlets and Inlets	Includes: devices used to supply or remove air from a space or building. See: 08 90 00 for louvers and vents. 09 58 00 for air outlets and inlets integral with integrated ceiling assemblies. 09 69 00 for air outlets and inlets integral with access flooring. 23 38 00 for ventilation hoods.
23 37 13	Diffusers, Registers, and Grilles	
23 37 13.43	Security Registers and Grilles	Includes: registers and grilles used in detention facilities.
23 37 16	Fabric Air Distribution Devices	
23 37 23	HVAC Gravity Ventilators	
23 37 23.13	HVAC Gravity Dome Ventilators	
23 37 23.16	HVAC Gravity Louvered-Penthouse Ventilators	
23 37 23.19	HVAC Gravity Upblast Ventilators	
23 38 00	Ventilation Hoods	Includes: ventilation hoods and equipment with fire-suppression systems for commercial kitchens. May Include: laboratory fumes hoods connected to HVAC system. See: 11 53 13 for recirculating laboratory fume hoods. 21 20 00 for fire extinguishing systems. See Also: 11 44 00 for commercial-kitchen hoods. 11 53 13 for laboratory fume hoods.
23 38 13	Commercial-Kitchen Hoods	
23 38 13.13	Listed Commercial-Kitchen Hoods	
23 38 13.16	Standard Commercial-Kitchen Hoods	
23 38 16	Fume Hoods	

NUMBER	TITLE	EXPLANATION
23 40 00	HVAC Air Cleaning Devices	Includes: devices of a variety of types to clean distribution air. Alternate Terms/Abbreviations: HEPA: high-efficiency particulate air (filter). ULPA: ultra-low penetration air (filter). SULPA: super ultra-low penetration air (filter). See: 23 31 00 for HVAC ducts and casings. 23 35 13 for sawdust collection systems. 44 11 00 for particulate control equipment.
23 41 00	Particulate Air Filtration	
23 41 13	Panel Air Filters	
23 41 16	Renewable-Media Air Filters	
23 41 19	Washable Air Filters	
23 41 23	Extended Surface Filters	
23 41 33	High-Efficiency Particulate Filtration	
23 41 43	Ultra-Low Penetration Filtration	
23 41 46	Super Ultra-Low Penetration Filtration	
23 42 00	Gas-Phase Air Filtration	
23 42 13	Activated-Carbon Air Filtration	
23 42 16	Chemically-Impregnated Adsorption Air Filtration	
23 42 19	Catalytic-Adsorption Air Filtration	
23 43 00	Electronic Air Cleaners	
23 43 13	Washable Electronic Air Cleaners	
23 43 16	Agglomerator Electronic Air Cleaners	
23 43 23	Self-Contained Electronic Air Cleaners	
23 50 00	Central Heating Equipment	Includes: boilers, feedwater equipment, and draft accessories; furnaces; heaters; heat exchangers; and breechings, chimneys, and stacks. See: 23 05 93 for testing, adjusting, and balancing for HVAC. 23 07 16 for HVAC equipment insulation. 23 09 00 for instrumentation and control for HVAC. 23 11 00 for facility fuel piping. 23 21 00 for hydronic piping and pumps. 23 22 00 for steam and condensate piping and pumps. 33 61 00 for site hydronic energy distribution. 33 63 00 for site steam energy distribution. 42 10 00 for process heating equipment.

NUMBER	TITLE	EXPLANATION
23 51 00	Breechings, Chimneys, and Stacks	Includes: double-wall and single-wall chimneys, breechings, and stacks. See: 04 50 00 for refractory masonry. 22 34 00 for fuel-fired domestic water heaters. 23 24 16 for internal-combustion engine exhaust piping. 23 52 00 for heating boilers. 23 54 00 for furnaces. 23 55 00 for fuel-fired heaters. 40 15 00 for combustion system gas piping. 40 47 00 for refractories.
23 51 13	Draft Control Devices	
23 51 13.11	Draft Control Fans	Includes: draft-inducer fans, venturi-draft-inducer fans, mechanical-draft vent fans, vent exhaust fans, and combustion-air fans.
23 51 13.13	Draft-Induction Fans	
23 51 13.16	Vent Dampers	
23 51 13.19	Barometric Dampers	
23 51 16	Fabricated Breechings and Accessories	
23 51 19	Fabricated Stacks	
23 51 23	Gas Vents	
23 51 33	Insulated Sectional Chimneys	
23 51 43	Flue-Gas Filtration Equipment	
23 51 43.13	Gaseous Filtration	
23 51 43.16	Particulate Filtration	
23 52 00	Heating Boilers	Includes: gas and oil fuel-fired, solid fuel-fired, and electric boilers and generators for both hot water and steam systems. See: 23 11 00 for facility fuel piping. 23 25 00 for HVAC water treatment. 23 51 00 for breechings, chimneys, and stacks. 23 53 00 for heating boiler feedwater equipment. 42 11 00 for process boilers.
23 52 13	Electric Boilers	
23 52 16	Condensing Boilers	
23 52 16.13	Stainless-Steel Condensing Boilers	
23 52 16.16	Aluminum Condensing Boilers	
23 52 17	Low Mass Boilers	
23 52 19	Pulse Combustion Boilers	
23 52 23	Cast-Iron Boilers	

NUMBER	TITLE	EXPLANATION
23 52 33	Water-Tube Boilers	
23 52 33.13	Finned Water-Tube Boilers	
23 52 33.14	Flexible Water-Tube Boilers	Includes: flexible water-tube boilers with burners, controls and accessories, suitable for use in steam or hot water systems fired by natural gas, propane, or fuel oil. Includes controls, hot water or steam boiler trim, and natural gas, propane, or fuel oil-fired burner.
23 52 33.16	Steel Water-Tube Boilers	
23 52 33.19	Copper Water-Tube Boilers	
23 52 39	Fire-Tube Boilers	
23 52 39.13	Scotch Marine Boilers	
23 52 39.16	Steel Fire-Tube Boilers	
23 52 83	Boiler Blowdown Systems	
23 53 00	Heating Boiler Feedwater Equipment	Includes: equipment to remove oxygen and other dissolved gases in boiler feed. See: 23 52 00 for heating boilers.
23 53 13	Boiler Feedwater Pumps	
23 53 16	Deaerators	
23 54 00	Furnaces	Includes: gas and oil fuel-fired, and electric furnaces. See: 23 11 00 for facility fuel piping. 23 51 00 for breechings, chimneys, and stacks. 42 14 00 for industrial furnaces.
23 54 13	Electric-Resistance Furnaces	
23 54 16	Fuel-Fired Furnaces	
23 54 16.13	Gas-Fired Furnaces	
23 54 16.16	Oil-Fired Furnaces	
23 55 00	Fuel-Fired Heaters	Includes: gas and oil fuel-fired duct, radiant, and unit heaters. See: 23 11 00 for facility fuel piping. 23 51 00 for breechings, chimneys, and stacks. 23 82 39 for unit heaters.
23 55 13	Fuel-Fired Duct Heaters	
23 55 13.13	Oil-Fired Duct Heaters	
23 55 13.16	Gas-Fired Duct Heaters	
23 55 23	Gas-Fired Radiant Heaters	
23 55 23.13	Low-Intensity Gas-Fired Radiant Heaters	
23 55 23.16	High-Intensity Gas-Fired Radiant Heaters	

DIVISION **23**

NUMBER	TITLE	EXPLANATION
23 55 33	Fuel-Fired Unit Heaters	
23 55 33.13	Oil-Fired Unit Heaters	
23 55 33.16	Gas-Fired Unit Heaters	

23 56 00 Solar Energy Heating Equipment

Includes: equipment powered by solar energy.

See: 22 33 30 for residential, collector-to-tank, solar-electric domestic water heaters.

26 31 00 for photovoltaic collectors.

48 14 00 for solar energy electrical power generation equipment.

23 56 13	Heating Solar Collectors	
23 56 13.13	Heating Solar Flat-Plate Collectors	
23 56 13.16	Heating Solar Concentrating Collectors	
23 56 13.19	Heating Solar Vacuum-Tube Collectors	
23 56 16	Packaged Solar Heating Equipment	
23 56 23	Solar Air-Heating Panels	

23 57 00 Heat Exchangers for HVAC

Includes: equipment used to transfer heat from one medium to another.

See: 22 35 00 for domestic water heat exchangers.

23 21 00 for hydronic piping and pumps.

23 22 00 for steam and condensate piping and pumps.

23 72 00 for air-to-air energy recovery equipment.

42 13 00 for industrial heat exchangers and recuperators.

23 57 13	Steam-to-Steam Heat Exchangers	
23 57 16	Steam-to-Water Heat Exchangers	
23 57 19	Liquid-to-Liquid Heat Exchangers	
23 57 19.13	Plate-Type, Liquid-to-Liquid Heat Exchangers	
23 57 19.16	Shell-Type, Liquid-to-Liquid Heat Exchangers	
23 57 33	Direct Geoexchange Heat Exchangers	

Includes: continuous sealed underground closed-loop heat exchanger with one refrigerant supply tube and one refrigerant return tube.

See: 23 71 00 for thermal storage of energy.

48 16 00 for geothermal energy electrical power generation equipment.

NUMBER	TITLE	EXPLANATION
23 60 00	Central Cooling Equipment	Includes: building space cooling equipment and equipment to reject heat from the refrigeration process. May Include: freezing equipment for ice rinks. See: 23 05 93 for testing, adjusting, and balancing for HVAC. 23 07 16 for HVAC equipment insulation. 23 09 00 for instrumentation and control for HVAC. 23 11 00 for facility fuel piping. 23 21 00 for hydronic piping and pumps. 23 76 00 for evaporative air-cooling equipment. 33 61 00 for site hydronic energy distribution. 42 20 00 for process cooling equipment. See Also: 13 18 00 for ice rinks.
23 61 00	Refrigerant Compressors	Includes: various types of compressors used in the refrigeration process. See: 28 44 00 for refrigerant detection and alarm.
23 61 13	Centrifugal Refrigerant Compressors	
23 61 13.13	Non-Condensable Gas Purge Equipment	
23 61 16	Reciprocating Refrigerant Compressors	
23 61 19	Scroll Refrigerant Compressors	
23 61 23	Rotary-Screw Refrigerant Compressors	
23 62 00	Packaged Compressor and Condenser Units	Includes: compressors used in the refrigeration process in combination with condensing units used to reject heat from the refrigeration process. See: 28 44 00 for refrigerant detection and alarm.
23 62 13	Packaged Air-Cooled Refrigerant Compressor and Condenser Units	
23 62 23	Packaged Water-Cooled Refrigerant Compressor and Condenser Units	
23 62 46	Packaged Variable-Refrigerant-Flow Air-Conditioning Systems	
23 63 00	Refrigerant Condensers	Includes: condensing units used to reject heat from the refrigeration process. See: 28 44 00 for refrigerant detection and alarm. 42 23 00 for process condensers and evaporators.
23 63 13	Air-Cooled Refrigerant Condensers	

NUMBER	TITLE	EXPLANATION
23 63 23	Water-Cooled Refrigerant Condensers	
23 63 33	Evaporative Refrigerant Condensers	
23 64 00	Packaged Water Chillers	Includes: various types of chillers used for building space cooling. See: 42 22 00 for process chillers and coolers. 42 22 13 for centrifugal process chillers and coolers. 42 22 16 for reciprocating process chillers and coolers. 42 22 23 for rotary process chillers and coolers.
23 64 13	Absorption Water Chillers	
23 64 13.13	Direct-Fired Absorption Water Chillers	
23 64 13.16	Indirect-Fired Absorption Water Chillers	
23 64 16	Centrifugal Water Chillers	
23 64 16.13	Air-Cooled Centrifugal Water Chillers	
23 64 16.16	Water-Cooled Centrifugal Water Chillers	
23 64 19	Reciprocating Water Chillers	
23 64 23	Scroll Water Chillers	
23 64 23.13	Air-Cooled Scroll Water Chillers	
23 64 23.16	Water-Cooled Scroll Water Chillers	
23 64 26	Rotary-Screw Water Chillers	
23 64 26.13	Air-Cooled, Rotary-Screw Water Chillers	
23 64 26.16	Water-Cooled, Rotary-Screw Water Chillers	
23 64 33	Modular Water Chillers	Includes: water- and air-cooled, packaged, modular water chillers.
23 64 33.13	Air-Cooled, Modular Water Chillers	
23 64 33.16	Water-Cooled, Modular Water Chillers	
23 65 00	Cooling Towers	Includes: factory- and field-assembled cooling towers used to reject heat from the refrigeration process. See: 31 00 00 for earthwork. 35 00 00 for waterway and marine construction. 42 21 00 for process cooling towers.
23 65 13	Forced-Draft Cooling Towers	
23 65 13.13	Open-Circuit, Forced-Draft Cooling Towers	
23 65 13.16	Closed-Circuit, Forced-Draft Cooling Towers	
23 65 14	Induced-Draft Cooling Towers	
23 65 14.13	Open-Circuit, Induced-Draft Counterflow Cooling Towers	Includes: open-circuit cooling tower in which airflow and condenser-water flow are in opposing directions.

NUMBER	TITLE	EXPLANATION
23 65 14.14	Open-Circuit, Induced-Draft Crossflow Cooling Towers	Includes: open-circuit cooling tower in which airflow is horizontal across cascading condenser-water flow and fan draws air through the tower.
23 65 14.16	Closed-Circuit, Induced-Draft Counterflow Cooling Towers	Includes: closed-circuit cooling tower in which airflow is drawn through condenser-water flowing through closed-circuit piping system.
23 65 14.17	Closed-Circuit, Induced-Draft Combined-Flow Cooling Towers	Includes: closed-circuit cooling tower in which airflow is drawn through sides of tower condenser-water flowing through closed-circuit piping system; induced-draft fan is mounted on top of tower.
23 65 16	Natural-Draft Cooling Towers	
23 65 23	Field-Erected Cooling Towers	
23 65 33	Liquid Coolers	

23 70 00 Central HVAC Equipment

Includes: equipment that serves a total HVAC system.

See: 23 05 93 for testing, adjusting, and balancing for HVAC.
 23 07 16 for HVAC equipment insulation.
 23 09 00 for instrumentation and control for HVAC.
 23 11 00 for facility fuel piping.
 23 21 00 for hydronic piping and pumps.
 33 61 00 for site hydronic energy distribution.

23 71 00 Thermal Storage

Includes: equipment to store thermal energy for use in heating and/or cooling and with charging or discharging this energy at a controllable rate.

See: 23 57 33 for direct geexchange heat exchanger.
 48 16 00 for geothermal energy electrical power generation equipment.

23 71 13	Thermal Heat Storage
23 71 13.13	Room Storage Heaters for Thermal Storage
23 71 13.16	Heat-Pump Boosters for Thermal Storage
23 71 13.19	Central Furnace Heat-Storage Units
23 71 13.23	Pressurized-Water Thermal Storage Tanks
23 71 16	Chilled-Water Thermal Storage
23 71 19	Ice Storage
23 71 19.13	Internal Ice-on-Coil Thermal Storage
23 71 19.16	External Ice-on-Coil Thermal Storage
23 71 19.19	Encapsulated-Ice Thermal Storage
23 71 19.23	Ice-Harvesting Thermal Storage

NUMBER	TITLE	EXPLANATION
23 71 19.26	Ice-Slurry Thermal Storage	
23 72 00	Air-to-Air Energy Recovery Equipment	Includes: air-to-air type energy recovery units. See: 23 57 00 for heat exchangers for HVAC.
23 72 13	Heat-Wheel Air-to-Air Energy-Recovery Equipment	
23 72 16	Heat-Pipe Air-to-Air Energy-Recovery Equipment	
23 72 19	Fixed-Plate Air-to-Air Energy-Recovery Equipment	
23 72 23	Packaged Air-to-Air Energy-Recovery Units	
23 73 00	Indoor Central-Station Air-Handling Units	Includes: indoor units consisting of fans, coils, dampers, control devices, and other accessories. Assembly of standard produced components or assembly of custom designed components. May Include: filters with unit. See: 23 81 00 for decentralized unitary HVAC equipment. See Also: 23 40 00 for HVAC air cleaning devices.
23 73 13	Modular Indoor Central-Station Air-Handling Units	
23 73 23	Custom Indoor Central-Station Air-Handling Units	
23 73 33	Indoor Indirect Fuel-Fired Heating and Ventilating Units	
23 73 33.13	Indoor Indirect Oil-Fired Heating and Ventilating Units	
23 73 33.16	Indoor Indirect Gas-Fired Heating and Ventilating Units	
23 73 39	Indoor, Direct Gas-Fired Heating and Ventilating Units	
23 74 00	Packaged Outdoor HVAC Equipment	Includes: outdoor units consisting of fans, coils, dampers, control devices and other accessories. Units may be for single-zone, multizone, and variable-air-volume applications. Units may be mounted on the roof or at grade level. Assembly of standard, readily available stock components. May Include: filters with unit. See Also: 23 40 00 for HVAC air cleaning devices.
23 74 13	Packaged, Outdoor, Central-Station Air-Handling Units	
23 74 16	Packaged Rooftop Air-Conditioning Units	
23 74 16.11	Packaged, Small-Capacity, Rooftop Air-Conditioning Units	Includes: six tons and smaller units. Rooftop heating, cooling, and ventilating units, and roof-mounting curbs and accessories.

NUMBER	TITLE	EXPLANATION
23 74 16.12	Packaged, Intermediate-Capacity, Rooftop Air-Conditioning Units	Includes: 7.5- to twenty-five-ton units. Rooftop heating, cooling, and ventilating units, and roof-mounting curbs and accessories.
23 74 16.13	Packaged, Large-Capacity, Rooftop Air-Conditioning Units	Includes: units larger than twenty-five tons. Rooftop heating, cooling, and ventilating units, and roof-mounting curbs and accessories.
23 74 23	Packaged, Outdoor, Heating-Only Makeup-Air Units	
23 74 23.13	Packaged, Direct-Fired, Outdoor, Heating-Only Makeup-Air Units	
23 74 23.16	Packaged, Indirect-Fired, Outdoor, Heating-Only Makeup-Air Units	
23 74 33	Dedicated Outdoor-Air Units	
23 75 00	Custom-Packaged Outdoor HVAC Equipment	Includes: outdoor unitary equipment that may consist of heat recovery, service vestibules, boilers, chillers, and space for other mechanical equipment. Units may be mounted on the roof or at grade level. Units may be for single-zone, multizone, and variable-air-volume applications; and can have multiple separate air handlers incorporated into a single unit. Assembly of non-standard customized or custom-built components, or a combination of standard components with non-standard customized or custom-built components.
23 75 13	Custom-Packaged, Outdoor, Central-Station Air-Handling Units	
23 75 16	Custom-Packaged, Rooftop Air-Conditioning Units	Includes: rooftop heating, cooling, and ventilating units and roof-mounting curbs and accessories.
23 75 23	Custom-Packaged, Outdoor, Heating and Ventilating Makeup-Air Units	
23 75 33	Custom-Packaged, Outdoor, Heating and Cooling Makeup Air-Conditioners	
23 76 00	Evaporative Air-Cooling Equipment	Includes: equipment used to reject heat from the refrigeration process by evaporation. Alternate Terms/Abbreviations: evaporative air coolers: swamp coolers. See: 23 60 00 for central cooling equipment. 23 81 00 for decentralized unitary HVAC equipment. 42 23 00 for process condensers and evaporators.
23 76 13	Direct Evaporative Air Coolers	
23 76 16	Indirect Evaporative Air Coolers	
23 76 19	Combined Direct and Indirect Evaporative Air Coolers	

NUMBER	TITLE	EXPLANATION
23 80 00	Decentralized HVAC Equipment	Includes: equipment that serves a portion of an HVAC system. See: 23 05 93 for testing, adjusting, and balancing for HVAC. 23 07 16 for HVAC equipment insulation. 23 09 00 for instrumentation and control for HVAC. 23 21 00 for hydronic piping and pumps.
23 81 00	Decentralized Unitary HVAC Equipment	Includes: equipment types that contain all the components of the refrigeration process within a single package. Air- and water-source heat pumps. See: 23 73 00 for indoor, central-station air-handling units. 23 76 00 for evaporative air-cooling equipment.
23 81 13	Packaged Terminal Air-Conditioners	
23 81 13.11	Packaged Terminal Air-Conditioners, Through-Wall Units	Includes: heating and cooling units; with electric heat, hydronic heat, or heat-pump configurations.
23 81 13.12	Packaged Terminal Air-Conditioners, Freestanding Units	Includes: heating and cooling units; with electric heat, hydronic heat, or heat-pump configurations.
23 81 13.13	Packaged Terminal Air-Conditioners, Outdoor, Wall-Mounted Units	Includes: heating and cooling units; with electric heat, hydronic heat, or heat-pump configurations.
23 81 16	Room Air-Conditioners	
23 81 19	Self-Contained Air-Conditioners	
23 81 19.13	Small-Capacity Self-Contained Air-Conditioners	
23 81 19.16	Large-Capacity Self-Contained Air-Conditioners	
23 81 23	Computer-Room Air-Conditioners	
23 81 23.11	Small-Capacity, Computer-Room Air-Conditioners, Floor Mounted Units	Includes: six tons (21 kW) and smaller units. Cooling-only units.
23 81 23.12	Large-Capacity, Computer-Room Air-Conditioners, Floor-Mounted Units	Includes: seven tons (25 kW) and larger units. Cooling-only units.
23 81 23.13	Computer-Room Air-Conditioners, Ceiling Mounted Units	Includes: cooling-only units.
23 81 23.14	Computer-Room Air-Conditioners, Console Units	Includes: cooling-only units.
23 81 23.16	Computer-Room Air-Conditioners, Rack Mounted, Space-Cooling Units	Includes: cooling-only units.

NUMBER	TITLE	EXPLANATION
23 81 23.18	Computer-Room, Rack-Cooling Equipment	Includes: cooling-only units.
23 81 26	Split-System Air-Conditioners	
23 81 26.13	Small-Capacity Split-System Air-Conditioners	
23 81 26.16	Large-Capacity Split-System Air-Conditioners	
23 81 29	Variable Refrigerant Flow HVAC Systems	
23 81 43	Air-Source Unitary Heat Pumps	
23 81 46	Water-Source Unitary Heat Pumps	
23 81 49	Ground-Source Unitary Heat Pumps	
23 82 00	Convection Heating and Cooling Units	Includes: terminal units with water, steam, and electric coils.
		See: 05 58 19 for custom formed metal heating/cooling unit covers.
		23 55 33 for fuel-fired unit heaters.
		42 12 00 for process heaters.
23 82 13	Valance Heating and Cooling Units	
23 82 14	Chilled Beams	
23 82 16	Air Coils	
23 82 16.11	Hydronic Air Coils	Includes: air coils with hot- and chilled-water transfer medium.
23 82 16.12	Steam Air Coils	Includes: air coils with steam transfer medium.
23 82 16.13	Refrigerant Air Coils	Includes: air coils with refrigerant transfer medium.
23 82 16.14	Electric-Resistance Air Coils	Includes: air coils with electric-resistance element.
23 82 19	Fan Coil Units	
23 82 23	Unit Ventilators	
23 82 26	Induction Units	
23 82 29	Radiators	
23 82 33	Convectors	
23 82 36	Finned-Tube Radiation Heaters	
23 82 39	Unit Heaters	
23 82 39.13	Cabinet Unit Heaters	
23 82 39.16	Propeller Unit Heaters	
23 82 39.19	Wall and Ceiling Unit Heaters	
23 82 41	Water-to-Water Heat Pumps	

NUMBER	TITLE	EXPLANATION
23 83 00	Radiant Heating and Cooling Units	Includes: electric cables and hydronic piping used for radiant heating for space heating or snow and ice control. See: 21 05 33 for heat tracing for fire-suppression piping. 22 05 33 for heat tracing for plumbing piping. 23 05 33 for heat tracing for HVAC piping. 23 55 23 for gas-fired radiant heaters. 32 13 00 for rigid paving. 40 41 00 for process piping and equipment heat tracing.
23 83 13	Radiant-Heating Electric Cables	
23 83 13.16	Radiant-Heating Electric Mats	
23 83 16	Radiant Heating and Cooling Hydronic Piping	
23 83 23	Radiant-Heating Electric Panels	
23 83 26	Radiant Heating and Cooling Hydronic Panels	
23 83 33	Electric Radiant Heaters	
23 84 00	Humidity Control Equipment	Includes: equipment that either adds moisture or removes moisture from a medium in order to control the humidity. See: 13 11 00 for swimming pools. 13 18 00 for ice rinks. 42 23 19 for process humidifiers. 42 31 00 for gas dryers and dehumidifiers.
23 84 13	Humidifiers	
23 84 13.13	Heated-Pan Humidifiers	
23 84 13.16	Wetted-Element Humidifiers	
23 84 13.19	Atomizing Humidifiers	
23 84 13.23	Direct-Steam-Injection Humidifiers	
23 84 13.26	Jacketed, Steam Humidifiers	
23 84 13.29	Self-Contained Steam Humidifiers	
23 84 13.33	Portable Humidifiers	
23 84 16	Mechanical Dehumidification Units	
23 84 16.13	Outdoor, Mechanical Dehumidification Units	
23 84 16.16	Indoor, Mechanical Dehumidification Units	
23 84 16.33	Portable Dehumidifiers	
23 84 19	Desiccant Dehumidification Units	

NUMBER	TITLE	EXPLANATION
DIVISION 25 – INTEGRATED AUTOMATION		
25 00 00	Integrated Automation	May be used as Division level Section title. Alternate Terms/Abbreviations: integrated automation: building automation. See: 02 41 19 for selective demolition of existing integrated automation systems. 07 84 00 for firestopping. 07 92 00 for joint sealants. 08 31 00 for access doors and panels. 09 91 00 for field painting. 31 23 33 for trenching and backfilling. Division 40 for process integration.
25 01 00	Operation and Maintenance of Integrated Automation	Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of integrated automation systems. Level 4 Numbering Recommendation: following numbering is recommended for the creation of Level 4 titles: .51-.59 for maintenance. .61-.69 for repair. .71-.79 for rehabilitation. .81-.89 for replacement. .91-.99 for restoration.
25 01 10	Operation and Maintenance of Integrated Automation Network Equipment	
25 01 20	Operation and Maintenance of Integrated Equipment	
25 01 30	Operation and Maintenance of Integrated Automation Instrumentation and Terminal Devices	
25 01 90	Diagnostic Systems for Integrated Automation	

NUMBER	TITLE	EXPLANATION
25 05 00	Common Work Results for Integrated Automation	Includes: subjects common to multiple titles in Division 25. See: 01 80 00 for performance requirements of subjects common to multiple titles. 05 45 00 for metal support assemblies. 13 48 00 for sound, vibration, and seismic control. 27 05 26 for grounding and bonding for communications systems. 27 05 26 for grounding and bonding for communications. 27 05 28 for pathways for communications systems. 27 05 28 for pathways for communications. 27 05 48 for vibration and seismic controls for communications. 27 05 53 for identification for communications. 27 15 01 for conductors and cables for electronic safety and security. 28 05 48 for vibration and seismic controls for electronic safety and security. 28 05 53 for identification for electronic safety and security.
25 05 05	Selective Demolition for Integrated Automation	See: 02 41 19 for selective demolition for whole structure demolition.
25 05 13	Conductors and Cables for Integrated Automation	
25 05 26	Grounding and Bonding for Integrated Automation	
25 05 28	Pathways for Integrated Automation	
25 05 28.29	Hangers and Supports for Integrated Automation	
25 05 28.33	Conduits and Backboxes for Integrated Automation	
25 05 28.36	Cable Trays for Integrated Automation	
25 05 28.39	Surface Raceways for Integrated Automation	
25 05 48	Vibration and Seismic Controls for Integrated Automation	
25 05 53	Identification for Integrated Automation	
25 06 00	Schedules for Integrated Automation	Includes: schedules of items common to multiple titles in Division 25. Notes: a schedule may be included on drawings, in the project manual, or a project book.
25 06 11	Schedules for Integrated Automation Network	
25 06 12	Schedules for Integrated Automation Network Gateways	
25 06 13	Schedules for Integrated Automation Control and Monitoring Network	
25 06 14	Schedules for Integrated Automation Local Control Units	
25 06 30	Schedules for Integrated Automation Instrumentation and Terminal Devices	

NUMBER	TITLE	EXPLANATION
25 08 00	Commissioning of Integrated Automation	Includes: commissioning of items common to multiple titles in Division 25. See: 01 91 00 for commissioning of subjects common to multiple divisions.
25 10 00	Integrated Automation Network Equipment	Includes: hardware and software for integrated automation network (building automation). See: 40 90 00 for instrumentation and control for process systems.
25 11 00	Integrated Automation Network Devices	Includes: hardware devices for integrated automation network. See: 11 28 00 for office equipment. 40 95 00 for process control hardware.
25 11 13	Integrated Automation Network Servers	
25 11 16	Integrated Automation Network Routers, Bridges, Switches, Hubs, and Modems	
25 11 19	Integrated Automation Network Operator Workstations	
25 12 00	Integrated Automation Network Gateways	Includes: hardware and/or software interfaces that allows the building automation system to command and/or process data from dissimilar equipment and systems in a transparent fashion.
25 12 13	Hardwired Integration Network Gateways	
25 12 16	Direct-Protocol Integration Network Gateways	
25 12 19	Neutral-Protocol Integration Network Gateways	
25 12 23	Client-Server Information/Database Integration Network Gateways	
25 13 00	Integrated Automation Control and Monitoring Network	Includes: networked supervisory controllers and integration panels to allow monitoring and control of building subsystems. See: 27 21 00 for data communications network equipment.
25 13 13	Integrated Automation Control and Monitoring Network Supervisory Control	
25 13 16	Integrated Automation Control and Monitoring Network Integration Panels	
25 13 19	Integrated Automation Control and Monitoring Network Interoperability	

NUMBER	TITLE	EXPLANATION
25 14 00	Integrated Automation Local Control Units	Includes: microprocessor-based controllers and/or panels that permit the monitoring, control and data gathering from various pieces of building equipment and devices. See: 27 22 00 for data communications hardware.
25 14 13	Integrated Automation Remote Control Panels	
25 14 16	Integrated Automation Application-Specific Control Panels	
25 14 19	Integrated Automation Terminal Control Units	
25 14 23	Integrated Automation Field Equipment Panels	
25 15 00	Integrated Automation Software	Includes: software requirements include alarm management, data display, report routines, graphics, applications, password protection, internet/intranet connectivity, and software licenses and upgrades. See: 27 25 00 for data communications software. 27 34 13 for toll fraud equipment and software. 27 34 16 for telemanagement software. 40 96 00 for process control software.
25 15 13	Integrated Automation Software for Network Gateways	
25 15 16	Integrated Automation Software for Control and Monitoring Networks	
25 15 19	Integrated Automation Software for Local Control Units	
25 30 00	Integrated Automation Instrumentation and Terminal Devices	Includes: because the instrumentation and terminal devices for most integrated automation systems depends on the seamless interaction among subsystems, it is recommended that all instrumentation and terminal devices be included in the following titles to prevent conflict between the requirements of these interdependent systems. Alternate Terms/Abbreviations: KW: kilowatt. UPS: uninterruptible power supply. See: 23 09 13 for instrumentation and control devices for HVAC. 25 50 00 for integrated automated facility controls. 25 90 00 for integrated automation control sequences. 40 90 00 for instrumentation and control for process systems. See Also: 23 09 13 for instrumentation and control devices for HVAC.

NUMBER	TITLE	EXPLANATION
25 31 00	Integrated Automation Instrumentation and Terminal Devices for Facility Equipment	See: Division 11 for facility equipment.
25 32 00	Integrated Automation Instrumentation and Terminal Devices for Conveying Equipment	See: Division 14 for conveying equipment.
25 33 00	Integrated Automation Instrumentation and Terminal Devices for Fire-Suppression Systems	See: Division 21 for fire-suppression systems.
25 34 00	Integrated Automation Instrumentation and Terminal Devices for Plumbing	See: Division 22 for plumbing.
25 35 00	Integrated Automation Instrumentation and Terminal Devices for HVAC	See: 22 15 00 for general service compressed air systems. 23 09 00 for instrumentation and control for HVAC. 23 09 13 for actuators and operators for HVAC. 23 09 13 for sensors and transmitters for HVAC. 23 09 13 for control valves for HVAC. 23 09 13 for control dampers for HVAC. Division 23 for HVAC.
25 35 13	Integrated Automation Actuators and Operators	
25 35 16	Integrated Automation Sensors and Transmitters	
25 35 19	Integrated Automation Control Valves	
25 35 23	Integrated Automation Control Dampers	
25 35 26	Integrated Automation Compressed Air Supply	
25 36 00	Integrated Automation Instrumentation and Terminal Devices for Electrical Systems	See: 26 09 00 for instrumentation and controls for electrical systems. 26 09 13 for electrical power monitoring and control. 26 09 23 for lighting control devices. 26 33 00 for battery equipment. 26 50 00 for lighting. Division 26 for electrical systems.
25 36 13	Integrated Automation Power Meters	
25 36 16	Integrated Automation KW Transducers	Alternate Terms/Abbreviations: KW: kilowatt
25 36 19	Integrated Automation Current Sensors	

DIVISION 25

NUMBER	TITLE	EXPLANATION
25 36 23	Integrated Automation Battery Monitors	
25 36 26	Integrated Automation Lighting Relays	
25 36 29	Integrated Automation UPS Monitors	
25 37 00	Integrated Automation Instrumentation and Terminal Devices for Communications Systems	See: Division 27 for communication systems.
25 38 00	Integrated Automation Instrumentation and Terminal Devices for Electronic Safety and Security Systems	See: Division 28 for electronic safety and security.
25 50 00	Integrated Automation Facility Controls	Includes: hardware and/or software that allows the building automation system to monitor and control equipment and systems specified in other Divisions. See: 23 09 13 for instrumentation and control devices for HVAC. 25 30 00 for integrated automation instrumentation and terminal devices. 25 90 00 for integrated automation control sequences. 40 90 00 for instrumentation and control for process systems.
25 51 00	Integrated Automation Control of Facility Equipment	See: Division 11 for facility equipment.
25 52 00	Integrated Automation Control of Conveying Equipment	See: Division 14 for conveying equipment.
25 53 00	Integrated Automation Control of Fire-Suppression Systems	See: Division 21 for fire-suppression systems.
25 54 00	Integrated Automation Control of Plumbing	See: Division 22 for plumbing.
25 55 00	Integrated Automation Control of HVAC	See: 23 09 23 for direct-digital control system for HVAC. Division 23 for HVAC.
25 56 00	Integrated Automation Control of Electrical Systems	See: Division 26 for electrical systems.

NUMBER	TITLE	EXPLANATION
25 57 00	Integrated Automation Control of Communications Systems	See: Division 27 for communication systems.
25 58 00	Integrated Automation Control of Electronic Safety and Security Systems	See: Division 28 for electronic safety and security.
25 90 00	Integrated Automation Control Sequences	Includes: description of how the equipment of a facility is intended to operate on an integrated control basis. See: 23 09 13 for instrumentation and control devices for HVAC. 25 30 00 for integrated automation instrumentation and terminal devices. 25 50 00 for integrated automated facility controls. 40 90 00 for instrumentation and control for process systems.
25 91 00	Integrated Automation Control Sequences for Facility Equipment	See: Division 11 for facility equipment.
25 92 00	Integrated Automation Control Sequences for Conveying Equipment	See: Division 14 for conveying equipment.
25 93 00	Integrated Automation Control Sequences for Fire-Suppression Systems	See: Division 21 for fire-suppression systems.
25 94 00	Integrated Automation Control Sequences for Plumbing	See: Division 22 for plumbing.
25 95 00	Integrated Automation Control Sequences for HVAC	See: 23 09 23 for direct-digital control system for HVAC. Division 23 for HVAC.
25 96 00	Integrated Automation Control Sequences for Electrical Systems	See: Division 26 for electrical systems.
25 97 00	Integrated Automation Control Sequences for Communications Systems	See: Division 27 for communication systems.

DIVISION 25

NUMBER	TITLE	EXPLANATION
25 98 00	Integrated Automation Control Sequences for Electronic Safety and Security Systems	See: Division 28 for electronic safety and security.

NUMBER	TITLE	EXPLANATION
DIVISION 26 – ELECTRICAL		
26 00 00	Electrical	May be used as Division level Section title. See: 02 41 19 for selective demolition of existing electrical systems. 03 30 00 for cast-in-place concrete equipment bases. 07 84 00 for firestopping. 07 92 00 for joint sealants. 08 31 00 for access doors and panels. 09 91 00 for field painting. 31 23 33 for trenching and backfilling.

26 01 00	Operation and Maintenance of Electrical Systems	Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of electrical systems. Level 4 Numbering Recommendation: following numbering is recommended for the creation of Level 4 titles: .51-.59 for maintenance. .61-.69 for repair. .71-.79 for rehabilitation. .81-.89 for replacement. .91-.99 for restoration. Definitions: medium-voltage: 2400 V to 69 kV. Low-voltage: 600 V and less.
26 01 10	Operation and Maintenance of Medium-Voltage Electrical Distribution	
26 01 20	Operation and Maintenance of Low-Voltage Electrical Distribution	
26 01 26	Maintenance Testing of Electrical Systems	
26 01 30	Operation and Maintenance of Facility Electrical Power Generating and Storing Equipment	
26 01 40	Operation and Maintenance of Electrical Protection Systems	
26 01 40.13	Operation and Maintenance of Lightning Protection Systems	
26 01 50	Operation and Maintenance of Lighting	
26 01 50.51	Luminaire Relamping	
26 01 50.81	Luminaire Replacement	

NUMBER	TITLE	EXPLANATION
26 05 00	Common Work Results for Electrical	Includes: subjects common to multiple titles in Division 26. Raceway and boxes include conduit, tubing, surface raceways, and electrical boxes. Medium-voltage: 2400 V to 69 kV. Low-voltage: 600 V and less. Control-voltage: 50 V and less. Alternate Terms/Abbreviations: EMT: electrical metallic tubing. Notes: definitions: medium-voltage: 2400 V to 69 kV. Low-voltage: 600 V and less. Control-voltage: 50 V and less. See: 01 80 00 for performance requirements of subjects common to multiple titles. 05 35 00 for raceway decking assemblies. 05 45 16 for electrical metal supports. 13 48 00 for sound, vibration, and seismic control. 25 05 13 for conductors and cables for integrated automation. 25 05 26 for grounding and bonding for integrated automation. 25 05 28 for pathways for integrated automation. 25 05 48 for vibration and seismic control for integrated automation. 25 05 53 for identification for integrated automation. 27 05 26 for grounding and bonding for communications systems. 27 05 28 for pathways for communications systems. 27 05 46 for utility poles for communications systems. 27 05 48 for vibration and seismic controls for communications. 27 05 53 for identification for communications. 27 15 01 for conductors and cables for electronic safety and security. 28 05 48 for vibration and seismic controls for electronic safety and security. 28 05 53 for identification for electronic safety and security. 33 71 16 for electrical utility poles. 33 71 19 for electrical utility underground ducts and manholes.
26 05 05	Selective Demolition for Electrical	See: 02 41 19 for selective demolition for whole structure demolition.
26 05 13	Medium-Voltage Cables	
26 05 13.13	Medium-Voltage Open Conductors	
26 05 13.16	Medium-Voltage, Single- and Multi-Conductor Cables	
26 05 19	Low-Voltage Electrical Power Conductors and Cables	
26 05 19.13	Undercarpet Electrical Power Cables	
26 05 19.23	Manufactured Wiring Assemblies	
26 05 23	Control-Voltage Electrical Power Cables	
26 05 26	Grounding and Bonding for Electrical Systems	
26 05 29	Hangers and Supports for Electrical Systems	
26 05 33	Raceway and Boxes for Electrical Systems	
26 05 33.13	Conduit for Electrical Systems	

NUMBER	TITLE	EXPLANATION
26 05 33.16	Boxes for Electrical Systems	
26 05 33.23	Surface Raceways for Electrical Systems	
26 05 36	Cable Trays for Electrical Systems	
26 05 39	Underfloor Raceways for Electrical Systems	
26 05 43	Underground Ducts and Raceways for Electrical Systems	
26 05 44	Sleeves and Sleeve Seals for Electrical Raceways and Cabling	Includes: penetration methods into and through fire-rated and unrated walls, ceilings, and floors. Sealing of conduit penetrations against water is included. Excludes firestopping requirements.
26 05 46	Poles for Electrical Systems	See Also: 33 71 16 for electrical utility poles.
26 05 48	Vibration and Seismic Controls for Electrical Systems	
26 05 48.16	Seismic Controls for Electrical Systems	Includes: restraint channel bracings, restraint cables, seismic-restraint accessories, and mechanical and adhesive anchor bolts.
26 05 53	Identification for Electrical Systems	
26 05 73	Power System Studies	
26 05 73.13	Short-Circuit Studies	
26 05 73.16	Coordination Studies	
26 05 73.19	Arc-Flash Hazard Analysis	
26 05 73.23	Load Flow Studies	
26 05 73.26	Stability Studies	
26 05 73.29	Harmonic-Analysis Studies	
26 05 76	Photometric Studies	
26 05 83	Wiring Connections	
26 06 00	Schedules for Electrical	Includes: schedules of items common to multiple titles in Division 26. Notes: a schedule may be included on drawings, in the project manual, or a project book.
26 06 10	Schedules for Medium-Voltage Electrical Distribution	
26 06 20	Schedules for Low-Voltage Electrical Distribution	
26 06 20.13	Electrical Switchboard Schedule	
26 06 20.16	Electrical Panelboard Schedule	
26 06 20.19	Electrical Motor-Control Center Schedule	
26 06 20.23	Electrical Circuit Schedule	
26 06 20.26	Wiring Device Schedule	
26 06 30	Schedules for Facility Electrical Power Generating and Storing Equipment	
26 06 40	Schedules for Electrical Protection Systems	
26 06 50	Schedules for Lighting	
26 06 50.13	Lighting Panelboard Schedule	
26 06 50.16	Lighting Fixture Schedule	

NUMBER	TITLE	EXPLANATION
26 08 00	Commissioning of Electrical Systems	Includes: commissioning of items common to multiple titles in Division 26. See: 01 91 00 for commissioning of subjects common to multiple Divisions.
26 09 00	Instrumentation and Control for Electrical Systems	Includes: instrumentation and control associated with electrical systems. See: 13 50 00 for special instrumentation. 25 36 00 for integrated automation instrumentation and terminal devices for electrical systems. 25 56 00 for integrated automation control of electrical systems. 25 96 00 for integrated automation control sequences for electrical systems. 33 09 70 for instrumentation and control for electrical utilities.
26 09 13	Electrical Power Monitoring	
26 09 15	Peak Load Controllers	
26 09 16	Electrical Control Components	
26 09 17	Programmable Controllers	
26 09 19	Enclosed Contactors	
26 09 23	Lighting Control Devices	Includes: clock and calendar, photoelectric switches, occupancy sensors, and light-leveling control devices. Control- and low-voltage lighting control devices connected through computers. Addressable lighting control devices and lighting components (ballasts) connected through computers. See: 11 61 00 for theater and stage equipment. 26 50 00 for lighting. 26 55 61 for theatrical lighting. See Also: 11 61 00 for theatrical lighting controls.
26 09 26	Lighting Control Panelboards	
26 09 33	Central Dimming Controls	
26 09 33.13	Multichannel Remote-Controlled Dimmers	
26 09 33.16	Remote-Controlled Dimming Stations	
26 09 36	Modular Dimming Controls	
26 09 36.13	Manual Modular Dimming Controls	
26 09 36.16	Integrated Multipreset Modular Dimming Controls	
26 09 36.19	Standalone Multipreset Modular Dimming	Includes: wall-box multiscene dimming controls and multipreset modular dimming controls, commonly known as architectural dimmers.
26 09 43	Network Lighting Controls	

NUMBER	TITLE	EXPLANATION
26 09 43.13	Digital-Network Lighting Controls	
26 09 43.16	Addressable Luminaire Lighting Controls	
26 09 43.19	Wireless Network Lighting Controls	
26 09 43.23	Relay-Based Lighting Controls	Includes: passive, network-type luminaire control systems using relays. Lighting control panels having mechanically held relays for switching, networked control panels interoperability with building automation systems, low-voltage and programmable signal sources, internet-protocol-based digital signal sources, and field-mounted sensors.

26 09 61 Theatrical Lighting Controls

26 10 00 Medium-Voltage Electrical Distribution

Includes: substations, transformers, switchgear, and circuit protection devices to distribute medium-voltage electrical power from the facility service point to the point of delivery.

Definitions: medium-voltage: 2400 V to 69 kV.

See: 26 05 13 for medium-voltage cables.

26 20 00 for low-voltage electrical distribution.

26 30 00 for facility electrical power generating and storing equipment.

33 71 00 for electrical utility distribution.

26 11 00 Substations

Includes: assembly of switches, circuit breakers, buses, and transformers to switch circuits and convert power from one voltage to another.

See: 33 72 00 for utility substations.

34 21 16 for traction power substations.

26 11 13 Primary Unit Substations

26 11 16 Secondary Unit Substations

Includes: factory assembled and test units consisting of medium-voltage primary incoming section, transformer section, and low-voltage secondary motor-control-center section, and low-voltage secondary switchboard section.

26 11 16.11 Secondary Unit Substations with Switchgear Secondary

Includes: factory-assembled and test units consisting of medium-voltage primary incoming section, transformer section, and low-voltage secondary switchgear section.

26 11 16.12 Secondary Unit Substations with Switchboard Secondary

Includes: factory-assembled and test units consisting of medium-voltage primary incoming section, transformer section, and low-voltage secondary switchboard section.

NUMBER	TITLE	EXPLANATION
26 11 16.13	Secondary Unit Substations with Motor Control Center Secondary	Includes: factory-assembled and test units consisting of medium-voltage primary incoming section, transformer section, and low-voltage secondary motor-control-center section, and low-voltage secondary switchboard section.
26 12 00	Medium-Voltage Transformers	Includes: transformers for medium-voltage applications. See: 26 22 00 for low-voltage transformers. 33 73 00 for utility transformers. 34 21 23 for traction power transformer-rectifier units.
26 12 13	Liquid-Filled, Medium-Voltage Transformers	
26 12 16	Dry-Type, Medium-Voltage Transformers	
26 12 19	Pad-Mounted, Liquid-Filled, Medium-Voltage Transformers	
26 13 00	Medium-Voltage Switchgear	Includes: switchgear for medium-voltage applications. See: 26 23 00 for low-voltage switchgear. 33 77 00 for medium-voltage utility switchgear. 34 21 19 for traction power switchgear.
26 13 13	Medium-Voltage Circuit Breaker Switchgear	
26 13 16	Medium-Voltage Fusible Interrupter Switchgear	
26 13 19	Medium-Voltage Vacuum Interrupter Switchgear	
26 13 23	Medium-Voltage Metal-Enclosed Switchgear	
26 13 26	Medium-Voltage Metal-Clad Switchgear	
26 13 29	Medium-Voltage Compartmentalized Switchgear	
26 15 00	Medium-Voltage Enclosed Bus Assemblies	
26 15 13	Medium-Voltage Busways	
26 15 16	Medium-Voltage Cablebus	
26 16 00	Medium-Voltage Metering	
26 18 00	Medium-Voltage Circuit Protection Devices	Includes: circuit protection devices for medium-voltage applications. See: 26 28 00 for low-voltage circuit protective devices. 26 41 23 for lightning protection surge arresters and suppressors. 33 77 00 for medium-voltage utility circuit protection devices.
26 18 13	Medium-Voltage Cutouts	
26 18 16	Medium-Voltage Fuses	

NUMBER	TITLE	EXPLANATION
26 18 19	Medium-Voltage Lightning Arresters	
26 18 23	Medium-Voltage Surge Arresters	
26 18 26	Medium-Voltage Reclosers	
26 18 29	Medium-Voltage Enclosed Bus	
26 18 33	Medium-Voltage Enclosed Fuse Cutouts	
26 18 36	Medium-Voltage Enclosed Fuses	
26 18 39	Medium-Voltage Motor Controllers	

26 20 00 Low-Voltage Electrical Distribution

Includes: overhead power systems, transformers, switchgear, switchboards, panelboards, enclosed bus assemblies, power distribution units, controllers, wiring devices, and circuit protection devices to distribute low-voltage electrical power from the point of voltage transformation to the point of use. Typical voltages: 120, 208, 230, 240, 277, 460, and 480.

Definitions: low-voltage: 600 V and less.

See: 26 05 19 for low-voltage electrical power conductors and cables.

26 10 00 for medium-voltage electrical distribution.

26 30 00 for facility electrical power generating and storing equipment.

26 21 00 Low-Voltage Electrical Service Entrance

See: 26 05 19 for low-voltage electrical power conductors and cables.

26 05 46 for utility poles for electrical systems.

33 71 13 for site electrical transmission towers.

33 71 16 for electrical utility poles.

26 21 13	Low-Voltage Overhead Electrical Service Entrance
26 21 16	Low-Voltage Underground Electrical Service Entrance

26 22 00 Low-Voltage Transformers

Includes: transformers for low-voltage applications.

See: 26 12 00 for medium-voltage transformers.

26 22 13	Low-Voltage Distribution Transformers
26 22 16	Low-Voltage Buck-Boost Transformers
26 22 17	Low-Voltage Transformers for Nonlinear Loads

Includes: enclosed dry-type transformers for nonlinear power loads, similar to computers and other electronic power supplies, with primaries and secondaries rated 600 volts and less.

DIVISION **26**

NUMBER	TITLE	EXPLANATION
26 22 18	Low-Voltage Transformers for Load Centers	Includes: small dry-type transformers with integral circuit breaker load center, with primary and secondary voltages of 600 volts or less.
26 22 19	Control and Signal Transformers	
26 23 00	Low-Voltage Switchgear	Includes: switchgear for low-voltage applications. See: 26 13 00 for medium-voltage switchgear.
26 23 13	Paralleling Low-Voltage Switchgear	
26 24 00	Switchboards and Panelboards	Includes: switchboards, panelboards, and control centers. See: 26 27 16 for electrical cabinets and enclosures. 26 29 13 for enclosed controllers. 26 29 23 for variable-frequency motor controllers.
26 24 13	Switchboards	
26 24 16	Panelboards	
26 24 16.16	Electronically Operated Circuit-Breaker Panelboards	Includes: manual and programmable, electrical branch circuit panelboards using low-voltage or digital control circuits. Includes electrically operated, molded-case circuit breakers.
26 24 19	Motor-Control Centers	
26 25 00	Low-Voltage Enclosed Bus Assemblies	Includes: busway, step bus, and tap boxes. See: 33 72 26 for utility substation bus assemblies.
26 25 13	Low-Voltage Busways	
26 25 16	Low-Voltage Cablebus Systems	
26 27 00	Low-Voltage Distribution Equipment	Includes: wiring devices includes receptacles, switches, dimmers, and finish plates. See: 26 24 00 for switchboards and panelboards. 33 71 73 for utility electric meters. See Also: 11 11 36 for vehicle charging unit.
26 27 13	Electricity Metering	

NUMBER	TITLE	EXPLANATION
26 27 16	Electrical Cabinets and Enclosures	
26 27 19	Multi-Outlet Assemblies	
26 27 23	Indoor Service Poles	
26 27 26	Wiring Devices	
26 27 33	Power Distribution Units	Includes: circuit protection devices for low-voltage applications. Enclosed switches and transfer switches.
		See: 26 24 16 for panelboards.
26 27 73	Door Chimes	
26 28 00	Low-Voltage Circuit Protective Devices	Includes: circuit protection devices for low-voltage applications. Enclosed switches and transfer switches.
		See: 26 18 00 for medium-voltage circuit protection devices.
26 28 13	Fuses	
26 28 16	Enclosed Switches and Circuit Breakers	
26 28 16.13	Enclosed Circuit Breakers	
26 28 16.16	Enclosed Switches	
26 29 00	Low-Voltage Controllers	Includes: contactors and motor controllers.
		May Include: fuses.
		Alternate Terms/Abbreviations: enclosed controllers: motor controllers.
		See: 26 24 19 for motor-control centers. 26 28 13 for fuses.
26 29 13	Enclosed Controllers	
26 29 13.13	Across-the-Line Motor Controllers	
26 29 13.16	Reduced-Voltage Motor Controllers	
26 29 23	Variable-Frequency Motor Controllers	
26 29 33	Controllers for Fire Pump Drivers	
26 29 33.13	Full-Service Controllers for Fire Pump Electric-Motor Drivers	
26 29 33.16	Limited-Service Controllers for Fire Pump Electric-Motor Drivers	
26 29 33.19	Controllers for Fire Pump Diesel Engine Drivers	

NUMBER	TITLE	EXPLANATION
26 30 00	Facility Electrical Power Generating and Storing Equipment	Includes: equipment to generate and store electrical power for a single facility. See: 48 10 00 for electrical power generation equipment.
26 31 00	Photovoltaic Collectors	Includes: solar cells to convert sunlight to electricity. See: 07 31 00 for solar collector roof shingles. 22 33 30 for residential, collector-to-tank, solar-electric domestic water heaters. 23 56 00 for solar energy heating equipment. 42 12 23 for solar process heaters. 42 13 26 for industrial solar radiation heat exchangers. 48 14 00 for solar energy electrical power generation equipment.
26 32 00	Packaged Generator Assemblies	Includes: generators, frequency changers, and rotary converters and uninterruptible power units. See: 23 11 00 for facility fuel piping. 23 24 00 for internal-combustion engine piping. 48 11 00 for fossil fuel plant electrical power generation equipment. 48 13 00 for hydroelectric plant electrical power generation equipment. 48 15 00 for wind energy electrical power generation equipment.
26 32 13	Engine Generators	
26 32 13.13	Diesel-Engine-Driven Generator Sets	
26 32 13.16	Gas-Engine-Driven Generator Sets	
26 32 13.26	Gas-Turbine Engine-Driven Generators	Alternate Terms/Abbreviations: microturbines. See: 48 11 23 for fossil fuel electrical power plant gas turbines.
26 32 16	Steam-Turbine Generators	
26 32 19	Hydro-Turbine Generators	
26 32 23	Wind Energy Equipment	
26 32 23.13	Horizontal-Axis Wind Turbines	Includes: small-scale facility-mounted units. See Also: 48 15 13 for wind turbines for power plant wind turbines.

NUMBER	TITLE	EXPLANATION
26 32 23.16	Vertical-Axis Wind Turbines	Includes: small-scale facility-mounted units. See Also: 48 15 13 for wind turbines for power plant wind turbines.
26 32 26	Frequency Changers	
26 32 29	Rotary Converters	
26 32 33	Rotary Uninterruptible Power Units	
26 32 36	Resistive Load Banks	
26 33 00	Battery Equipment	Includes: batteries, battery racks, battery chargers, static power converters, uninterruptible power supplies, and accessories. May Include: battery-operated emergency light fixtures. See: 25 36 23 for integrated automation battery monitors. 26 31 00 for photovoltaic collectors. 33 72 33 for electrical utility substation. 48 17 13 for electrical power generation batteries. See Also: 26 52 13 for emergency lighting incorporating batteries.
26 33 13	Batteries	
26 33 16	Battery Racks	
26 33 19	Battery Units	
26 33 23	Central Battery Equipment	
26 33 23.13	Central Battery Equipment for Emergency Lighting	
26 33 33	Static Power Converters	
26 33 43	Battery Chargers	
26 33 46	Battery Monitoring	
26 33 53	Static Uninterruptible Power Supply	
26 35 00	Power Filters and Conditioners	Includes: capacitors, chokes and inductors, filters, power factor controllers, and voltage regulators. Alternate Terms/Abbreviations: EMI: electromagnetic interference. RFI: radio frequency interference. Power factor correction equipment: power factor controllers. See: 08 34 46 for RFI shielding doors. 08 56 46 for RFI shielding windows. 13 49 00 for radiation protection. 26 18 23 for medium-voltage surge arresters. 28 41 00 for radiation detection and alarm. 40 91 16 for electromagnetic process measurement devices.

DIVISION **26**

NUMBER	TITLE	EXPLANATION
26 35 13	Capacitors	
26 35 16	Chokes and Inductors	
26 35 23	Electromagnetic-Interference Filters	
26 35 26	Harmonic Filters	
26 35 33	Power Factor Correction Equipment	
26 35 33.13	Medium-Voltage Power Factor Correction Equipment	
26 35 33.16	Low-Voltage Power Factor Correction Equipment	
26 35 36	Slip Controllers	
26 35 43	Static-Frequency Converters	
26 35 46	Radio-Frequency-Interference Filters	
26 35 53	Voltage Regulators	
26 36 00	Transfer Switches	
		Includes: switches transfer from one source of electricity to another.
26 36 13	Manual Transfer Switches	
26 36 23	Automatic Transfer Switches	
26 40 00	Electrical Protection	
26 41 00	Facility Lightning Protection	
		Includes: wiring and equipment for lightning protection.
		Notes: other building elements may be used in lightning protection systems in lieu of air terminals.
		See: 26 18 19 for medium-voltage lightning arresters.
		33 79 00 for site grounding.
		33 79 93 for site lightning protection.
		See Also: 02 30 00 for subsurface investigation for testing of electrical conductance of soils.
26 41 13	Lightning Protection for Structures	
26 41 13.13	Lightning Protection for Buildings	
26 41 16	Lightning Prevention and Dissipation	
26 41 19	Early Streamer Emission Lightning Protection	
26 41 23	Lightning Protection Surge Arresters and Suppressors	
26 43 00	Surge Protective Devices	
		Includes: devices to protect against voltage surges on electrical distribution systems.
26 43 13	Surge Protective Devices for Low-Voltage Electrical Power Circuits	

NUMBER	TITLE	EXPLANATION
26 50 00	Lighting	Includes: luminaries, lighting equipment, ballasts, dimming controls, and lighting accessories. Fluorescent, high intensity discharge, incandescent, mercury vapor, neon, and sodium vapor lighting. Alternate Terms/Abbreviations: HID: high intensity discharge. See: 10 84 00 for gas lighting. 25 36 26 for integrated automation lighting relays. 26 09 23 for lighting controls. 26 20 00 for low-voltage electrical transmission.
26 51 00	Interior Lighting	Includes: lighting for interior locations, except for emergency lighting, lighting in hazardous locations, and special purpose lighting. Chandeliers, troffers. See: 09 54 16 for luminous ceilings. 09 58 00 for integrated ceiling assemblies. 10 14 33 for illuminated panel signage.
26 51 13	Incandescent Interior Lighting	
26 51 16	Fluorescent Interior Lighting	
26 51 19	LED Interior Lighting	
26 51 23	HID Interior Lighting	
26 52 00	Safety Lighting	Includes: equipment for exitway lighting and other emergency applications, including emergency battery units, fixtures with integral batter power supplies. See: 26 52 13 for exit signs.
26 52 13	Emergency and Exit Lighting	See Also: 10 14 00 for signage.
26 52 13.13	Emergency Lighting	
26 52 13.16	Exit Signs	
26 54 00	Classified Location Lighting	Includes: lighting for application in areas classified as hazardous. See: 26 55 33 for hazard warning lighting.
26 54 13	Incandescent Classified Location Lighting	
26 54 16	Fluorescent Classified Location Lighting	

NUMBER	TITLE	EXPLANATION
26 54 19	LED Classified Location Lighting	
26 55 00	Special Purpose Lighting	Includes: lighting equipment for specialized applications. Alternate Terms/Abbreviations: healthcare lighting: medical lighting. See: 11 13 26 for loading dock lights. 11 59 00 for exhibit and display equipment. 11 61 00 for theater and stage equipment. 11 70 00 for healthcare equipment. 13 10 00 for swimming pools. 13 12 00 for fountains. 13 14 00 for aquatic park structures. 13 17 00 for tubs and pools. 26 54 00 for classified location lighting. 34 40 00 for transportation signals. 35 13 13 for navigation signals. See Also: 11 61 00 for theatrical lighting.
26 55 23	Outline Lighting	
26 55 29	Underwater Lighting	
26 55 33	Hazard Warning Lighting	
26 55 36	Obstruction Lighting	
26 55 39	Helipad Lighting	See: 34 43 00 for airfield signaling and control equipment.
26 55 53	Security Lighting	
26 55 59	Display Lighting	
26 55 61	Theatrical Lighting	
26 55 63	Detention Lighting	
26 55 68	Athletic Field Lighting	
26 55 70	Healthcare Lighting	
26 55 83	Broadcast Lighting	

NUMBER	TITLE	EXPLANATION
26 56 00	Exterior Lighting	Includes: lighting equipment for exterior locations, except for special purpose and signal lighting. Airfield general exterior lighting. Alternate Terms/Abbreviations: athletic lighting: sports lighting. See: 10 14 33 for illuminated panel signage. 11 13 26 for loading dock lights. 11 68 23 for exterior court athletic equipment. 32 94 00 for planting accessories. 34 41 13 for traffic signals. 34 42 13 for railway signals. 34 43 13 for airfield signals. 34 43 16 for airfield landing equipment. 34 71 00 for roadway construction. 34 72 00 for railway construction. 34 73 00 for airfield construction. 34 75 00 for roadway equipment.
26 56 13	Lighting Poles and Standards	
26 56 17	Fluorescent Exterior Lighting	
26 56 18	Incandescent Exterior Lighting	
26 56 19	LED Exterior Lighting	
26 56 21	HID Exterior Lighting	

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NUMBER	TITLE	EXPLANATION
DIVISION 27 – COMMUNICATIONS		
27 00 00	Communications	May be used as Division level Section title. See: 02 41 19 for selective demolition of existing communications systems. 07 84 00 for firestopping. 07 92 00 for joint sealants. 08 31 00 for access doors and panels. 09 91 00 for field painting. 25 37 00 for integrated automation instrumentation and terminal devices for communications. 25 57 00 for integrated automation control of communications. 25 97 00 for integrated automation control sequences for communications. 31 23 33 for trenching and backfilling.
27 01 00	Operation and Maintenance of Communications Systems	Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of communications. Level 4 Numbering Recommendation: following numbering is recommended for the creation of Level 4 titles: .51-.59 for maintenance. .61-.69 for repair. .71-.79 for rehabilitation. .81-.89 for replacement. .91-.99 for restoration.
27 01 10	Operation and Maintenance of Structured Cabling and Enclosures	
27 01 20	Operation and Maintenance of Data Communications	
27 01 30	Operation and Maintenance of Voice Communications	
27 01 40	Operation and Maintenance of Audio-Video Communications	
27 01 50	Operation and Maintenance of Distributed Communications and Monitoring	

NUMBER	TITLE	EXPLANATION
27 05 00	Common Work Results for Communications	Includes: subjects common to multiple titles in Division 27. Alternate Terms/Abbreviations: T1: high speed digital network. DSL: digital subscriber line. See: 01 80 00 for performance requirements of subjects common to multiple titles. 05 45 19 for communications metal supports. 13 48 00 for sound, vibration, and seismic control. 25 05 13 for conductors and cables for integrated automation. 25 05 26 for grounding and bonding for integrated automation. 25 05 28 for pathways for integrated automation. 25 05 48 for vibration and seismic control for integrated automation. 25 05 53 for identification for integrated automation. 26 05 43 for underground ducts and raceways for electrical. 26 05 46 for utility poles for electrical. 27 05 26 for grounding and bonding for communications systems. 27 05 28 for pathways for communications systems. 27 15 01 for conductors and cables for electronic safety and security. 28 05 48 for vibration and seismic controls for electronic safety and security. 28 05 53 for identification for electronic safety and security. 33 81 19 for site communications utility poles. 33 82 43 for grounding and bonding for site communications distribution.
27 05 05	Selective Demolition for Communications	See: 02 41 19 for selective demolition for whole structure demolition.
27 05 11	Cameras for Audio-Visual Systems	
27 05 13	Communications Services	
27 05 13.13	Dialtone Services	
27 05 13.23	T1 Services	
27 05 13.33	DSL Services	
27 05 13.43	Cable Services	
27 05 13.53	Satellite Services	
27 05 26	Grounding and Bonding for Communications Systems	
27 05 28	Pathways for Communications Systems	
27 05 29	Hangers and Supports for Communications Systems	
27 05 33	Conduits and Backboxes for Communications Systems	
27 05 36	Cable Trays for Communications Systems	
27 05 39	Surface Raceways for Communications Systems	
27 05 43	Underground Ducts and Raceways for Communications Systems	

NUMBER	TITLE	EXPLANATION
27 05 44	Sleeves and Sleeve Seals for Communications Pathways and Cabling	Includes: penetration methods into and through fire-rated and unrated walls, ceilings, and floors. Sealing of conduit penetrations against water is included. Does Not Include: firestopping requirements.
27 05 46	Utility Poles for Communications Systems	
27 05 48	Vibration and Seismic Controls for Communications Systems	
27 05 48.16	Seismic Controls for Communications Systems	Includes: restraint channel bracings, restraint cables, seismic-restraint accessories, and mechanical and adhesive anchor bolts.
27 05 53	Identification for Communications Systems	
27 06 00	Schedules for Communications	Includes: schedules of items common to multiple titles in Division 27. Notes: a schedule may be included on drawings, in the project manual, or a project book.
27 06 10	Schedules for Structured Cabling and Enclosures	
27 06 20	Schedules for Data Communications	
27 06 30	Schedules for Voice Communications	
27 06 40	Schedules for Audio-Video Communications	
27 06 50	Schedules for Distributed Communications and Monitoring	
27 08 00	Commissioning of Communications	See: 01 91 00 for commissioning of subjects common to multiple divisions.
27 10 00	Structured Cabling	Includes: backbone and horizontal cabling for a facility. See: 27 05 13 for communications services. 27 13 00 for communications backbone cabling. 27 15 00 for communications horizontal cabling. 27 16 00 for communications connecting cords, devices and adapters.
27 11 00	Communications Equipment Room Fittings	Includes: equipment to fit out a communications equipment room or space.
27 11 13	Communications Entrance Protection	
27 11 16	Communications Cabinets, Racks, Frames, and Enclosures	

NUMBER	TITLE	EXPLANATION
27 11 19	Communications Termination Blocks and Patch Panels	
27 11 23	Communications Cable Management and Ladder Rack	
27 11 26	Communications Rack Mounted Power Protection and Power Strips	

27 13 00 **Communications Backbone Cabling**

Includes: inside and outside: copper, fiber, and coaxial cable plant.

Alternate Terms/Abbreviations: optical fiber: fiber optic. T1: high-speed digital network. DSL: digital subscriber line.

See: 27 05 13 for communications services.
 27 11 00 for communications equipment room work.
 27 15 00 for communications horizontal cabling.
 27 15 13 for communications copper horizontal cabling.
 27 15 23 for communications optical fiber horizontal cabling.
 27 15 33 for communications coaxial horizontal cabling.
 27 16 00 for communications connecting cords, devices and adapters.
 33 82 00 for site communications distribution.
 33 82 13 for site copper communications distribution cabling.
 33 82 23 for site optical fiber communications distribution cabling.
 33 82 33 for site coaxial communications distribution cabling.

27 13 13	Communications Copper Backbone Cabling
27 13 13.13	Communications Copper Cable Splicing and Terminations
27 13 23	Communications Optical Fiber Backbone Cabling
27 13 23.13	Communications Optical Fiber Splicing and Terminations
27 13 33	Communications Coaxial Backbone Cabling
27 13 33.13	Communications Coaxial Splicing and Terminations
27 13 43	Communications Services Cabling
27 13 43.13	Dialtone Services Cabling
27 13 43.23	T1 Services Cabling
27 13 43.33	DSL Services Cabling
27 13 43.43	Cable Services Cabling
27 13 43.53	Satellite Services Cabling

NUMBER	TITLE	EXPLANATION
27 15 00	Communications Horizontal Cabling	Includes: horizontal cable plant from communications equipment room to workstation or drop location. Alternate Terms/Abbreviations: CCTV: closed circuit television. IF/RF: intermediate frequency/radio frequency. Optical fiber: fiber optic. See: 27 11 00 for communications equipment room work. 27 13 00 for communications backbone cabling. 27 13 13 for communications copper backbone cabling. 27 13 23 for communications optical fiber backbone cabling. 27 13 33 for communications coaxial backbone cabling. 27 16 00 for communications connecting cords, devices and adapters. 27 20 00 for data communications. 27 30 00 for voice communications. 27 40 00 for audio-video communications. 27 51 13 for paging systems. 27 52 13 for patient monitoring and telemetry systems. 27 52 23 for nurse call/code blue systems. 27 53 19 for internal cellular, paging, and antenna systems.
27 15 01	Communications Horizontal Cabling Applications	Notes: additional Level 4 applications may be added as needed by project.
27 15 01.11	Conductors and Cables for Electronic Safety and Security	
27 15 01.13	Video Surveillance Communications Conductors and Cables	
27 15 01.15	Access Control Communications Conductors and Cables	
27 15 01.16	Voice Communications Horizontal Cabling	
27 15 01.17	Intrusion Detection Communications Conductors and Cables	
27 15 01.19	Fire Alarm Communications Conductors and Cables	
27 15 01.23	Audio-Video Communications Horizontal Cabling	
27 15 01.39	Patient Monitoring and Telemetry Communications Horizontal Cabling	
27 15 01.43	Nurse Call and Intercom Communications Horizontal Cabling	
27 15 01.46	Paging Communications Horizontal Cabling	
27 15 01.49	Intermediate Frequency/Radio Frequency Communications Horizontal Cabling	
27 15 01.53	Antennas Communications Horizontal Cabling	
27 15 13	Communications Copper Horizontal Cabling	
27 15 23	Communications Optical Fiber Horizontal Cabling	
27 15 33	Communications Coaxial Horizontal Cabling	
27 15 43	Communications Faceplates and Connectors	

NUMBER	TITLE	EXPLANATION
27 16 00	Communications Connecting Cords, Devices, and Adapters	Includes: patch cords, station cords and other adapters. See: 27 11 00 for communications equipment room work. 27 13 00 for communications backbone cabling. 27 15 00 for communications horizontal cabling.
27 16 13	Communications Custom Cable Assemblies	
27 16 16	Communications Media Converters, Adapters, and Transceivers	
27 16 19	Communications Patch Cords, Station Cords, and Cross Connect Wire	
27 20 00	Data Communications	See: 27 15 00 for data communications horizontal cabling.
27 21 00	Data Communications Network Equipment	Includes: switching and routing equipment for data communications. Alternate Terms/Abbreviations: CSU/DSU: channel service unit/data service unit. LAN: local area network. WAN: wide area network. See: 25 13 00 for integrated automation control and monitoring network. 27 22 00 for data communications hardware. 27 24 00 for data communications peripheral data equipment. 27 25 00 for data communications software. 27 26 00 for data communications programming and integration services. 28 05 31 for communications equipment for electronic safety and security.
27 21 13	Data Communications Firewalls	
27 21 16	Data Communications Routers, CSU/DSU, Multiplexers, Codecs, and Modems	
27 21 26	Data Communications Network Management	
27 21 29	Data Communications Switches and Hubs	
27 21 33	Data Communications Wireless Access Points	

NUMBER	TITLE	EXPLANATION
27 22 00	Data Communications Hardware	Includes: computer equipment for data communications. See: 11 28 00 for office equipment. 25 14 00 for integrated automation local control units. 27 21 00 for data communications network equipment. 27 24 00 for data communications peripheral data equipment. 27 25 00 for data communications software. 27 26 00 for data communications programming and integration services. 40 95 00 for process control hardware.
27 22 13	Data Communications Mainframes	
27 22 16	Data Communications Storage and Backup	
27 22 19	Data Communications Servers	
27 22 23	Data Communications Desktops	
27 22 26	Data Communications Laptops	
27 22 29	Data Communications Handhelds	
27 24 00	Data Communications Peripheral Data Equipment	Includes: additional equipment for data communications. See: 11 28 00 for office equipment. 27 21 00 for data communications network equipment. 27 22 00 for data communications hardware. 27 25 00 for data communications software. 27 26 00 for data communications programming and integration services. 27 41 00 for audio-video systems. 27 41 23 for audio-video accessories. 40 95 00 for process control hardware.
27 24 13	Printers	
27 24 16	Scanners	
27 24 19	External Drives	
27 24 23	Audio-Video Devices	
27 24 26	Virtual Reality Equipment	
27 24 29	Disaster Recovery Equipment	

NUMBER	TITLE	EXPLANATION
27 25 00	Data Communications Software	Includes: software for data communications. Alternate Terms/Abbreviations: CRM: customer relationship management. VPN: virtual private network. See: 25 15 00 for integrated automation software. 27 21 00 for data communications network equipment. 27 22 00 for data communications hardware. 27 24 00 for data communications peripheral data equipment. 27 26 00 for data communications programming and integration services. 27 34 13 for toll fraud equipment and software. 27 34 16 for telemanagement software. 40 96 00 for process control software.
27 25 13	Virus Protection Software	
27 25 16	Application Suites	
27 25 19	Email Software	
27 25 23	Graphics/Multimedia Software	
27 25 26	Customer Relationship Management Software	
27 25 29	Operating System Software	
27 25 33	Database Software	
27 25 37	Virtual Private Network Software	
27 25 39	Internet Conferencing Software	
27 26 00	Data Communications Programming and Integration Services	Includes: custom programming and integration requirements. See: 27 21 00 for data communications network equipment. 27 22 00 for data communications hardware. 27 24 00 for data communications peripheral data equipment. 27 25 00 for data communications software. See Also: 28 05 45 for systems integration and unified systems.
27 26 13	Web Development	
27 26 16	Database Development	
27 26 19	Application Development	
27 26 23	Network Integration Requirements	
27 26 26	Data Communications Integration Requirements	

NUMBER	TITLE	EXPLANATION
27 30 00	Voice Communications	See: 27 15 00 for voice communications horizontal cabling.
27 31 00	Voice Communications Switching and Routing Equipment	Includes: switching and routing equipment for voice communications. Alternate Terms/Abbreviations: PBX: private branch exchange. IP: internet protocol. See: 27 32 00 for voice communications terminal equipment. 27 33 00 for voice communications messaging. 27 34 00 for call accounting. 27 35 00 for call management.
27 31 13	PBX/ Key Systems	
27 31 23	Internet Protocol Voice Switches	
27 32 00	Voice Communications Terminal Equipment	Includes: telephones and other equipment associated with use of telephone lines. Alternate Terms/Abbreviations: TTY: teletypewriter, also called TDD: telecommunication device for the deaf. Facsimiles: fax machines. See: 10 17 00 for telephone specialties. 10 25 16 for modular service walls. 14 20 00 for elevators. 27 31 00 for voice communications switching and routing equipment. 27 33 00 for voice communications messaging. 27 33 26 for facsimile servers. 27 34 00 for call accounting. 27 35 00 for call management.
27 32 13	Telephone Sets	
27 32 16	Wireless Transceivers	
27 32 23	Elevator Telephones	
27 32 26	Ring-Down Emergency Telephones	
27 32 29	Facsimiles and Modems	
27 32 36	TTY Equipment	
27 32 43	Radio Communications Equipment	

NUMBER	TITLE	EXPLANATION
27 33 00	Voice Communications Messaging	Includes: voice mail, fax servers and other automated response systems. See: 27 31 00 for voice communications switching and routing equipment. 27 32 00 for voice communications terminal equipment. 27 32 29 for facsimiles and modems. 27 34 00 for call accounting. 27 35 00 for call management.
27 33 16	Voice Mail and Auto Attendant	
27 33 23	Interactive Voice Response	
27 33 26	Facsimile Servers	
27 34 00	Call Accounting	Includes: hardware and software that facilitates call accounting. See: 27 25 00 for data communications software. 27 31 00 for voice communications switching and routing equipment. 27 32 00 for voice communications terminal equipment. 27 33 00 for voice communications messaging. 27 35 00 for call management.
27 34 13	Toll Fraud Equipment and Software	
27 34 16	Telemanagement Software	
27 35 00	Call Management	Includes: call center equipment and systems. See: 27 31 00 for voice communications switching and routing equipment. 27 32 00 for voice communications terminal equipment. 27 33 00 for voice communications messaging. 27 34 00 for call accounting.
27 35 13	Digital Voice Announcers	
27 35 16	Automatic Call Distributors	
27 35 19	Call Status and Management Displays	
27 35 23	Dedicated 911 Systems	

NUMBER	TITLE	EXPLANATION
27 40 00	Audio-Video Communications	See: 27 15 00 for audio-video communications horizontal cabling.
27 41 00	Audio-Video Systems	Includes: standalone and integrated audio-video systems and equipment that generally support a single room or space in a facility with many components that function as one system. In the case of CATV, the integrated audio-video system connects to a cable distribution system. Source and production equipment, switches, amplifiers, control interfaces, microphones, and presentation devices. Broadcast studio audio-video equipment. May Include: audio-visual furniture. Alternate Terms/Abbreviations: CATV: community antenna television: community access television: cable television: cable TV. MATV: master antenna television. See: 11 25 00 for hospitality equipment. 11 28 00 for office equipment. 11 50 00 for educational equipment. 11 52 00 for audio-visual equipment. 11 61 00 for theater and stage equipment. 27 15 00 for communications horizontal cabling. 27 24 23 for audio-video devices. 27 51 00 for distributed audio-video communications systems. See Also: 12 56 52 for audio-visual furniture.
27 41 13	Architecturally Integrated Audio-Video Equipment	
27 41 16	Integrated Audio-Video Systems and Equipment	
27 41 16.25	Integrated Audio-Video Systems and Equipment for Restaurants and Bars	
27 41 16.28	Integrated Audio-Video Systems and Equipment for Conference Rooms	
27 41 16.29	Integrated Audio-Video Systems and Equipment for Board Rooms	
27 41 16.51	Integrated Audio-Video Systems and Equipment for Classrooms	
27 41 16.52	Integrated Audio-Video Systems and Equipment for Religious Facilities	
27 41 16.61	Integrated Audio-Video Systems and Equipment for Theaters	
27 41 16.62	Integrated Audio-Video Systems and Equipment for Auditoriums	
27 41 16.63	Integrated Audio-Video Systems and Equipment for Stadiums and Arenas	
27 41 19	Portable Audio-Video Equipment	
27 41 23	Audio-Video Accessories	
27 41 33	Master Antenna Television Systems	
27 41 43	Audio-Video Conferencing	

NUMBER	TITLE	EXPLANATION
27 42 00	Electronic Digital Systems	Includes: hardware, software, and equipment for digital signage systems. See: 10 13 13 for self-contained electronic directories. 10 14 63 for self-contained electronic message signage.
27 42 13	Point of Sale Systems	
27 42 16	Transportation Information Display Systems	
27 42 19	Public Information Systems	
27 50 00	Distributed Communications and Monitoring Systems	
27 51 00	Distributed Audio-Video Communications Systems	Includes: paging, public address, sound masking, and intercommunications systems that service all or part of a facility. See: 11 20 00 for commercial equipment. 11 28 00 for office equipment. 11 30 00 for residential equipment. 11 50 00 for educational equipment. 11 52 00 for audio-visual equipment. 11 70 00 for healthcare equipment. 27 15 00 for paging communications horizontal cabling. 27 40 00 for audio-video communications.
27 51 13	Paging Systems	
27 51 13.13	Overhead Paging Systems	
27 51 16	Public Address Systems	
27 51 19	Sound Masking Systems	
27 51 23	Intercommunications and Program Systems	Includes: paging, public address, sound masking, access control, and intercommunications systems that service all or part of a facility.
27 51 23.20	Commercial Intercommunications and Program Systems	
27 51 23.30	Residential Intercommunications and Program Systems	
27 51 23.50	Educational Intercommunications and Program Systems	
27 51 23.63	Detention Intercommunications and Program Systems	
27 51 23.70	Healthcare Intercommunications and Program Systems	
27 51 26	Assistive Listening Systems	
27 51 29	Emergency Communications Systems	Includes: communication rescue assistance equipment to aid first responders, area of refuge communication systems, and other life safety communications systems typically powered on emergency power networks. See Also: 28 49 00 for electronic personal protection systems.

NUMBER	TITLE	EXPLANATION
27 51 29.13	Rescue Assistance Signal Systems	
27 52 00	Healthcare Communications and Monitoring Systems	Includes: communications and monitoring systems for healthcare applications. See: 11 70 00 for healthcare equipment. 27 15 00 for nurse call communications horizontal cabling. 27 15 00 for patient monitoring and telemetry communications horizontal cabling.
27 52 13	Patient Monitoring and Telemetry Systems	
27 52 16	Telemedicine Systems	
27 52 19	Healthcare Imaging Systems	
27 52 23	Nurse Call/Code Blue Systems	
27 53 00	Distributed Systems	Includes: clocks, interior antenna systems and tracking systems. Alternate Terms/Abbreviations: RF: radio frequency. See: 10 25 16 for modular service walls. 10 74 13 for exterior clocks. 12 46 19 for interior clocks.
27 53 13	Clock Systems	
27 53 13.13	Wireless Clock Systems	
27 53 16	Infrared and Radio Frequency Tracking Systems	
27 53 19	Internal Cellular, Paging, and Antenna Systems	

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NUMBER	TITLE	EXPLANATION
DIVISION 28 – ELECTRONIC SAFETY AND SECURITY		
28 00 00	Electronic Safety and Security	May be used as Division level Section title. See: 02 41 19 for selective demolition of existing electronic safety and security systems. 07 84 00 for firestopping. 08 31 00 for access doors and panels. 08 42 00 for entrances. 08 56 53 for security windows. 08 70 00 for hardware. 08 87 23 for safety and security films. 10 14 63 for electronic message signage. 11 12 00 for parking control equipment. 11 97 00 for security equipment. 11 98 00 for detention equipment. 12 59 00 for systems furniture. 25 38 00 for integrated automation instrumentation and terminal devices for electronic safety and security. 25 58 00 for integrated automation control of electronic safety and security. 25 98 00 for integrated automation control sequences for electronic safety and security. 26 55 53 for security lighting. 27 11 00 for communications equipment room fittings. 27 15 01 for communications horizontal cabling applications. 27 20 00 for data communications. 27 30 00 for voice communications. 27 40 00 for audio-video communications. 27 50 00 for distributed communications and monitoring systems. 32 31 13.53 for high-security chain link fences and gates. 33 79 00 for site grounding. 33 80 00 for communications utilities. 34 40 00 for transportation signaling and control equipment. 40 63 00 for control system equipment.

NUMBER	TITLE	EXPLANATION
28 01 00	Operation and Maintenance of Electronic Safety and Security	Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of electronic safety and security. Level 4 Numbering Recommendation: Level 4 titles for electronic safety and security are provided as examples for user's creation of more detailed titles for other subjects as required. Following Level 4 numbering is recommended: .51-.59 for maintenance. .61-.69 for repair. .71-.79 for rehabilitation. .81-.89 for replacement. .91-.99 for restoration.
28 01 10	Operation and Maintenance of Access Control	
28 01 10.51	Maintenance and Administration of Access Control	
28 01 10.71	Revisions and Upgrades of Access Control	
28 01 20	Operation and Maintenance of Video Surveillance	
28 01 20.51	Maintenance and Administration of Video Surveillance	
28 01 20.71	Revisions and Upgrades of Video Surveillance	
28 01 30	Operation and Maintenance of Security Detection, Alarm, and Monitoring	
28 01 30.51	Maintenance and Administration of Security Detection, Alarm, and Monitoring	
28 01 30.71	Revisions and Upgrades of Security Detection, Alarm, and Monitoring	
28 01 50	Operation and Maintenance of Specialized Systems	
28 01 50.51	Maintenance and Administration of Specialized Systems	
28 01 50.71	Revisions and Upgrades of Specialized Systems	
28 01 60	Operation and Maintenance of Mass Notification	
28 01 60.51	Maintenance and Administration of Mass Notification	
28 01 60.71	Revisions and Upgrades of Mass Notification	
28 01 70	Operation and Maintenance of Life Safety	
28 01 70.51	Maintenance and Administration of Life Safety	
28 01 70.71	Revisions and Upgrades of Life Safety	
28 01 80	Operation and Maintenance of Fire Detection and Alarm	
28 01 80.51	Maintenance and Administration of Fire Detection and Alarm	
28 01 80.71	Revisions and Upgrades of Fire Detection and Alarm	
28 05 00	Common Work Results for Electronic Safety and Security	See: 27 11 00 for communications equipment room fittings. 27 15 01 for communications horizontal cabling applications. 27 20 00 for data communications. 27 30 00 for voice communications. 27 40 00 for audio-video communications.
28 05 05	Selective Security Systems Demolition	

NUMBER	TITLE	EXPLANATION
28 05 07	Power Sources for Electronic Safety and Security	Includes: electronic low-voltage power sources specifically for electronic safety and security purposes, also covering UL294 and UL294B approved power supplies for access control. Alternate Terms/Abbreviations: UL294. PoE: Power over Ethernet See Also: 26 20 00 for low-voltage electrical distribution.
28 05 07.11	Power Sources for Access Control	Includes: UL294 and UL294B approved power supplies for access control.
28 05 07.13	Power Sources for Video Surveillance	
28 05 07.15	Power Sources for Intrusion Detection	
28 05 07.17	Power Sources for Fire Detection and Alarm	
28 05 07.19	Solar Power Sources for Electronic Safety and Security	See: 48 14 13: for solar energy collectors.
28 05 07.21	PoE Power Sources for Electronic Safety and Security	Includes: UL294B mid-span injectors or network switches used exclusively for access control. Alternate Terms/Abbreviations: UL294, UL294B. PoE: Power over Ethernet. See: 28 05 07 for power sources for access control.
28 05 07.23	Uninterruptible Power Supply	See Also: 26 33 53 for static uninterruptible power supply.
28 05 07.25	Power Source Monitoring	See Also: 26 33 46 for battery monitoring.
28 05 07.27	Power Source Monitoring Appliances	
28 05 07.29	Power Source Monitoring Software	
28 05 09	Surge Protection for Electronic Safety and Security	See Also: 26 43 13 for surge protective devices for low-voltage electrical power circuits.
28 05 09.11	Surge Protection for Access Control	
28 05 09.13	Surge Protection for Video Surveillance	
28 05 09.15	Surge Protection for Intrusion Detection	
28 05 09.17	Surge Protection for Fire Detection and Alarm	
28 05 11	Cyber Security Requirements for Electronic Safety and Security	

DIVISION 28

NUMBER	TITLE	EXPLANATION
28 05 13	Servers, Workstations and Storage for Electronic Safety and Security	Includes: storage hardware, software, and services supporting security, life safety, and fire alarm multiple requirements. See: 27 21 00 for data communications network equipment. 27 22 16 data communications storage and backup. 27 24 00 for data communications peripheral data equipment. See Also: 27 22 16 for data communications storage and backup.
28 05 13.11	System Requirements for Servers, Workstations and Storage	Includes: requirements on redundancy and resiliency. Notes: this Section is not intended to duplicate or supplant Division 01 General Requirements content.
28 05 15	Servers for Electronic Safety and Security	See Also: 27 22 19 for data communications servers.
28 05 17	Workstations for Electronic Safety and Security	See Also: 27 22 23 for data communications desktops.
28 05 19	Storage Appliances for Electronic Safety and Security	Alternate Terms/Abbreviations: DVR: digital video recorder. NVR: network video recorder. See Also: 27 42 13 for of sale systems.
28 05 19.11	Digital Video Recorders	
28 05 19.13	Hybrid Digital Video Recorders	
28 05 19.15	Network Video Recorders	
28 05 21	Network Attached Storage for Electronic Safety and Security	Alternate Terms/Abbreviations: NAS: network attached storage.
28 05 23	Storage Area Network for Electronic Safety and Security	Alternate Terms/Abbreviations: SAN: storage area network
28 05 25	Cloud Based Storage for Electronic Safety and Security	
28 05 27	Archival Systems for Electronic Safety and Security	
28 05 27.11	Digital Tape Storage	
28 05 27.13	Storage Media	
28 05 29	Storage Management Software for Electronic Safety and Security	See Also: 27 25 00 for data communications software.
28 05 31	Communications Equipment for Electronic Safety and Security	See Also: 27 00 00 for data communications.

NUMBER	TITLE	EXPLANATION
28 05 33	Safety and Security Network Communications Equipment	Includes: data communications network equipment is to be dedicated for security use only. See Also: 27 22 00 for data communications hardware.
28 05 33.11	Security Data Communications Firewalls	See Also: 27 21 13 for data communications firewalls.
28 05 33.13	Security Data Communications Routers	See Also: 27 21 16 for data communications routers, CSU/DSU, multiplexers, codecs, and modems.
28 05 33.15	Security Data Communications Power-over-Ethernet Switches	See Also: 27 21 29 for data communications switches and hubs.
28 05 33.17	Security Data Communications Non-Power-over-Ethernet Switches	See Also: 27 21 29 for data communications switches and hubs.
28 05 33.19	Security Data Communications Media Converters	See Also: 27 16 16 for communications media converters, adapters, and transceivers.
28 05 35	Security Data Communications Wireless Transmission Equipment	See Also: 27 21 33 for data communications wireless access points.
28 05 35.11	Microwave Transmission Equipment	See Also: 33 83 16 for microwave transmitters and receivers.
28 05 37	Security Voice Communications Distributed Antenna System	Includes: voice communications systems serving a dedicated security function in a standalone security system, and/or when there is no other voice communication distributed antenna system being specified under Division 27. See Also: 27 53 19 for internal cellular, paging, and antenna systems.
28 05 37.11	Bi-directional Amplifiers	
28 05 37.13	Repeaters	
28 05 37.15	Remote Repeaters	

DIVISION 28

NUMBER	TITLE	EXPLANATION
28 05 39	Antennas	Includes: antennas used when the section for security voice communications distributed antenna system is being utilized. See: 27 15 01 for antennas communications horizontal cabling. See Also: 27 53 19 for internal cellular, paging, and antenna systems.
28 05 39.11	Donor Antennas	
28 05 39.13	Multi-band Antennas	
28 05 39.15	Band Specific Antennas	
28 05 39.17	Radiating Coaxial Cable	
28 05 39.19	Other Antennas	
28 05 41	Signal Controllers	Includes: components used when the section for security voice communications distributed antenna system is being utilized. See: 27 15 01 for antennas communications horizontal cabling.
28 05 41.11	Splitters	
28 05 41.13	Directional Couplers	
28 05 43	Security Voice Communications Private Mobile Radio System	Includes: mobile radio equipment used when the section for security voice communications distributed antenna system is being utilized. See Also: 27 32 43 for radio communications equipment.
28 05 43.11	Base Station	
28 05 43.13	Handheld Radio	
28 05 44	Sleeves and Sleeve Seals for Electronic Safety and Security Pathways and Cabling	Includes: penetration methods into and through fire-rated and unrated walls, ceilings, and floors. Sealing of conduit penetrations against water is included. Does Not Include: firestopping requirements.
28 05 45	Systems Integration and Unified Systems	Includes: requirements to integrate various Division 28 and other subsystems into a functional integrated system. See Also: 27 26 00 for data communications programming and integration services.
28 05 45.11	Mechanical	
28 05 45.13	Electrical	
28 05 45.15	Information	

NUMBER	TITLE	EXPLANATION
28 05 48	Vibration and Seismic Controls for Electronic Safety and Security	
28 05 48.16	Seismic Controls for Electronic Safety and Security	Includes: restraint channel bracings, restraint cables, seismic-restraint accessories, and mechanical and adhesive anchor bolts.
28 05 53	Identification for Electronic Safety and Security	
28 06 00	Schedules for Electronic Safety and Security	Includes: schedules of items common to multiple titles in Division 28. Notes: a schedule may be included on drawings, in the project manual, or a project book.
28 06 10	Schedules for Access Control	See Also: 08 06 71 for door hardware schedule.
28 06 20	Schedules for Video Surveillance	
28 06 30	Schedules for Security Detection, Alarm, and Monitoring	
28 06 40	Schedules for Detention Monitoring and Control	
28 06 50	Schedules for Information Management and Display	
28 06 60	Schedules for Mass Notification	
28 06 70	Schedules for Life Safety	
28 06 80	Schedules for Fire Detection and Alarm	
28 08 00	Commissioning of Electronic Safety and Security	Includes: commissioning of items common to multiple titles in Division 28. See: 01 91 00 for commissioning of subjects common to multiple Divisions.
28 08 11	Testing for Baseline Performance Criteria	Includes: parameters that provide the minimum acceptable performance for testing and qualification prior to commissioning, validating all aspects have correct signal strength, consistent signal transmission, perform basic function reliably, are programmed correctly and accurately, etc.

NUMBER	TITLE	EXPLANATION
28 10 00	Access Control	Includes: control and monitoring equipment and systems for controlling access to a facility. See: 08 34 53 for security doors and frames. 08 42 33 for revolving door entrances. 08 71 00 for door hardware. 08 74 00 for non-integrated access control hardware. 11 12 00 for parking control equipment. 11 14 00 for pedestrian control equipment. 11 15 00 for security control equipment. 11 98 00 for detention equipment. 26 05 33 for raceway and boxes for electrical. 27 05 28 for pathways for communications. 27 05 33 for conduits and backboxes for communications systems. 27 15 01 for access control communications conductors and cables. 28 05 07 for power sources for access control. 28 05 13 for servers, workstations, and storage for electronic safety and security. 28 05 31 for communications equipment for electronic safety and security. 28 20 00 for video surveillance. 28 31 00 for intrusion detection. 28 46 00 for fire detection and alarm. 28 51 00 for information management and presentation. 28 52 00 for detention security systems. 34 50 00 for transportation fare collection equipment. 41 24 36 for process metal detectors.
28 11 00	Access Control Global Applications	
28 12 00	Special Requirements for Access Control Systems	
28 12 01	System Requirements for Access Control Systems	Notes: this Section is not intended to duplicate or supplant Division 01 General Requirements content.
28 13 00	Access Control Software and Database Management	See Also: 28 05 13 for servers, workstations, and storage. 28 23 00 video management system. 28 31 00 intrusion detection. 28 51 00 information management and presentation.
28 13 11	Access Control Software	
28 13 11.11	Access Control Operating Systems	
28 13 11.13	Access Control Application Services	
28 13 11.15	Access Control Mobile Applications	

NUMBER	TITLE	EXPLANATION
28 13 13	Access Control Software Interfaces	See Also: 28 16 00 for access control interfaces.
28 13 15	Access Control Cloud Services	
28 14 00	Access Control System Hardware	Includes: access control hardware that functions at the system level, i.e., set back from and capable of supporting multiple openings.
28 14 11	Access Control Network Controllers	
28 14 13	Access Control Door Controllers	
28 14 15	Access Control Inputs and Outputs	
28 14 17	Access Control Printers and Encoders	
28 14 19	Access Control Enclosures	
28 15 00	Integrated Access Control Hardware Devices	Includes: electronic devices that control access and are integrated with another system or systems.
		See Also: 08 06 71 for door hardware schedule. 28 06 10 for schedules for access control.
28 15 11	Integrated Credential Readers and Entry Management	
28 15 11.11	Standard Card Readers	
28 15 11.13	Keypads	
28 15 11.15	Biometric Identity Devices	
28 15 11.17	Combination Reader Devices	Includes: multi-technology readers (e.g. proximity, keypad, Transportation Worker Identification Credential (TWIC) readers with proximity, PIN code and biometric).
28 15 13	Access Control Credentials	
28 15 13.11	Access Control Cards	
28 15 13.13	Wireless Key Fobs	
28 15 15	Electrified Locking Devices and Accessories	Includes: electrified locking devices and accessories (mortise, cylindrical, exit devices, electric strikes, electromagnetic locks).
28 15 15.11	Direct Interface Integrated Locking Devices	Includes: devices incorporating multiple technologies, such as Wiegand and RS-485.
28 15 15.13	Wireless Integrated Locking Devices	
28 15 15.15	Intelligent Ethernet Integrated Locking Devices	
28 15 15.17	Status Monitoring and Egress Devices	Includes: door position switches, REX motion sensors, push buttons, automatic operators.
		Alternate Terms/Abbreviations: REX: Request to Exit.

DIVISION 28

NUMBER	TITLE	EXPLANATION
28 15 17	Egress Management Devices	
28 15 17.11	Magnetic Lock Delayed Egress Devices	
28 15 17.13	Panic Hardware Delayed Egress Devices	
28 15 19	Access Control Remote Devices	
28 15 19.11	Smart Phone Interface Requirements	Alternate Terms/Abbreviations: NFC: near field communications.
28 15 19.13	Wireless Access Control Devices	Includes: transponders and corresponding receivers.
28 15 21	Telephone Entry Systems	
28 15 23	Intercom Entry Systems	See Also: 11 98 00 for detention equipment. 27 50 00 for distributed communications and monitoring systems. 27 51 00 for distributed audio-video communications systems. 27 51 23 for intercommunications and program systems.
28 15 23.11	Audio Intercom (Analog)	
28 15 23.13	Audio-Video Intercom (Analog)	
28 15 23.15	Audio Intercom (IP)	
28 15 23.17	Audio-Video Intercom (IP)	
28 15 25	Electronic Key Management Systems	See Also: 08 79 13 for key storage equipment. 10 41 16 for emergency key cabinets. 11 12 13 for parking key and card control units.
28 15 27	Access Control Electronic Turnstiles and Mobility Systems	Includes: turnstile systems incorporating an electronic access control requirement. See: 11 14 13 for turnstiles.
28 16 00	Access Control Interfaces	See Also: 28 05 45 for systems integration and unified systems.
28 16 11	Access Control Interfaces to Access Control Hardware	See: 08 34 53 for security doors and frames. 08 42 33 for revolving door entrances. 08 71 00 for door hardware. 08 74 00 for non-integrated access control hardware.
28 16 13	Access Control Interfaces to Intrusion Detection	
28 16 15	Access Control Interfaces to Video Surveillance	
28 16 17	Access Control Interfaces to Fire Alarm	

NUMBER	TITLE	EXPLANATION
28 16 19	Access Control Interfaces to Enterprise Software	Includes: access control interfaces to enterprise software business application packages.
28 16 21	Access Control Interfaces to Mechanical Systems	Includes: interfaces to HVAC. Alternate Terms/Abbreviations: HVAC: heating, ventilation, and air conditioning.
28 16 23	Access Control Interfaces to Electrical Systems	Includes: interfaces to lighting.
28 16 25	Access Control Interfaces to Electronic Key Management System	
28 16 27	Access Control Interfaces to Elevator Controls	See Also: 14 28 00 for elevator equipment and controls.
28 16 29	Access Control Interfaces to Parking Equipment	See Also: 11 12 00 for parking control equipment.
28 16 31	Access Control Interfaces to Perimeter Security Systems	
28 17 00	Access Control Identification Management Systems	
28 17 11	Visitor Management Systems	
28 17 13	Self Check in and Kiosk Visitor Systems	
28 17 15	Contractor and Vendor Management Systems	
28 17 17	Mobile Access Identification Management Systems	
28 18 00	Security Access Detection Equipment	
28 18 11	Security Access Metal Detectors	
28 18 13	Security Access X-ray Equipment	
28 18 15	Security Access Explosive Detection Equipment	
28 18 17	Security Access Sniffing Equipment	
28 19 00	Access Control Vehicle Identification Systems	
28 19 11	Loading Dock Access Management Systems	See: 11 13 00 for loading dock equipment.
28 19 13	Parking Garage Access Management Systems	See: 11 12 00 for parking control equipment.
28 19 15	Perimeter Vehicle Access Management Systems	See: 28 31 21 for area and perimeter intrusion detection.

NUMBER	TITLE	EXPLANATION
28 20 00	Video Surveillance	Includes: video surveillance cameras and supporting hardware and video management systems. See: 26 05 33 for raceway and boxes for electrical. 27 05 28 for pathways for communication systems. 27 05 33 for conduits and backboxes for communications systems. 27 15 01 for video surveillance communications conductors and cables. 28 05 07 for power sources for video surveillance. 28 05 13 for servers, workstations, and storage for electronic safety and security. 28 05 31 for communications equipment for electronic safety and security. 28 10 00 for access control. 28 30 00 for security detection, alarm, and monitoring. 28 51 00 for information management and presentation. 28 52 00 for detention security systems. See Also: 26 55 53 for security lighting.
28 21 00	Surveillance Cameras	Includes: any type of video surveillance camera.
28 21 11	Analog Camera	
28 21 13	IP Cameras	Includes: high-definition and mega-pixel cameras.
28 21 13.11	Panoramic IP Cameras	Includes: 180- and 360-degree cameras.
28 21 13.13	Analytic Packages for IP Cameras	
28 21 15	Specialty Cameras	Includes: thermal and combination cameras.
28 21 17	Camera Housings	Includes: housings not integral to the camera.
28 21 19	Camera Mounts	Includes: mounts not integral to the camera.
28 21 21	Illuminators	Includes: visible and infrared camera illuminators.

NUMBER	TITLE	EXPLANATION
28 23 00	Video Management System	Includes: video management platforms and attendant software, licenses, and interfaces. See Also: 28 05 13 for servers, workstations, and storage. 28 51 19 for control room monitoring equipment, including displays and video walls.
28 23 11	Video Management System Analytics	Includes: separate stand-alone analytic equipment.
28 23 13	Video Management System Interfaces	Includes: definition of interfaces with external systems for alarm call-up triggers and events.
28 25 00	Video Surveillance Positioning Equipment	Includes: specialty pan/tilt controllers for camera housings, camera poles and camera lowering equipment.
28 30 00	Security Detection, Alarm, and Monitoring	Includes: hardware, software, and equipment for building, area, and perimeter intrusion systems and perimeter protection systems, alarm operations and monitoring centers, asset and personnel tracking system, and audio monitoring equipment. See: 26 05 33 for raceway and boxes for electrical. 27 05 28 for pathways for communication systems. 27 05 33 for conduits and backboxes for communications systems. 27 15 01 for intrusion detection communications conductors and cables. 28 05 07 for power sources for intrusion detection. 28 05 13 for servers, workstations, and storage for electronic safety and security. 28 05 31 for communications equipment for electronic safety and security. 28 10 00 for access control. 28 20 00 for video surveillance. 28 51 00 for information management and presentation. 28 52 00 for detention security systems.
28 31 00	Intrusion Detection	
28 31 11	Building Intrusion Detection	Includes: hardware, software, and equipment for burglary systems. Alternate Terms/Abbreviations: GUI: graphical user interface.
28 31 11.11	Building Intrusion Detection Control, GUI, and Logic Systems	

DIVISION **28**

NUMBER	TITLE	EXPLANATION
28 31 11.13	Building Intrusion Detection Remote Devices and Sensors	
28 31 21	Area and Perimeter Intrusion Detection	
28 31 21.11	Fiber Optic Area and Perimeter Security Systems	
28 31 21.13	Microwave Area and Perimeter Security Systems	
28 31 21.15	Integrated Video Area and Perimeter Security Systems	
28 31 21.17	Fixed Optical Beam Area and Perimeter Security Systems	
28 31 21.19	LIDAR Area and Perimeter Security Systems	Alternate Terms/Abbreviations: LIDAR: light detection and ranging.
28 31 21.21	Coaxial Area and Perimeter Security Systems	
28 31 21.23	Area and Perimeter Intrusion Detection Devices and Sensors	Includes: detecting area and perimeter devices and sensors provisioned as individual components.
28 31 31	Intrusion Detection Interfaces	
28 31 31.11	Intrusion Detection Interfaces to Security Monitoring and Control	
28 31 31.13	Intrusion Detection Interfaces to Access Control System	
28 31 31.15	Intrusion Detection Interfaces to Video Surveillance	
28 31 31.17	Intrusion Detection Interfaces to Information Management and Presentation	
28 31 31.19	Intrusion Detection Interfaces to Mass Notification	
28 31 31.21	Intrusion Detection Interfaces to Life Safety	
28 31 31.23	Intrusion Detection Interfaces to Fire Alarm	
28 31 31.25	Intrusion Detection Interfaces to Temperature Controls	Includes: interfaces to food service equipment, freezer units, process controllers, environmental control processors.
		See Also: 11 41 00 for food service equipment. 23 09 13 for temperature sensors for HVAC.
28 33 00	Security Monitoring and Control	Includes: requirements for security operations centers, monitoring and control software and services, asset and personnel tracking systems.
		See Also: 22 09 63 for medical gas alarms.
28 33 11	Electronic Structural Monitoring Systems	
28 33 13	Security Operations and Monitoring Centers	
28 33 15	Security Monitoring and Control Software	
28 33 17	Security Monitoring and Control Services	
28 35 00	Tracking Systems	Alternate Terms/Abbreviations: RF: radio frequency. GPS: global positioning systems.
28 35 11	Asset Tracking Systems	See Also: 27 53 16 for infrared and radio frequency tracking systems.
28 35 11.11	RF Asset Tracking Systems	

NUMBER	TITLE	EXPLANATION
28 35 11.13	GPS Asset Tracking Systems	
28 35 11.15	Cellular Asset Tracking Systems	
28 35 13	Personnel Tracking Systems	See Also: 27 53 16 for infrared and radio frequency tracking systems.
28 35 13.11	RF Personnel Tracking Systems	
28 35 13.13	GPS Personnel Tracking Systems	
28 35 13.15	Cellular Personnel Tracking Systems	
28 37 00	Audio Monitoring	
28 37 11	Audio Monitoring Devices	
28 37 13	Gunshot Detection Systems	
28 40 00	Life Safety	Includes: systems and equipment that monitor and report the need for assistance or potentially life threatening situations. See: 26 05 33 for raceway and boxes for electrical. 27 05 28 for pathways for communication systems. 27 05 33 for conduits and backboxes for communications systems. 27 15 01 for fire alarm communications conductors and cables. 28 05 07 for power sources for fire detection and alarm. 28 05 13 for servers, workstations, and storage for electronic safety and security. 28 05 31 for communications equipment for electronic safety and security. 28 10 00 for access control. 28 20 00 for video surveillance. 28 30 00 for security detection, alarm, and monitoring. 28 51 00 for information management and presentation.
28 41 00	Radiation Detection and Alarm	Includes: systems and equipment for the detection and notification of radioactive presence. Alternate Terms/Abbreviations: GUI: graphical user interface. See: 11 77 00 for radiology equipment. 13 49 00 for radiation protection. 28 40 00 for security monitoring and control. 28 60 00 for mass notification. 28 80 00 for fire detection and alarm.
28 41 11	Radiation Detection and Alarm Control, GUI, and Logic Systems	
28 41 13	Radiation Detection and Alarm Integrated Audio Evacuation Systems	
28 41 15	Radiation Detection Sensors	

NUMBER	TITLE	EXPLANATION
28 41 17	Radiation Dosimeters	
28 42 00	Gas Detection and Alarm	Includes: systems and equipment for the detection and notification of gas presence. Alternate Terms/Abbreviations: GUI: graphical user interface. See: 23 10 00 for facility fuel systems. 28 40 00 for security monitoring and control. 28 43 00 for fuel-oil detection and alarm. 28 60 00 for mass notification. 28 80 00 for fire detection and alarm. 33 50 00 for fuel distribution utilities. 40 10 00 for fuel gases process piping.
28 42 11	Gas Detection and Alarm Control, GUI, and Logic Systems	
28 42 13	Gas Detection and Alarm Integrated Audio Evacuation Systems	
28 42 15	Gas Detection Sensors	
28 43 00	Fuel Oil Detection and Alarm	Includes: systems and equipment for the detection and notification of fuel oil presence. Alternate Terms/Abbreviations: GUI: graphical user interface. See: 23 11 13 for facility fuel-oil piping. 23 11 13 for facility fuel-oil pumping. 23 12 13 for facility fuel-oil pumps. 23 13 00 for facility fuel storage tanks. 28 40 00 for security monitoring and control. 28 42 00 for gas detection and alarm. 28 60 00 for mass notification. 28 80 00 for fire detection and alarm. 33 52 13 for fuel-oil distribution utilities. 40 21 00 for liquid fuel process piping.
28 43 11	Fuel Oil Detection and Alarm Control, GUI, and Logic Systems	
28 43 13	Fuel Oil Detection and Alarm Integrated Audio Evacuation Systems	
28 43 15	Fuel Oil Detection Sensors	

NUMBER	TITLE	EXPLANATION
28 44 00	Refrigerant Detection and Alarm	Includes: systems and equipment for the detection and notification of refrigerant presence. Alternate Terms/Abbreviations: GUI: graphical user interface. See: 23 23 00 for refrigerant piping. 23 61 00 for refrigerant compressors. 23 62 00 for packaged compressor and condenser units. 23 63 00 for refrigerant condensers. 28 40 00 for security monitoring and control. 28 60 00 for mass notification. 28 80 00 for fire detection and alarm.
28 44 11	Refrigerant Detection and Alarm Control, GUI, and Logic Systems	
28 44 13	Refrigerant Detection and Alarm Integrated Audio Evacuation Systems	
28 44 15	Refrigerant Detection Sensors	
28 45 00	Water Detection and Alarm	
28 45 11	Water Detection and Alarm Control, GUI, and Logic Systems	
28 45 13	Water Detection Sensors	
28 45 15	Building Envelope Water Detection and Alarm	
28 46 00	Fire Detection and Alarm	Includes: monitoring and recording of fire protection devices and fire-suppression systems; audio and/or visual warning systems for inhabitants of structures and for firefighting authorities; activation of elevators and air handling systems to a predetermined fire mode, and other fire protection functions. See: 10 44 00 for fire protection specialties. 27 15 01 for fire alarm communications conductors and cables. 28 32 00 for radiation detection and alarm. 28 33 00 for gas detection and alarm. 28 35 00 for refrigerant detection and alarm. 28 43 00 for fuel-oil detection and alarm. Division 21 for fire-suppression systems.
28 46 10	Detection and Initiation	Includes: interfaces from external systems.
28 46 11	Fire Sensors and Detectors	
28 46 11.11	Linear Heat Detection Sensors	
28 46 11.13	Spot-type Heat Detection Sensors	
28 46 11.15	Duct Smoke Detection Sensors	
28 46 11.17	Beam Smoke Detection Sensors	
28 46 11.19	Video Smoke Detection Sensors	
28 46 11.21	Carbon-monoxide Detection Sensors	

DIVISION 28

NUMBER	TITLE	EXPLANATION
28 46 11.23	Combination Sensors	Includes: sensors which combine two or more sensor technologies into one sensing device.
28 46 11.25	Air Sampling Systems	
28 46 11.27	Other Sensors	
28 46 12	Other Initiating Devices	
28 46 12.11	Fire Alarm Pull Stations	
28 46 12.13	Fire Alarm Level Detector Switches	
28 46 12.15	Fire Alarm Flow Switches	
28 46 12.17	Fire Alarm Pressure Sensors	
28 46 12.19	Sprinkler Supervisory Devices	
28 46 21	Fire Alarm	
28 46 21.11	Addressable Fire-alarm Systems	
28 46 21.13	Conventional Fire-alarm Systems	
28 46 21.15	Protected Premise Systems	
28 46 21.17	Proprietary Systems	
28 46 21.19	Central Station Systems	
28 46 21.21	Remote Station Systems	
28 46 21.23	Public Emergency Alarm Reporting Systems	
28 46 22	Network Command Centers, Printers, and Peripherals for Fire Systems	
28 46 23	Fire Alarm Notification Appliances	
28 46 23.11	Fire Alarm Horns and Strobes	
28 46 23.13	Fire Alarm Speakers	
28 46 23.15	Fire Alarm Low Frequency Sounders	
28 46 23.17	Wide Band Sounders	Includes: devices used in directional evacuation systems.
28 46 24	Fire Detection and Alarm Emergency Control Function Interfaces	Includes: interfaces to remote monitoring, access control video surveillance, intrusion detection, elevator control. See: 14 28 16 elevator controls. 28 10 00 for access control. 28 20 00 for video surveillance. 28 30 00 for security detection, alarm, and monitoring. 28 50 00 for specialized systems.
28 46 24.11	Fire Detection and Alarm Interfaces to Remote Monitoring	
28 46 24.13	Fire Detection and Alarm Interfaces to Access Control Hardware	
28 46 24.15	Fire Detection and Alarm Interfaces to Access Control System	
28 46 24.17	Fire Detection and Alarm Interfaces to Intrusion Detection	
28 46 24.19	Fire Detection and Alarm Interfaces to Video Surveillance	
28 46 24.21	Fire Detection and Alarm Interfaces to Elevator Control	
28 46 24.23	Other Fire Detection and Alarm Interfaces	

NUMBER	TITLE	EXPLANATION
28 47 00	Mass Notification	Includes: servers, workstations, software, and system integration. Notes: mass notification system (MNS) is a life safety system and is integrated with fire alarm systems. MNS may also be integrated with a variety of other systems, including A/V, integrated security management system, controls systems, depending on the application. See Also: 28 05 15 servers for electronic safety and security. 28 05 17 workstations for electronic safety and security. 28 05 45 for systems integration and unified systems.
28 47 11	System Requirements for Mass Notification	Includes: common requirements, redundancy, and resiliency. Notes: this Section is not intended to duplicate or supplant Division 01 General Requirements content.
28 47 13	Mass Notification Software	Includes: operating systems, server and client, integration platform, and integration plug-ins or add-ons.
28 47 15	Mass Notification Control Panels	
28 47 17	Notification Devices and Methods	
28 47 17.11	Notification Appliances and Methods Indoors	Includes: emergency voice communications (speakers) and audio-visual notification appliances (chime strobe).
28 47 17.13	Notification Appliances and Methods Outdoors	Includes: emergency voice communications (Big Voice) and audio-visual notifications (chime/strobe).
28 47 17.15	Notification Appliances and Methods Distributed Recipient	Includes: SMS, VOIP, smart phone apps, visual electronic signage (LED/LCD), ADA assistive technologies. Alternate Terms/Abbreviations: SMS: short message service: text message. VOIP: voice over internet protocol. ADA: Americans with Disabilities Act.
28 47 17.17	Notification Appliances and Methods Public Methods	Includes: broadcast TV, broadcast radio, internet media channels.
28 47 21	Mass Notification System Interfaces	Includes: fire alarm, security management systems, energy management systems, control systems, network, and telephone.

DIVISION **28**

NUMBER	TITLE	EXPLANATION
28 47 21.11	Notification Interfaces to Access Control Systems	
28 47 21.13	Notification Interfaces to Video Surveillance Systems	
28 47 21.15	Notification Interfaces to Security Detection, Alarm, and Monitoring	
28 47 21.17	Notification Interfaces to Security Monitoring and Control	
28 47 21.19	Notification Interfaces to Life Safety Systems	
28 47 21.21	Notification Interfaces to Fire Detection and Alarm	
28 47 21.23	Notification Interfaces to Detention Systems	
28 47 21.25	Notification Interfaces to Emergency Response Systems	
28 47 21.27	Notification Interfaces to Network and Facilities Monitoring	
28 47 21.29	Notification Interfaces to Web Intelligence	
28 47 21.31	Notification Interfaces to Traffic Management and Monitoring Systems	
28 47 21.33	Notification Interfaces to Smart Phones	
28 47 21.35	Notification Interfaces to Other External Subsystems	

28 48 00 Emergency Response Systems

See Also: 27 32 26 ring-down emergency telephones.
27 32 43 radio communications equipment.

28 48 23 Computer Aided Dispatch

28 49 00 Electronic Personal Protection Systems

Includes: systems and equipment that monitor and annunciate the need for assistance.

See: 10 43 00 for emergency aid specialties.

28 49 11 Electronic Personal Safety Detection Systems
28 49 13 Electronic Personal Safety Alarm Annunciation and Control Systems
28 49 15 Electronic Personal Safety Interfaces to Remote Monitoring
28 49 17 Electronic Personal Safety Emergency Aid Devices

28 50 00 Specialized Systems

Includes: hybrid and special purpose systems.

See Also: 28 10 00 for access control.
28 20 00 for video surveillance.
28 30 00 for security detection, alarm, and monitoring.
28 33 00 for security monitoring and control.
28 40 00 for life safety.
28 46 00 for fire detection and alarm.
28 47 00 for mass notification.

NUMBER	TITLE	EXPLANATION
28 51 00	Information Management and Presentation	See Also: 28 10 00 for access control. 28 20 00 for video surveillance. 28 30 00 for security detection, alarm, and monitoring. 28 33 00 for security monitoring and control. 28 40 00 for life safety. 28 46 00 for fire detection and alarm. 28 47 00 for mass notification.
28 51 11	Information Management and Presentation Architecture	
28 51 13	Data and Information Management	Includes: rules engines, decision trees, data fusion, response planning, data retention, and audit trail requirements.
28 51 15	Information Interfaces to Connected Subsystems	Includes: any connected subsystem or remote data structure. See Also: 26 09 13 for electrical power monitoring. 26 33 46 for battery monitoring. 27 50 00 for distributed communications and monitoring systems. 27 52 00 for healthcare communications and monitoring systems. 28 10 00 for access control. 28 20 00 for video surveillance. 28 30 00 for security detection, alarm, and monitoring. 28 33 00 for security monitoring and control. 28 40 00 for life safety. 28 46 00 for fire detection and alarm. 28 47 00 for mass notification.
28 51 15.11	Information Interfaces to Access Control Systems	
28 51 15.13	Information Interfaces to Video Surveillance Systems	
28 51 15.15	Information Interfaces to Security Detection, Alarm, and Monitoring	
28 51 15.17	Information Interfaces to Security Monitoring and Control	Includes: interfaces to asset tracking and personnel monitoring systems.
28 51 15.19	Information Interfaces to Mass Notification Systems	
28 51 15.21	Information Interfaces to Life Safety Systems	Includes: interfaces to panic/alert systems.
28 51 15.23	Information Interfaces to Fire Detection and Alarm	
28 51 15.25	Information Interfaces to Detention Systems	
28 51 15.27	Information Interfaces to Emergency Response Systems	
28 51 15.29	Information Interfaces to License Plate Recognition Systems	
28 51 15.31	Information Interfaces to Facial Recognition Systems	
28 51 15.33	Information Interfaces to Remote Data Structures	Includes: government watch lists, criminal history databases, license plate databases, identity management systems, etc.

DIVISION 28

NUMBER	TITLE	EXPLANATION
28 51 15.35	Information Interfaces to Network and Facilities Monitoring	See Also: 25 10 00 integrated automation network equipment.
28 51 15.37	Information Interfaces to Crowd Sourcing	
28 51 15.39	Information Interfaces to Web Intelligence	
28 51 15.41	Information Interfaces to Traffic Management and Monitoring Systems	See: 27 42 16 for transportation information display systems. 34 41 00 for roadway signaling and control equipment.
28 51 15.43	Information Interfaces to Environmental Monitoring Systems	
28 51 15.45	Information Interfaces to Directly Connected Sensors	
28 51 15.47	Information Interfaces to Other Safety and Security Subsystems	
28 51 15.49	Information Interfaces to Other External Subsystems	See Also: 26 09 13 for electrical power monitoring. 26 33 46 for battery monitoring. 27 50 00 for distributed communications and monitoring systems. 27 52 00 for healthcare communications and monitoring systems.
28 51 17	Data Presentation	Includes: logical presentation of data and alarms to user, and user interaction.
28 51 17.11	Maps	
28 51 17.13	Visualization	
28 51 17.15	Events/Alarms/Incidents	
28 51 17.17	Video Workspace	
28 51 17.19	Dispatch/Notification	
28 51 17.21	Review and Investigation	
28 51 17.23	Reporting	
28 51 19	Control Room and Monitoring Equipment	See Also: 11 52 00 for audio-visual equipment. 27 41 00 for audio-video systems. 27 42 00 for electronic digital systems.
28 51 19.11	Display/Workstations	
28 51 19.13	Video Walls	
28 51 19.15	Audio Equipment	
28 51 19.17	Remote Displays	

NUMBER	TITLE	EXPLANATION
28 52 00	Detention Security Systems	Includes: electronic equipment and systems for detention applications. See: 05 59 63 for detention enclosures. 11 98 00 for detention equipment. 13 42 63 for detention cell modules. 26 55 63 for detention lighting. 27 51 23 for detention intercommunications and program systems. 28 10 00 for access control. 28 20 00 for video surveillance. 28 31 00 for intrusion detection. 32 31 13 for high-security chain link fences and gates.
28 52 11	Detention Security System Requirements	Includes: electronic equipment and systems specific to detention monitoring and control.
28 52 13	Detention Monitoring and Control System and Interfaces	Includes: interfaces to subsystems that may not be detention specific.
28 52 13.11	Detention Interfaces to Access Control Systems	See: 28 10 00 for access control.
28 52 13.13	Detention Interfaces to Video Surveillance Systems	See: 28 20 00 for video surveillance.
28 52 13.15	Detention Interfaces to Security Detection, Alarm, and Monitoring	See: 28 30 00 for security detection, alarm, and monitoring.
28 52 13.17	Detention Interfaces to Security Monitoring and Control	See: 28 40 00 for security monitoring and control.
28 52 13.19	Detention Interfaces to Duress Alarm Systems	See: 28 40 00 for life safety.
28 52 13.21	Detention Interfaces to Distributed Television Systems	
28 52 13.23	Detention Interfaces to Perimeter Security Systems	See: 28 31 21 for area and perimeter intrusion detection.
28 52 15	Detention Video Systems	Includes: video based systems that provide specific detention support services.
28 52 15.11	Detention Interfaces to Video Interrogation Systems	
28 52 15.13	Detention Interfaces to Video Arraignment Systems	
28 52 15.15	Detention Interfaces to Video Visitation Systems	

DIVISION **28**

NUMBER	TITLE	EXPLANATION
28 52 17	Detention Communication Systems	
28 52 17.11	Detention Interfaces to Intercom and Paging Systems	
28 52 17.13	Detention Interfaces to Local Intercom Systems	
28 52 17.15	Detention Interfaces to Inmate Television Systems	
28 52 19	Detention Personal Protection Systems	

SPECIFICATIONS GROUP

SITE AND INFRASTRUCTURE SUBGROUP

Division 30 – Reserved for Future Expansion

DIVISION 31 – EARTHWORK

DIVISION 32 – EXTERIOR IMPROVEMENTS

DIVISION 33 – UTILITIES

DIVISION 34 – TRANSPORTATION

DIVISION 35 – WATERWAY AND MARINE CONSTRUCTION

Division 36 – Reserved for Future Expansion

Division 37 – Reserved for Future Expansion

Division 38 – Reserved for Future Expansion

Division 39 – Reserved for Future Expansion

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NUMBER	TITLE	EXPLANATION
DIVISION 31 – EARTHWORK		
31 00 00	Earthwork	May be used as Division level Section title.

31 01 00 Maintenance of Earthwork

Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of earthwork.

Level 4 Numbering Recommendation: Level 4 titles for earthwork are provided as example for user's creation of more detailed titles for other subjects as required. Following Level 4 numbering is recommended:

.51-.59 for maintenance.

.61-.69 for repair.

.71-.79 for rehabilitation.

.81-.89 for replacement.

.91-.99 for restoration.

31 01 10	Maintenance of Clearing
31 01 20	Maintenance of Earth Moving
31 01 40	Maintenance of Shoring and Underpinning
31 01 50	Maintenance of Excavation Support and Protection
31 01 60	Maintenance of Special Foundations and Load Bearing Elements
31 01 62	Maintenance of Driven Piles
31 01 62.61	Driven Pile Repairs
31 01 63	Maintenance of Bored and Augered Piles
31 01 63.61	Bored and Augered Pile Repairs
31 01 70	Maintenance of Tunneling and Mining
31 01 70.61	Tunnel Leak Repairs

31 05 00 Common Work Results for Earthwork

Includes: subjects common to multiple titles in Division 31.

See: 01 80 00 for performance requirements of subjects common to multiple titles.

02 05 19 for geosynthetics for existing conditions.

32 05 13 for soils for exterior improvements.

32 05 16 for aggregates for exterior improvements.

32 05 19 for geosynthetics for exterior improvements.

32 05 23 for cement and concrete for exterior improvements.

31 05 05	Selective Demolition for Earthwork
	See: 02 41 19 for selective demolition for whole structure demolition.
31 05 13	Soils for Earthwork
31 05 16	Aggregates for Earthwork
31 05 19	Geosynthetics for Earthwork

DIVISION **31**

NUMBER	TITLE	EXPLANATION
31 05 19.13	Geotextiles for Earthwork	
31 05 19.16	Geomembranes for Earthwork	
31 05 19.19	Geogrids for Earthwork	
31 05 19.23	Geosynthetic Clay Liners	
31 05 19.26	Geocomposites	
31 05 19.29	Geonets	
31 05 23	Cement and Concrete for Earthwork	

31 06 00 Schedules for Earthwork

Includes: schedules of items common to multiple titles in Division 31.

Notes: a schedule may be included on drawings, in the project manual, or a project book.

31 06 10	Schedules for Clearing	
31 06 20	Schedules for Earth Moving	
31 06 20.13	Trench Dimension Schedule	
31 06 20.16	Backfill Material Schedule	
31 06 40	Schedules for Shoring and Underpinning	
31 06 50	Schedules for Excavation Support and Protection	
31 06 60	Schedules for Special Foundations and Load Bearing Elements	
31 06 60.13	Driven Pile Schedule	
31 06 60.16	Caisson Schedule	
31 06 70	Schedules for Tunneling and Mining	

31 08 00 Commissioning of Earthwork

Includes: commissioning of items common to multiple titles in Division 31.

See: 01 91 00 for commissioning of subjects common to multiple Divisions.

31 08 13	Pile Load Testing	
31 08 13.13	Dynamic Pile Load Testing	
31 08 13.16	Static Pile Load Testing	

31 09 00 Geotechnical Instrumentation and Monitoring of Earthwork

Includes: geotechnical instrumentation and monitoring of items common to multiple titles in Division 31.

See: 00 31 32 for geotechnical data available to bidders, proposers, or contractors before construction begins.
 02 32 00 for geotechnical investigations performed by the contractor as part of the project.
 02 32 23 for groundwater monitoring before construction.
 31 60 00 for special foundations and load-bearing elements.
 33 11 53 for monitoring wells.

31 09 13	Geotechnical Instrumentation and Monitoring	
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NUMBER	TITLE	EXPLANATION
31 09 13.13	Groundwater Monitoring During Construction	
31 09 16	Foundation Performance Instrumentation	
31 09 16.26	Bored and Augered Pile Load Tests	

31 10 00 Site Clearing

Includes: removal of vegetation at the site, including stripping of sod and soil, and tree pruning for site clearing.

See: 01 56 39 for temporary tree and plant protection.
 32 01 90 for pruning for maintenance of shrubs and trees.
 32 91 19 for landscape grading.
 32 91 19 for topsoil placement and grading.
 32 96 00 for transplanting.

31 11 00 Clearing and Grubbing

31 12 00 Selective Clearing

31 13 00 Selective Tree and Shrub Removal and Trimming

31 13 13 Selective Tree and Shrub Removal
 31 13 16 Selective Tree and Shrub Trimming

31 14 00 Earth Stripping and Stockpiling

31 14 13 Soil Stripping and Stockpiling
 31 14 13.13 Soil Stripping
 31 14 13.16 Soil Stockpiling
 31 14 13.23 Topsoil Stripping and Stockpiling
 31 14 16 Sod Stripping and Stockpiling
 31 14 16.13 Sod Stripping
 31 14 16.16 Sod Stockpiling

31 20 00 Earth Moving

Includes: grading, excavating, embankments, and sedimentation and erosion control. Earth moving for foundations, structures, pavement, railways, ditches, channels, culverts, drains, and utilities.

See: 31 30 00 for earthwork methods.
 35 24 00 for dredging.
 41 61 00 for mobile earth moving equipment.

31 21 00 Off-Gassing Mitigation

Includes: mitigation of gases from underground sources.

See: 00 31 24 for environmental assessment information.
 02 24 00 for environmental assessment.
 02 50 00 for site remediation.
 02 56 19 for gas containment.
 02 80 00 for facility remediation.

DIVISION 31

NUMBER	TITLE	EXPLANATION
31 21 13	Radon Mitigation	
31 21 13.13	Radon Venting	
31 21 16	Methane Mitigation	
31 21 16.13	Methane Venting	
31 22 00	Grading	Includes: earth moving to reshape contours. See: 31 14 00 for earth stripping and stockpiling. 32 71 00 for constructed wetlands. 32 91 19 for landscape grading.
31 22 13	Rough Grading	
31 22 16	Fine Grading	
31 22 16.13	Roadway Subgrade Reshaping	
31 22 19	Finish Grading	
31 22 19.13	Spreading and Grading Topsoil	
31 23 00	Excavation and Fill	Includes: excavation and fill for foundations, structures, pavement, railways, ditches, channels, culverts, drains, and utilities. May Include: foundation and underslab drainage. Alternate Terms/Abbreviations: flowable fill: controlled density fill: unshrinkable fill: flowcrete: liquid dirt: fly ash slurry: flowable mortar: plastic soil cement: soil replacement: lean mix backfill: controlled low strength material. See: 31 21 00 for off-gassing mitigation. 31 24 00 for embankments. 31 71 00 for tunnel excavation. 33 05 07 for trenchless installation of utility piping. 33 05 97 for identification and signage for utilities. See Also: 33 41 00 for subdrainage.
31 23 13	Subgrade Preparation	
31 23 16	Excavation	
31 23 16.13	Trenching	
31 23 16.16	Structural Excavation for Minor Structures	
31 23 16.26	Rock Removal	
31 23 19	Dewatering	
31 23 23	Fill	
31 23 23.13	Backfill	
31 23 23.23	Compaction	
31 23 23.33	Flowable Fill	
31 23 23.43	Geofoam	
31 23 33	Trenching and Backfilling	

NUMBER	TITLE	EXPLANATION
31 24 00	Embankments	Includes: soil and rock embankments for support or containment and protection purposes. See: 31 34 00 for embankments with soil reinforcement. 31 35 00 for embankments with slope protection. 34 71 00 for roadway construction. 34 72 00 for railway construction. 35 41 13 for embankment levees. 35 73 00 for embankment dams.
31 24 13	Roadway Embankments	
31 24 16	Railway Embankments	
31 25 00	Erosion and Sedimentation Controls	Includes: construction of permanent erosion checks and controls. See: 22 14 00 for facility storm drainage. 31 34 00 for erosion and sedimentation control with soil reinforcement. 31 35 00 for erosion and sedimentation control with slope protection. 33 40 00 for storm drainage utilities.
31 25 14	Stabilization Measures for Erosion and Sedimentation Control	
31 25 14.13	Hydraulically-Applied Erosion Control	
31 25 14.16	Rolled Erosion Control Mats and Blankets	
31 25 24	Structural Measures for Erosion and Sedimentation Control	
31 25 24.13	Rock Barriers	
31 25 34	Retention Measures for Erosion and Sedimentation Controls	
31 25 34.13	Rock Basins	
31 30 00	Earthwork Methods	Includes: earthwork methods used for multiple purposes. See: 31 20 00 for earth moving.
31 31 00	Soil Treatment	Includes: soil treatment to resist invasion of adjoining structure by pests and growth of vegetation. See: 06 05 73 for wood treatment. 10 81 00 for pest control devices. 31 32 16 for chemical treatment soil stabilization. 32 91 00 for planting preparation.
31 31 13	Rodent Control	
31 31 13.16	Rodent Control Bait Systems	

DIVISION **31**

NUMBER	TITLE	EXPLANATION
31 31 13.19	Rodent Control Traps	
31 31 13.23	Rodent Control Electronic Systems	
31 31 13.26	Rodent Control Repellents	
31 31 16	Termite Control	
31 31 16.13	Chemical Termite Control	
31 31 16.16	Termite Control Bait Systems	
31 31 16.19	Termite Control Barriers	
31 31 19	Vegetation Control	
31 31 19.13	Chemical Vegetation Control	
31 31 33	Sanitary Vacuum System Piping	Includes: systems with onsite pneumatic vacuum valve sumps, where conveyance is under vacuum to a centralized vacuum station.

31 32 00 Soil Stabilization

Includes: addition and integration of material in and on soil to stabilize and strengthen soil material and the construction of which it is a part. Fully grouted, passive (non-tensioned) soil nails used to resist failure forces on slopes.

Alternate Terms/Abbreviations: soil mixing stabilization: soil modification: soil conditioning. Shotcrete: gunite.

See: 02 55 00 for remediation soil stabilization.

03 37 13 for shotcrete.

31 05 19 for geosynthetics for earthwork.

31 33 33 for shotcrete for rock stabilization.

31 34 19 for geosynthetic soil reinforcement.

31 35 19 for geosynthetic slope protection.

31 74 19 for shotcrete tunnel lining.

32 91 13 for soil preparation.

32 91 16 for planting soil stabilization.

31 32 13	Soil Mixing Stabilization
31 32 13.13	Asphalt Soil Stabilization
31 32 13.16	Cement Soil Stabilization
31 32 13.19	Lime Soil Stabilization
31 32 13.23	Fly-Ash Soil Stabilization
31 32 13.26	Lime-Fly-Ash Soil Stabilization
31 32 16	Chemical Treatment Soil Stabilization
31 32 16.13	Polymer Emulsion Soil Stabilization
31 32 17	Water Injection Soil Stabilization
31 32 19	Geosynthetic Soil Stabilization and Layer Separation
31 32 19.13	Geogrid Soil Stabilization
31 32 19.16	Geotextile Soil Stabilization
31 32 19.19	Geogrid Layer Separation
31 32 19.23	Geotextile Layer Separation
31 32 23	Pressure Grouting Soil Stabilization
31 32 23.13	Cementitious Pressure Grouting Soil Stabilization
31 32 23.16	Chemical Pressure Grouting Soil Stabilization
31 32 33	Shotcrete Soil Slope Stabilization
31 32 36	Soil Nailing

NUMBER	TITLE	EXPLANATION
31 32 36.13	Driven Soil Nailing	
31 32 36.16	Grouted Soil Nailing	
31 32 36.19	Corrosion-Protected Soil Nailing	
31 32 36.23	Jet-Grouted Soil Nailing	
31 32 36.26	Launched Soil Nailing	

31 33 00 **Rock Stabilization**

Includes: various technologies to make an unstable rock slope more stable.

Alternate Terms/Abbreviations: shotcrete: gunite.

See: 03 37 13 for shotcrete.

31 32 33 for shotcrete for soil slope stabilization.

31 51 00 for excavation anchor tiebacks.

31 68 13 for rock foundation anchors.

31 72 13 for tunnel rock reinforcement and initial support.

31 74 19 for shotcrete tunnel lining.

31 33 13	Rock Bolting and Grouting
31 33 23	Rock Slope Netting
31 33 26	Rock Slope Wire Mesh
31 33 33	Shotcrete Rock Slope Stabilization
31 33 43	Vegetated Rock Slope Stabilization

31 34 00 **Soil Reinforcement**

Includes: various technologies to reinforce the strength and stability of soils.

Alternate Terms/Abbreviations: soil reinforcement: reinforced earth.

See: 31 05 19 for geosynthetics for earthwork.

31 25 00 for erosion and sedimentation control.

31 32 19 for geosynthetic soil stabilization and layer separation.

31 35 19 for geosynthetic slope protection.

32 32 34 for reinforced soil retaining walls.

31 34 19	Geosynthetic Soil Reinforcement
31 34 19.13	Geogrid Soil Reinforcement
31 34 19.16	Geotextile Soil Reinforcement
31 34 23	Fiber Soil Reinforcement
31 34 23.13	Geosynthetic Fiber Soil Reinforcement

NUMBER	TITLE	EXPLANATION
31 35 00	Slope Protection	Includes: various technologies to protect slopes from erosion. See: 03 30 00 for CIP concrete. 03 40 00 for precast concrete. 04 22 00 for concrete unit masonry. 07 17 00 for bentonite waterproofing. 31 05 19 for geosynthetics for earthwork. 31 25 00 for erosion and sedimentation control. 31 32 19 for geosynthetic soil stabilization and layer separation. 31 34 19 for geosynthetic soil reinforcement. 31 56 00 for slurry walls. 31 56 13 for bentonite slurry walls. 35 42 35 for slope protection bank protection. 35 43 35 for slope protection scour protection.
31 35 19	Geosynthetic Slope Protection	
31 35 19.13	Geogrid Slope Protection	
31 35 19.16	Geotextile Slope Protection	
31 35 19.19	Slope Protection with Mulch Control Netting	
31 35 23	Slope Protection with Slope Paving	
31 35 23.13	Cast-in-Place Concrete Slope Paving	
31 35 23.16	Precast Concrete Slope Paving	
31 35 23.19	Concrete Unit Masonry Slope Paving	
31 35 26	Containment Barriers	
31 35 26.13	Clay Containment Barriers	
31 35 26.16	Geomembrane Containment Barriers	
31 35 26.23	Bentonite Slurry Trench	
31 36 00	Gabions	Includes: wire baskets filled with rock used for a variety of purposes. Alternate Terms/Abbreviations: gabion boxes: gabion baskets. See: 32 32 36 for gabion retaining walls. 35 31 19 for gabion revetments. 35 31 26 for gabion jetties. 35 31 29 for gabion groins. 35 42 36 for gabion bank protection. 35 43 36 for gabion scour protection.
31 36 13	Gabion Boxes	
31 36 19	Gabion Mattresses	
31 36 19.13	Vegetated Gabion Mattresses	

NUMBER	TITLE	EXPLANATION
31 37 00	Riprap	Includes: cover of rock used to stabilize slopes subject to water erosion intended to be permanent. See: 35 42 37 for riprap bank protection. 35 43 37 for riprap scour protection.
31 37 13	Machined Riprap	
31 37 16	Non-Machined Riprap	
31 37 16.13	Rubble-Stone Riprap	
31 37 16.16	Concrete Unit Masonry Riprap	
31 37 16.19	Sacked Sand-Cement Riprap	
31 40 00	Shoring and Underpinning	Includes: underpinning, needling, grouting, and other methods of strengthening or shoring structures and earth to protect and stabilize existing construction. See: 31 50 00 for excavation support and protection.
31 41 00	Shoring	Includes: temporary support of structural components and earth. Alternate Terms/Abbreviations: trench shielding; trench boxing. See: 31 51 00 for anchor tiebacks. 31 53 00 for cribbing and walers. 35 31 16 for steel sheet piling seawalls. 35 42 13 for steel sheet piling bank protection.
31 41 13	Timber Shoring	
31 41 16	Sheet Piling	
31 41 16.13	Steel Sheet Piling	
31 41 16.16	Plastic Sheet Piling	
31 41 19	Metal Hydraulic Shoring	
31 41 19.13	Aluminum Hydraulic Shoring	
31 41 23	Pneumatic Shoring	
31 41 33	Trench Shielding	
31 43 00	Concrete Raising	Includes: variety of technologies to raise sunken concrete. Alternate Terms/Abbreviations: pressure grouting; slabjacking; mudjacking. See: 02 43 16 for structure raising. 31 46 00 for needle beams. 31 48 00 for underpinning. 32 01 29 for concrete pavement raising.

DIVISION **31**

NUMBER	TITLE	EXPLANATION
31 43 13	Pressure Grouting	
31 43 13.13	Concrete Pressure Grouting	
31 43 13.16	Polyurethane Pressure Grouting	
31 43 16	Compaction Grouting	
31 43 19	Mechanical Jacking	

31 45 00 Vibroflotation and Densification

Includes: activities requiring vibrator penetration at high depth to densify, reinforce, and stabilize loose soil or soft soils.

Alternate Terms/Abbreviations: vibroflotation: vibro compaction.

31 45 13	Vibroflotation
31 45 16	Densification

31 46 00 Needle Beams

Includes: horizontal shoring members placed under or through a wall to support the structural load while performing work under the base of the wall.

See: 31 43 00 for concrete raising.
31 48 00 for underpinning.

31 46 13	Cantilever Needle Beams
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31 48 00 Underpinning

Includes: replacing a structural element's foundation while the structural element remains in place.

Alternate Terms/Abbreviations: micropiles: screw piles.

See: 31 43 00 for concrete raising.
31 46 00 for needle beams.
31 63 33 for drilled micropiles.

31 48 13	Underpinning Piers
31 48 19	Bracket Piers
31 48 23	Jacked Piers
31 48 33	Micropile Underpinning

31 50 00 Excavation Support and Protection

Includes: permanent and temporary support of excavations and excavation enclosures, such as cofferdams, for construction below water and in water.

See: 31 23 19 for dewatering.
31 40 00 for shoring and underpinning.

NUMBER	TITLE	EXPLANATION
31 51 00	Anchor Tiebacks	Includes: anchor assemblies to support excavations by grouted prestressed ties to earth or rock. Tendons, equipment, corrosion protection grout, drilling, anchor fabrication, grouting, and testing. See: 31 33 13 for rock bolting and grouting. 31 41 00 for shoring. 31 68 00 for foundation anchors.
31 51 13	Excavation Soil Anchors	
31 51 16	Excavation Rock Anchors	
31 52 00	Cofferdams	Includes: watertight enclosures that allow construction operations below the surface of water. See: 03 37 26 for underwater placed concrete. 31 64 00 for caissons. 35 53 00 for underwater construction.
31 52 13	Sheet Piling Cofferdams	
31 52 16	Timber Cofferdams	
31 52 19	Precast Concrete Cofferdams	
31 53 00	Cribbing and Walers	Includes: structural members to line and support the lining of an excavation. Alternate Terms/Abbreviations: walers: whalers. See: 31 41 00 for shoring.
31 53 13	Timber Cribwork	
31 54 00	Ground Freezing	Includes: stabilizing earthwork by freezing. See: 31 66 19 for refrigerated foundations.
31 56 00	Slurry Walls	Includes: subsurface barriers constructed of trenches filled with a slurry mixture to impede or stop groundwater flow. See: 07 17 00 for bentonite waterproofing. 31 35 26 for bentonite slurry trench containment barrier. 31 66 16 for slurry diaphragm foundation walls.

DIVISION **31**

NUMBER	TITLE	EXPLANATION
31 56 13	Bentonite Slurry Walls	
31 56 13.13	Soil-Bentonite Slurry Walls	
31 56 13.16	Cement-Bentonite Slurry Walls	
31 56 13.19	Slag-Cement-Bentonite Slurry Walls	
31 56 13.23	Soil-Cement-Bentonite Slurry Walls	
31 56 13.26	Pozzolan-Bentonite Slurry Walls	
31 56 13.29	Organically-Modified Bentonite Slurry Walls	
31 56 16	Attipulgite Slurry Walls	
31 56 16.13	Soil-Attipulgite Slurry Walls	
31 56 19	Slurry-Geomembrane Composite Slurry Walls	
31 56 23	Lean Concrete Slurry Walls	
31 56 26	Bio-Polymer Trench Drain	

31 60 00 Special Foundations and Load-Bearing Elements

31 62 00 Driven Piles

Includes: driven deep pile foundations of various materials for support of structures.

May Include: reinforcement in cast-in-place concrete piles and concrete-filled steel piles.

Alternate Terms/Abbreviations: pressure-injected footings: compacted-expanded base-concrete piles.

See: 03 30 00 for CIP concrete.
 03 40 00 for precast concrete.
 05 12 00 for structural steel framing.
 06 13 00 for heavy timber.
 31 63 00 for bored piles.
 31 66 13 for special piles.

31 62 13	Concrete Piles
31 62 13.13	Cast-in-Place Concrete Piles
31 62 13.16	Concrete Displacement Piles
31 62 13.19	Precast Concrete Piles
31 62 13.23	Prestressed Concrete Piles
31 62 13.26	Pressure-Injected Footings
31 62 16	Steel Piles
31 62 16.13	Sheet Steel Piles
31 62 16.16	Steel H Piles
31 62 16.19	Unfilled Tubular Steel Piles
31 62 19	Timber Piles
31 62 23	Composite Piles
31 62 23.13	Concrete-Filled Steel Piles
31 62 23.16	Wood and Cast-in-Place Concrete Piles

NUMBER	TITLE	EXPLANATION
31 63 00	Bored Piles	Includes: bored and deep augured pile and drilled shaft and caisson construction for support of structures. Test and production piles. May Include: pile, shaft, or caisson reinforcement. Alternate Terms/Abbreviations: micropiles: screw piles. Fixed end caisson piles: rock socketed piles. See: 03 30 00 for cast-in-place concrete. 31 23 19 for dewatering. 31 40 00 for shoring and underpinning. 31 48 33 for micropile underpinning. 31 50 00 for excavation support and protection. 31 62 00 for driven piles. 31 66 13 for special piles. 35 53 00 for underwater construction. See Also: 03 20 00 for concrete reinforcement.
31 63 13	Bored and Augered Test Piles	
31 63 16	Auger Cast Grout Piles	
31 63 19	Bored and Socketed Piles	
31 63 19.13	Rock Sockets for Piles	
31 63 23	Bored Concrete Piles	
31 63 23.13	Bored and Belled Concrete Piles	
31 63 23.16	Bored Friction Concrete Piles	
31 63 26	Drilled Caissons	
31 63 26.13	Fixed End Caisson Piles	
31 63 26.16	Concrete Caissons for Marine Construction	
31 63 29	Drilled Concrete Piers and Shafts	
31 63 29.13	Uncased Drilled Concrete Piers	
31 63 29.16	Cased Drilled Concrete Piers	
31 63 33	Drilled Micropiles	
31 64 00	Caissons	Includes: sunken or driven load-bearing enclosures to protect excavation for foundations, promote substructure construction, and become part of permanent structure. Casing and reinforcement. See: 31 52 00 for cofferdams. 31 63 26 for drilled caissons. 35 53 00 for underwater construction.
31 64 13	Box Caissons	
31 64 16	Excavated Caissons	
31 64 19	Floating Caissons	
31 64 23	Open Caissons	

DIVISION **31**

NUMBER	TITLE	EXPLANATION
31 64 26	Pneumatic Caissons	
31 64 29	Sheeted Caissons	

31 66 00 Special Foundations

Includes: foundations of a special design, including load-bearing wall products and systems, crib stabilized, reinforced, and unit walls.

May Include: retaining walls.

See: 03 30 00 for cast-in-place concrete structures.

31 54 00 for ground freezing.

31 56 00 for slurry walls.

31 62 00 for driven piles.

31 63 00 for bored piles.

31 64 00 for caissons.

32 32 26 for metal crib retaining walls.

32 32 43 for soldier beam retaining walls.

See Also: 32 32 00 for retaining walls.

31 66 13	Special Piles
31 66 13.13	Rammed Aggregate Piles
31 66 15	Helical Foundation Piles
31 66 16	Special Foundation Walls
31 66 16.13	Anchored Foundation Walls
31 66 16.23	Concrete Cribbing Foundation Walls
31 66 16.26	Metal Cribbing Foundation Walls
31 66 16.33	Manufactured Modular Foundation Walls
31 66 16.43	Mechanically Stabilized Earth Foundation Walls
31 66 16.46	Slurry Diaphragm Foundation Walls
31 66 16.53	Soldier-Beam Foundation Walls
31 66 16.56	Permanently-Anchored Soldier-Beam Foundation Walls
31 66 19	Refrigerated Foundations

31 68 00 Foundation Anchors

Includes: anchor assemblies to fasten deep foundations, walls, and similar elements to rock or other material. Tendons, equipment, corrosion protection grout, drilling, anchor fabrication, grouting, and testing.

See: 31 33 13 for rock bolting and grouting.

31 51 00 for excavation anchor tiebacks.

31 68 13	Rock Foundation Anchors
31 68 16	Helical Foundation Anchors

NUMBER	TITLE	EXPLANATION
31 70 00	Tunneling and Mining	Includes: tunnel boring, bracing and jacking work, including excavating, linings and casings, grouting support systems, boring machines, control and spoil removal systems. See: 03 30 00 for cast-in-place concrete. 31 23 19 for dewatering. 31 40 00 for shoring and underpinning. 31 52 00 for cofferdams. 33 05 07 for trenchless installation of utility piping.
31 71 00	Tunnel Excavation	Includes: moles and tunnel-boring machines. Rock excavation by boring machine, drill, and blast. Robotic processes. See: 31 23 00 for excavation and fill. 31 50 00 for excavation support and protection. 31 72 00 for tunnel support systems. 33 05 07 for trenchless installation of utility piping.
31 71 13	Shield Driving Tunnel Excavation	
31 71 16	Tunnel Excavation by Drilling and Blasting	
31 71 19	Tunnel Excavation by Tunnel Boring Machine	
31 71 23	Tunneling by Cut and Cover	
31 72 00	Tunnel Support Systems	Includes: supports put in place as tunnel excavation proceeds. See: 31 33 00 for rock stabilization. 31 71 00 for tunnel excavation. 31 74 00 for tunnel construction linings.
31 72 13	Rock Reinforcement and Initial Support	
31 72 16	Steel Ribs and Lagging	
31 73 00	Tunnel Grouting	Includes: stabilizing tunnel excavation by grouting. See: 03 60 00 for grouting.
31 73 13	Cement Tunnel Grouting	
31 73 16	Chemical Tunnel Grouting	

NUMBER	TITLE	EXPLANATION
31 74 00	Tunnel Construction	Includes: lining of tunnel after tunnel excavation. Alternate Terms/Abbreviations: shotcrete: gunite. See: 03 30 00 for cast-in-place concrete. 03 40 00 for precast concrete. 31 72 00 for tunnel support systems. 31 77 00 for submersible tube tunnels.
31 74 13	Cast-in-Place Concrete Tunnel Lining	
31 74 16	Precast Concrete Tunnel Lining	
31 74 19	Shotcrete Tunnel Lining	
31 75 00	Shaft Construction	Includes: support systems for vertical shafts. See: 03 30 00 for cast-in-place concrete. 03 40 00 for precast concrete.
31 75 13	Cast-in-Place Concrete Shaft Lining	
31 75 16	Precast Concrete Shaft Lining	
31 77 00	Submersible Tube Tunnels	Includes: construction of tunnels by installation of fabricated tunnel tubes. See: 31 74 00 for tunnel construction.
31 77 13	Trench Excavation for Submerged Tunnels	
31 77 16	Tube Construction (Outfitting Tunnel Tubes)	
31 77 19	Floating and Laying Submerged Tunnels	

NUMBER	TITLE	EXPLANATION
DIVISION 32 – EXTERIOR IMPROVEMENTS		
32 00 00	Exterior Improvements	May be used as Division level Section title.

32 01 00 Operation and Maintenance of Exterior Improvements

Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of exterior improvements.

Alternate Terms/Abbreviations: pavement raising: pavement jacking. Reused: recycled.

Level 4 Numbering Recommendation: Level 4 titles for exterior improvements are provided as examples for user's creation of more detailed titles for other subjects as required. Following Level 4 numbering is recommended:

.51-.59 for maintenance.

.61-.69 for repair.

.71-.79 for rehabilitation.

.81-.89 for replacement.

.91-.99 for restoration.

See: 32 12 16 for seal coats.

32 01 11	Paving Cleaning
32 01 11.51	Rubber and Paint Removal from Paving
32 01 11.52	Rubber Removal from Paving
32 01 11.53	Paint Removal from Paving
32 01 13	Flexible Paving Surface Treatment
32 01 13.61	Slurry Seal (Latex Modified)
32 01 13.62	Asphalt Surface Treatment
32 01 16	Flexible Paving Rehabilitation
32 01 16.71	Cold Milling Asphalt Paving
32 01 16.72	Asphalt Paving Reuse
32 01 16.73	In Place Cold Reused Asphalt Paving
32 01 16.74	In Place Hot Reused Asphalt Paving
32 01 16.75	Heater Scarifying of Asphalt Paving
32 01 17	Flexible Paving Repair
32 01 17.61	Sealing Cracks in Asphalt Paving
32 01 17.62	Stress-Absorbing Membrane Interlayer
32 01 19	Rigid Paving Surface Treatment
32 01 19.61	Sealing of Joints in Rigid Paving
32 01 19.62	Patching of Rigid Paving
32 01 23	Base Course Reconditioning
32 01 26	Rigid Paving Rehabilitation
32 01 26.71	Grooving of Concrete Paving

DIVISION **32**

NUMBER	TITLE	EXPLANATION
32 01 26.72	Grinding of Concrete Paving	
32 01 26.73	Milling of Concrete Paving	
32 01 26.74	Concrete Overlays	
32 01 26.75	Concrete Paving Reuse	
32 01 29	Rigid Paving Repair	
32 01 29.61	Partial Depth Patching of Rigid Paving	
32 01 29.62	Concrete Paving Raising	
32 01 29.63	Subsealing and Stabilization	
32 01 30	Operation and Maintenance of Site Improvements	
32 01 30.13	Snow Removal	
32 01 80	Operation and Maintenance of Irrigation	
32 01 90	Operation and Maintenance of Planting	
		See: 31 13 00 for selective tree and shrub removal and trimming.
		32 90 00 for maintenance of new planting as part of construction contract.
		32 91 00 for planting preparation.
		32 96 00 for transplanting.
32 01 90.13	Fertilizing	
32 01 90.16	Amending Soils	
32 01 90.19	Mowing	
32 01 90.23	Pruning	
32 01 90.26	Watering	
32 01 90.29	Topsoil Preservation	
32 01 90.33	Tree and Shrub Preservation	

32 05 00 Common Work Results for Exterior Improvements

Includes: subjects common to multiple titles in Division 32.

See: 01 80 00 for performance requirements of subjects common to multiple titles.

02 05 19 for geosynthetics for existing conditions.

31 05 13 for soils for earthwork.

31 05 16 for aggregates for earthwork.

31 05 19 for geosynthetics for earthwork.

31 05 23 for cement and concrete for earthwork.

32 05 05 Selective Demolition for Exterior Improvements

See: 02 41 19 for selective demolition for whole structure demolition.

32 05 13	Soils for Exterior Improvements
32 05 16	Aggregates for Exterior Improvements
32 05 19	Geosynthetics for Exterior Improvements
32 05 19.13	Geotextiles for Exterior Improvements
32 05 19.16	Geomembranes for Exterior Improvements
32 05 19.19	Geogrids for Exterior Improvements
32 05 23	Cement and Concrete for Exterior Improvements
32 05 33	Common Work Results for Planting

NUMBER	TITLE	EXPLANATION
32 06 00	Schedules for Exterior Improvements	Includes: schedules of items common to multiple titles in Division 32. Notes: a schedule may be included on drawings, in the project manual, or a project book.
32 06 10	Schedules for Bases, Ballasts, and Paving	
32 06 10.13	Pedestrian Walkway Schedule	
32 06 30	Schedules for Site Improvements	
32 06 30.13	Retaining Wall Schedule	
32 06 80	Schedules for Irrigation	
32 06 80.13	Irrigation Piping Schedule	
32 06 90	Schedules for Planting	
32 06 90.13	Planting Schedule	
32 08 00	Commissioning of Exterior Improvements	Includes: commissioning of items common to multiple titles in Division 32. See: 01 91 00 for commissioning of subjects common to multiple Divisions.
32 10 00	Bases, Ballasts, and Paving	Includes: granular, concrete, treated, synthetic, and other materials that comprise the structure; paving, and surfacing for walks, parking lots, roads, streets, highways, rail transportation, airfields, and athletic playing surfaces. Curbs, gutters, and paving specialties. See: 03 10 00 for concrete forming. 03 20 00 for concrete reinforcement. 03 30 00 for cast-in-place concrete. 34 10 00 for guideways/railways. 34 71 00 for roadway construction. 34 72 00 for railway construction.

NUMBER	TITLE	EXPLANATION
32 11 00	Base Courses	Includes: bound and unbound base and subbase courses that separate flexible, rigid, or unit paving courses from the underlying foundation or subgrade. Bound and unbound base and subbase courses that separate rail track supports from the underlying foundation or subgrade.
		See: 31 20 00 for subgrade and roadbed. 31 24 13 for roadway embankments. 31 24 16 for railway embankments. 32 01 23 for base course reconditioning. 32 12 00 for flexible paving. 32 13 00 for rigid paving. 32 14 00 for unit paving. 34 11 00 for rail tracks. 34 71 00 for roadway construction. 34 72 00 for railway construction.
32 11 13	Subgrade Modifications	
32 11 13.13	Lime-Treated Subgrades	
32 11 13.16	Bituminous-Treated Subgrades	
32 11 16	Subbase Courses	
32 11 16.13	Sand-Clay Subbase Courses	
32 11 16.16	Aggregate Subbase Courses	
32 11 23	Aggregate Base Courses	Includes: crushed aggregate, caliche, and lime rock base courses.
32 11 23.13	Sand-Clay Base Courses	
32 11 23.23	Base Course Drainage Layers	
32 11 26	Asphaltic Base Courses	
32 11 26.13	Plant Mix Asphaltic Base Courses	
32 11 26.16	Road Mix Asphaltic Base Courses	
32 11 26.19	Bituminous-Stabilized Base Courses	
32 11 29	Lime Treated Base Courses	
32 11 29.13	Lime-Fly Ash-Treated Base Courses	
32 11 33	Cement-Treated Base Courses	
32 11 33.13	Portland Cement-Stabilized Base Courses	
32 11 36	Concrete Base Courses	
32 11 36.11	Soil-Cement Concrete Base Courses	
32 11 36.13	Lean Concrete Base Courses	
32 11 36.16	Plain Cement Concrete Base Courses	
32 11 36.19	Hydraulic Cement Concrete Base Courses	
32 11 39	Shell Base Course	

NUMBER	TITLE	EXPLANATION
32 12 00	Flexible Paving	Includes: pavements above base course, including conventional and modified pavements for a variety of applications. Asphaltic scratch or leveling courses, bituminous binder courses, and bituminous concrete wearing courses. Tack coats and prime coats. May Include: base courses. Alternate Terms/Abbreviations: paving: bituminous paving: bituminous concrete: blacktop. Porous paving: pervious paving. See: 32 01 13 for flexible paving surface treatment. 32 01 16 for flexible paving rehabilitation. 32 01 17 for flexible paving repair. 32 13 00 for rigid paving. 32 14 00 for unit paving. 32 14 43 for porous unit paving. 32 16 13 for curbs and gutters. 41 66 19 for mobile paving equipment. See Also: 32 11 00 for base courses.
32 12 13	Preparatory Coats	
32 12 13.13	Tack Coats	
32 12 13.16	Asphaltic Tack Coats	
32 12 13.19	Prime Coats	
32 12 13.23	Asphaltic Prime Coats	
32 12 16	Asphalt Paving	
32 12 16.13	Plant-Mix Asphalt Paving	
32 12 16.16	Road-Mix Asphalt Paving	
32 12 16.19	Cold-Mix Asphalt Paving	
32 12 16.23	Reinforced Asphalt Paving	
32 12 16.26	Fiber-Modified Asphalt Paving	
32 12 16.29	Polymer-Modified Asphalt Paving	
32 12 16.33	Granulated Rubber-Modified Asphalt Paving	
32 12 16.36	Athletic Asphalt Paving	
32 12 19	Asphalt Paving Wearing Courses	
32 12 19.13	Road-Mix Asphalt Paving Wearing Courses	
32 12 19.16	Resin-Modified Asphalt Paving Wearing Courses	
32 12 19.19	Porous Friction Asphalt Paving Wearing Courses	
32 12 33	Flexible Paving Surface Treatments	
32 12 36	Seal Coats	
32 12 36.13	Asphaltic Seal and Fog Coats	
32 12 36.16	Coal Tar Seal Coats	
32 12 36.19	Coal Tar Seal Coats with Unvulcanized Rubber	
32 12 36.23	Fuel-Resistant Sealers	
32 12 43	Porous Flexible Paving	
32 12 73	Asphalt Paving Joint Sealants	

NUMBER	TITLE	EXPLANATION
32 13 00	Rigid Paving	Includes: rigid cement concrete pavements above base course, including conventional modified pavements for a variety of applications. May Include: base courses. See: 03 35 23 for exposed aggregate concrete finishing. 03 35 33 for stamped concrete finishing. 03 37 23 for roller-compacted concrete. 32 01 19 for rigid paving surface treatment. 32 01 26 for rigid paving rehabilitation. 32 01 29 for rigid paving repair. 32 12 00 for flexible paving. 32 14 00 for unit paving. 32 16 13 for curbs and gutters. 41 66 19 for mobile paving equipment. See Also: 32 11 00 for base courses.
32 13 13	Concrete Paving	
32 13 13.13	Exposed Aggregate Concrete Paving	
32 13 13.16	Power-Compacted Concrete Paving	
32 13 13.17	Roller-Compacted Concrete Paving	
32 13 13.19	Prestressed Concrete Paving	
32 13 13.23	Concrete Paving Surface Treatment	
32 13 13.26	Stained Concrete Paving	
32 13 13.33	Plain Concrete Paving	Alternate Terms/Abbreviations: unreinforced concrete paving.
32 13 16	Decorative Concrete Paving	
32 13 16.13	Patterned Concrete Paving	
32 13 16.19	Imprinted Concrete Paving	
32 13 16.23	Stamped Concrete Paving	
32 13 43	Pervious Concrete Paving	Alternate Terms/Abbreviations: porous concrete.
32 13 73	Concrete Paving Joint Sealants	
32 13 73.13	Fuel-Resistant Concrete Paving Joint Sealants	
32 13 73.16	Field-Molded Concrete Paving Joint Sealants	
32 13 73.19	Compression Concrete Paving Joint Sealants	

NUMBER	TITLE	EXPLANATION
32 14 00	Unit Paving	Includes: pavers laid on a prepared surface or base. May Include: base courses. Alternate Terms/Abbreviations: porous unit paving; pervious unit paving. See: 32 12 00 for flexible paving. 32 12 43 for porous flexible paving. 32 13 00 for rigid paving. 32 16 13 for curbs and gutters. 32 92 00 for grasses. 32 93 13 for ground covers. See Also: 32 11 00 for base courses.
32 14 13	Precast Concrete Unit Paving	
32 14 13.13	Interlocking Precast Concrete Unit Paving	
32 14 13.16	Precast Concrete Unit Paving Slabs	
32 14 13.19	Porous Precast Concrete Unit Paving	
32 14 16	Brick Unit Paving	
32 14 23	Asphalt Unit Paving	
32 14 26	Wood Paving	
32 14 29	Rubber Paving	
32 14 33	Plastic Paving	
32 14 33.13	Permeable Plastic Paving	
32 14 40	Stone Paving	
32 14 43	Porous Unit Paving	
32 15 00	Aggregate Surfacing	Includes: aggregate-only surfacing and binding agents for aggregate surfacing.
32 15 13	Cinder Surfacing	
32 15 40	Crushed Stone Surfacing	
32 15 50	Aggregate-Turf Pavement	Includes: soil-bound stone, gravel, or sand with seedbed soil.
32 16 00	Curbs, Gutters, Sidewalks, and Driveways	See: 03 30 00 for cast-in-place concrete. 03 40 00 for precast concrete. 32 12 00 for flexible paving. 32 13 00 for rigid paving. 32 14 00 for unit paving.
32 16 13	Curbs and Gutters	
32 16 13.13	Cast-in-Place Concrete Curbs and Gutters	
32 16 13.16	Steel Faced Curbs	
32 16 13.23	Precast Concrete Curbs and Gutters	

DIVISION 32

NUMBER	TITLE	EXPLANATION
32 16 13.33	Asphalt Curbs	
32 16 13.43	Stone Curbs	
32 16 23	Sidewalks	
32 16 33	Driveways	
32 17 00	Paving Specialties	Includes: special construction to enhance the safety and functionality of paved surfaces. See: 26 56 00 for exterior lighting. 34 71 13 for vehicle barriers. 34 71 16 for impact attenuating devices. 34 71 19 for vehicle delineators.
32 17 13	Parking Bumpers	Includes: bumpers to facilitate parking of vehicles. See: 03 48 26 for precast concrete parking bumpers.
32 17 13.13	Metal Parking Bumpers	
32 17 13.16	Plastic Parking Bumpers	
32 17 13.19	Precast Concrete Parking Bumpers	
32 17 13.23	Rubber Parking Bumpers	
32 17 13.26	Wood Parking Bumpers	
32 17 16	Manufactured Traffic-Calming Devices	Includes: preformed rubber or plastic traffic-calming devices; including speed bumps, speed humps, and speed cushions; applied to pavement. Does Not Include: site-formed or cut calming items integral with the paving.
32 17 23	Pavement Markings	Includes: informational and directional markings on pavement. See: 10 14 00 for signage. See Also: 09 91 00 for painting pavement markings.
32 17 23.13	Painted Pavement Markings	Includes: game and athletic marking.
32 17 23.23	Raised Pavement Markings	
32 17 23.33	Plastic Pavement Markings	
32 17 26	Tactile Warning Surfacing	Includes: modified walking surface textures and colors to provide information and warning about conditions to visually impaired. See: 10 14 00 for signage.

NUMBER	TITLE	EXPLANATION
32 17 29	Pavement Embedded Warning Lights	See: 34 41 33 for public and large private roadway installations.
32 17 43	Pavement Snow Melting Systems	
32 18 00	Athletic and Recreational Surfacing	Includes: surfaces for exterior athletic and recreational activities. Alternate Terms/Abbreviations: athletic surfacing: sports surfacing. See: 09 64 66 for wood athletic flooring. 09 65 66 for resilient athletic flooring. 09 67 66 for fluid-applied athletic flooring. 11 68 13 for playground equipment. 11 68 16 for play structures. 11 68 23 for exterior court athletic equipment. 11 68 33 for athletic field equipment. 13 28 66 for demountable athletic surfaces. 32 31 13 for recreational and tennis court fences and gates. 32 31 13 for chain link backstops for softball and baseball.
32 18 13	Synthetic Grass Surfacing	
32 18 16	Synthetic Resilient Surfacing	
32 18 16.13	Playground Protective Surfacing	
32 18 23	Athletic Surfacing	
32 18 23.13	Baseball Field Surfacing	
32 18 23.16	Natural Baseball Field Surfacing	
32 18 23.19	Synthetic Baseball Field Surfacing	
32 18 23.23	Field Sport Surfacing	
32 18 23.26	Natural Field Sport Surfacing	
32 18 23.29	Synthetic Field Sport Surfacing	
32 18 23.33	Running Track Surfacing	
32 18 23.36	Natural Running Track Surfacing	
32 18 23.39	Synthetic Running Track Surfacing	
32 18 23.43	Recreational Court Surfacing	
32 18 23.53	Tennis Court Surfacing	
32 18 23.56	Natural Tennis Court Surfacing	
32 18 23.59	Synthetic Tennis Court Surfacing	
32 18 23.63	Equestrian Surfacing	

32 30 00 Site Improvements

Includes: variety of exterior improvements.

NUMBER	TITLE	EXPLANATION
32 31 00	Fences and Gates	Includes: fences for protective, security, decorative, recreational, and right-of-way applications. Protective fence for bridge parapets and sidewalks. Gate operators. See: 01 56 26 for temporary fencing. 02 58 13 for snow fencing. 10 22 13 for wire mesh partitions. 11 68 00 for play field equipment and structures. 13 19 13 for kennel enclosures and gates.
32 31 11	Gate Operators	
32 31 13	Chain Link Fences and Gates	
32 31 13.23	Recreational Court Fences and Gates	
32 31 13.26	Tennis Court Fences and Gates	
32 31 13.29	Tennis Court Wind Breaker	
32 31 13.33	Chain Link Backstops	
32 31 13.53	High-Security Chain Link Fences and Gates	
32 31 16	Welded Wire Fences and Gates	
32 31 17	Expanded Metal Fences and Gates	
32 31 19	Decorative Metal Fences and Gates	
32 31 19.53	Decorative Metal Security Fences and Gates	Includes: decorative metallic-coated-steel security fences, including posts, rails, pickets, swing gates, horizontal-slide gates, gate operators and controls, and hardware.
32 31 23	Plastic Fences and Gates	
32 31 26	Wire Fences and Gates	
32 31 26.11	Wire Fences with Wood Posts	
32 31 26.13	Wire Fences with Steel Posts	
32 31 29	Wood Fences and Gates	
32 31 32	Composite Fences and Gates	
32 31 53	Cattle Guards	
32 31 56	Wild Life Deterrent Fence	

NUMBER	TITLE	EXPLANATION
32 32 00	Retaining Walls	Includes: retaining walls of any construction associated with exterior construction. See: 03 30 00 for cast-in-place concrete. 03 40 00 for precast concrete. 04 22 33 for interlocking concrete unit masonry. 06 13 00 for heavy timber. 31 34 00 for soil reinforcement. 31 36 00 for gabions. 31 66 16 for metal cribbing foundation walls. 35 31 19 for gabion revetments. 35 31 26 for gabion jetties. 35 31 29 for gabion groins. 35 42 36 for gabion bank protection. 35 43 36 for gabion scour protection. See Also: 03 30 00 for cast-in-place concrete retaining walls. 03 40 00 for precast concrete retaining walls. 33 41 33 for retaining wall drainage.
32 32 13	Cast-in-Place Concrete Retaining Walls	
32 32 16	Precast Concrete Retaining Walls	
32 32 19	Unit Masonry Retaining Walls	
32 32 23	Segmental Retaining Walls	
32 32 23.13	Segmental Concrete Unit Masonry Retaining Walls	
32 32 23.16	Manufactured Modular Walls	
32 32 26	Metal Crib Retaining Walls	
32 32 29	Timber Retaining Walls	
32 32 34	Reinforced Soil Retaining Walls	
32 32 36	Gabion Retaining Walls	
32 32 43	Soldier-Beam Retaining Walls	
32 32 53	Stone Retaining Walls	
32 33 00	Site Furnishings	Includes: furniture and fixtures for exterior applications. See: 12 60 00 for multiple seating. 12 68 00 for interior seat and table assemblies. 12 92 00 for interior planters and artificial plants. 32 90 00 for planting. 32 93 00 for live plants. 32 94 33 for planters. See Also: 32 94 33 for manufactured planters.
32 33 13	Site Bicycle Racks	
32 33 14	Site Bicycle Lockers	
32 33 15	Site Bicycle Shelters	
32 33 23	Site Trash and Litter Receptacles	

NUMBER	TITLE	EXPLANATION
32 33 33	Site Manufactured Planters	
32 33 43	Site Seating and Tables	
32 33 43.13	Site Seating	
32 33 43.53	Site Tables	

32 34 00 **Fabricated Bridges**

Includes: fabricated bridges for foot and vehicular traffic.

See: 05 12 33 for structural steel for bridges.

05 16 33 for bridge cabling.

34 81 00 for bridge machinery.

34 82 00 for bridge specialties.

32 34 13	Fabricated Pedestrian Bridges
32 34 23	Fabricated Roadway Bridges
32 34 33	Fabricated Railway Bridges

32 35 00 **Screening Devices**

Includes: self-supporting walls for noise control, safety, protection, and privacy.

Alternate Terms/Abbreviations: sound barriers: highway sound barriers.

See: 06 49 19 for wood screens and exterior wood shutters.

10 71 13 for exterior sun control devices.

10 82 13 for exterior grilles and screens.

34 73 19 for jet blast barriers.

32 35 13	Screens and Louvers
32 35 16	Sound Barriers
32 35 23	Fire Barriers
32 35 26	Blast Barriers

32 39 00 **Manufactured Site Specialties**

32 39 13	Manufactured Metal Bollards
32 39 13.11	Manufactured Metal Security Bollards
32 39 16	Manufactured Fire Pits
32 39 33	Artificial Rock Fabrications

32 70 00 **Wetlands**

Includes: construction and restoration of excavated basins with purposely placed wetland vegetation to enhance pollutant removal from stormwater runoff.

See: 02 50 00 for site remediation.

22 14 00 for facility storm drainage.

31 20 00 for earth moving.

33 40 00 for storm drainage utilities.

NUMBER	TITLE	EXPLANATION
32 71 00	Constructed Wetlands	
32 72 00	Wetlands Restoration	
32 80 00	Irrigation	Includes: pipe underground or surface systems with equipment, piping, sprinkler heads, valves, and controls. See: 11 92 00 for agricultural equipment. 11 93 00 for horticultural equipment. 13 34 13 for greenhouses. 22 11 00 for facility water distribution. 32 90 00 for planting. 33 14 00 for water utility transmission and distribution.
32 81 00	Irrigation Components	
32 81 13	Irrigation Intake Screens	
32 82 00	Irrigation Pumps	
32 84 00	Planting Irrigation	
32 84 13	Drip Irrigation	
32 84 23	Underground Sprinklers	
32 86 00	Agricultural Irrigation	
32 90 00	Planting	Includes: planting and transplanting of trees, shrubs, vines, and other herbaceous plants; temporary storage, layout, bed preparation, preparation of pits, pruning, planting, fertilizing, and backfilling, wrapping, staking and guying, protection, edging, and maintenance of new plantings when part of construction or landscaping contract. See: 01 56 39 for temporary tree and plant protection. 31 13 00 for selective tree and shrub removal and trimming. 31 14 13 for topsoil stripping and stockpiling. 31 22 19 for finish grading. 32 01 90 for maintenance for existing planting. 32 84 00 for planting irrigation.

NUMBER	TITLE	EXPLANATION
32 91 00	Planting Preparation	Includes: tillage and plant bed preparation, protection, and stabilization. Usually Includes: netting, blankets, and stabilizer mats for stabilizing soil and establishing plants. See: 31 14 13 for topsoil stripping and stockpiling. 31 22 19 for finish grading. 31 31 19 for vegetation control. 31 32 13 for soil mixing stabilization. 32 01 90 for fertilizing and amending soils for maintenance.
32 91 13	Soil Preparation	
32 91 13.13	Hydro-Punching	
32 91 13.16	Mulching	
32 91 13.19	Planting Soil Mixing	
32 91 13.23	Structural Soil Mixing	
32 91 13.26	Planting Beds	
32 91 16	Planting Soil Stabilization	
32 91 16.13	Blanket Planting Soil Stabilization	
32 91 16.16	Mat Planting Soil Stabilization	
32 91 16.19	Netting Planting Soil Stabilization	
32 91 19	Landscape Grading	
32 91 19.13	Topsoil Placement and Grading	
32 92 00	Turf and Grasses	Includes: application of seed, sod, or plants, fertilizer, lime, and mulch, and maintenance until acceptance. May Include: soil preparation and finish grading with topsoil, and planting accessories. See: 31 25 00 for erosion and sedimentation control. 32 14 43 for porous unit paving. See Also: 31 22 19 for finish grading. 32 91 00 for planting preparation. 32 94 00 for planting accessories.
32 92 13	Hydro-Mulching	
32 92 16	Plugging	
32 92 19	Seeding	
32 92 19.13	Mechanical Seeding	
32 92 19.16	Hydraulic Seeding	
32 92 23	Sodding	
32 92 26	Sprigging	
32 92 26.13	Stolonizing	

NUMBER	TITLE	EXPLANATION
32 93 00	Plants	Includes: planting pits, pruning, planting, supplements, backfilling, wrapping, staking, and guying, protection, and maintenance of new plantings as part of construction or landscaping contract. May Include: soil preparation and finish grading with topsoil, and planting accessories. See: 12 92 33 for interior planters. 31 13 00 for selective tree and shrub removal and trimming. 32 01 90 for pruning for maintenance of shrubs and trees. 32 14 43 for porous unit paving. 32 96 00 for transplanting. See Also: 31 22 19 for finish grading. 32 91 00 for planting preparation. 32 94 00 for planting accessories.
32 93 13	Ground Covers	
32 93 23	Plants and Bulbs	
32 93 33	Shrubs	
32 93 43	Trees	
32 93 93	Exterior Artificial Plants	
32 94 00	Planting Accessories	Includes: accessories to separate, house, and protect plants. May Include: interior planters, manufactured planters. See: 13 34 13 for greenhouses. 32 14 43 for porous unit paving. See Also: 12 92 33 for interior planters.
32 94 13	Landscape Edging	
32 94 16	Landscape Timbers	
32 94 19	Landscape Surfacing	
32 94 33	Planters	
32 94 43	Tree Grates	
32 94 46	Tree Grids	
32 95 00	Exterior Planting Support Structures	Includes: trellis, treillage, green wall, arbor, and climbing support netting.

NUMBER	TITLE	EXPLANATION
32 96 00	Transplanting	Includes: relocation of plant material. Application of antidesiccant and antitranspirant, excavation, replanting, pruning, watering, and maintenance of transplanted shrubs and trees. Alternate Terms/Abbreviations: transplanting: replanting. See: 31 13 00 for selective tree and shrub removal and trimming. 32 93 00 for plants.
32 96 13	Ground Cover Transplanting	
32 96 23	Plant and Bulb Transplanting	
32 96 33	Shrub Transplanting	
32 96 43	Tree Transplanting	

NUMBER	TITLE	EXPLANATION
DIVISION 33 – UTILITIES		
33 00 00	Utilities	May be used as Division level Section title. See: 02 41 13 for selective demolition of existing utilities. 03 30 00 for cast-in-place concrete. 09 90 00 for field painting. 31 23 33 for trenching and backfilling. 40 60 00 and 40 70 00 for process instrumentation and controls. 43 20 00 for process liquid pumps in specially constructed (non-packaged) pumping stations. 43 40 00 for fabricated process storage tanks at a facility.

33 01 00 Operation and Maintenance of Utilities

Includes: maintenance, repair, rehabilitation, replacement, restoration, preservation, etc. of utilities.

Level 4 Numbering Recommendation: Level 4 titles for utilities are provided as examples for user's creation of more detailed titles for other subjects as required. Following Level 4 numbering is recommended:

.51-.59 for maintenance.

.61-.69 for repair.

.71-.79 for rehabilitation.

.81-.89 for replacement.

.91-.99 for restoration.

See: 09 97 00 for surface preparation and painting of existing water storage tanks.

33 01 10	Operation and Maintenance of Water Utilities	
33 01 10.51	Water Utility Intake Cleaning and Maintenance	Includes: cleaning of water intake structures and intake conduits, including removal of sediment and aquatic organisms.
33 01 10.52	Reservoir Maintenance	
33 01 10.53	Flushing of Water Utility Piping	
33 01 10.54	Cleaning of Water Utility Piping	Includes: cleaning of transmission and distribution system piping by methods other than flushing. Includes dewatering and removal of sediment and other matter.

DIVISION 33

NUMBER	TITLE	EXPLANATION
33 01 10.56	Cleaning of Water Utility Storage Tanks	
33 01 10.57	Disinfection of Water Supply Wells	
33 01 10.58	Disinfection of Water Utility Piping Systems	
33 01 10.59	Disinfection of Water Utility Storage Tanks	
33 01 10.81	Water Supply Well Abandonment	
33 01 10.91	Rehabilitation of Water Supply Wells	
33 01 10.92	Redevelopment of Water Supply Wells	
33 01 12	Inspection and Testing of Water Utilities	
33 01 12.11	Leakage Detection	
33 01 12.13	Electrical Continuity Testing	
33 01 30	Operation and Maintenance of Sewer Utilities	
33 01 30.11	Television Inspection of Sewers	
33 01 30.13	Dye Testing of Sewers	
33 01 30.16	Smoke Testing	
33 01 30.23	Pipe Bursting	
33 01 30.41	Cleaning of Sewers	
33 01 30.42	Cleaning of Manholes	
33 01 30.43	Removal of Protruding Service Connection	
33 01 30.44	Chemical Treatment of Roots in Sewer Lines	
33 01 30.61	Packer Injection Grouting	
33 01 30.72	Cured-in-Place Pipe Lining	Includes: cured-in-place lining by water, steam, and ultraviolet inversion methods.
33 01 30.73	Cured-in-Place Point Repair	
33 01 30.74	Cured-in-Place Lateral Lining	
33 01 30.77	Cured-in-Place Lateral Shortliners	
33 01 30.79	Fold-and-Form Pipe Lining	
33 01 30.81	Manhole Rehabilitation	
33 01 30.82	Epoxy Lining for Concrete Manholes	
33 01 30.84	Manhole Inserts	
33 01 30.86	Manhole Rim Adjustment	
33 01 34	Operation and Maintenance of Onsite Wastewater Disposal Systems	
33 01 34.13	Septic Tank Solids Removal	
33 01 34.14	Septic Tank Maintenance	Includes: probiotic additives, chlorine tablets, and other work results associated with routine maintenance of septic tanks.
33 01 34.19	Septic Tank Abandonment	
33 01 34.33	Maintenance of Onsite Sand Filters	
33 01 50	Operation and Maintenance of Hydrocarbon Utilities	
33 01 50.53	Cleaning of Hydrocarbon Conveyance Piping	
33 01 50.56	Cleaning of Hydrocarbon Storage Tanks	
33 01 50.73	Relining of Hydrocarbon Conveyance Piping	
33 01 50.76	Relining of Steel Hydrocarbon Storage Tanks	
33 01 50.81	Hydrocarbon Well Abandonment	
33 01 50.91	Rehabilitation of Hydrocarbon Wells	
33 01 60	Operation and Maintenance of Hydronic and Steam Energy Utilities	
33 01 70	Operation and Maintenance of Electrical Utilities	
33 01 70.51	Electrical Utility Visual Inspection	
33 01 70.52	Electrical Utility Thermographic Surveys	
33 01 70.53	Electrical Utility Insulation Testing	
33 01 70.54	Electrical Utility Insulating Oil/Fluid Analysis	
33 01 80	Operation and Maintenance of Communications Utilities	

NUMBER	TITLE	EXPLANATION
33 01 80.51	Communication Utility Visual Inspection	
33 01 80.52	Communication Utility Thermographic Surveys	

33 05 00 Common Work Results for Utilities

Includes: subjects common to multiple titles in Division 33.

Alternate Terms/Abbreviations: ABS: acrylonitrile butadiene styrene. PVC: polyvinyl chloride. PCCP: prestressed concrete cylinder pipe. BWP: bar-wrapped concrete pressure pipe steel cylinder type or bar-wrapped pipe. HDPE: high-density polyethylene.

See: 01 80 00 for performance requirements of subjects common to multiple titles.

26 05 53 for identification for electrical systems.

31 23 33 for trenching and backfilling.

33 41 00 for subdrainage.

33 42 13 for stormwater culverts.

40 05 00 for common work results for exposed process piping.

40 05 00 for valves used in utilities and process service.

33 05 04	Selective Demolition for Utilities
33 05 05	Buried Piping Installation
33 05 05.11	Installing Bell-and-Spigot Joints
33 05 05.13	Installing Push-on Joints
33 05 05.15	Installing Butt-and-Strap Joints
33 05 05.16	Installing Mechanical Joints
33 05 05.18	Installing Restrained Joints
33 05 05.19	Fusing Joints
33 05 05.31	Hydrostatic Testing
33 05 05.33	Infiltration and Exfiltration Testing
33 05 05.36	Vacuum Testing
33 05 05.41	Air Testing
33 05 05.43	Mandrel Testing
33 05 05.46	Lamp Testing
33 05 07	Trenchless Installation of Utility Piping
33 05 07.13	Utility Directional Drilling
33 05 07.23	Utility Boring and Jacking
33 05 07.24	Casing Piping for Utility Jacking
33 05 07.33	Tunnelling
33 05 07.36	Microtunnelling
33 05 07.43	Utility Impact Moling
33 05 07.53	Cable Trenching and Plowing
33 05 09	Piping Specials for Utilities
33 05 09.13	Couplings for Utility Piping
33 05 09.23	Adapters for Utility Piping
33 05 09.33	Thrust Restraint for Utility Piping
33 05 09.43	Tapping Sleeves and Saddles for Utility Piping
33 05 11	Aluminum Utility Pipe
33 05 12	Brass Utility Pipe
33 05 16	Cast-Iron Utility Pipe
33 05 17	Copper Utility Pipe and Tubing

DIVISION 33

NUMBER	TITLE	EXPLANATION
33 05 19	Ductile-Iron Utility Pipe	
33 05 21	High-Alloy Nickel Utility Pipe and Tubing	
33 05 23	Stainless Steel Utility Pipe and Tubing	
33 05 23.13	Duplex and Super-Duplex Stainless Steel Utility Pipe	
33 05 23.23	300- and 400-Series Stainless Steel Utility Pipe	
33 05 23.33	Special-Fabrication Stainless Steel Utility Pipe and Tubing	
33 05 24	Steel Utility Pipe	
33 05 24.13	Steel Pipe and Tubing for Air and Gas Service	
33 05 24.23	Steel Pipe for Water Service	
33 05 24.26	Steel Pipe and Tubing for Utility Liquid Service	
33 05 24.33	Steel Pipe for Bulk Materials Conveyance	
33 05 27	Corrugated Metal Utility Pipe	
33 05 27.13	Corrugated Metal Culvert Pipe	
33 05 27.23	Corrugated Metal Slotted Pipe	
33 05 27.33	Corrugated Metal Culvert Arch Pipe	
33 05 31	Thermoplastic Utility Pipe	
33 05 31.11	Polyvinyl Chloride Gravity Sewer Pipe	
33 05 31.13	Polyvinyl Chloride Pressure Pipe	
33 05 31.16	Polyvinyl Chloride Pressure Pipe for Water Transmission and Distribution (AWWA C900 and AWWA C905)	
33 05 31.19	Fusible Polyvinyl Chloride Pipe	
33 05 31.21	Ribbed Polyvinyl Chloride Gravity Pipe	
33 05 31.23	Dual-Wall Polyvinyl Chloride Profile Pipe	
33 05 31.26	Polyvinyl Chloride Sleeve Pipe	
33 05 31.27	Polyvinyl Chloride Perforated Drainage Pipe	
33 05 31.29	Polyvinyl Chloride Electrical Duct	
33 05 31.33	ABS Utility Pipe	
33 05 33	Polyethylene Utility Pipe	
33 05 33.13	Corrugated-Wall, Smooth Interior HDPE Gravity Pipe	
33 05 33.23	Polyethylene Pressure Pipe and Tubing (AWWA C901 and AWWA C906)	
33 05 33.33	Corrugated, Perforated HDPE Drainage Pipe	
33 05 33.36	Corrugated, Non-Perforated HDPE Drainage Pipe	
33 05 33.53	HDPE Gas Utility Pipe	
33 05 33.56	Medium-Density Polyethylene Gas Utility Pipe	
33 05 33.63	HDPE Communication Duct	
33 05 36	Fiberglass-Reinforced Plastic Utility Pipe	
33 05 36.13	Fiberglass-Reinforced Plastic Pipe for Utility Air, Gas, and Odor Service	
33 05 36.23	Fiberglass-Reinforced Plastic Pipe for Utility Liquid Service	
33 05 36.33	Fiberglass-Reinforced Plastic Pipe for Electrical Service	
33 05 37	Centrifugally Cast Fiberglass-Reinforced Polymer Mortar Utility Pipe	
33 05 39	Concrete Pipe	
33 05 39.11	Concrete Pressure Pipe, Steel-Cylinder Type (AWWA C300)	
33 05 39.13	Prestressed Concrete Pressure Pipe, Steel-Cylinder Type (AWWA C301)	
33 05 39.16	Reinforced Concrete Pressure Pipe, Non-cylinder Type (AWWA C302)	
33 05 39.17	Concrete Pressure Pipe, Bar-Wrapped Steel-Cylinder Type (AWWA C303)	
33 05 39.23	Reinforced Concrete Low-Head Pressure Utility Pipe	
33 05 39.33	Non-Reinforced Concrete Sewer and Culvert Pipe	
33 05 39.36	Non-Reinforced Concrete Irrigation and Drainage Pipe	
33 05 39.41	Reinforced Concrete Pipe for Sewers and Culverts	
33 05 39.43	Perforated Concrete Utility Pipe	
33 05 39.46	Reinforced Concrete Elliptical Pipe for Sewers and Culverts	
33 05 39.49	Reinforced Concrete Box Sections for Sewers and Culverts	
33 05 39.63	Concrete Pipe Closures	
33 05 39.66	Concrete Pipe Thrust Restraints	
33 05 39.69	Concrete Pipe Wall Fittings	

NUMBER	TITLE	EXPLANATION
33 05 41	Vitrified Clay Utility Pipe	
33 05 43	Ceramic Drainage Tile	
33 05 46	Asbestos Cement Utility Pipe	See: 02 82 33 for removal of asbestos cement pipe.
33 05 61	Concrete Manholes	Includes: standard manholes, doghouse manholes, manhole barrels, flat-top sections, cone-top sections, manhole benches, and accessories such as rubber boots and manhole steps.
33 05 63	Concrete Vaults and Chambers	
33 05 66	Concrete Distribution Chambers	Includes: leech field distribution chambers and leachate distribution or collection chambers, and accessories.
33 05 69	Concrete Trenchways	
33 05 71	Cleanouts	
33 05 73	Polyethylene Manholes	
33 05 76	Fiberglass Manholes	
33 05 77	Fiberglass Metering Manholes	Includes: manhole, integral flume for open-channel flow metering or mounting for closed-pipe flow meter installation, and accessories.
33 05 81	Metallic Castings for Utility Structures	Includes: metal covers for manholes, chambers, and vaults, metallic steps for manholes, and other metallic accessories for buried utility structures.
33 05 83	Non-Metallic Accessories for Utility Structures	Includes: fiberglass covers for manholes, chambers, and vaults, polypropylene steps for manholes, and other non-metallic accessories for buried utility structures.
33 05 93	Common Motor Requirements for Utilities	
33 05 96	Vibration and Seismic Controls for Utilities	
33 05 97	Identification and Signage for Utilities	See: 26 05 53 for identification requirements for electrical systems in a facility.
33 05 97.13	Signage for Utilities	
33 05 97.16	Markers for Utility Identification	
33 05 97.23	Underground Utility Warning Tape	
33 05 97.33	Utility Identification Flags	
33 06 00	Schedules for Utilities	Includes: schedules of items common to multiple titles in Division 33. Notes: a schedule may be included on drawings, in the project manual, or a project book.

DIVISION 33

NUMBER	TITLE	EXPLANATION
33 06 05	Schedules for Utility Piping	
33 06 10	Schedules for Water Utilities	
33 06 30	Schedules for Sanitary Sewerage Utilities	
33 06 40	Schedules for Stormwater Utilities	
33 06 40.13	Stormwater Drainage System Schedules	
33 06 50	Schedules for Hydrocarbon Utilities	
33 06 60	Schedules for Hydronic and Steam Energy Utilities	
33 06 70	Schedules for Electrical Utilities	
33 06 80	Schedules for Communications Utilities	

33 08 00 Commissioning of Utilities

Includes: commissioning of items common to multiple titles in Division 33.

See: 01 91 00 for commissioning of subjects common to multiple Divisions.

33 08 10	Commissioning of Water Utilities
33 08 30	Commissioning of Sanitary Sewerage Utilities
33 08 40	Commissioning of Stormwater Utilities
33 08 50	Commissioning of Hydrocarbon Utilities
33 08 60	Commissioning of Hydronic and Steam Energy Utilities
33 08 70	Commissioning of Electrical Utilities
33 08 80	Commissioning of Communications Utilities

33 09 00 Instrumentation and Control for Utilities

Includes: instrumentation and control common to multiple titles in Division 33.

Notes: radio frequency (RF) communications between RF-capable meter registers and data collection/repeater units can employ various communication technologies, including narrow band (single, fixed radio frequency), spread spectrum, direct-sequence spread spectrum, frequency-hopping spread spectrum, cellular phone data networks, Bluetooth, ZigBee, Wavenis, and other wireless RF communication technologies.

See: 13 50 00 for special instrumentation.

21 09 00 for instrumentation and control for fire-suppression systems.

22 09 00 for instrumentation and control for plumbing.

23 09 00 for instrumentation and control for HVAC.

26 09 00 for instrumentation and control for electrical systems.

33 19 00 for water utility metering equipment.

33 59 00 for hydrocarbon utility metering equipment.

40 60 00 and 40 70 00 for instrumentation and control for process systems.

48 09 00 for instrumentation and control for electrical power generation.

NUMBER	TITLE	EXPLANATION
33 09 01	Utility Meter Registers	
33 09 01.13	Mechanical Registers	
33 09 01.23	Pulse Registers	
33 09 01.33	Encoder Registers	
33 09 02	Touch-based Utility Meter Reading Equipment	
33 09 03	Handheld Radio Frequency Utility Meter Reading Equipment	Alternate Terms/Abbreviations: handheld: walk-by.
33 09 04	Mobile Radio Frequency Utility Meter Reading Equipment	Alternate Terms/Abbreviations: mobile: drive-by.
33 09 05	Fixed-Network Radio Frequency Utility Meter Reading Equipment	Includes: a series of antennas, towers, collectors, repeaters, or other permanently installed infrastructure to collect transmissions of meter readings from enabled meter registers and relay the data to a central computer, without a person in the field to collect such data. May Include: star network systems (data transmitted to a central data collector by repeaters), mesh network systems (where meter registers serve as repeaters), and hybrid networks (networks that include both mobile and fixed meter reading and transmitting equipment).
33 09 06	Satellite Radio Frequency Utility Meter Reading Equipment	
33 09 07	Power Line Communication Utility Meter Reading Equipment	
33 09 08	Wireless Local Area Network Utility Meter Reading Equipment	Alternate Terms/Abbreviations: WLAN: wireless local area network: wi-fi.
33 09 09	Meter Data Acquisition and Hosting Software	
33 09 10	Instrumentation and Control for Water Utilities	
33 09 30	Instrumentation and Control for Sanitary Sewerage Utilities	
33 09 40	Instrumentation and Control for Stormwater Utilities	
33 09 50	Instrumentation and Control for Hydrocarbon Utilities	
33 09 53	Leak Detection for Hydrocarbon Systems	
33 09 60	Instrumentation and Control for Hydronic and Steam Energy Utilities	
33 09 70	Instrumentation and Control for Electrical Utilities	
33 09 80	Instrumentation and Control for Communications Utilities	

NUMBER	TITLE	EXPLANATION
33 10 00	Water Utilities	Includes: public and private water sources, distribution, and storage for potable uses, firefighting, and irrigation for a site, for multiple facilities, or municipalities. See: 21 11 00 for fire-suppression water service for a building or structure. 22 11 00 for water distribution for a building or structure. 32 80 00 for irrigation. 33 01 10 for maintenance of water utilities, including cleaning and disinfecting. 33 09 10 for instrumentation and control for water utilities. 43 20 00 for pumps used in non-packaged, field-assembled water pumping systems. 43 30 00 for gas and liquid purification equipment. 46 00 00 for water treatment equipment, including chemical feed systems used in water distribution systems.
33 11 00	Groundwater Sources	Includes: dug, bored, driven, or drilled wells, for water supply and aquifer recharging, and monitoring wells. Well casings, well screens, riser pipes, well pumps, and accessories. Wells for a single site or facility and for multiple facilities and municipalities. See: 02 32 23 for groundwater monitoring before construction. 02 70 00 for wells used for groundwater extraction for remediation purposes. 28 20 00 for video surveillance. 31 09 13 for groundwater monitoring during construction. 31 23 19 for groundwater systems for dewatering of excavations. 32 80 00 for irrigation. 33 01 10 for abandonment, rehabilitation, redevelopment, and disinfection of water supply wells. 33 09 10 for instrumentation and control for water supply wells. 40 70 00 for instrumentation for wells.
33 11 11	Test Wells	
33 11 13	Potable Water Supply Wells	
33 11 16	Irrigation Wells	
33 11 19	Aquifer Recharging Wells	
33 11 33	Vertically Suspended Dry-Location Centrifugal Well Pumps	
33 11 36	Submersible Centrifugal Well Pumps	
33 11 39	Jet-Type Well Pumps	
33 11 53	Monitoring Wells	

NUMBER	TITLE	EXPLANATION
33 12 00	Surface Water Sources	Includes: work results associated with surface water sources, including reservoirs and intakes in streams and lakes, for public, institutional, or private water supply. See: 03 00 00 for concrete construction. 28 20 00 for video surveillance. 31 05 19 for geosynthetics for reservoir liners. 31 23 00 for excavation and fill for reservoirs. 31 24 00 for embankments for reservoirs. 31 37 00 for riprap for reservoirs, rivers, and streams. 31 51 00 for anchoring systems for reservoir floating covers. 32 31 00 for fencing and gates for site security. 33 01 10 for intake structure cleaning and maintenance. 33 01 10 for reservoir maintenance. 33 05 00 for pipe used in water intakes. 33 16 00 for rain water harvesting cisterns. 33 40 00 for conveyance of stormwater drained from covered reservoirs. 35 53 00 for waterways and marine construction, including dams and associated equipment, shoreline protection, waterway scour protection, and underwater construction. 40 05 59 for hydraulic gates. 46 21 00 for water screening equipment and trash raking equipment.
33 12 11	Intake Structure Systems	
33 12 13	Intake Cribs	
33 12 16	Intake Drop and Riser Shafts	See: 33 01 10 for intake cleaning and maintenance. 33 05 00 for intake tubes. 35 53 00 for underwater construction.
33 12 31	Reservoir Weight-tensioned, Defined Sump Floating Covers	
33 12 31.13	Reservoir Weight-tensioned, Defined Sump Polyethylene-reinforced Floating Covers	
33 12 31.23	Reservoir Weight-tensioned, Defined Sump Polypropylene-reinforced Floating Covers	
33 12 33	Reservoir Mechanically-tensioned Floating Covers	
33 12 33.13	Reservoir Mechanically-tensioned Polyethylene-reinforced Floating Covers	
33 12 33.23	Reservoir Mechanically-tensioned Polypropylene-reinforced Floating Covers	
33 12 36	Reservoir Short-term Floating Covers	
33 12 39	Reservoir Ball Floating Covers	
33 12 43	Floats for Reservoir Floating Covers	
33 12 46	Appurtenances for Reservoir Floating Covers	May Include: hatches, air vents, walkways, and other appurtenances not otherwise specified with the floating cover system.
33 12 53	Reservoir Flexible Baffle Curtains	

NUMBER	TITLE	EXPLANATION
33 14 00	Water Utility Transmission and Distribution	Includes: water transmission and distribution piping and equipment for public or private services, whether for a single site or facility, or multiple facilities, including municipalities. See: 13 47 13 for cathodic protection. 21 11 00 for facility fire-suppression water service piping and appurtenances for a building or structure. 22 11 00 for facility water distribution for a building or structure. 28 20 00 for video surveillance. 32 80 00 for irrigation. 33 01 10 for flushing, cleaning, and disinfecting of water transmission and distribution piping and equipment. 33 05 00 for pipe used in water transmission and distribution. 33 05 09 for piping specialties, including couplings, adapters, thrust restraints, and tapping sleeves. 33 05 63 for vaults and chambers. 33 09 00 for registers and meter reading equipment for water meters. 33 16 00 for water utility storage tanks. 33 19 00 for water utility metering equipment. 40 05 76 for hot-tapping devices, including tapping sleeves and line stops for process piping. 40 05 78 for valves used in water utility service, including various types of isolation, throttling, and check valves, pump control valves, backflow preventers of various types, surge valves, altitude valves, air release valves, air/vacuum valves, and combination air valves for water service, and other valves. 46 30 00 for chemical feed systems used in water transmission, storage, and distribution systems.
33 14 11	Water Utility Transmission Piping	
33 14 13	Public Water Utility Distribution Piping	
33 14 16	Site Water Utility Distribution Piping	
33 14 17	Site Water Utility Service Laterals	
33 14 19	Valves and Hydrants for Water Utility Service	Notes: this is an alternate location for specifying water utility valves and hydrants. Valves and hydrants are indicated in detail under 40 05 00.
33 14 23	Enclosures for Water Utility Piping and Valves	
33 14 23.11	Heated, Insulated Aluminum Enclosures for Water Utility Piping and Valves	
33 14 23.13	Heated, Insulated Fiberglass Enclosures for Water Utility Piping and Valves	
33 14 23.16	Unheated, Insulated Aluminum Enclosures for Water Utility Piping and Valves	
33 14 23.17	Unheated, Insulated Fiberglass Enclosures for Water Utility Piping and Valves	
33 14 23.23	Fiberglass Artificial Rock Enclosures for Water Utility Piping and Valves	
33 14 23.26	Plastic Artificial Rock Enclosures for Water Utility Piping and Valves	
33 14 23.33	Enclosures for Water Utility Air Valves	

NUMBER	TITLE	EXPLANATION
33 14 43	Packaged Pumping Systems for Water Utility Service	Includes: factory-assembled systems purchased as a package. Field-assembled engineered systems not purchased as a package are indicated in detail under 43 20 00 (pumps) and 40 00 00 (exposed process piping). Packaged pumping systems include a variety of skid-mounted pumps, suction and discharge piping and valves, and controls, including control panels, variable-speed drives, two-speed drives, constant-speed units, and combinations thereof. Typical water utility service application is pressure-boosting. Pumps may be oriented vertically or horizontally on the skid. Pump types may include: between-bearings centrifugal pumps (split-case axial pumps), overhung (end suction) centrifugal pumps, and vertically suspended centrifugal pumps (vertical turbine pumps).

NUMBER	TITLE	EXPLANATION
33 16 00	Water Utility Storage Tanks	Includes: water storage tanks for a single facility or site, and for multiple facilities and municipalities. Includes tanks for storing treated water and for rain water harvesting (cisterns). Alternate Terms/Abbreviations: AWWA: American Water Works Association. See: 03 00 00 for concrete used in constructing foundations and other tank elements. 05 50 00 for metal ladders, stairs, platforms, handrails, and railings. 09 97 13 for storage tank coatings. 13 20 00 for special-purpose rooms installed in pillars of elevated storage tanks. 13 47 13 for cathodic protection for metallic storage tanks. 21 41 00 for storage tanks for fire-suppression water. 22 12 00 for potable water storage tanks for buildings or structures. 22 14 53 for rain water storage tanks within a structure. 28 20 00 for video surveillance. 33 01 10 for disinfection of water storage tanks. 33 34 00 for onsite wastewater disposal. 33 41 00 for subdrainage. 33 56 00 for hydrocarbon storage tanks. 33 80 00 for communication utility equipment installed on water storage tanks. 34 43 13 for air traffic hazard lighting installed on elevated water storage tanks and antennas. 40 05 73 for altitude valves. 40 72 00 for level-measuring instrumentation. 43 40 00 for fabricated storage tanks used at a process facility. 46 30 00 for chemical feed systems used with water storage tanks. 46 41 00 for equipment used for mixing water storage tank contents.
33 16 11	Elevated Composite Water Storage Tanks	Notes: composite tanks are constructed using concrete and steel.
33 16 13	Elevated Pillar-Type Steel Water Storage Tanks	Alternate Terms/Abbreviations: fluted-column water storage tanks. Notes: typically includes tank installed atop a single, large-diameter, corrugated-wall pillar.

NUMBER	TITLE	EXPLANATION
33 16 16	Multi-Column Elevated Steel Water Storage Tanks	Includes: tanks with ellipsoidal, torical, dome, or flat bottoms, and ellipsoidal, torical, or cone-shaped roofs.
33 16 19	Elevated Spheroid Steel Water Storage Tanks	
33 16 23	Ground-Level Steel Water Storage Tanks	Includes: tanks with ellipsoidal, torical, dome, or cone-shaped roofs, typically with flat bottoms. Includes welded or bolted construction. Alternate Terms/Abbreviations: ground-level storage tanks where the tank height is greater than its diameter: standpipes. Ground-level storage tanks where the tank diameter is greater than its height: reservoirs.
33 16 31	Ground-Level AWWA D110 Type I Prestressed Concrete Water Storage Tanks	
33 16 32	Ground-Level AWWA D110 Type II Prestressed Concrete Water Storage Tanks	
33 16 33	Ground-Level AWWA D110 Type III Prestressed Concrete Water Storage Tanks	May Include: roofs constructed as clear span free-standing concrete domes, or with column-supported conventionally reinforced flat slab roofs.
33 16 36	Ground-Level Reinforced Concrete Water Storage Tanks	
33 16 53	Ground-Level Metal Water Storage Cisterns	
33 16 56	Ground-Level Plastic Water Storage Cisterns	
33 16 61	Underground Concrete Water Storage Cisterns	Includes: precast and specially fabricated cast-in-place concrete tanks, typically used for rain water harvesting. May Include: integral inlet filters, discharge pump system, and appurtenances.
33 16 63	Underground Non-Metallic Water Storage Cisterns	Includes: polyethylene and fiberglass tanks, typically used for rain water harvesting. May Include: integral inlet filters, discharge pump system, and appurtenances.
33 16 66	Underground Plastic Modular Water Storage Cisterns	
33 16 93	Water Storage Tank Interior Curtain Walls	Includes: divider walls of various materials such as geomembranes, precast concrete, shotcrete, concrete masonry units, and others, used for baffling and separation inside a water storage tank.
33 16 96	Water Storage Tank Appurtenances	Includes: hatches, overflow weirs/boxes and discharge piping, vents, fall-prevention equipment, and other appurtenances not specified with the tank.

NUMBER	TITLE	EXPLANATION
33 19 00	Water Utility Metering Equipment	Includes: equipment for metering water distributed to multiple facilities, including municipal water metering systems. Notes: water meters indicated under 33 19 00 are uniquely adapted to utility water service metering. General types of process flow meters are indicated in detail under 40 71 00. See: 33 01 10 for flushing, cleaning, and disinfecting of water transmission and distribution piping and equipment. 33 09 00 for registers and meter reading equipment for water utilities. 33 14 00 for water utility transmission and distribution. 40 05 00 for valves used in water utility service. 40 71 00 for process flow meters.
33 19 13	Positive Displacement Water Meters	Includes: meters with bronze, brass, or plastic bodies. Often used for metering residences and small commercial properties.
33 19 13.13	Oscillating Piston Water Meters	
33 19 13.23	Nutating Disc Water Meters	
33 19 16	Velocity-Type Water Meters	
33 19 16.13	Single-Jet Water Meters	
33 19 16.16	Multi-Jet Water Meters	
33 19 16.23	Turbine Water Meters	Includes: meters with bronze, cast-iron, or ductile-iron bodies.
33 19 16.26	Fire-Type Turbine Water Meters	
33 19 16.29	Portable Fire Hydrant-Type Turbine Water Meters	
33 19 16.33	Propeller Water Meters	
33 19 19	Compound Water Meters	
33 19 23	Magnetic Water Meters	
33 19 26	Ultrasonic Water Meters	
33 19 29	Pre-paid Water Meters	

33 30 00 Sanitary Sewerage

Includes: wastewater sewerage for a facility site and for multiple facilities, for both sanitary sewage and combined sewage.

See: 22 13 00 for sanitary sewerage for a structure.
33 09 30 for instrumentation and control for sanitary sewerage.
44 40 00 for water pollution control equipment.
46 00 00 for wastewater treatment equipment, including packaged treatment plants.

NUMBER	TITLE	EXPLANATION
33 31 00	Sanitary Sewerage Piping	Includes: sanitary waste piping systems, with piping, valves, piping specialties, interceptors, separators, and other components serving a facility site or multiple facilities. Alternate Terms/Abbreviations: SDGS: small-diameter gravity sewers. STEP: septic tank effluent pump. Notes: gray-water system may refer to either small-diameter gravity sewers or septic tank effluent pump systems. Because they convey sanitary sewage and are constructed equal to sanitary sewerage, combined sewers are included under sanitary sewerage gravity piping and force main piping. See: 13 47 13 for cathodic protection. 22 13 13 for facility sanitary sewers. 33 05 71 for manholes, chambers, and cleanouts. 33 31 23 for sanitary sewerage force main piping. 40 05 78 for air valves used in wastewater service.
33 31 11	Public Sanitary Sewerage Gravity Piping	
33 31 13	Site Sanitary Sewerage Gravity Piping	
33 31 16	Site Sanitary Sewerage Gravity Service Laterals	Notes: service laterals are typically the buried sanitary line between the building or structure and the main collector sewer.
33 31 19	Small-Diameter Sanitary Sewerage Gravity Piping	Notes: small-diameter gravity sewer systems have onsite primary treatment, usually via septic tanks.
33 31 23	Sanitary Sewerage Force Main Piping	
33 31 26	Sanitary Pressure Sewer Piping	Includes: systems without onsite primary treatment where conveyance is under pressure. Properties in such systems are often serviced via grinder pumps.
33 31 29	STEP System Piping	Includes: systems with onsite primary treatment, usually provided via septic tanks, where conveyance is under pressure by septic tank effluent pumps (STEP).

NUMBER	TITLE	EXPLANATION
33 32 00	Sanitary Sewerage Equipment	Includes: equipment specialized for sanitary sewerage systems, such as packaged pumping systems and vacuum systems. Alternate Terms/Abbreviations: wastewater pumps: sewage pumps: sewage ejectors. STEP: septic tank effluent pump. See: 33 37 00 for overflow control equipment. 43 20 00 for pumps used for field-assembled systems used in wastewater conveyance. 46 20 00 for screening, trash-raking, and grit removal equipment for preliminary treatment.
33 32 11	Field-Erected Wastewater Pumping Stations	Includes: public and private field-erected pumping stations for conventional wastewater conveyance, typically for smaller pumping stations utilizing submersible pumps in a wet well with or without a separate dry valve chamber, or self-priming pumps. May Include: instrumentation and pump controls.
33 32 13	Packaged Wastewater Pumping Stations	Includes: factory-assembled systems, purchased as a package, for conventional wastewater pumping. Field-assembled engineered systems not purchased as a package are indicated in detail under 43 20 00 (pumps) and 40 00 00 (exposed process piping). Packaged pumping systems include a variety of skid-mounted pumps, wet well structures, and discharge piping and valves, and controls including control panels, variable-speed drives, two-speed drives, constant-speed units, and combinations thereof. May be used for public or private sewerage systems. Pumps may vary with the package. Pump types may include: dry-pit overhung (end suction) centrifugal pumps, submersible overhung (end suction) centrifugal pumps, and other pump types.
33 32 16	Packaged Wastewater Grinder Pump Assemblies	Includes: packaged sump/holding tank equipped with centrifugal or semi-positive displacement pump(s), controls, and alarm panel, serving one or more buildings or structures, installed either in a basement or buried outdoors. Pumps include integral cutting mechanisms. May discharge to either a conventional gravity or pressurized wastewater collection system.
33 32 19	Septic Tank Effluent Pumps	
33 32 26	Pneumatic Sewage Ejectors	
33 32 33	Wastewater Collection Vacuum Stations	

NUMBER	TITLE	EXPLANATION
33 32 36	Pneumatic Wastewater Vacuum Valve Sumps	Includes: the packaged sump/holding tank assembly equipped with an air pressure sensor and vacuum valve, serving one or more buildings or structures connected to a vacuum wastewater collection system.
33 34 00	Onsite Wastewater Disposal	Includes: onsite wastewater treatment and disposal systems serving one or more buildings or structures. May Include: septic tanks and other onsite systems. Alternate Terms/Abbreviations: septic tank: interceptor tank. Drainage field: leach field: absorption field: drainfield. See: 33 01 34 for operation and maintenance of onsite wastewater disposal systems, including septic tank solids removal. 33 05 00 and 33 31 00 for piping for onsite wastewater disposal systems. 33 09 30 for instrumentation and control for onsite wastewater disposal systems. 33 32 11 for field-erected wastewater pumping stations. 46 07 53 for packaged wastewater treatment plants. 46 53 00 for rotating biological contactors, trickling filters, sequencing batch reactors, and other biological treatment systems used for wastewater treatment at a site.
33 34 13	Septic Tanks	Includes: horizontal and vertical septic tanks.
33 34 13.13	Concrete Septic Tanks	
33 34 13.23	Fiberglass Septic Tanks	
33 34 13.33	Polyethylene Septic Tanks	
33 34 16	Septic Tank Effluent Filters	Includes: removable, cartridge-type filters of one or more stages installed downstream of a septic tank to reduce solids discharge to the drainage field. Tube filters, textile media filters, and other filter media types.
33 34 16.13	Septic Tank Gravity Effluent Filters	
33 34 16.16	Septic Tank Pressurized Effluent Filters	
33 34 19	Septic Tank Accessories	Includes: risers, tank covers, vent pipes, vent pipe insect screens, vent covers, vent pipe odor filters, sludge samplers, and other septic tank accessories.

DIVISION 33

NUMBER	TITLE	EXPLANATION
33 34 23	Onsite Aerobic Equipment	Includes: aerobic treatment systems serving a single residence or small number of buildings or structures. Includes aerobic equipment retrofitted into septic tanks such as diaphragm, piston, or rotary vane air pumps. See Also: 46 53 00 for rotating biological contactors, trickling filters, sequencing batch reactors, and other biological treatment systems used for wastewater treatment at a site.
33 34 33	Onsite Intermittent Wastewater Filtration Systems	Includes: intermittently dosed filters using sand, gravel, anthracite, crushed glass, expanded shale, bottom ash, or other media. May be constructed in an excavation or above the pre-construction grade. Usually preceded by a septic tank. May discharge to a drainage field or infiltrate directly from the filter bed. See Also: 46 53 41 for intermittent sand filters used at wastewater treatment facilities serving multiple sites.
33 34 36	Onsite Recirculating Wastewater Filtration Systems	
33 34 51	Drainage Field System	Notes: this is an alternative location for specifying drainage fields as a system. Drainage fields are comprised of solid and perforated piping, aggregate such as crushed stone, geosynthetics, and engineered fill, and are typically specified under other divisions and shown as a system on drawings. May be conventional or mound systems. See Also: 33 05 43 for perforated PVC drainage pipe, perforated concrete drainage pipe, and ceramic drainage tile.
33 34 53	Distribution Chambers	See Also: 33 05 66 for concrete distribution chambers.
33 34 53.13	Drainage Field Concrete Distribution Chambers	
33 34 53.23	Drainage Field Fiberglass Distribution Chambers	
33 34 53.33	Drainage Field Polyethylene Distribution Chambers	
33 34 56	Drainage Field Dosing Chambers	
33 34 57	Drainage Field Infiltration Chambers	

NUMBER	TITLE	EXPLANATION
33 36 00	Wastewater Utility Storage Tanks	Includes: field-erected tanks for temporary storage of wastewater, including sanitary sewage and combined sewage. May be located at a remote site in the collection system or at a treatment facility. Alternate Terms/Abbreviations: ground-level: ground-supported. AWWA: American Water Works Association. Notes: treatment lagoons are typically shown on the drawings and specified as part of the work under other divisions, including 31 00 00 and 32 00 00. See: 03 00 00 for concrete used in constructing foundations and other tank elements. 05 50 00 for metal ladders, stairs, platforms, handrails, and railings. 09 97 13 for storage tank coatings. 13 47 13 for cathodic protection for metallic storage tanks. 33 16 00 for field-erected utility tanks used for storing potable water and rain water (cisterns). 33 37 46 for end-of-pipe in-water storage systems for sanitary sewer overflow and combined sewer overflow. 33 41 00 for subdrainage. 33 56 00 for hydrocarbon storage tanks. 40 72 00 for level-measuring instrumentation. 43 40 00 for fabricated storage tanks used at a process facility. 44 30 00 for odor control equipment. 46 30 00 for chemical feed systems used with wastewater storage tanks. 46 41 00 for equipment used for mixing wastewater storage tank contents. 46 46 00 for sediment removal equipment.
33 36 23	Ground-Level Steel Wastewater Storage Tanks	Includes: tanks with ellipsoidal, torical, dome, or cone-shaped roofs. Welded or bolted construction.
33 36 31	Ground-Level AWWA D110 Type I Prestressed Concrete Wastewater Storage Tanks	
33 36 32	Ground-Level AWWA D110 Type II Prestressed Concrete Wastewater Storage Tanks	
33 36 33	Ground-Level AWWA D110 Type III Prestressed Concrete Wastewater Storage Tanks	May Include: roofs constructed as clear span free-standing concrete domes, or with column-supported conventionally reinforced flat slab roofs.
33 36 36	Ground-Level Reinforced Concrete Wastewater Storage Tanks	
33 36 43	Underground Reinforced Concrete Wastewater Storage Tanks	

NUMBER	TITLE	EXPLANATION
33 36 53	Wastewater Storage Tunnel Systems	Notes: this is an alternate location for specifying tunnel systems used for temporary storage of wastewater. Primary location for tunnels and associated linings is 31 70 00.
33 37 00	Overflow Control	Includes: treatment equipment and physical control equipment intended for sanitary sewer overflow (SSO), combined sewer overflow (CSO), and stormwater applications. Alternate Terms/Abbreviations: swirl concentrators: vortex separators. See: 33 34 00 for onsite wastewater disposal. 33 41 00 for subdrainage. 40 70 00 for process instrumentation. 46 07 53 for packaged wastewater, combined sewer overflow, and stormwater treatment systems at a treatment facility. 46 43 66 for ballasted high-rate clarifier equipment. 46 43 67 for high rate clarification/thickening equipment. 46 46 26 for tipping sediment flushing tanks, flushing gates, water cannons, and nozzle systems.
33 37 13	Bending Weirs	
33 37 16	Spring-loaded Weirs	
33 37 23	Flow Regulators	
33 37 33	Swirl Concentrators	
33 37 36	Vortex-Type Hydrodynamic Separators	
33 37 43	Netting Systems	
33 37 46	End-of-Pipe In-water Storage Systems	

33 40 00 Stormwater Utilities

Includes: public or private stormwater drainage for surface water and/or groundwater for a single site or for multiple sites.

See: 01 57 13 for temporary control of erosion and sediment.
01 57 23 for temporary control of stormwater pollution.
22 14 00 for facility stormwater drainage within or under a structure.
31 25 00 for permanent erosion and sediment controls.
33 09 40 for instrumentation and control for stormwater utilities.
44 40 00 for water pollution control equipment.

NUMBER	TITLE	EXPLANATION
33 41 00	Subdrainage	Includes: subdrainage for intercepting and conveying groundwater away from a structure or to a drainage sump. Alternate Terms/Abbreviations: foundation drainage: footing drains: footer drains. See: 22 14 29 for sump pumps. 22 14 33 for packaged, pedestal drainage pump units. 22 14 36 for packaged, submersible, drainage pump units. 31 05 19 for geosynthetics for earthwork and drainage. 32 05 19 for geosynthetics for exterior improvements and drainage. 32 32 00 for retaining walls. 33 05 39 for perforated PVC, perforated corrugated metal, perforated concrete pipe, and ceramic drain tile pipe.
33 41 13	Foundation Drainage	
33 41 13.13	Foundation Drainage Piping	
33 41 13.16	Geocomposite Foundation Drainage	
33 41 16	Subdrainage Piping	
33 41 16.16	Geocomposite Subdrainage	
33 41 16.19	Pipe Underdrains	
33 41 19	Underslab Drainage	
33 41 19.13	Underslab Drainage Piping	
33 41 19.16	Geocomposite Underslab Drainage	
33 41 23	Drainage Layers	Notes: drainage layers may be used for surface drainage or as part of a vegetated roofing system.
33 41 23.16	Gravel Drainage Layers	
33 41 23.19	Geosynthetic Drainage Layers	
33 41 33	Retaining Wall Drainage	
33 41 43	Synthetic-Media French Drains	

NUMBER	TITLE	EXPLANATION
33 42 00	Stormwater Conveyance	Includes: piping and structures for stormwater conveyance, either for a single site or for multiple sites, including public and private stormwater conveyance. Includes culverts under transportation ways, embankments, and similar barriers. Alternate Terms/Abbreviations: trench drain: channel drain. Stormwater or gravity piping: storm sewers. Notes: drainage inlets such as catch basins for area drainage typically do not have traps to prevent release of in-pipe gases. See: 22 14 13 for facility stormwater drainage piping within a structure. 33 05 00 for manholes, chambers, and vaults used for stormwater conveyance, and associated manhole covers and similar accessories. 33 05 00 for various types of pipe used for stormwater conveyance. 33 44 00 for pumps, inlet filters, and other equipment used in stormwater utilities.
33 42 11	Stormwater Gravity Piping	Includes: buried stormwater gravity piping of various materials. Notes: this is an alternative location for specifying stormwater gravity piping. Various types of pipe are in 33 05 00.
33 42 13	Stormwater Culverts	Includes: culverts of various materials and cross-sectional shapes, such as circular, arch, elliptical, and other shapes. Notes: this is an alternative location for specifying stormwater culverts. Various types of pipe are in 33 05 00.
33 42 16	Stormwater Force Main Piping	
33 42 23	Headwalls and Endwalls for Stormwater Piping	Includes: headwalls of various materials, such as concrete and fiberglass, for storm sewer outfalls and culverts. May be fabricated or field-constructed. See Also: 03 00 00 for concrete.
33 42 26	End Sections for Stormwater Piping	
33 42 26.11	Aluminum End Sections for Stormwater Piping	
33 42 26.13	Steel End Sections for Stormwater Piping	
33 42 26.16	Corrugated Metal End Sections for Stormwater Piping	
33 42 26.23	Concrete End Sections for Stormwater Piping	

NUMBER	TITLE	EXPLANATION
33 42 31	Stormwater Area Drains and Inlets	Includes: catch basins, drop inlets, and similar flat-topped structures for collecting stormwater from an area, including concrete, cast-iron, fiberglass, and other materials.
33 42 33	Stormwater Curbside Drains and Inlets	Includes: curbside catch basins, drop inlets, and similar structures intended for installation as part of a curbing system for collecting stormwater from an area.
33 42 36	Stormwater Trench Drains	Includes: trench drains, channel drains, and similar modular, long inlets for collecting stormwater from an area, including concrete, cast-iron, fiberglass, and other materials. May be flat-bottomed or sloped-bottomed.
33 42 41	Gratings and Frames for Stormwater Drainage Inlets	Includes: flat, concave, and atrium-type gratings, and fitting-type gratings of various shapes and grating types for stormwater inlets.
33 42 41.11	Aluminum Gratings and Frames for Stormwater Drainage Inlets	
33 42 41.21	Brass Gratings and Frames for Stormwater Drainage Inlets	
33 42 41.31	Cast-Iron Gratings and Frames for Stormwater Drainage Inlets	
33 42 41.41	Chrome Gratings and Frames for Stormwater Drainage Inlets	
33 42 41.51	Steel Gratings and Frames for Stormwater Drainage Inlets	
33 42 41.61	Fiberglass Gratings and Frames for Stormwater Drainage Inlets	
33 44 00	Stormwater Utility Equipment	Includes: equipment such as pumps, filters, and other items utilized for stormwater conveyance and stormwater quality control. See: 22 14 33 for packaged, pedestal drainage pump units used within a building or structure. 22 14 36 for packaged, submersible drainage pump units serving a facility or structure. 33 32 00 for pumping stations used with sanitary sewage and combined sewage. 33 37 00 for overflow control equipment. 33 41 00 for subdrainage. 43 20 00 for pumps used for field-assembled systems for stormwater conveyance. 46 21 00 for screening and trash-raking equipment. 46 25 00 for oil and grease separators at a treatment facility.

DIVISION 33

NUMBER	TITLE	EXPLANATION
33 44 11	Field-Erected Stormwater Pumping Stations	Includes: public and private field-erected pumping stations for conventional stormwater conveyance, typically for smaller pumping stations utilizing submersible pumps in a wet well with or without a separate dry valve chamber, or self-priming pumps. May Include: instrumentation and pump controls.
33 44 13	Packaged Stormwater Pumping Stations	Includes: factory-assembled systems, purchased as a package, for conventional stormwater pumping. Field-assembled engineered systems not purchased as a package are indicated in detail under 43 20 00 (pumps) and 40 00 00 (exposed process piping). Packaged pumping systems include a variety of skid-mounted pumps, wet well structures, and discharge piping and valves, and controls including control panels, variable-speed drives, two-speed drives, constant-speed units, and combinations thereof. May be used for public or private stormwater systems. Pumps may vary with the package. Pump types may include: dry-pit overhung (end suction) centrifugal pumps, submersible overhung (end suction) centrifugal pumps, and other pump types.
33 44 21	Stormwater Downspout Filters	
33 44 23	Inline Stormwater Filters	
33 44 26	Stormwater Catch Basin Insert Filters	
33 44 27	Stormwater Trench Drain Insert Filters	
33 44 31	Stormwater Membrane Filtration	
33 44 33	Stormwater Trash and Debris Guards	
33 44 36	Oil and Stormwater Separators	Includes: ground-level and buried units of materials such as steel, fiberglass, and concrete. See Also: 46 25 00 for oil and grease separation equipment at a treatment facility.
33 44 43	Vortex-Type Hydrodynamic Separators	
33 44 53	Stormwater Area Drainage Pop-up Relief Valves	

NUMBER	TITLE	EXPLANATION
33 46 00	Stormwater Management	Includes: facilities for attenuating stormwater flow rate and for improving stormwater quality, and for infiltrating stormwater into the ground. Stormwater detention ponds, stormwater retention ponds, fire protection ponds, cooling water ponds, and similar impoundments. Green alternatives for stormwater infiltration and reuse. Alternate Terms/Abbreviations: leaching pit: infiltration well. Retention pond: stabilization pond. Notes: ponds and impoundments that do not employ fabricated elements are typically shown on the drawings and are specified under other Divisions such as 31 00 00 and 35 00 00. See: 21 40 00 for fire-suppression water storage for a building or structure. 31 23 00 for excavation and fill. 31 24 00 for embankments. 31 35 26 for containment barriers. 32 14 43 for porous unit paving. 32 31 00 for fences. 33 10 00 for water utilities, including surface water sources such as reservoirs. 33 16 00 for storage tanks used for rain water harvesting for reuse. 33 37 00 for overflow controls. 33 42 41 for gratings and frames on outlet structures for stormwater ponds. 35 73 00 for embankment dams. 40 05 59 for hydraulic gates.
33 46 11	Stormwater Ponds	Notes: alternate location for specifying stormwater ponds, typically under other Divisions.
33 46 11.13	Stormwater Detention Ponds	
33 46 11.23	Stormwater Retention Ponds	
33 46 11.33	Fire Protection Ponds	
33 46 11.43	Cooling Water Ponds	
33 46 16	Outlet Structures for Stormwater Ponds	Includes: structures for controlling the rate of discharge from a stormwater pond.
33 46 16.13	Stormwater Pond Concrete Outlet Structures	
33 46 16.23	Stormwater Pond Fiberglass Outlet Structures	

DIVISION **33**

NUMBER	TITLE	EXPLANATION
33 46 23	Modular Buried Stormwater Storage Units	Includes: fabricated modular units that together form an underground stormwater detention, retention, or infiltration. Typically non-metallic materials, including polyethylene or concrete. Accessories may include items such as sediment traps to improve stormwater quality. See Also: 33 16 66 for modular buried storage tanks used for rain water harvesting.
33 46 39	Rain Barrels	
33 46 53	Stormwater Leaching Pits	
33 46 56	Tree Box Filters	Includes: structures for access to underground storm drainage piping. Chambers and underground detention chambers. See: 22 05 73 for facility drainage manholes. 33 05 00 for common work results for utilities. 33 05 61 for concrete manholes. 33 05 63 for concrete vaults and chambers. 33 05 66 for concrete distribution chambers. 33 05 69 for concrete trenchways. 33 05 71 for cleanouts. 33 05 73 for polyethylene manholes. 33 05 76 for fiberglass manholes.
33 46 59	Concrete Modular Biofiltration Cells	
33 46 63	Swale Protection Prefilters	

33 50 00 Hydrocarbon Utilities

Includes: utilities for extracting, conveying, and storing liquid, gas, and other hydrocarbon transmission and distribution for a single site or facility or for multiple facilities. Includes various forms of petroleum, liquid and gaseous natural gas, propane, and other hydrocarbons.

See: 09 90 00 for coating systems.
33 05 97 for identification and signage for utilities.
33 09 50 for instrumentation and control for hydrocarbon utilities, including leak detection systems.
43 12 00 for gas compressors.
43 20 00 for liquid pumps.

NUMBER	TITLE	EXPLANATION
33 51 00	Hydrocarbon Sources	Includes: dug, bored, driven, or drilled wells, for production of hydrocarbons, and monitoring wells. Well casings, well screens, riser pipes, well pumps, and accessories. Wells for a single site or facility, and for commercial production. See: 02 24 00 for environmental assessments. 02 30 00 for subsurface investigations, including exploratory drilling. 02 32 23 for groundwater monitoring before construction. 02 70 00 for wells used for groundwater extraction for remediation purposes. 28 20 00 for video surveillance. 31 09 13 for groundwater monitoring during construction. 33 01 50 for abandonment and rehabilitation of hydrocarbon wells. 33 09 50 for instrumentation and control for hydrocarbon wells. 33 14 00 for water utility transmission and distribution. 35 50 00 for docks, piers, and appurtenances for marine transport of hydrocarbons. 35 52 00 for offshore platform construction. 40 05 00 for valves used in hydrocarbon production. 40 10 00 for exposed process piping for gaseous hydrocarbons. 40 21 00 for exposed process piping for liquid hydrocarbons. 40 70 00 for instrumentation for wells. 42 00 00 for hydrocarbon refinery process equipment. 43 12 00 for gas compressors. 43 20 00 for process liquid pumps. 44 40 00 for water pollution control, including spill containment. 46 00 00 for treatment of wastewater resulting from hydrocarbon production.
33 51 13	Appraisal Wells for Liquid Hydrocarbons	
33 51 16	Appraisal Wells for Gaseous Hydrocarbons	
33 51 23	Production Wells for Liquid Hydrocarbons	
33 51 26	Production Wells for Gaseous Hydrocarbons	
33 51 33	Liquid Injection Wells	
33 51 36	Gas Injection Wells	
33 51 39	Monitoring Wells	
33 51 53	Reciprocating Pumpjack	Alternate Terms/Abbreviations: pumpjack: horse: donkey pumper: nodding donkey: pumping unit: horsehead pump: rocking horse: beam pump: dinosaur: grasshopper pump: big Texan: thirsty bird: jack pump.
33 51 56	Submersible Liquid Hydrocarbon Pumps	

NUMBER	TITLE	EXPLANATION
33 52 00	Hydrocarbon Transmission and Distribution	Includes: work results for conveyance of hydrocarbons by a utility or private company, including transmission and distribution for a single site, facility, or multiple facilities. Includes handling of pipe, including pipe stacking and storing; welding of conveyance pipe; pigging; and farm tap installations. Alternate Terms/Abbreviations: pigging: in-line inspection (ILI): cleaning pig: gauge pig: smart pig: intelligent pig. See: 01 51 29 for temporary natural gas. 01 53 00 for temporary pigging. 01 57 13 for temporary erosion and sediment controls. 09 96 00 for specialized, field-applied pipe coatings. 13 47 13 for cathodic protection. 23 11 00 for fuel piping within, under, or closely associated with a building or structure. 23 12 00 for fuel pumps closely associated with a building or structure. 31 23 00 for excavation and fill, including trenching. 33 05 09 for tapping sleeves and saddles for utility piping. 33 56 53 for compressed gaseous hydrocarbon storage tanks. 40 05 76 for hot-tapping valves and appurtenances. 40 10 00 for exposed gas process piping at a hydrocarbon facility. 40 21 00 for exposed liquid process piping at a hydrocarbon facility. 44 40 00 for water pollution control, including spill containment.
33 52 13	Liquid Hydrocarbon Piping	
33 52 16	Gas Hydrocarbon Piping	
33 52 16.13	Steel Natural Gas Piping	
33 52 16.23	Medium Density Polyethylene Piping for Natural Gas	
33 52 16.26	High Density Polyethylene Piping for Natural Gas	
33 52 19	Hydrocarbon Piping Handling	
33 52 19.13	Hydrocarbon Piping Stringing, Storing, and Stacking	
33 52 23	Hydrocarbon Piping Welding	
33 52 23.13	Liquid Hydrocarbon Piping Welding	
33 52 23.16	Gas Hydrocarbon Piping Welding	
33 52 26	Hydrocarbon Piping Pigging	
33 52 26.13	Hydrocarbon Piping Pig Launchers	
33 52 26.16	Hydrocarbon Piping Pig Receivers	
33 52 26.23	Pigs for Hydrocarbon Piping	

NUMBER	TITLE	EXPLANATION
33 53 00	Hydrocarbon Utility Pressure Regulation	Includes: equipment to regulate pressure for various hydrocarbons applications. See: 40 05 67 for general-purpose pressure-regulating valves in process applications. 40 94 00 for self-contained pressure regulators used in process control applications. 43 13 23 for gas pressure regulators used in industrial processes. 43 22 59 for liquid pressure regulators used in industrial processes.
33 53 13	Liquid Hydrocarbon Pressure Regulation	
33 53 16	Gas Hydrocarbon Pressure Regulation	
33 54 00	Hydrocarbon Utility Safety Equipment	Includes: valves and leak detection used for protecting hydrocarbon utility piping, product within piping, the public, and adjacent property. See: 33 53 00 for hydrocarbon utility pressure regulation. 40 05 00 for general-purpose valves used in process applications. 40 70 00 for instrumentation used in general-purpose detection of liquid and gas leaks in process applications.
33 54 13	Liquid Hydrocarbon Leak Detection Equipment	
33 54 16	Gas Hydrocarbon Leak Detection Equipment	
33 54 23	Hydrocarbon Pipeline Venting	
33 54 23.13	Liquid Hydrocarbon Pipeline Venting	
33 54 23.16	Gas Hydrocarbon Pipeline Venting	

NUMBER	TITLE	EXPLANATION
33 56 00	Hydrocarbon Storage	Includes: storage tanks and equipment for bulk storage of hydrocarbon liquids and gases for a single facility or site, and for industrial/commercial facilities. Commercial bulk storage facilities. Alternate Terms/Abbreviations: ground-level: ground-supported. Aboveground storage tank: AST. Underground storage tank: UST. Notes: metallic tanks include steel, stainless steel, and other metals. Non-metallic tanks include concrete, polyethylene, and fiberglass. See: 03 00 00 for concrete used in constructing foundations and other tank elements. 05 50 00 for metal ladders, stairs, platforms, handrails, and railings. 09 97 13 for steel storage tank coatings. 09 97 23 for concrete storage tank and containment-dike coatings. 13 47 13 for cathodic protection of metallic storage tanks. 23 13 00 for facility fuel storage tanks for a building or structure. 28 43 00 for fuel oil detection and alarm. 33 16 00 for water utility storage tanks. 33 34 00 for onsite wastewater disposal. 33 41 00 for subdrainage. 33 52 00 for hydrocarbons transmission and distribution. 33 57 00 for hydrocarbon receiving/dispensing equipment. 34 73 00 for airfield construction. 43 40 00 for fabricated storage tanks used at a process facility. 43 42 26 for welded steel gas storage sphere. 44 40 00 for water pollution control equipment, including prefabricated containment berms/dikes, collapsible storage tanks, containment booms, and spill cleanup equipment.
33 56 13	Aboveground Hydrocarbon Storage Tanks	Includes: tanks elevated above the ground on supports, columns, or pillars.
33 56 21	Ground-Level Fixed Roof Hydrocarbon Storage Tanks	Includes: tanks with cone-shaped, domed, umbrella-type, or other fixed roofs. Typically used with liquids with high flashpoints.
33 56 21.13	Double-Wall Ground-Level Fixed Roof Metallic Hydrocarbon Storage Tanks	
33 56 21.16	Single-Wall Ground-Level Fixed Roof Metallic Hydrocarbon Storage Tanks	
33 56 21.23	Single-Wall Ground-Level Fixed Roof Non-Metallic Hydrocarbon Storage Tanks	
33 56 21.26	Double-Wall Ground-Level Fixed Roof Non-Metallic Hydrocarbon Storage Tanks	

NUMBER	TITLE	EXPLANATION
33 56 23	Ground-Level External Floating Roof Hydrocarbon Storage Tanks	Notes: external floating roof tanks have no fixed roof of any sort; typically used with liquids that have medium vapor pressure.
33 56 26	Ground-Level Internal Floating Roof Hydrocarbon Storage Tanks	Notes: internal floating roof tanks have an external fixed roof and an internal cover that floats on the liquid surface; typically used with liquids that have low vapor pressure.
33 56 33	Underground Hydrocarbon Storage	
33 56 33.13	Double-Wall Metallic Underground Hydrocarbon Storage Tanks	
33 56 33.23	Single-Wall Metallic Underground Hydrocarbon Storage Tanks	
33 56 33.26	Single-Wall Non-Metallic Underground Hydrocarbon Storage Tanks	
33 56 33.29	Double-Wall Non-Metallic Underground Hydrocarbon Storage Tanks	
33 56 33.43	Underground Hydrocarbon Storage Caverns	
33 56 43	Aviation Fuel Storage Tanks	
33 56 43.13	Ground-Level Aviation Fuel Storage Tanks	
33 56 43.53	Underground Aviation Fuel Storage Tanks	
33 56 53	Compressed Gaseous Hydrocarbon Storage Tanks	
33 56 60	Moveable Hydrocarbon Storage Tanks	Includes: skid-mounted, trailer-mounted, and other tanks not permanently installed.
		See Also: 44 41 41 for collapsible storage tanks used for pollution control.
33 56 93	Hydrocarbon Tank Linings	
33 56 96	Hydrocarbon Storage Tank Appurtenances	Includes: hatches, inlet and discharge piping, vents, nozzles, fall-prevention equipment, and other appurtenances not specified with the tank.
33 57 00	Bulk Hydrocarbons Receiving/Dispensing Equipment	Includes: tanker truck, rail, and marine transport off-loading assemblies, pantograph, controls, pumps, and transfer equipment for hydrocarbons transfer.
		See: 23 10 00 for fuel piping and pumps for a building or structure.
		23 12 16 for facility fuel-dispensing pumps.
		33 56 00 for hydrocarbons storage tanks.
		40 05 89 for quick-connect couplers.
		40 10 00 for gaseous hydrocarbons process piping.
		40 21 00 for liquid fuels process piping.
		40 22 00 for process piping for petroleum products.
		40 60 00 and 40 70 00 for instrumentation and controls associated with a process facility.
		44 41 13 for containment structures and spill cleanup.

NUMBER	TITLE	EXPLANATION
33 59 00	Hydrocarbon Utility Metering	Includes: metering for liquid and gas hydrocarbons. Metering may also include facilities for sampling and quality measurement. Metering equipment indicated herein is relatively unique to hydrocarbons production, transmission, and distribution. See: 28 43 00 for fuel oil detection and alarm. 33 09 00 for registers and meter reading equipment. 40 70 00 for general-purpose flow meters and analyzers used in process applications.
33 59 11	Diaphragm/Bellows Hydrocarbons Meters	
33 59 13	Hydrocarbons Rotary Meters	
33 59 16	Hydrocarbons Turbine Meters	
33 59 19	Hydrocarbons Orifice Meters	
33 59 23	Hydrocarbons Ultrasonic Flow Meters	
33 59 26	Hydrocarbons Coriolis Meters	
33 59 43	Liquid Hydrocarbon Sampling and Quality Analysis Instrumentation	
33 59 46	Gas Hydrocarbon Sampling and Quality Analysis Instrumentation	
33 60 00	Hydronic and Steam Energy Utilities	Includes: hydronic and steam energy distribution for a single facility or for multiple facilities. See: 13 47 13 for cathodic protection. 23 20 00 for hydronic, steam, and condensate piping and pumps for a building or structure. 33 09 60 for instrumentation and control for hydronic and steam energy utilities. 40 60 00 and 40 70 00 for instrumentation and control for process applications.
33 61 00	Hydronic Energy Distribution	Includes: chilled condenser and hot water piping, valves, piping specialties, pumps, and other components serving a single facility or for multiple facilities. See: 13 47 13 for cathodic protection. 23 21 00 for hydronic piping and pumps for a structure. 33 09 60 for instrumentation and control for steam and hydronic energy utilities. 33 46 11 for cooling water ponds.
33 61 13	Buried Hydronic Energy Distribution	
33 61 23	Ground-Level Hydronic Energy Distribution	
33 61 27	Prefabricated Energy Distribution	
33 61 29	Pre-Engineered, Field-Assembled Energy Distribution	
33 61 33	Hydronic Energy Distribution Metering	

NUMBER	TITLE	EXPLANATION
33 63 00	Steam Energy Distribution	Includes: steam and condensate piping, valves, piping specialties and other components serving a single facility or multiple facilities. Alternate Terms/Abbreviations: utilidor: utility tunnel: utility corridor. See: 13 47 13 for cathodic protection. 23 22 00 for steam and condensate piping and pumps within a building or structure. 40 11 00 for steam process piping.
33 63 13	Underground Steam and Condensate Distribution Piping	
33 63 13.13	Shallow-Trench Steam Distribution Piping	
33 63 13.16	Utilidor Steam Distribution Piping	
33 63 23	Ground-Level Steam and Condensate Distribution Piping	
33 63 33	Steam Energy Distribution Metering	

33 70 00 Electrical Utilities

Includes: substations, transformers, switchgear, circuit protection devices, utility structures, and wiring used to transmit and distribute extra-high-, high- and medium-voltage electrical power from the point of power generation to the supply points of facilities. Distribution at high- and medium-voltage levels within a facility to or between supply points.

See: 33 09 70 for instrumentation and control for electrical utilities.

33 72 33 for control house equipment.

33 80 00 for communication utilities.

48 10 00 for electrical power generation equipment.

NUMBER	TITLE	EXPLANATION
33 71 00	Electrical Utility Transmission and Distribution	Includes: utility structures and wiring to distribute extra-high-, high- and medium-voltage electrical power from the point of power generation to the supply points of facilities. Metering and termination equipment and utility company requirements related to electrical service entrances. Notes: definitions: extra-high-voltage: 345 kV to 765 kV. High-voltage: 115 kV to 230 kV. Medium-voltage: 2400 V to 69 kV. Alternate Terms/Abbreviations: EHV: extra-high-voltage. HV: high-voltage. See: 26 05 43 for facility underground ducts and raceways for electrical systems. 26 05 46 for facility utility poles for electrical systems. 26 16 00 for medium-voltage metering for a facility. 26 21 00 for low-voltage overhead electrical power systems. 26 27 13 for electricity metering for a facility. 33 79 16 for tower grounding. 33 79 19 for utilities grounding. 33 81 00 for communication structures. 34 23 13 for traction power poles. 48 19 00 for electrical power control equipment.
33 71 13	Electrical Utility Towers	
33 71 13.13	Precast Concrete Electrical Utility Towers	
33 71 13.23	Steel Electrical Utility Towers	
33 71 13.33	Wood Electrical Utility Towers	
33 71 16	Electrical Poles	Includes: crossarms and guys.
33 71 16.13	Precast Concrete Electrical Utility Poles	
33 71 16.23	Steel Electrical Utility Poles	
33 71 16.33	Wood Electrical Utility Poles	
33 71 19	Electrical Underground Ducts, Ductbanks, and Manholes	
33 71 19.13	Electrical Manholes and Handholes	
33 71 19.23	Trenched Electrical Underground Duct and Ductbanks	
33 71 19.33	Trenchless Electrical Underground Duct and Ductbanks	
33 71 23	Insulators and Fittings	Includes: conductor hardware, insulators, dampers, sleeves.
33 71 23.13	Suspension Insulators	
33 71 23.16	Post Insulators	
33 71 23.23	Potheads	
33 71 26	Transmission and Distribution Equipment	
33 71 26.13	Capacitor Banks	Includes: series and shunt capacitors.
33 71 26.16	Coupling Capacitors	
33 71 26.23	Current Transformers	

NUMBER	TITLE	EXPLANATION
33 71 26.26	Potential Transformers	
33 71 36	Extra-High-Voltage Transmission Wiring	
33 71 36.13	Overhead Extra-High-Voltage Transmission Wiring	Includes: marker balls.
33 71 39	High-Voltage Transmission Wiring	
33 71 39.13	Overhead High-Voltage Transmission Wiring	Includes: marker balls.
33 71 39.23	Underground High-Voltage Transmission Wiring	
33 71 39.33	Underwater High-Voltage Transmission Wiring	
33 71 49	Medium-Voltage Transmission Wiring	
33 71 49.13	Overhead Medium-Voltage Transmission Wiring	Includes: marker balls.
33 71 49.23	Underground Medium-Voltage Transmission Wiring	
33 71 49.33	Underwater Medium-Voltage Transmission Wiring	
33 71 53	Direct-Current Transmission	
33 71 73	Electrical Utility Services	
33 71 83	Transmission and Distribution Specialties	
33 72 00	Utility Substations	Includes: assembly of structures, switches, circuit breakers, buses, and transformers to switch circuits and convert power from transmission voltage to distribution voltage for utility-owned transmission and distribution substations. Alternate Terms/Abbreviations: control house: doghouse. Double-end break: double-side break. Vertical reach: pantograph: semi-pantograph: knee-type switch. See: 13 34 23 for fabricated substation control rooms. 26 11 00 for facility substations. 26 13 00 for medium-voltage switchgear. 33 73 00 for utility transformers. 33 75 00 for high-voltage switchgear. 33 77 00 for medium-voltage utility switchgear. 33 79 00 for site grounding and lighting protection. 33 79 23 for utility substation grounding. 34 21 16 for traction power substations.
33 72 13	Deadend Structures	
33 72 23	Structural Bus Supports	
33 72 23.13	Bus Support Insulators	
33 72 26	Substation Bus Assemblies	
33 72 26.13	Aluminum Substation Bus Assemblies	
33 72 26.16	Copper Substation Bus Assemblies	
33 72 33	Control House Equipment	
33 72 33.13	Relays	Includes: breaker failure (BFR), breaker reclosing, line protection, including remedial action schemes (RAS), load shedding, reactive, transformer, bus differential.

DIVISION **33**

NUMBER	TITLE	EXPLANATION
33 72 33.16	Substation Control Panels	Includes: control battery main (129 vdc) and auxiliary (48-24 vdc), battery chargers, station service transformers, engine generator and transfer switch.
33 72 33.23	Power-Line Carriers	
33 72 33.26	Substation Metering	
33 72 33.43	Substation DC Supply	Includes: control battery main (129 vdc) and auxiliary (48-24 vdc), battery chargers, station service transformers, engine generator and transfer switch.
33 72 43	Substation Control Wiring	Includes: local control and schematic relay wiring between relay devices.
33 72 53	Shunt Reactors	
33 72 63	Substation Disconnect Switches	
33 72 63.13	Vertical Break Disconnect Switches	
33 72 63.16	Double End Break Disconnect Switches	
33 72 63.19	Double End Break "Vee" Disconnect Switches	
33 72 63.21	Center Break Disconnect Switches	
33 72 63.23	Center Break "Vee" Disconnect Switches	
33 72 63.26	Single Side Break Disconnect Switches	
33 72 63.27	Vertical Reach Disconnect Switches	
33 72 63.29	Grounding for Substation Disconnect Switches	
33 72 63.31	Hook Sticks	
33 73 00	Utility Transformers	Includes: transformers for utility substation applications for transforming transmission voltage to distribution voltage. See: 26 12 00 for medium-voltage facility transformers. 26 13 00 for medium-voltage switchgear. 26 16 00 for medium-voltage metering. 26 18 00 for medium-voltage protection devices. 26 22 00 for low-voltage facility transformers. 33 72 00 for utility substations. 33 75 00 for high-voltage switchgear. 33 77 00 for medium-voltage utility switchgear. 34 21 23 for traction power transformer-rectifier units. 48 19 23 for electrical power generation transformers.
33 73 13	Liquid-Filled Utility Transformers	
33 73 23	Dry-Type Utility Transformers	

NUMBER	TITLE	EXPLANATION
33 74 00	Extra-High-Voltage (EHV) Switchgear and protection devices	Includes: switchgear and circuit protection devices for extra-high-voltage utility applications. Extra-high-voltage: 345 kV to 765 kV. Alternate Terms/Abbreviations: circuit breakers: disconnect switches. Gas-filled circuit breakers: SF6 circuit breakers. See: 33 72 00 for utility substations. 33 73 00 for utility transformers. 33 75 00 for high-voltage switchgear and protection devices. 33 77 00 for medium-voltage utility switchgear and protection devices. 33 79 23 for utility substation grounding. 34 21 19 for traction power switchgear.
33 74 13	Air EHV Circuit Breaker	
33 74 16	Oil EHV Circuit Breaker	
33 74 19	Gas EHV Circuit Breaker	
33 74 30	EHV Disconnect Switch	
33 74 36	EHV Utility Fuses	
33 74 39	EHV Surge Arresters	
33 74 40	EHV Rod Gap	
33 74 43	EVH Shunt Reactors	
33 74 43.13	EHV Shunt Reactor Banks	
33 74 43.16	EHV Current Limiting Reactors	
33 75 00	High-Voltage Switchgear and Protection Devices	Includes: switchgear and circuit protection devices for high-voltage utility applications. High-voltage: 115 kV to 230 kV. Alternate Terms/Abbreviations: circuit breakers: disconnect switches. Gas-filled circuit breakers: SF6 circuit breakers. See: 33 72 00 for utility substations. 33 73 00 for utility transformers. 33 74 00 for extra-high-voltage switchgear and protection devices. 33 77 00 for medium-voltage utility switchgear and protection devices. 33 79 23 for utility substation grounding. 34 21 19 for traction power switchgear.
33 75 13	Air High-Voltage Circuit Breaker	
33 75 16	Oil High-Voltage Circuit Breaker	
33 75 19	Gas High-Voltage Circuit Breaker	
33 75 23	Vacuum High-Voltage Circuit Breaker	
33 75 30	High-Voltage Disconnect Switch	

DIVISION **33**

NUMBER	TITLE	EXPLANATION
33 75 36	High-Voltage Utility Fuses	
33 75 39	High-Voltage Surge Arresters	
33 75 40	High-Voltage Rod Gap	

33 77 00 Medium-Voltage Utility Switchgear and Protection Devices

Includes: switchgear and circuit protection devices for medium-voltage utility applications. Medium-voltage: 2400 V to 69 kV.

Alternate Terms/Abbreviations: circuit breakers: disconnect switches. Gas circuit breakers: SF6 circuit breakers.

See: 26 13 00 for medium-voltage facility switchgear.
 26 16 00 for medium-voltage metering.
 26 18 00 for medium-voltage circuit protection devices.
 33 72 00 for utility substations.
 33 73 00 for utility transformers.
 33 75 00 for high-voltage utility switchgear and circuit protection devices.
 33 79 23 for utility substation grounding.
 34 21 19 for traction power switchgear.

33 77 13	Air Medium-Voltage Circuit Breaker
33 77 16	Oil Medium-Voltage Circuit Breaker
33 77 19	Gas Medium-Voltage Circuit Breaker
33 77 23	Vacuum Medium-Voltage Circuit Breaker
33 77 26	Medium-Voltage Utility Fusible Interrupter Switchgear
33 77 30	Medium-Voltage Disconnect Switch
33 77 33	Medium-Voltage Utility Cutouts
33 77 36	Medium-Voltage Utility Fuses
33 77 39	Medium-Voltage Utility Surge Arresters
33 77 40	Medium-Voltage Rod Gap
33 77 53	Medium-Voltage Utility Reclosers

33 78 00 Substation Converter Stations

33 79 00 Site Grounding

Includes: a conducting connection between an element and a ground electrode placed in the earth to provide a safe path for the dissipation of fault currents, lightning strikes, static discharges, EMI and RFI signals and interference.

See: 26 05 26 for grounding and bounding of electrical systems.
 26 41 00 for facility lightning protection.
 33 71 13 for electrical utility towers.
 33 81 13 for communications transmission towers.
 33 81 16 for antenna towers.
 34 73 16 for airfield grounding.

NUMBER	TITLE	EXPLANATION
33 79 13	Site Improvement Grounding	
33 79 13.13	Fence Grounding	
33 79 16	Tower Grounding	
33 79 16.13	Communications Tower Grounding	
33 79 16.16	Antenna Tower Grounding	
33 79 19	Utilities Grounding	
33 79 19.13	Electrical Utilities Grounding	
33 79 19.16	Communications Utilities Grounding	
33 79 23	Utility Substation Grounding	
33 79 83	Site Grounding Conductors	
33 79 93	Site Lightning Protection	
33 79 93.13	Lightning Arrestors	
33 79 93.16	Lightning Strike Counters	
33 79 93.19	Lightning Strike Warning Devices	

33 80 00 Communications Utilities

Includes: communications transmission and distribution for one site or multiple sites.

Alternate Terms/Abbreviations: OSP: outside plant. ISP: inside plant.

See: 26 00 00 for electrical associated with a building or structure.

27 10 00 for facility structured cabling.

27 21 00 for data communications network equipment.

33 09 80 for instrumentation and control for communication utilities.

33 70 00 for electrical utilities.

40 66 00 for process control-related network and communications equipment.

33 81 00 Communications Structures

Includes: structures for elevated and below-ground distribution of communications cables and signals.

See: 03 00 00 for concrete.

27 05 46 for facility utility poles for communications systems.

33 71 00 for electrical structures.

33 79 16 for tower grounding.

34 23 13 for traction power poles.

33 81 13	Monopole Communications Towers
33 81 16	Lattice Communications Towers
33 81 19	Guyed Communications Towers
33 81 21	Concealed Communications Towers
33 81 23	Broadcast Communications Towers
33 81 27	Aerial Cable Installation Hardware
33 81 29	Communications Vaults, Pedestals, and Enclosures
33 81 63	Cell Site Shelter Equipment

DIVISION **33**

NUMBER	TITLE	EXPLANATION
33 81 63.16	DC Rectifiers	
33 81 63.19	Backup Batteries	
33 81 63.23	Baseband Microwave Processing	
33 81 63.26	Base Station Transceivers	
33 81 63.43	Remote Monitoring	

33 82 00 Communications Transmission and Distribution

Includes: copper and optical fiber outside facility.

Alternate Terms/Abbreviations: ISDN: integrated services digital network.

See: 27 05 26 for grounding and bonding for facility communications systems.
 27 10 00 for facility structured cabling.
 27 13 00 for communications backbone cabling.
 27 13 13 for communications copper backbone cabling.
 27 13 23 for communications optical fiber backbone cabling.
 27 13 33 for communications coaxial backbone cabling.
 33 79 19 for utilities grounding.

33 82 13	Copper Communications Transmission and Distribution Cabling
33 82 13.13	Copper Splicing and Terminations
33 82 23	Optical Fiber Communications Transmission and Distribution Cabling
33 82 23.13	Optical Fiber Splicing and Terminations
33 82 33	Coaxial Communications Transmission and Distribution Cabling
33 82 33.13	Coaxial Splicing and Terminations
33 82 43	Grounding and Bonding for Communications Transmission and Distribution
33 82 46	Cable Pressurization Equipment
	Includes: waveguide dehydrators.

33 82 53	Cleaning, Lubrication, and Restoration Chemicals
33 82 63	Shelter Equipment
33 82 63.13	Transceivers
33 82 63.16	Control Panels
33 82 63.23	Power-Line Carriers
33 82 63.43	Backup Batteries

33 83 00 Wireless Communications Transmission and Distribution

Includes: external wireless communications systems.

Alternate Terms/Abbreviations: UHF: ultra-high frequency, 300 MHz to 3.0 GHz. VHF: very-high frequency, 30 MHz to 300 MHz.

See: 33 09 00 for wireless equipment used for meter reading and metered data transmissions.

33 83 13	Laser Transmitters and Receivers
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NUMBER	TITLE	EXPLANATION
33 83 16	Microwave Transmitters and Receivers	Includes: analog radio: order wire, service channel, and digital radio: wayside.
33 83 17	Microwave Antennas	
33 83 17.13	Microwave Waveguides	
33 83 19	Infrared Transmitters and Receivers	
33 83 23	Radio Antennas	